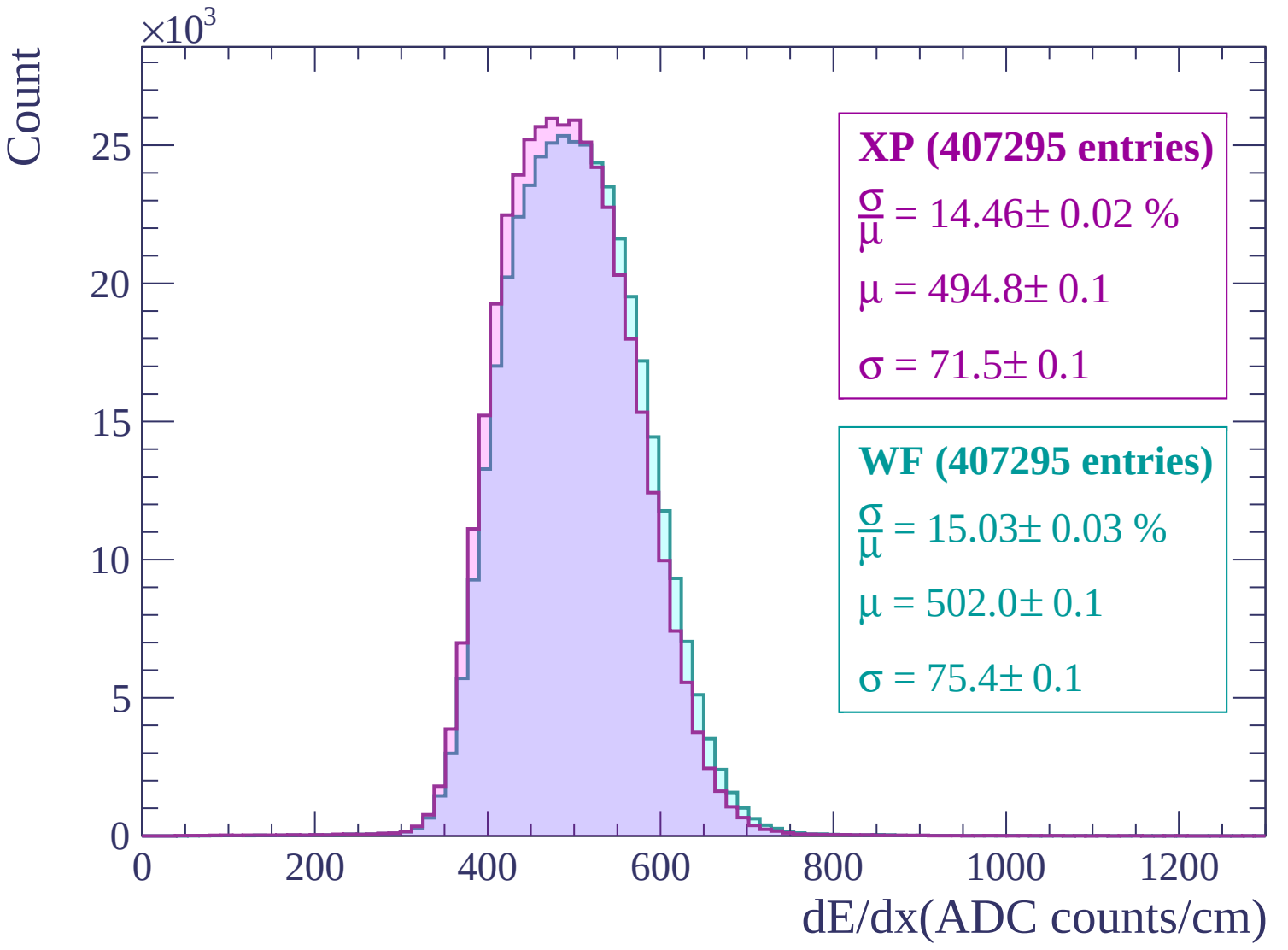
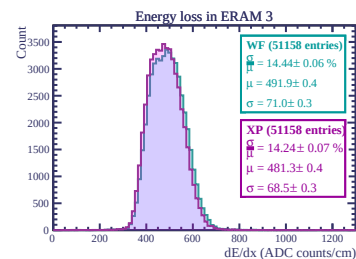
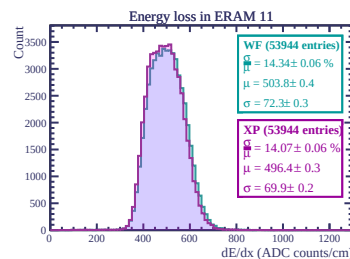
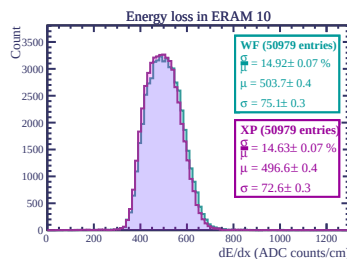
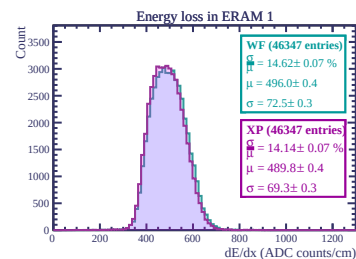
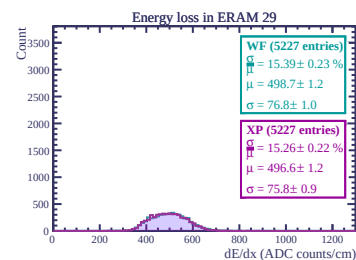
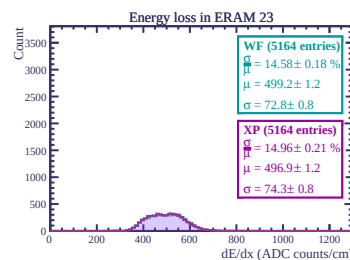
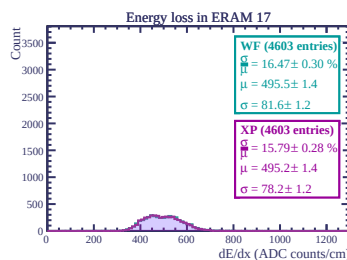
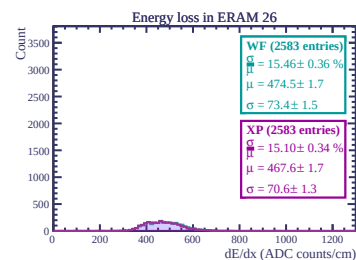
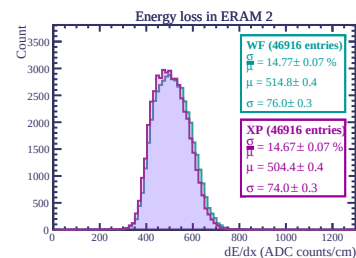
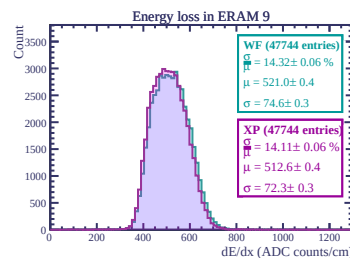
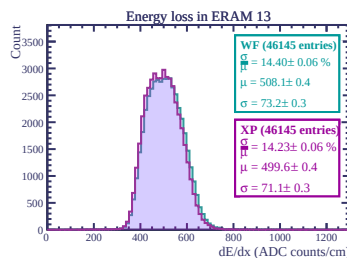
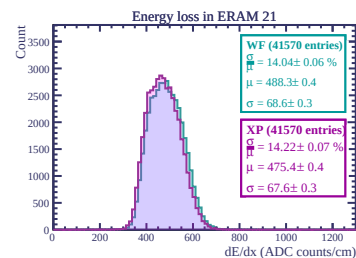
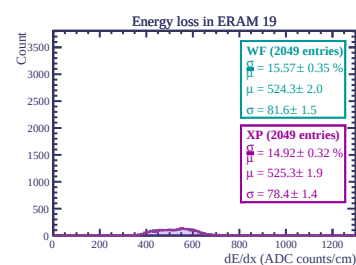
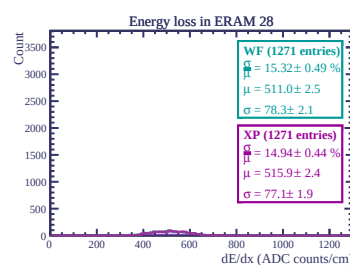
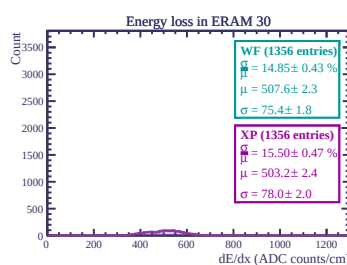
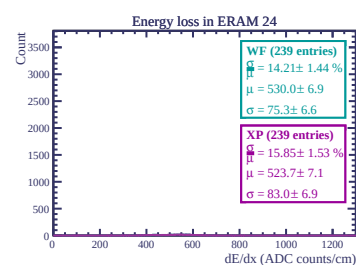
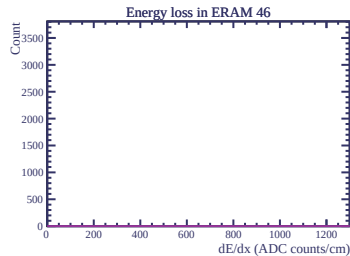
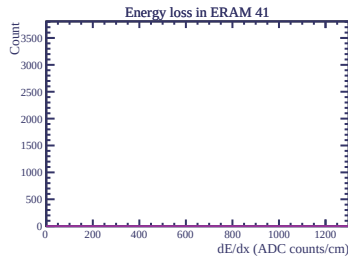
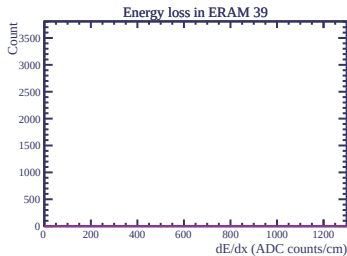
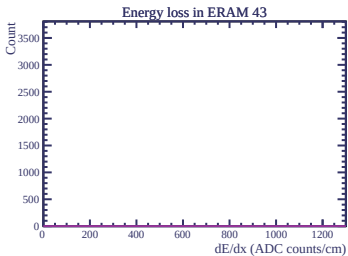
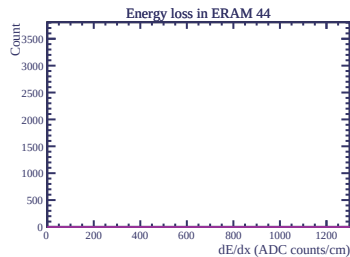
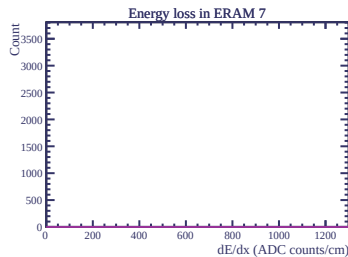
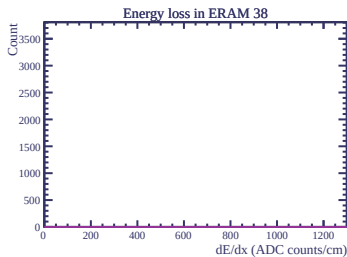
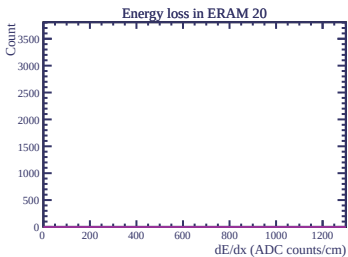
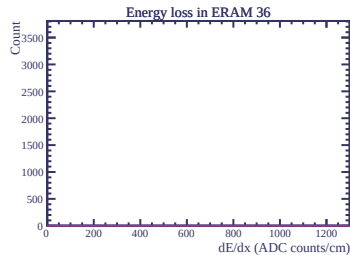
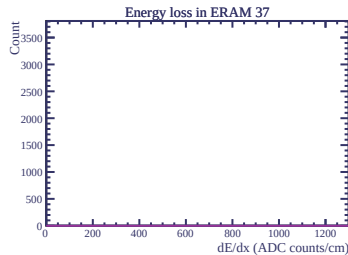
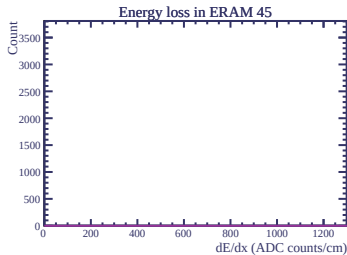
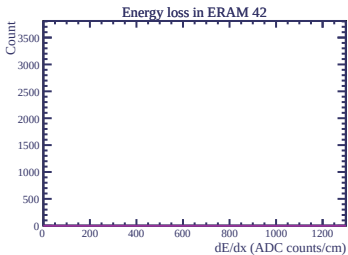
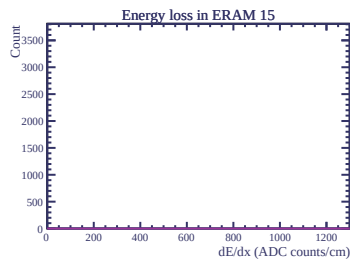
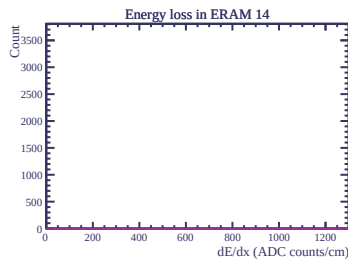
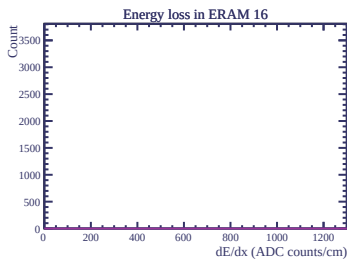
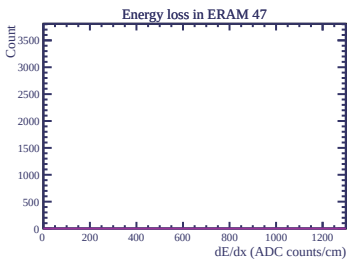
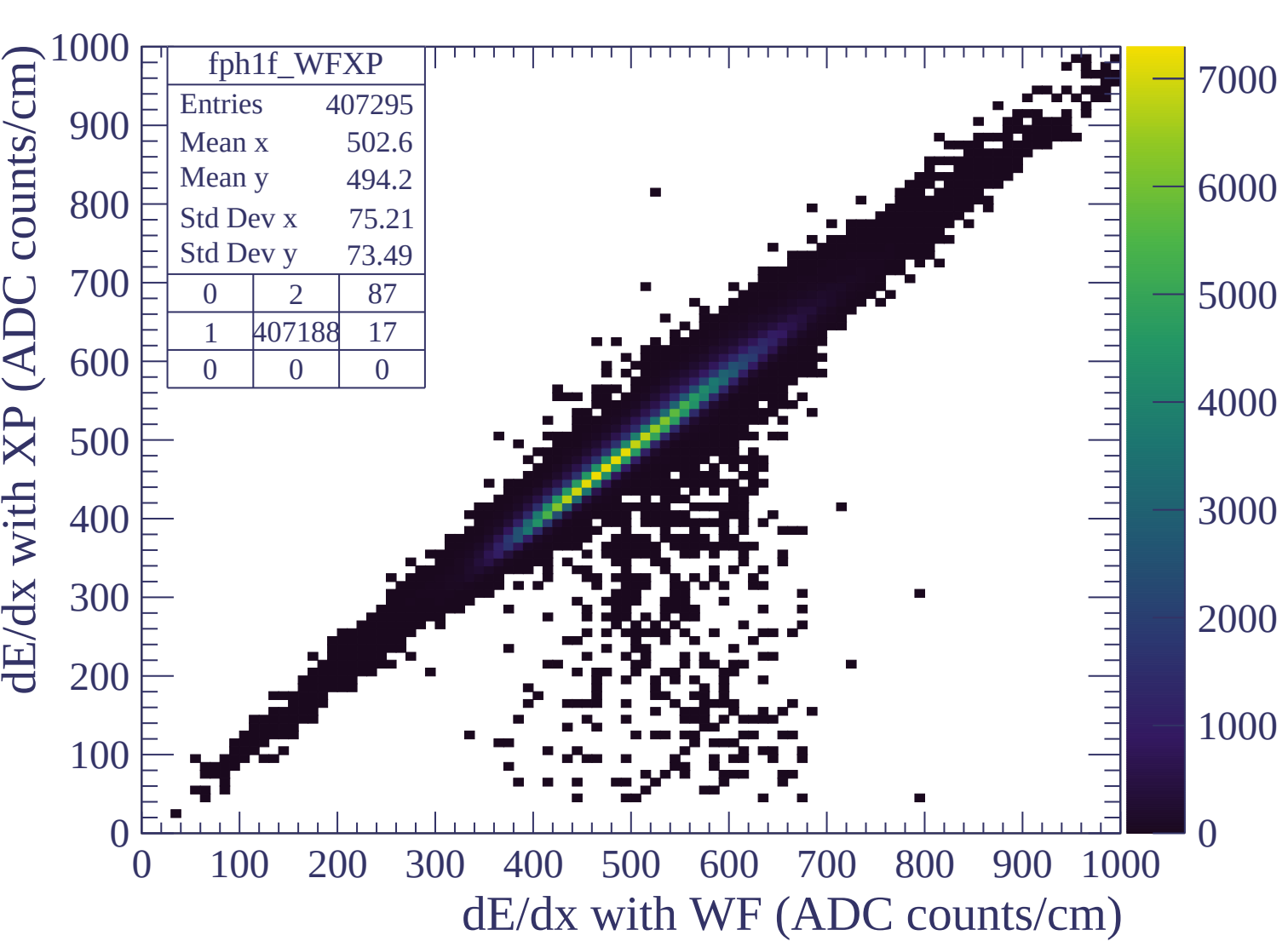
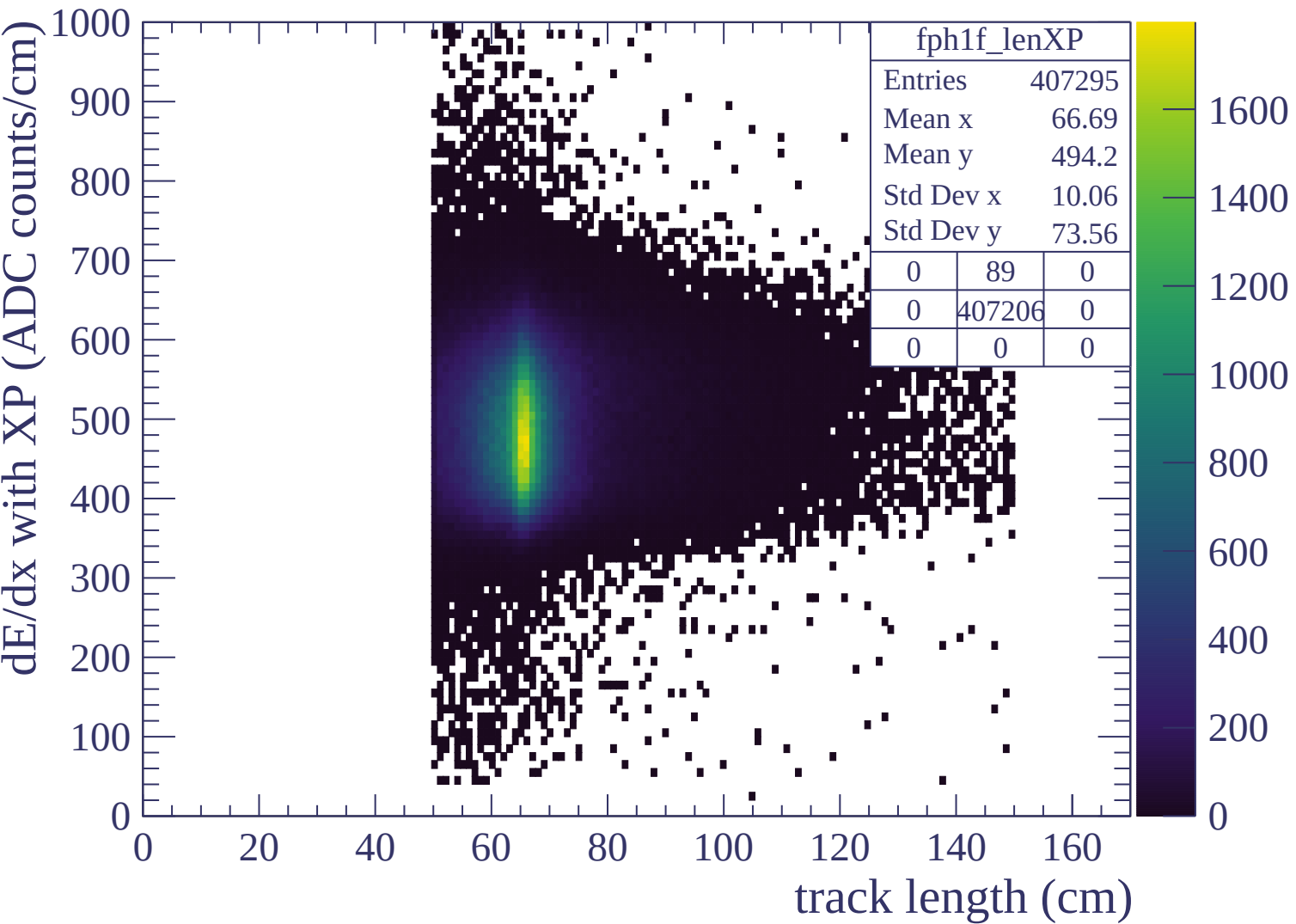


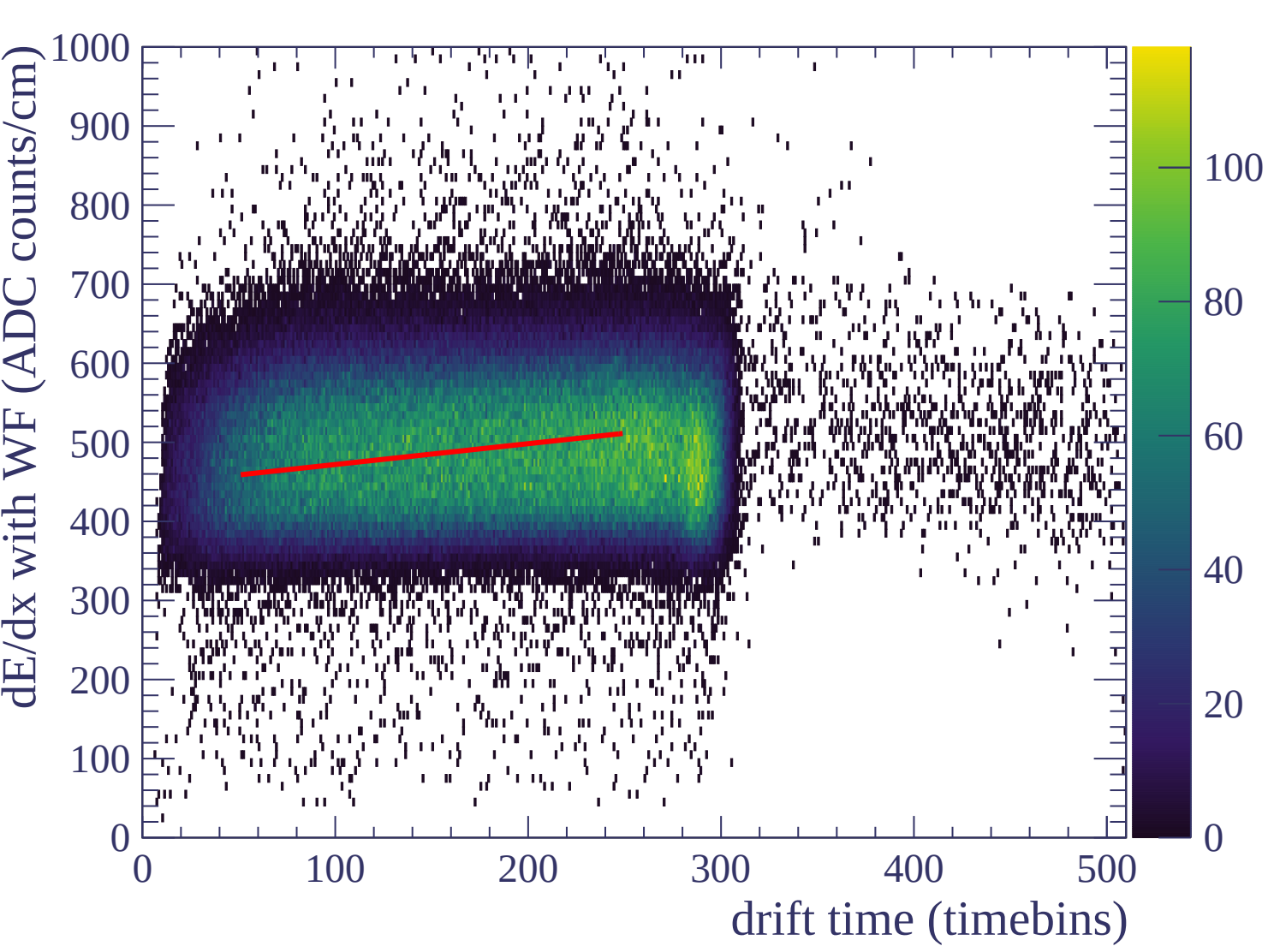


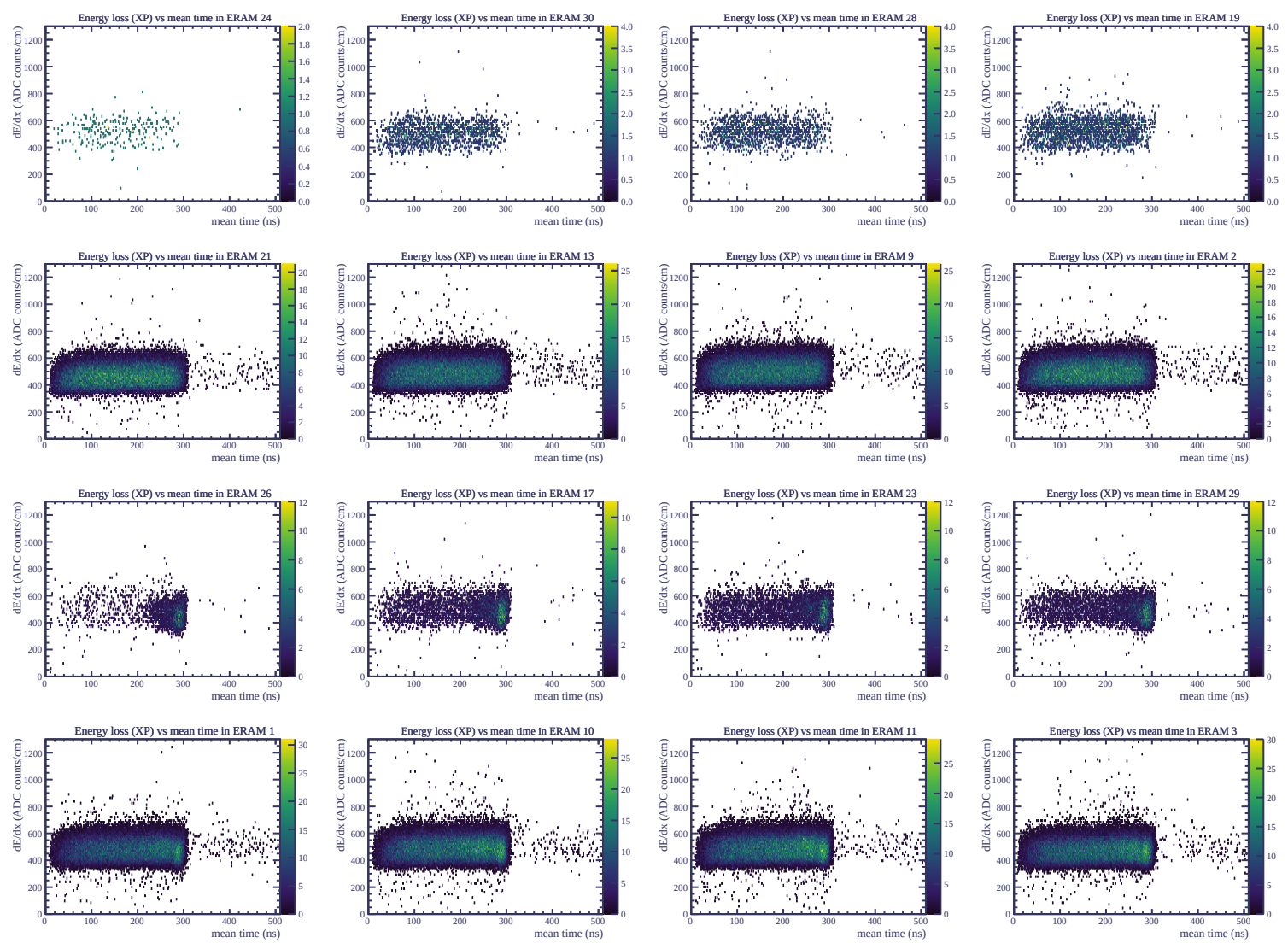


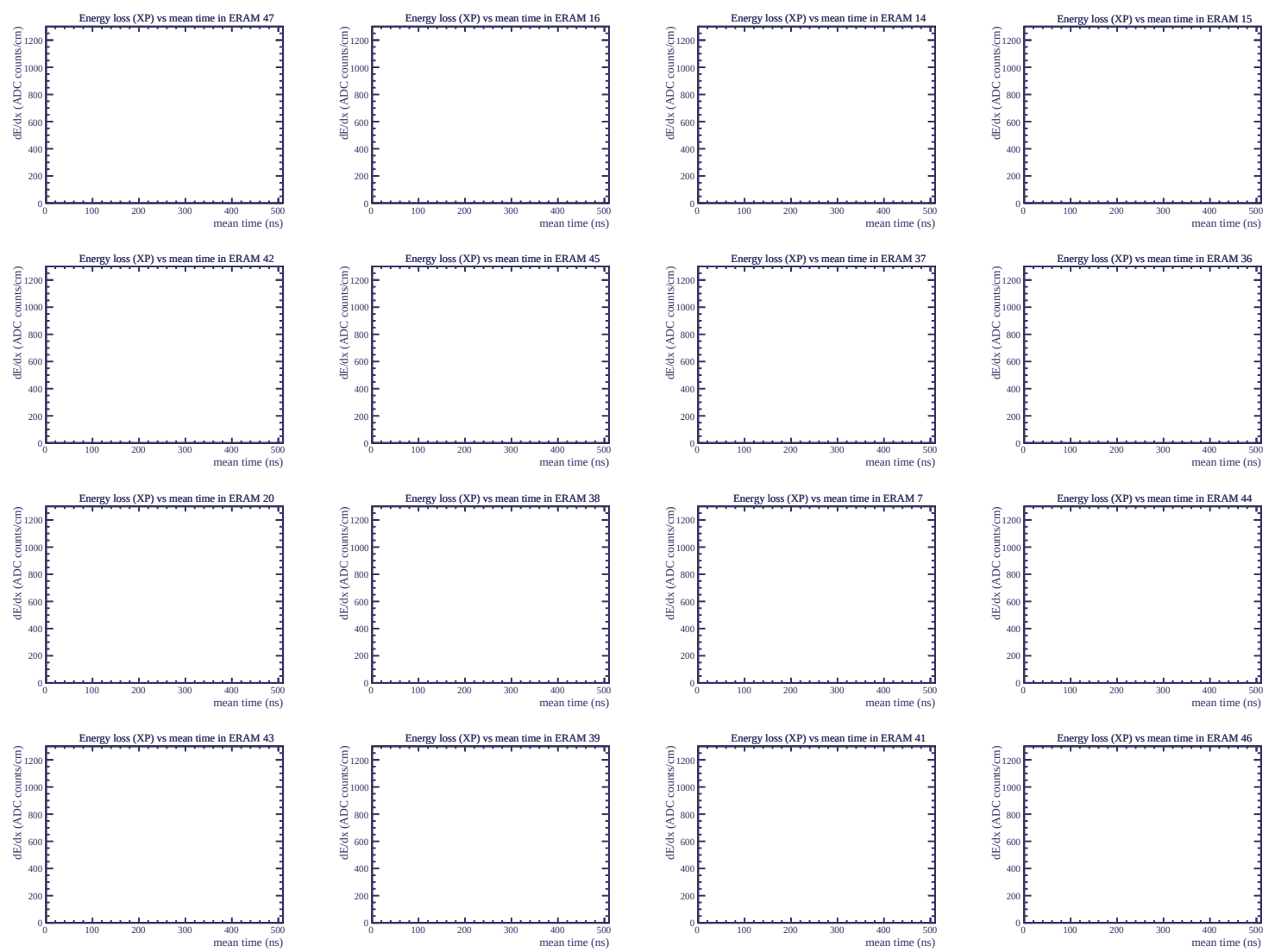




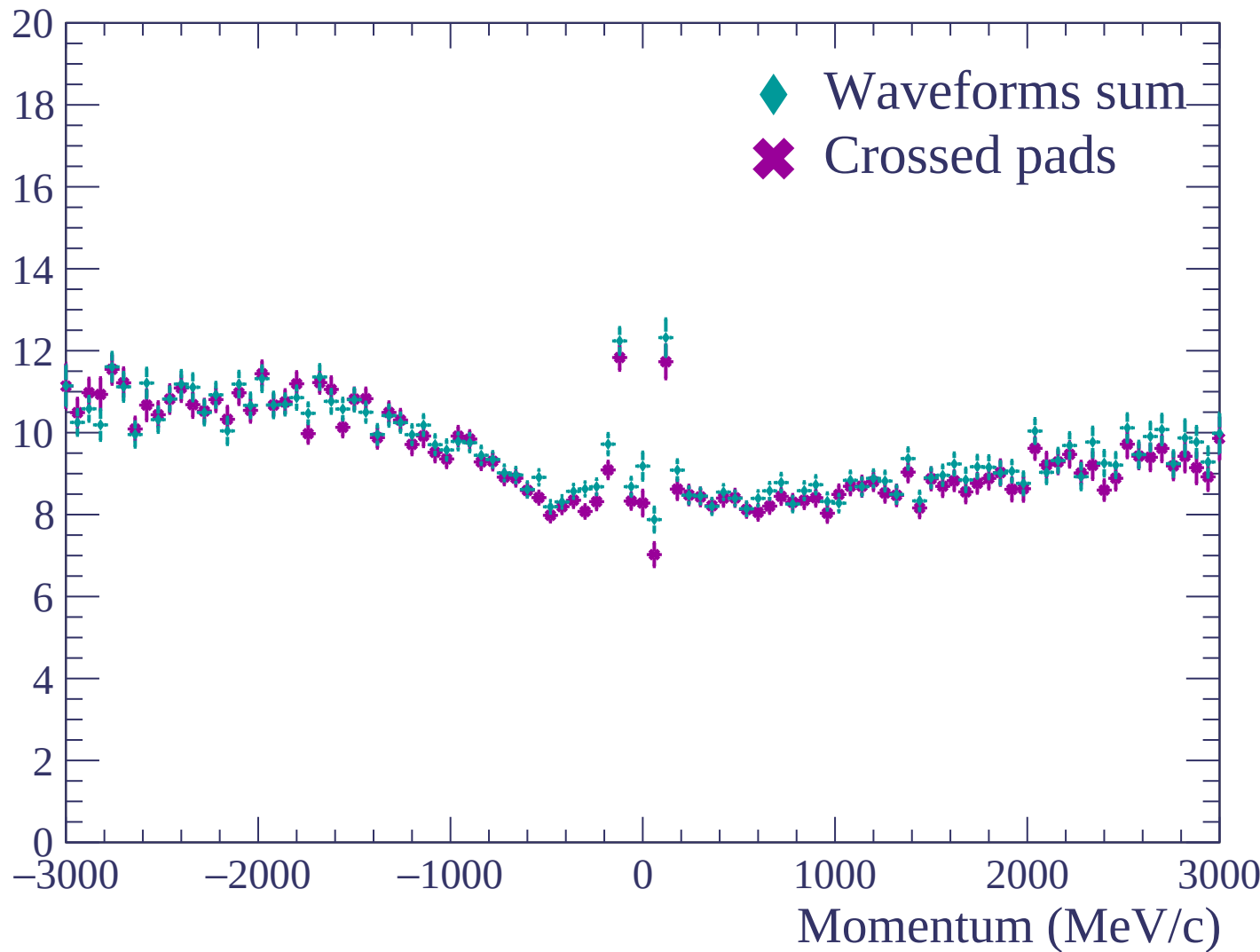


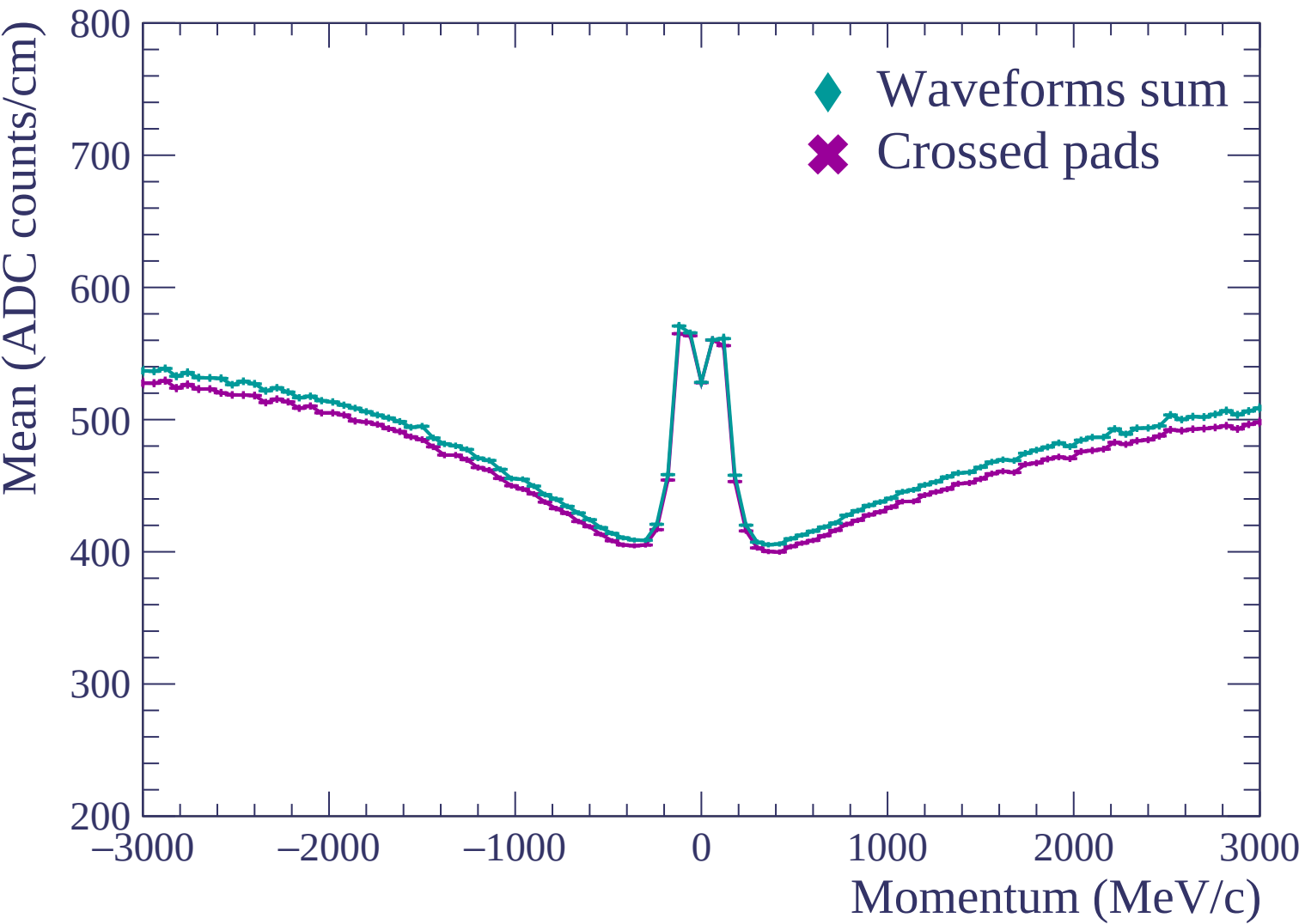


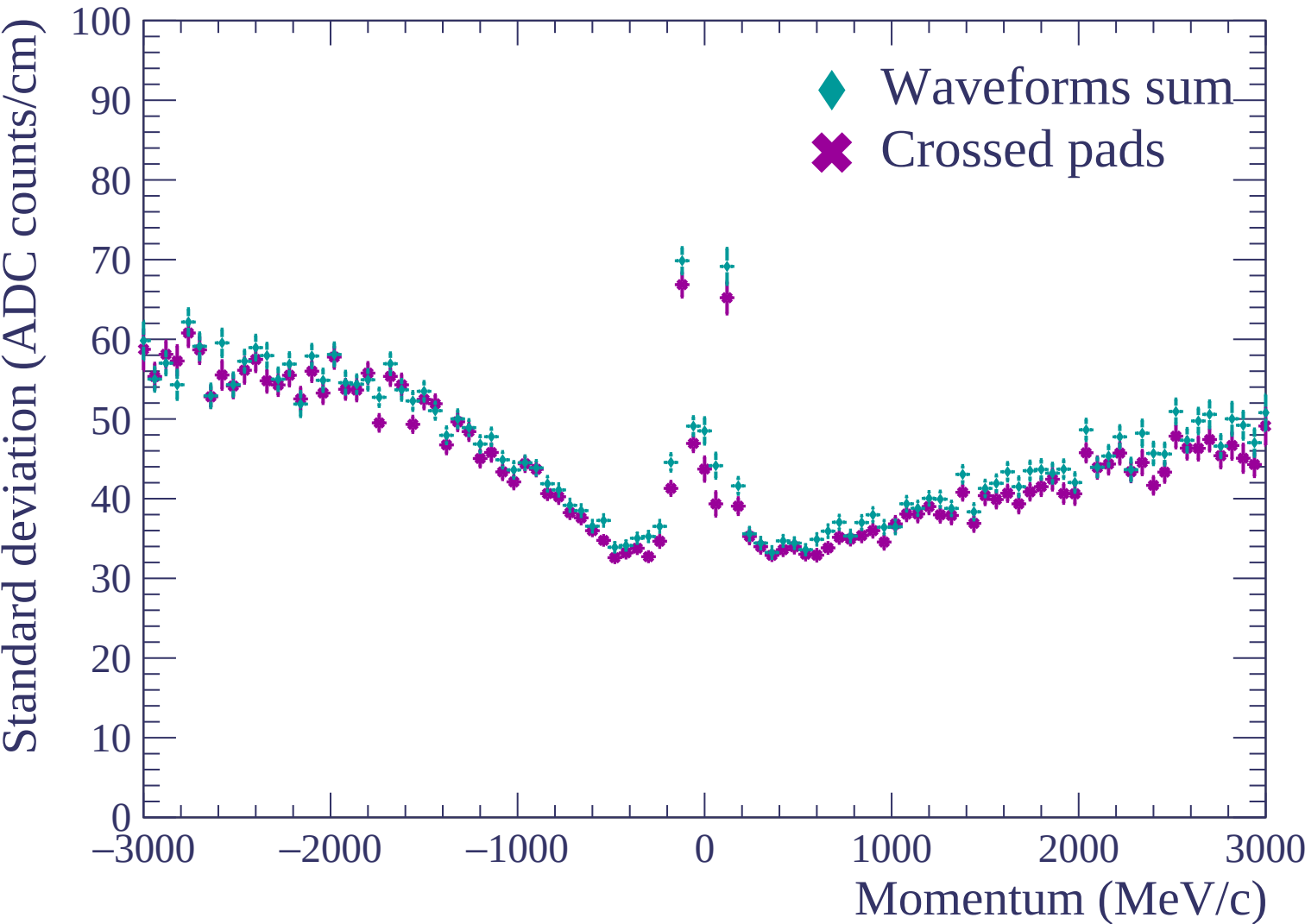


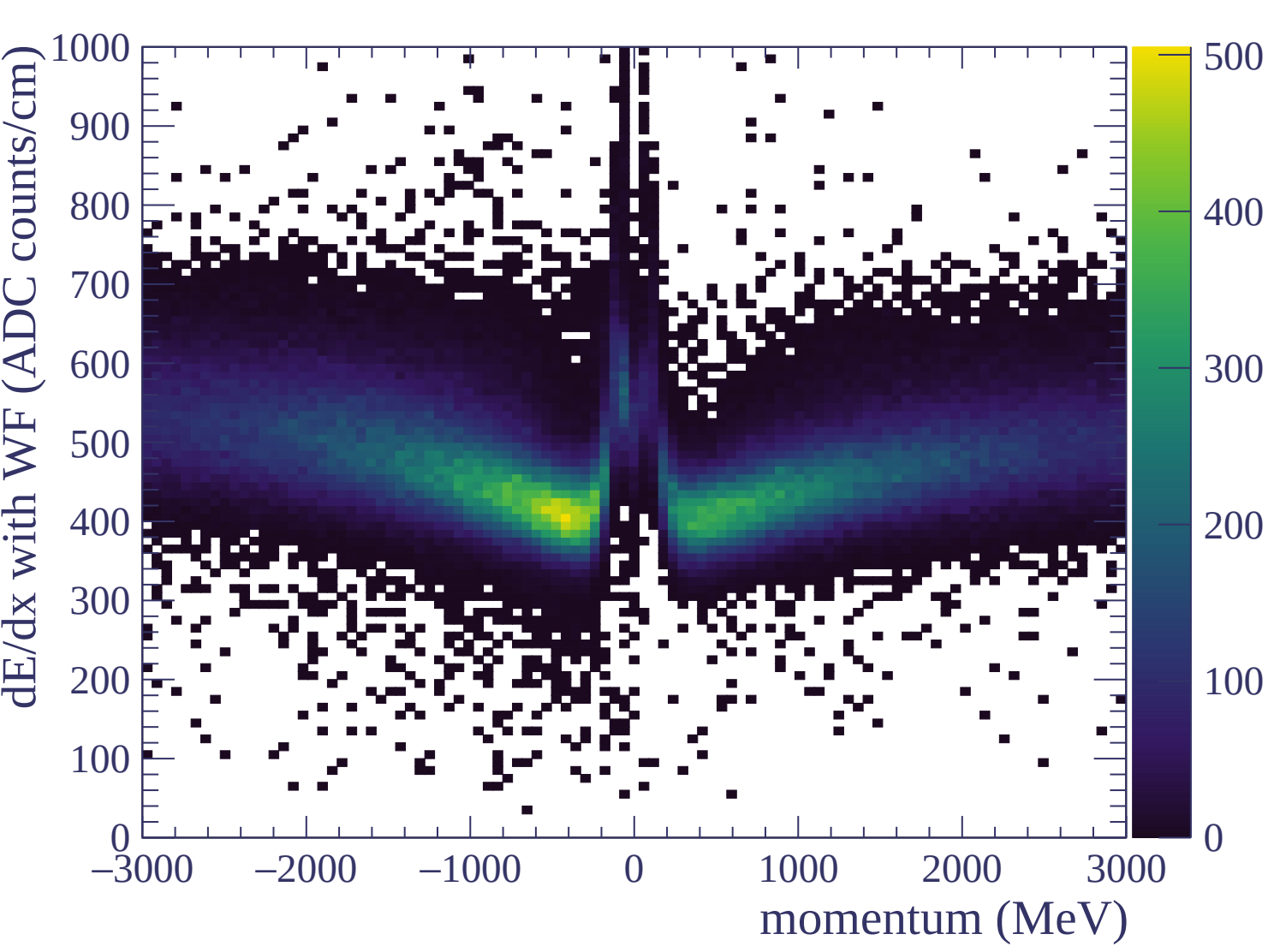


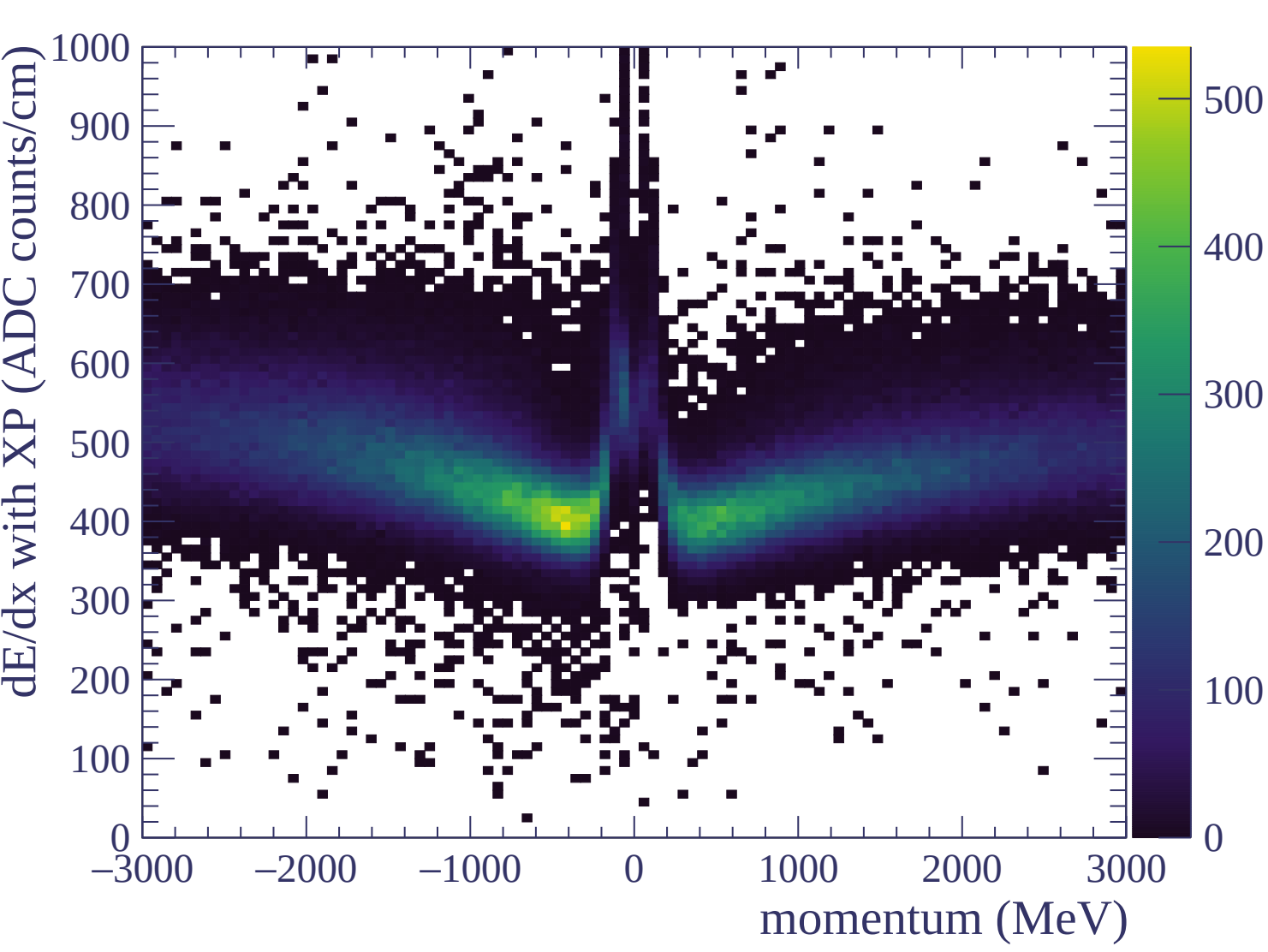
Resolution (%)

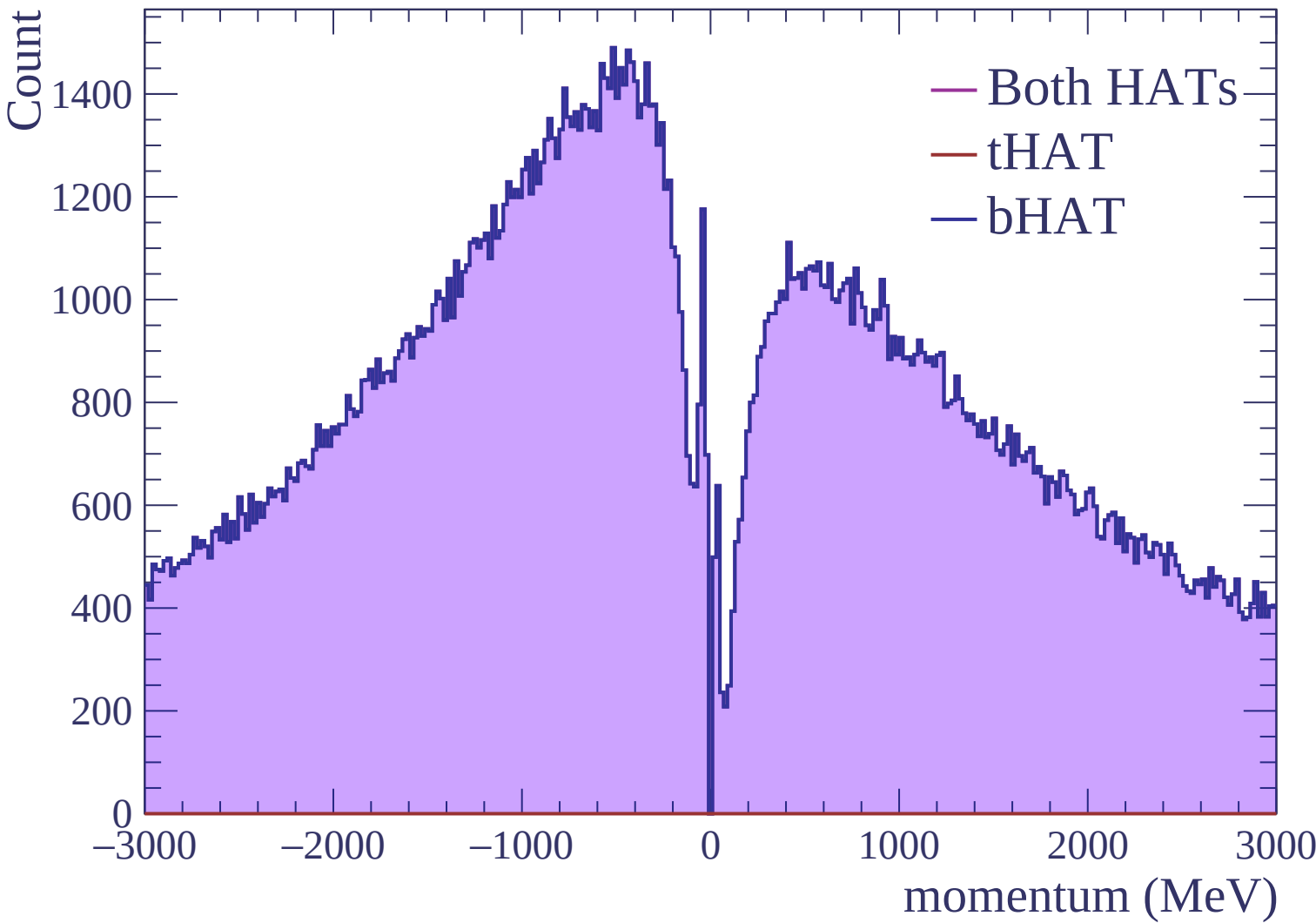






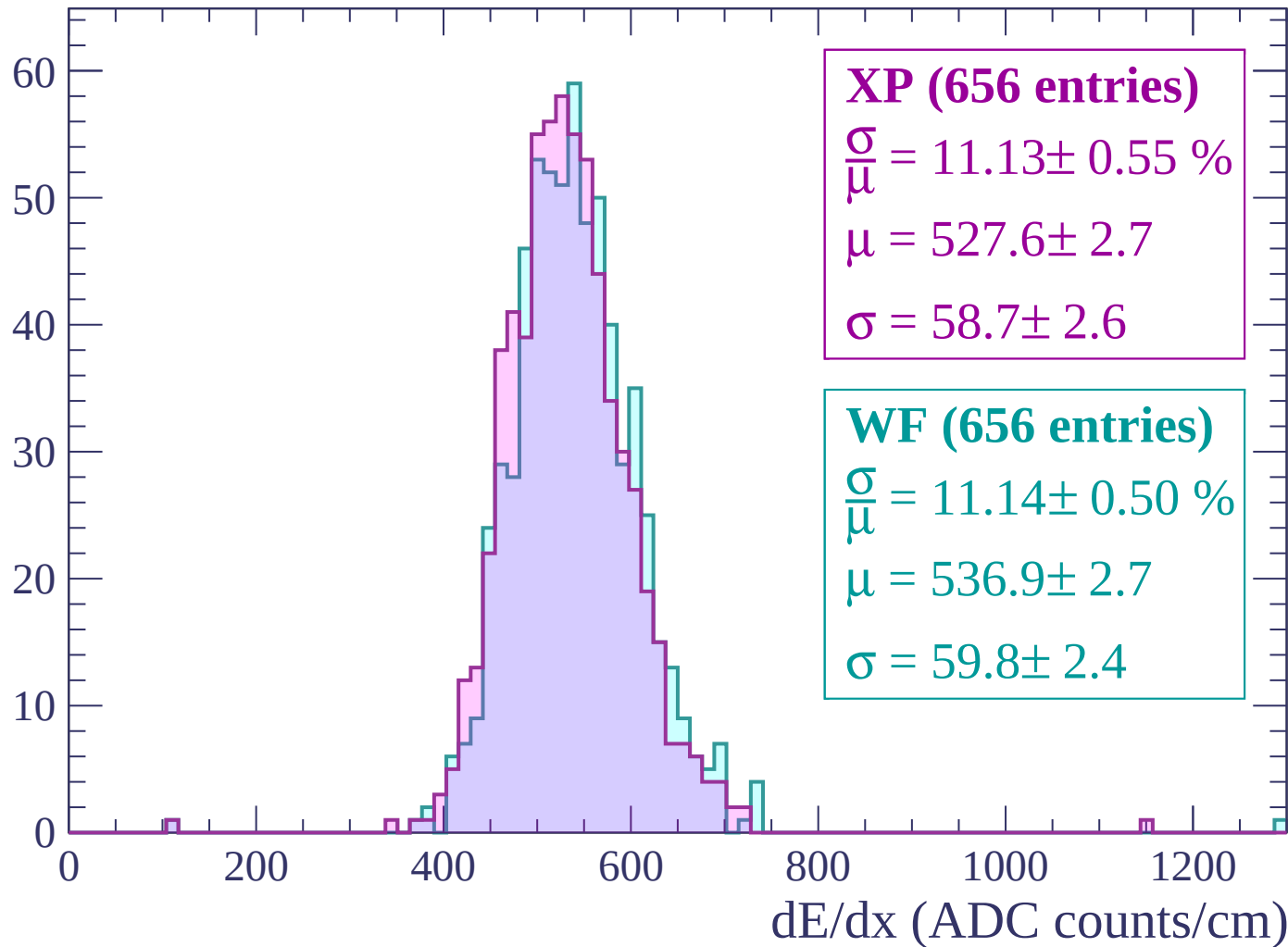






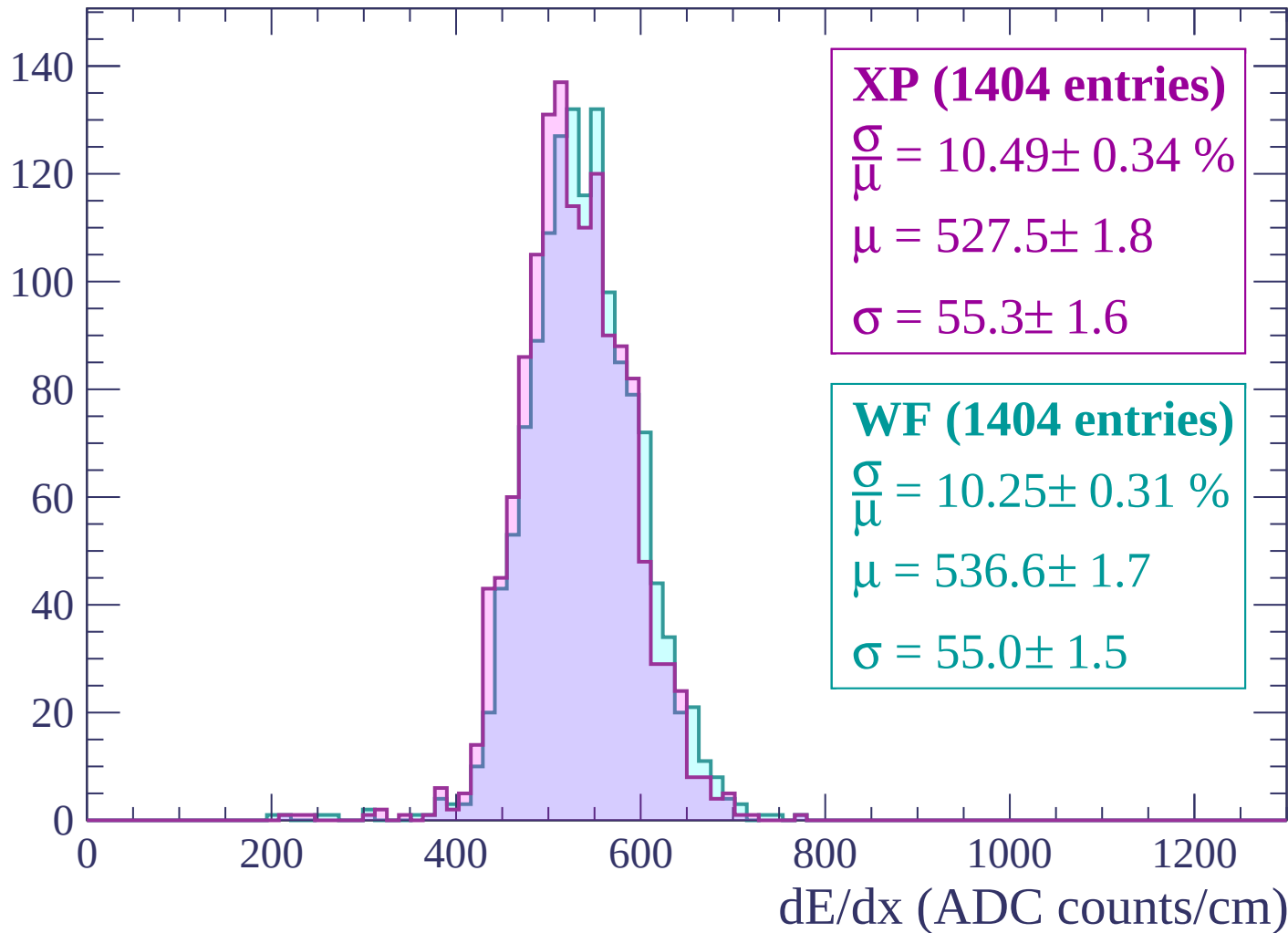
Energy loss | $-3000 < p < -2940$

Count



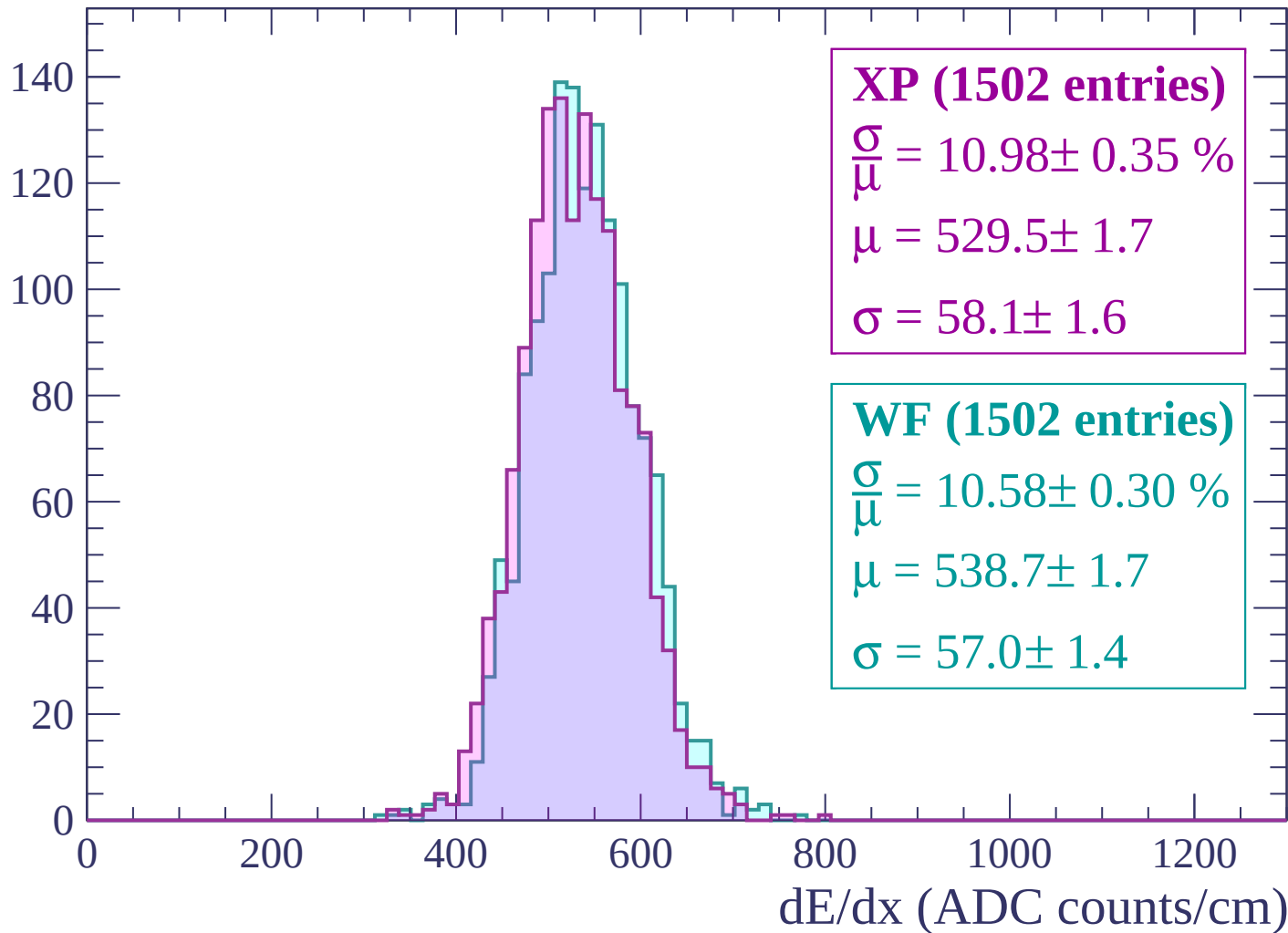
Energy loss | $-2940 < p < -2880$

Count



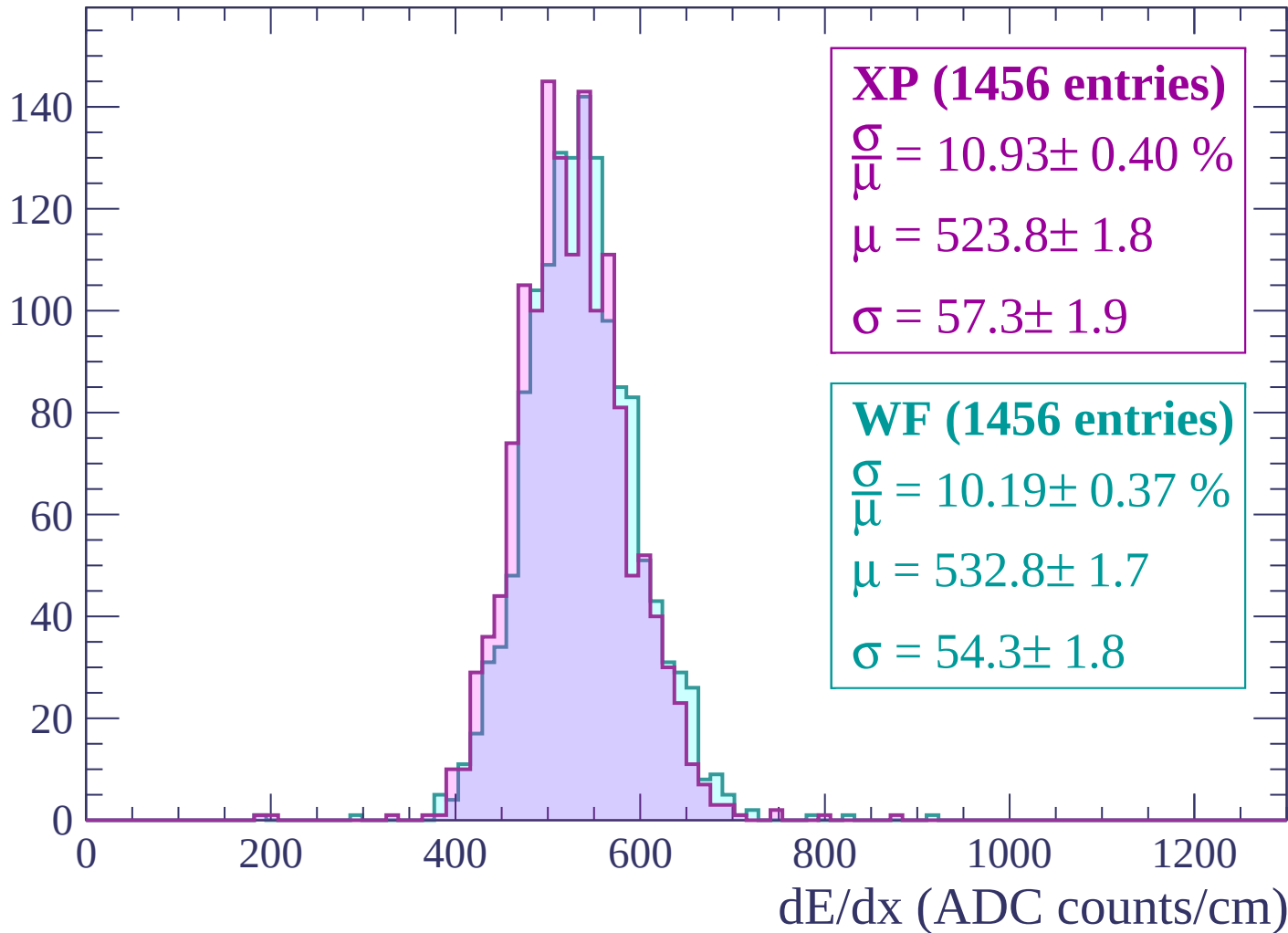
Energy loss | $-2880 < p < -2820$

Count



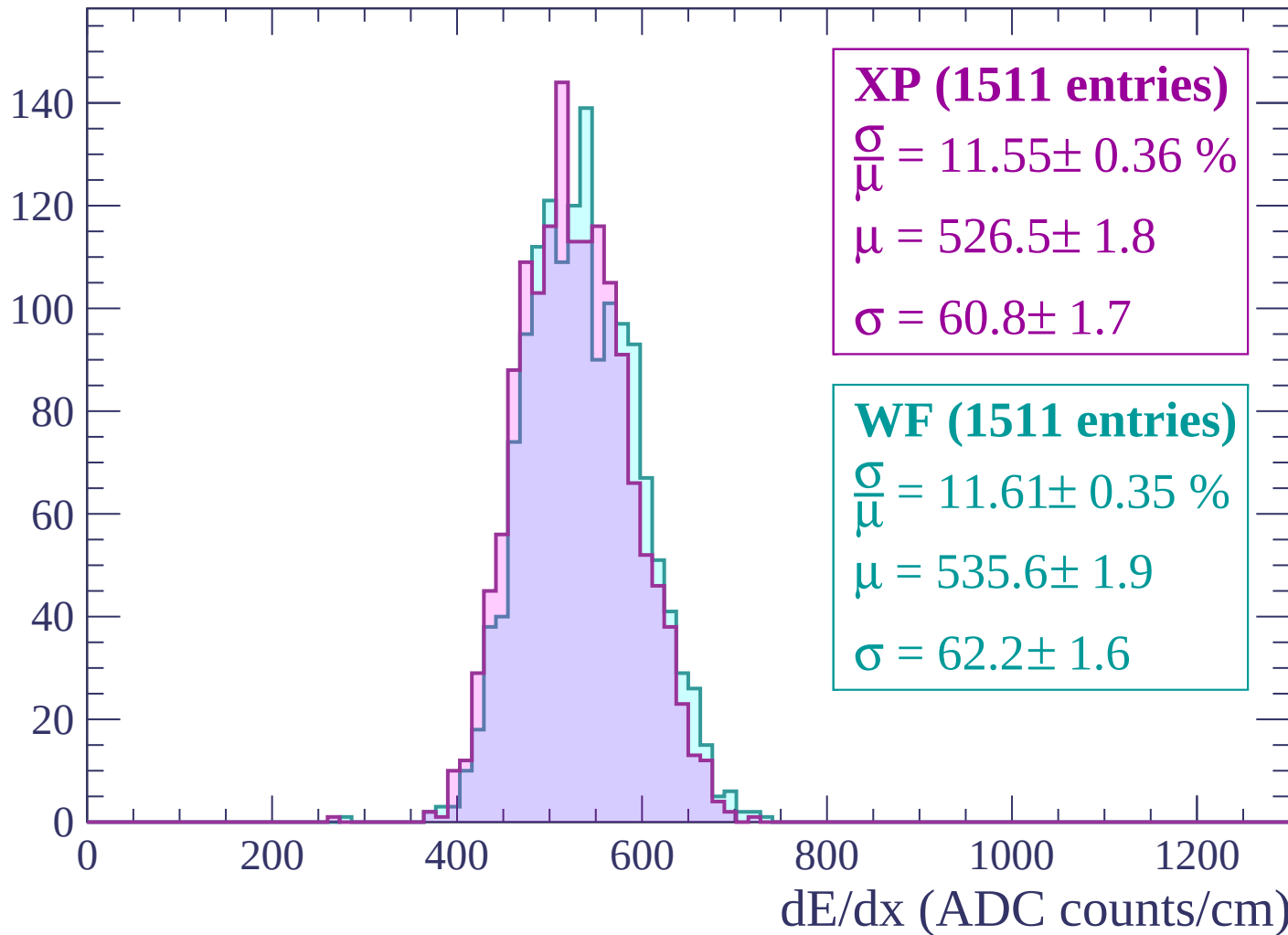
Energy loss | $-2820 < p < -2760$

Count



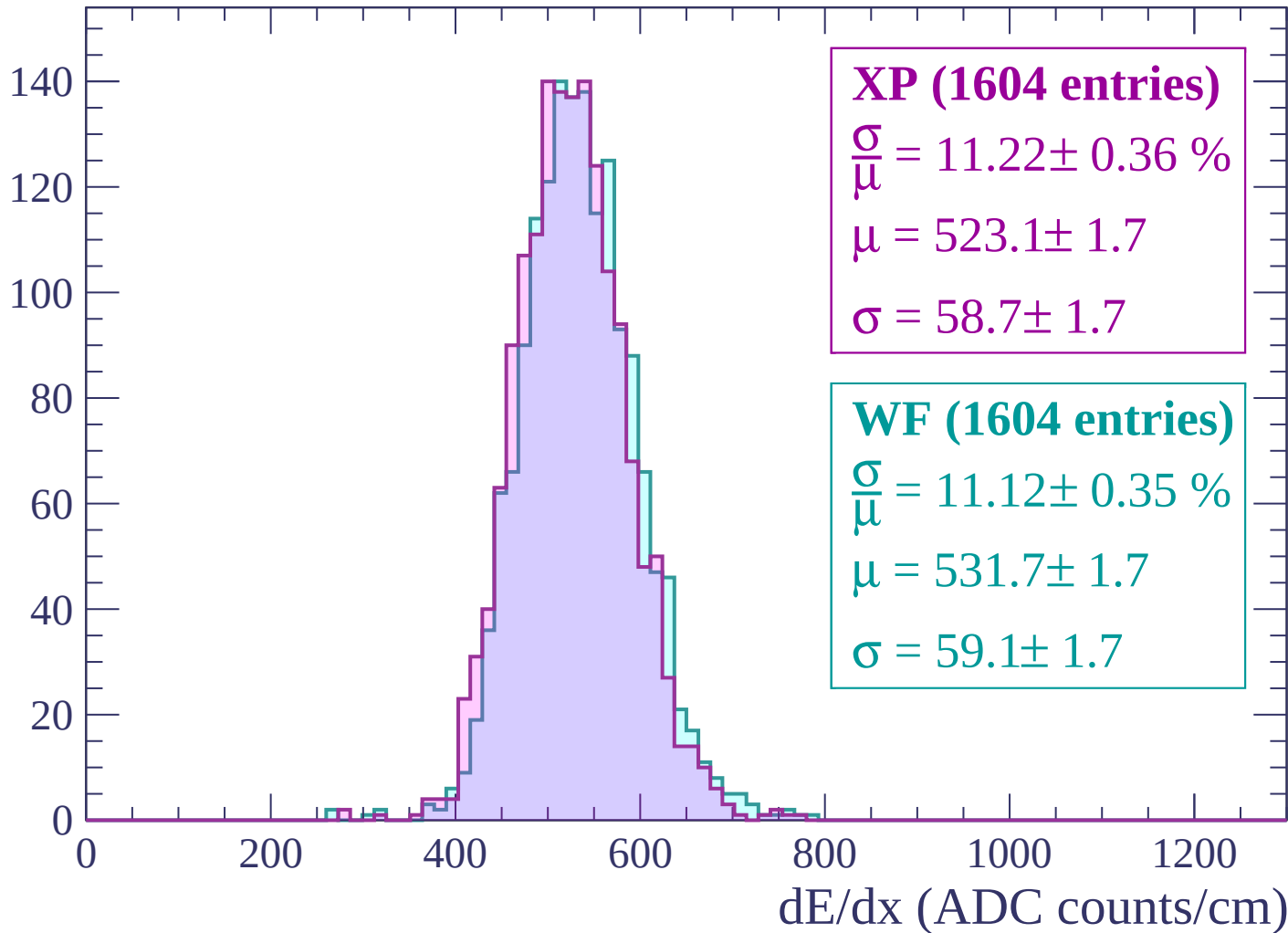
Energy loss | $-2760 < p < -2700$

Count



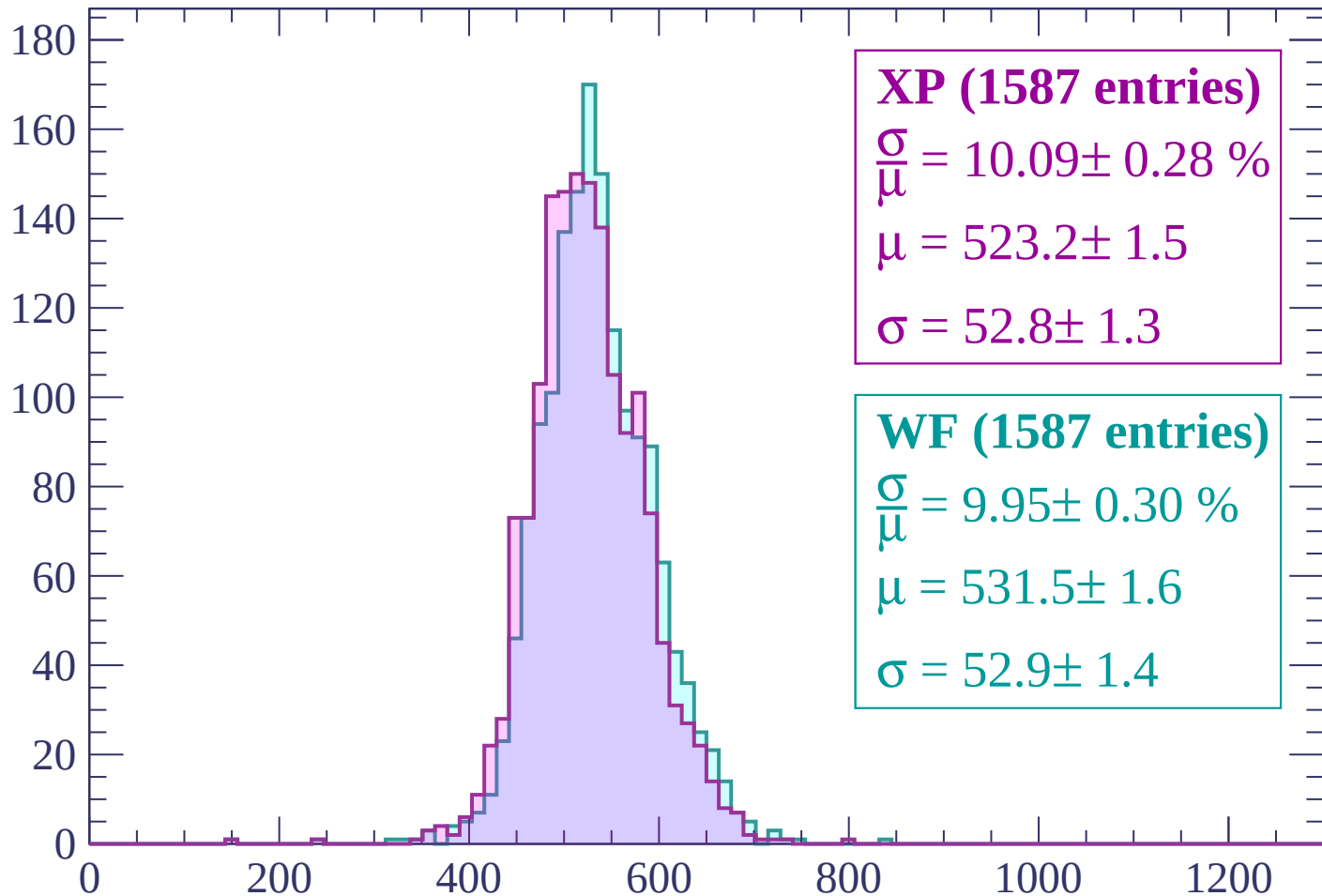
Energy loss | -2700 < p < -2640

Count



Energy loss | $-2640 < p < -2580$

Count



XP (1587 entries)

$$\frac{\sigma}{\mu} = 10.09 \pm 0.28 \%$$

$$\mu = 523.2 \pm 1.5$$

$$\sigma = 52.8 \pm 1.3$$

WF (1587 entries)

$$\frac{\sigma}{\mu} = 9.95 \pm 0.30 \%$$

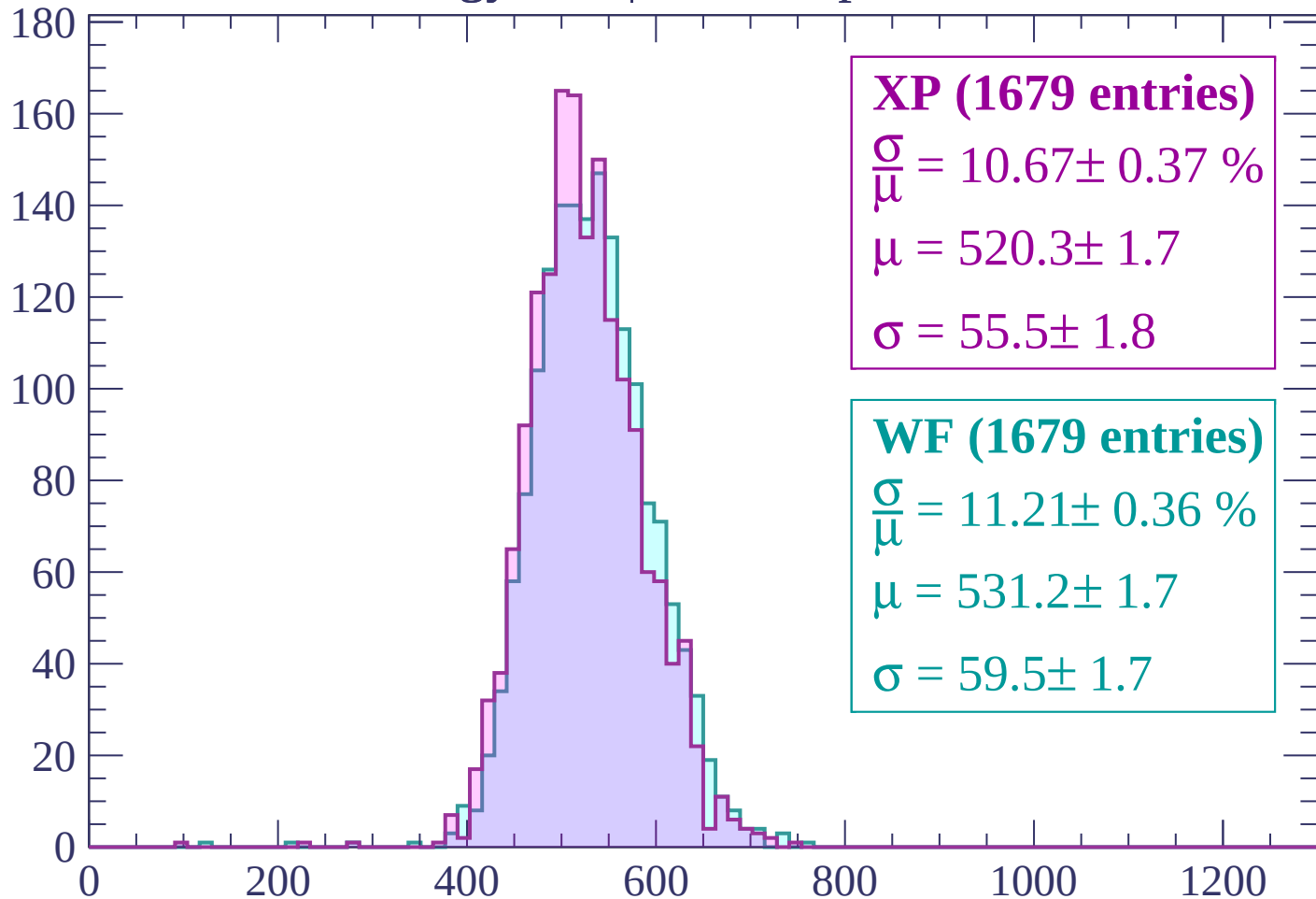
$$\mu = 531.5 \pm 1.6$$

$$\sigma = 52.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -2580 < p < -2520

Count



XP (1679 entries)

$$\frac{\sigma}{\mu} = 10.67 \pm 0.37 \%$$

$$\mu = 520.3 \pm 1.7$$

$$\sigma = 55.5 \pm 1.8$$

WF (1679 entries)

$$\frac{\sigma}{\mu} = 11.21 \pm 0.36 \%$$

$$\mu = 531.2 \pm 1.7$$

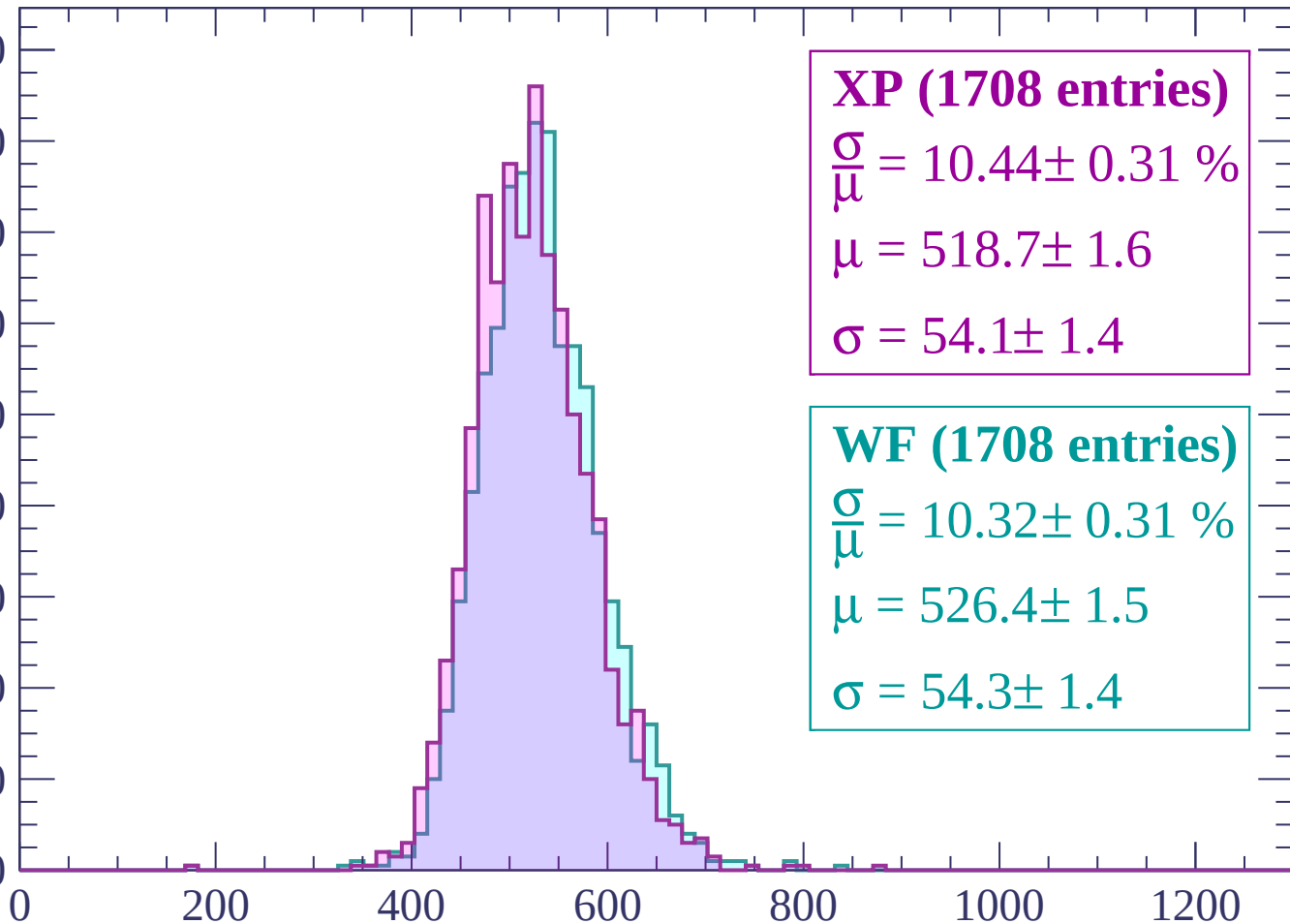
$$\sigma = 59.5 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | -2520 < p < -2460

Count

180
160
140
120
100
80
60
40
20
0



XP (1708 entries)

$$\frac{\sigma}{\mu} = 10.44 \pm 0.31 \%$$

$$\mu = 518.7 \pm 1.6$$

$$\sigma = 54.1 \pm 1.4$$

WF (1708 entries)

$$\frac{\sigma}{\mu} = 10.32 \pm 0.31 \%$$

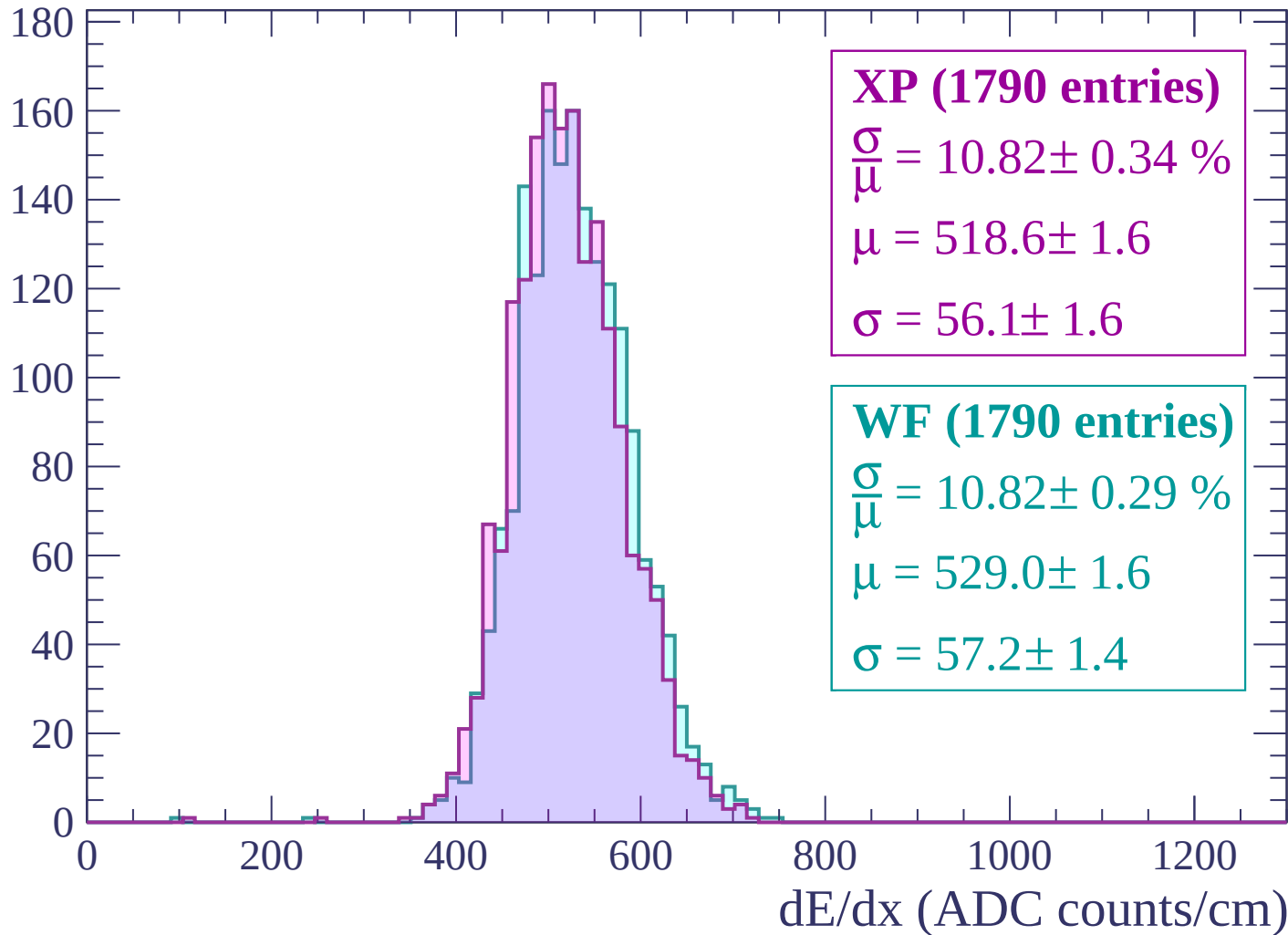
$$\mu = 526.4 \pm 1.5$$

$$\sigma = 54.3 \pm 1.4$$

dE/dx (ADC counts/cm)

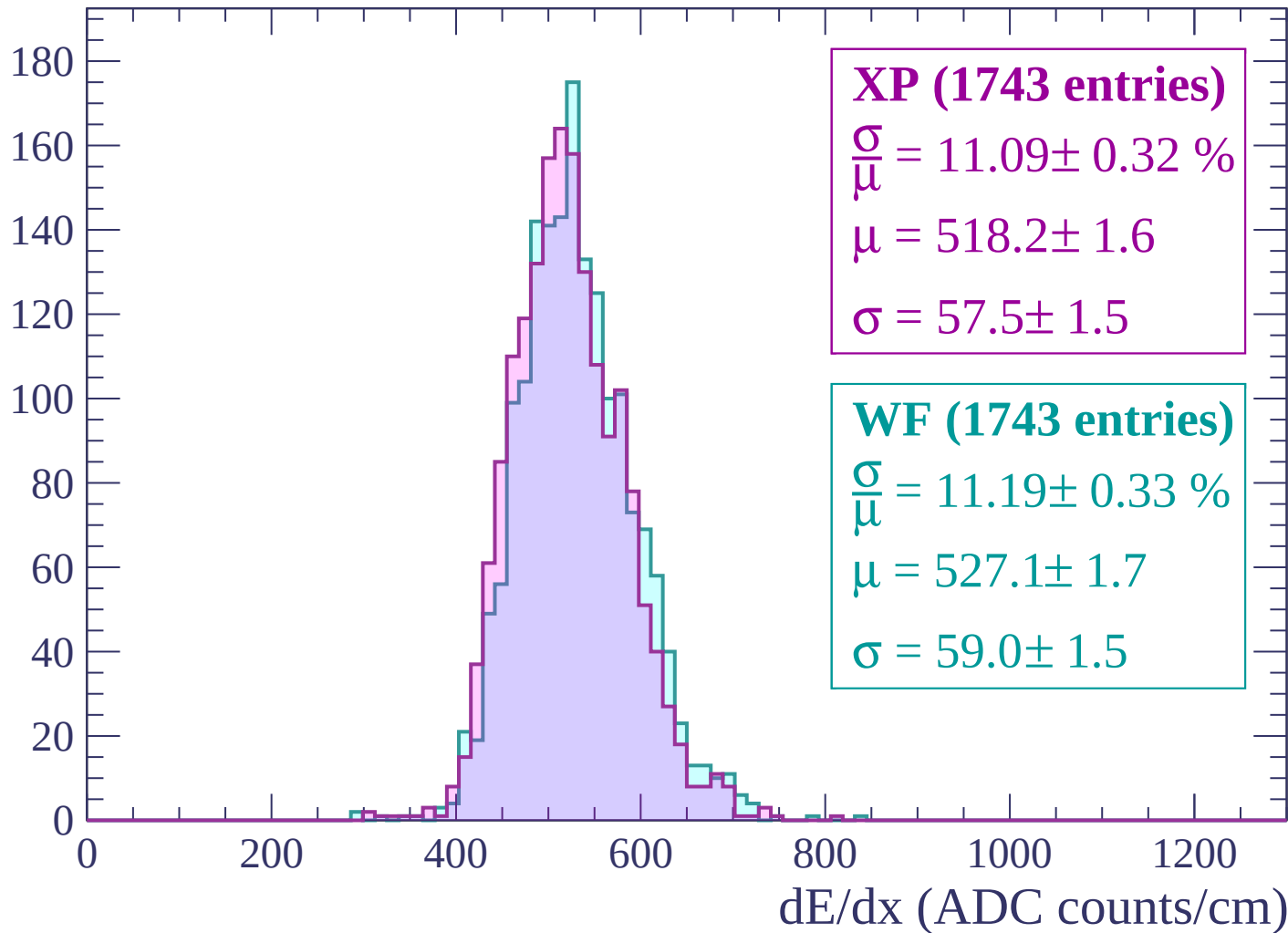
Energy loss | -2460 < p < -2400

Count



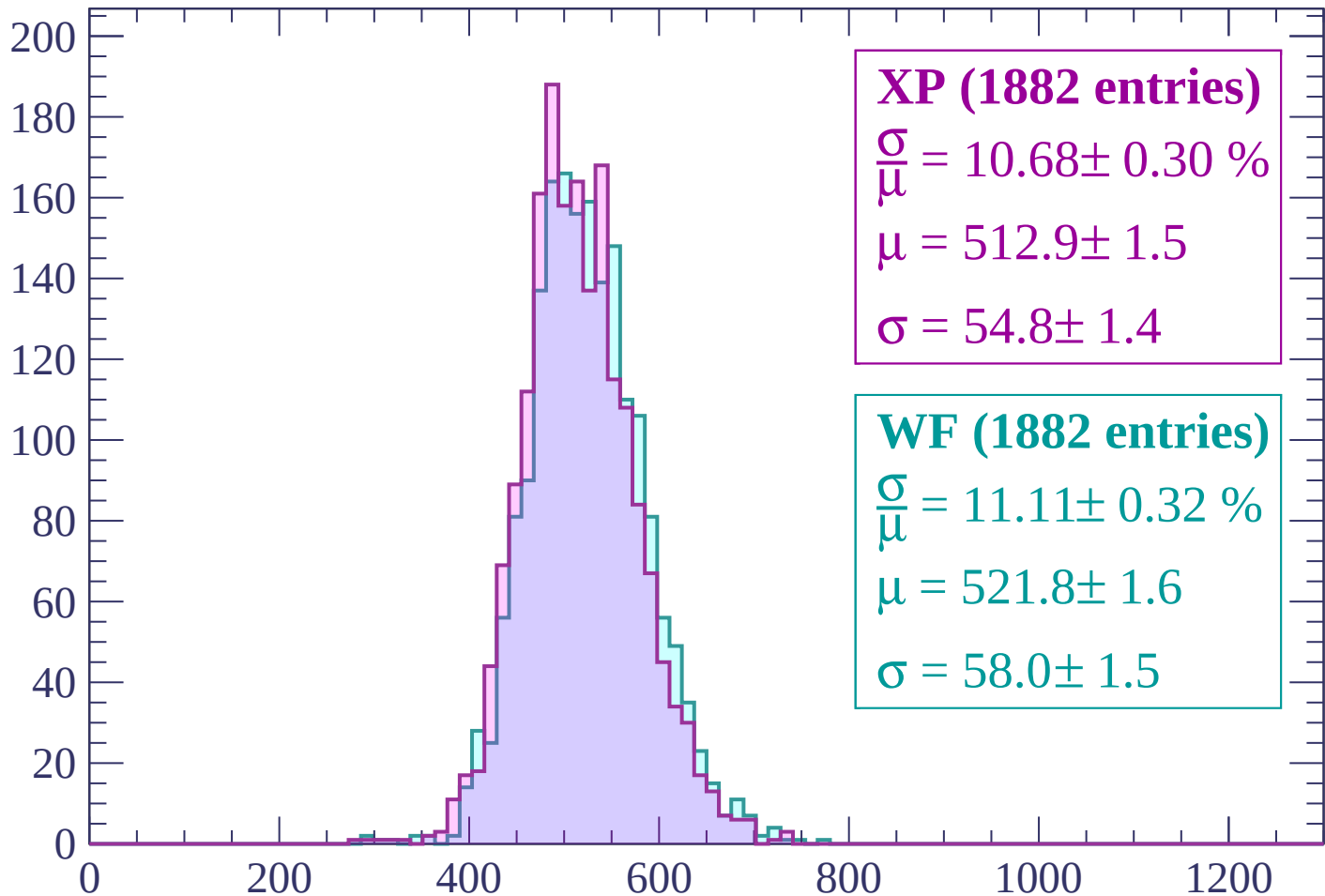
Energy loss | $-2400 < p < -2340$

Count



Energy loss | $-2340 < p < -2280$

Count



XP (1882 entries)

$\frac{\sigma}{\mu} = 10.68 \pm 0.30 \%$

$\mu = 512.9 \pm 1.5$

$\sigma = 54.8 \pm 1.4$

WF (1882 entries)

$\frac{\sigma}{\mu} = 11.11 \pm 0.32 \%$

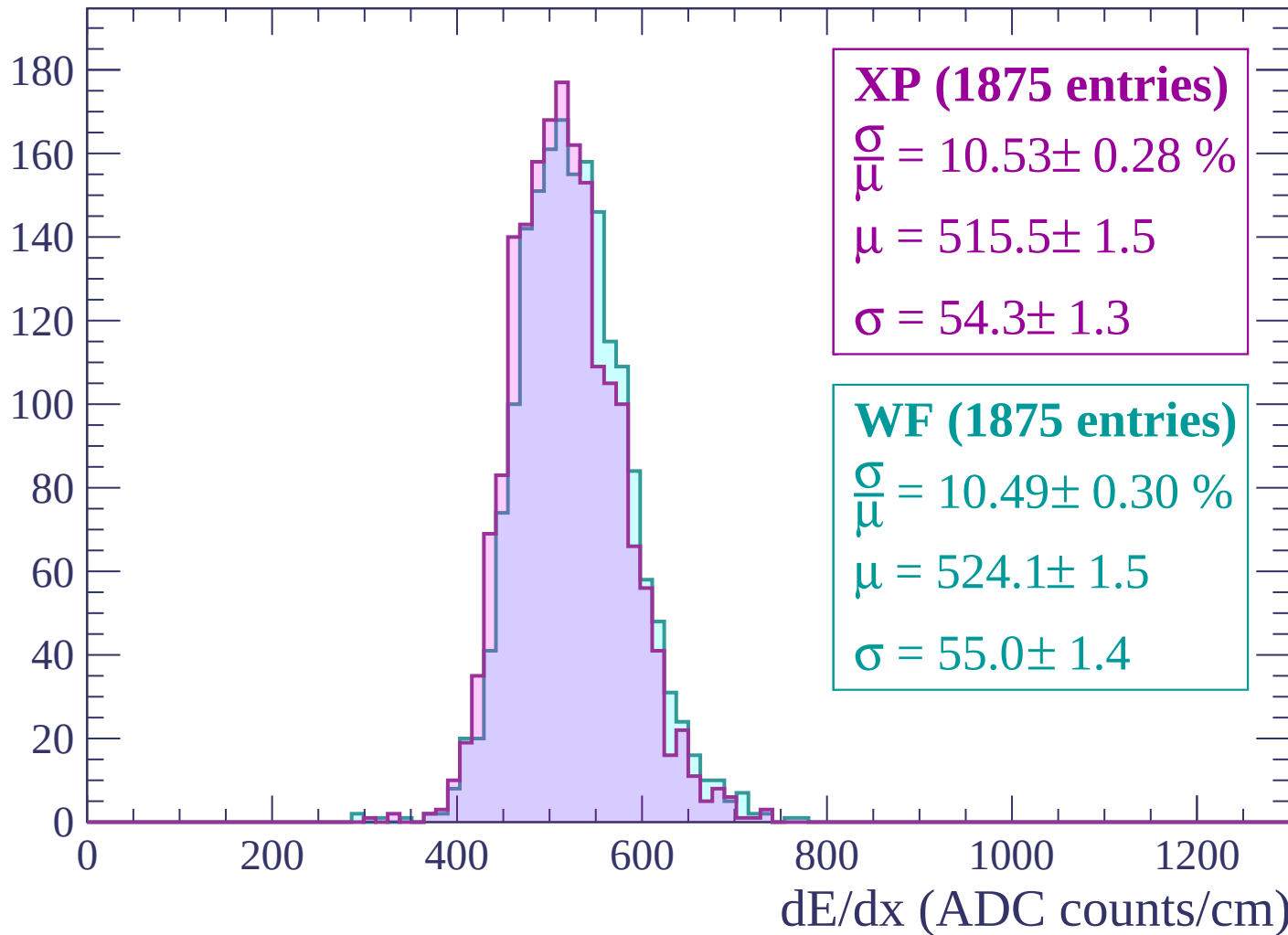
$\mu = 521.8 \pm 1.6$

$\sigma = 58.0 \pm 1.5$

dE/dx (ADC counts/cm)

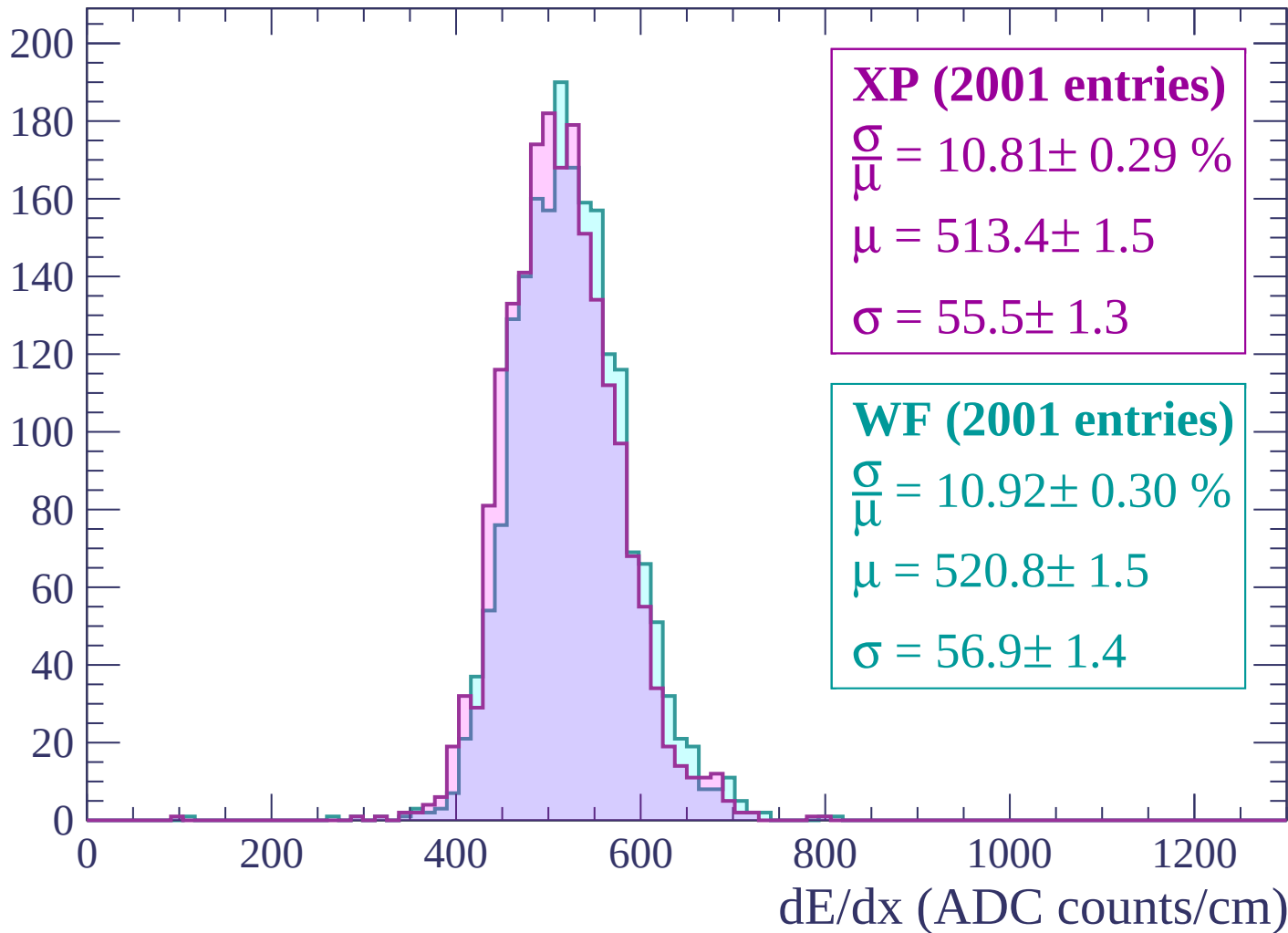
Energy loss | $-2280 < p < -2220$

Count



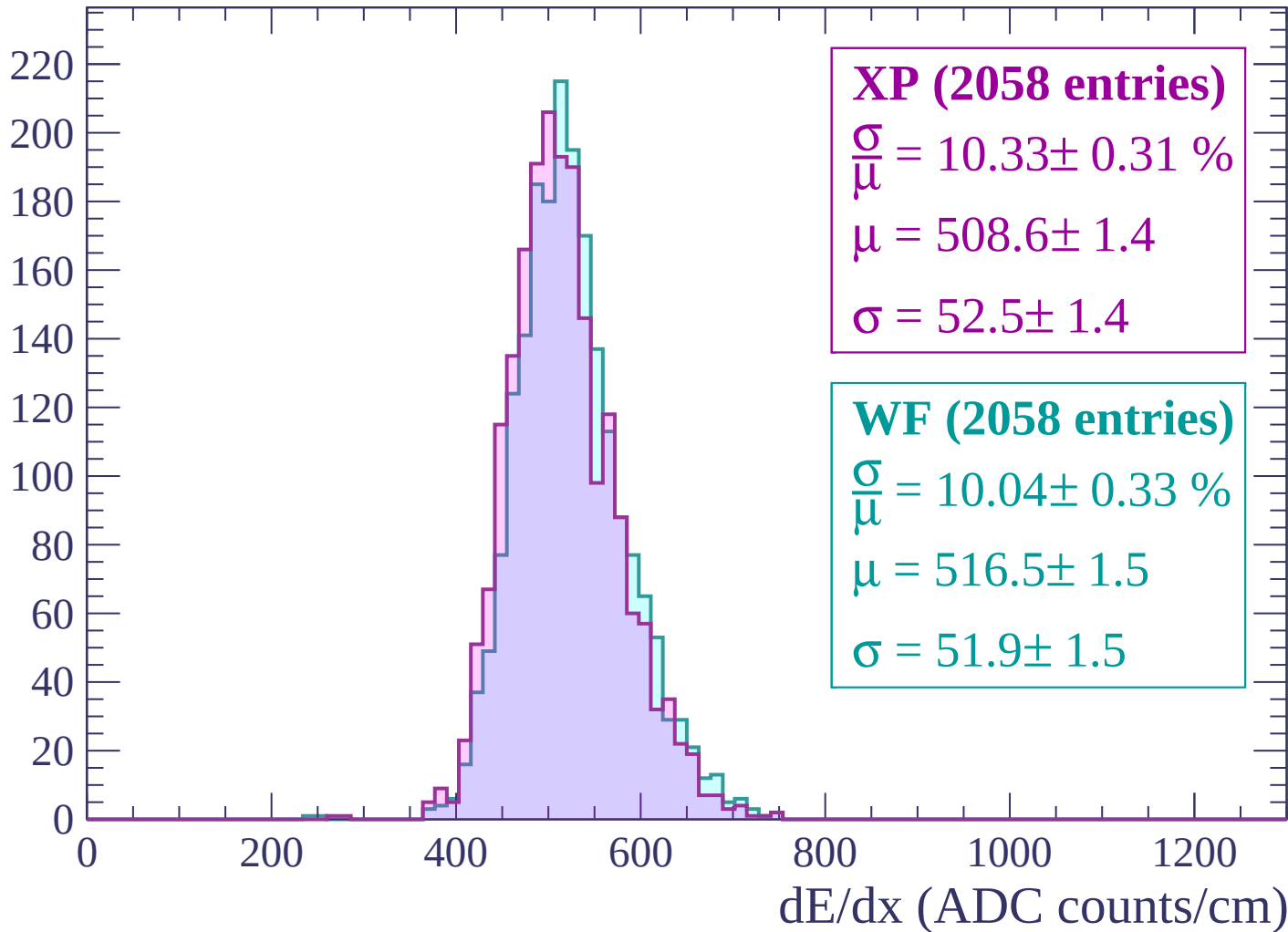
Energy loss | -2220 < p < -2160

Count



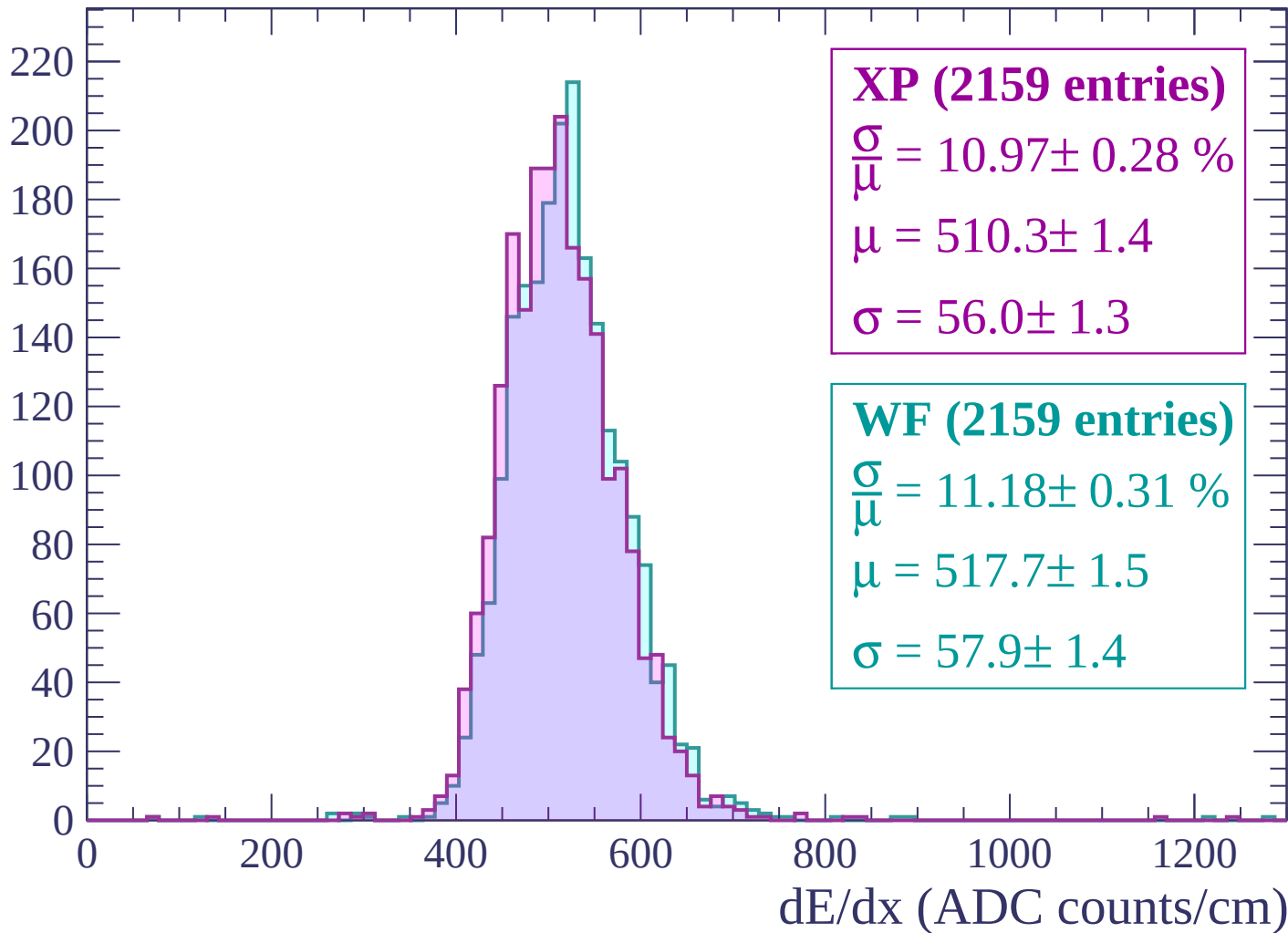
Energy loss | -2160 < p < -2100

Count



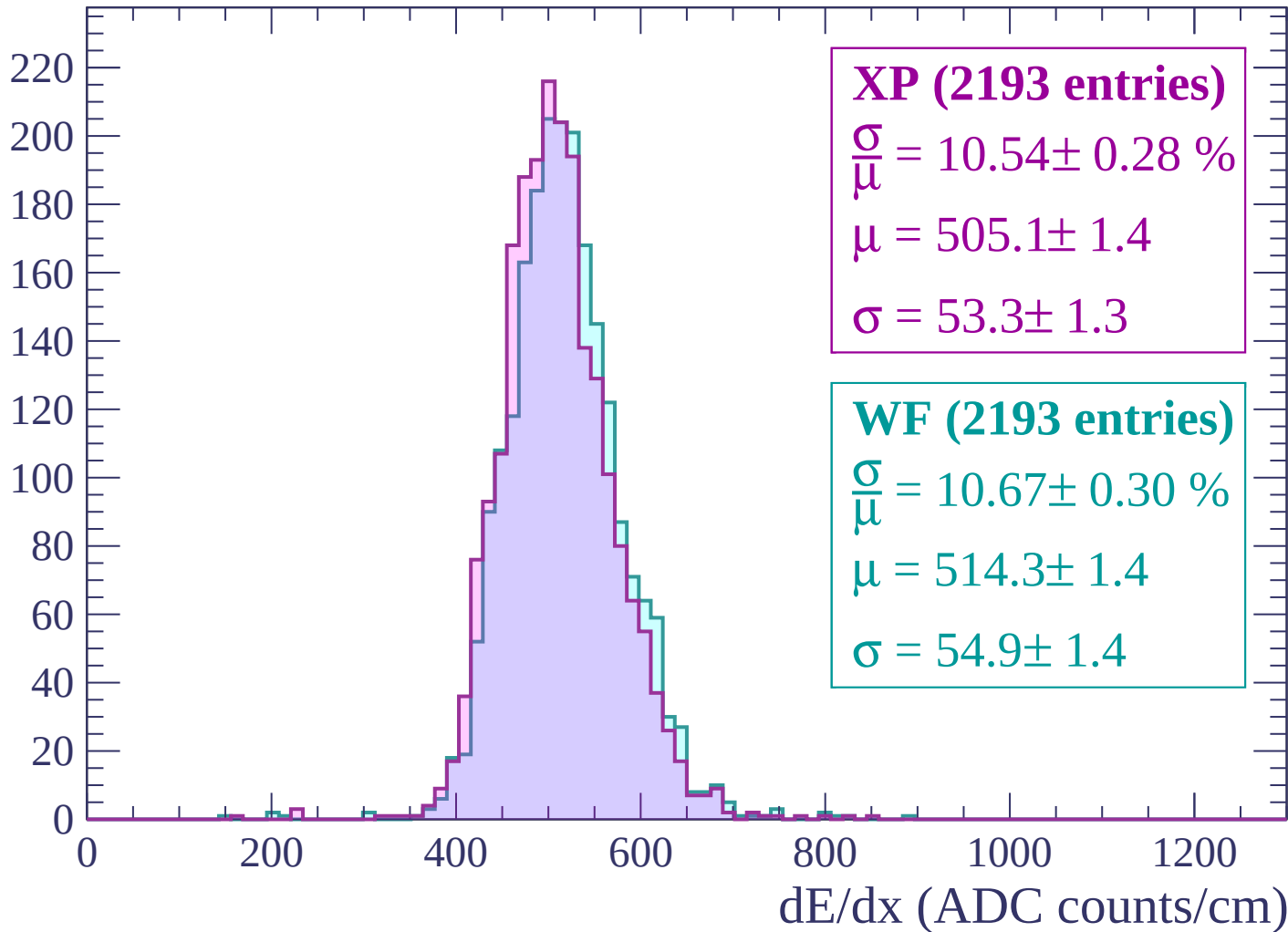
Energy loss | $-2100 < p < -2040$

Count



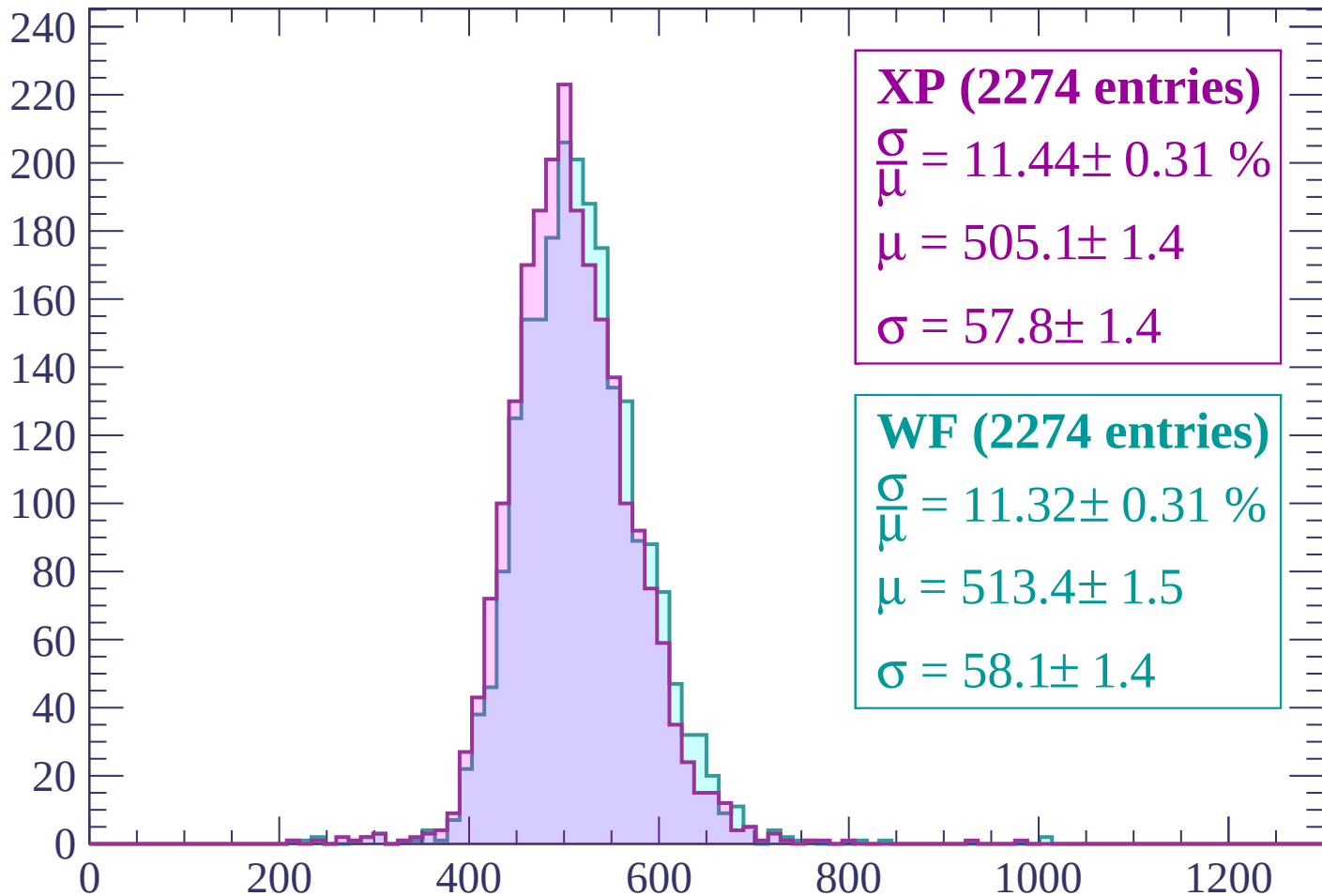
Energy loss | -2040 < p < -1980

Count



Energy loss | -1980 < p < -1920

Count



XP (2274 entries)

$$\frac{\sigma}{\mu} = 11.44 \pm 0.31 \%$$

$$\mu = 505.1 \pm 1.4$$

$$\sigma = 57.8 \pm 1.4$$

WF (2274 entries)

$$\frac{\sigma}{\mu} = 11.32 \pm 0.31 \%$$

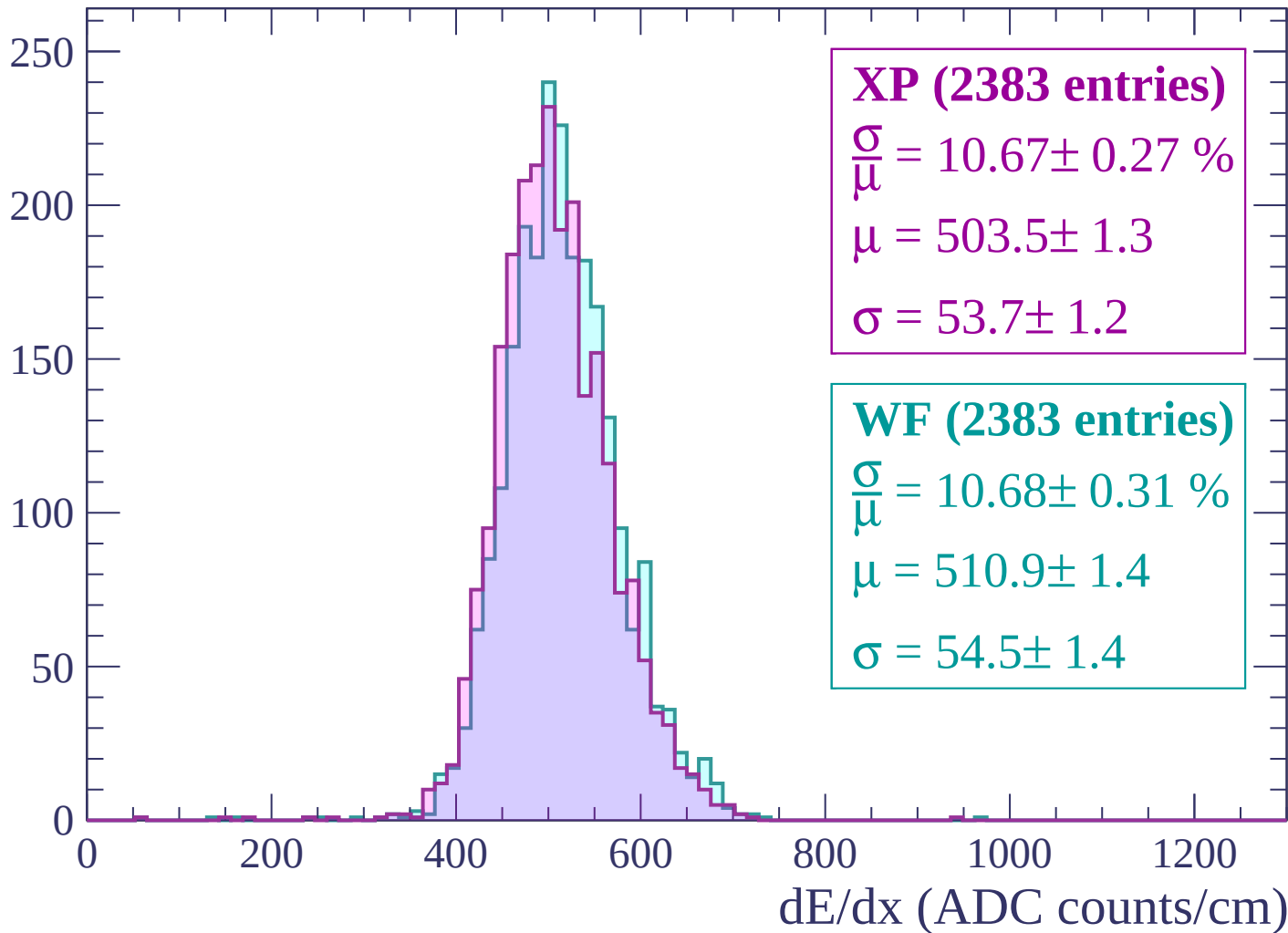
$$\mu = 513.4 \pm 1.5$$

$$\sigma = 58.1 \pm 1.4$$

dE/dx (ADC counts/cm)

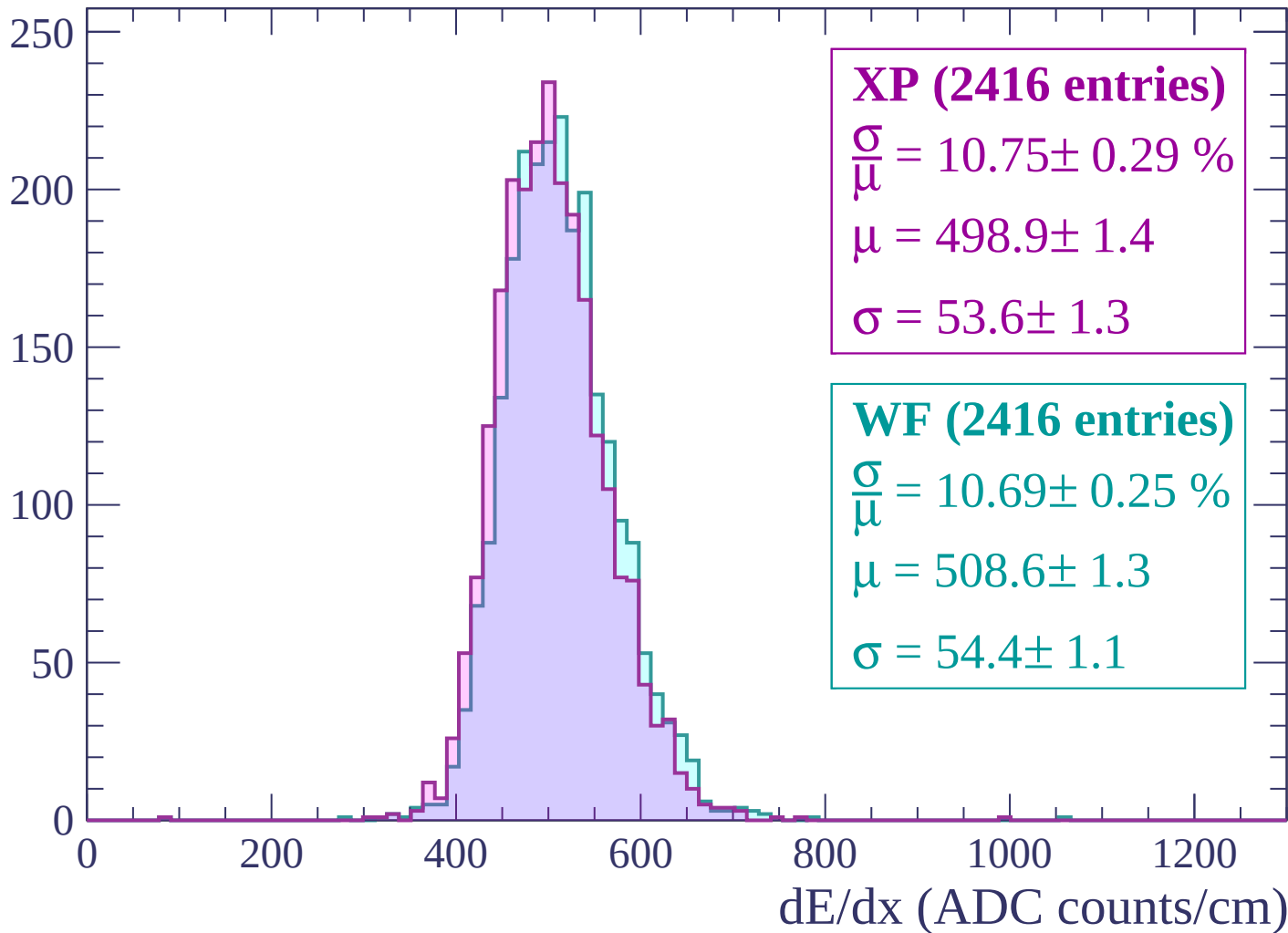
Energy loss | -1920 < p < -1860

Count



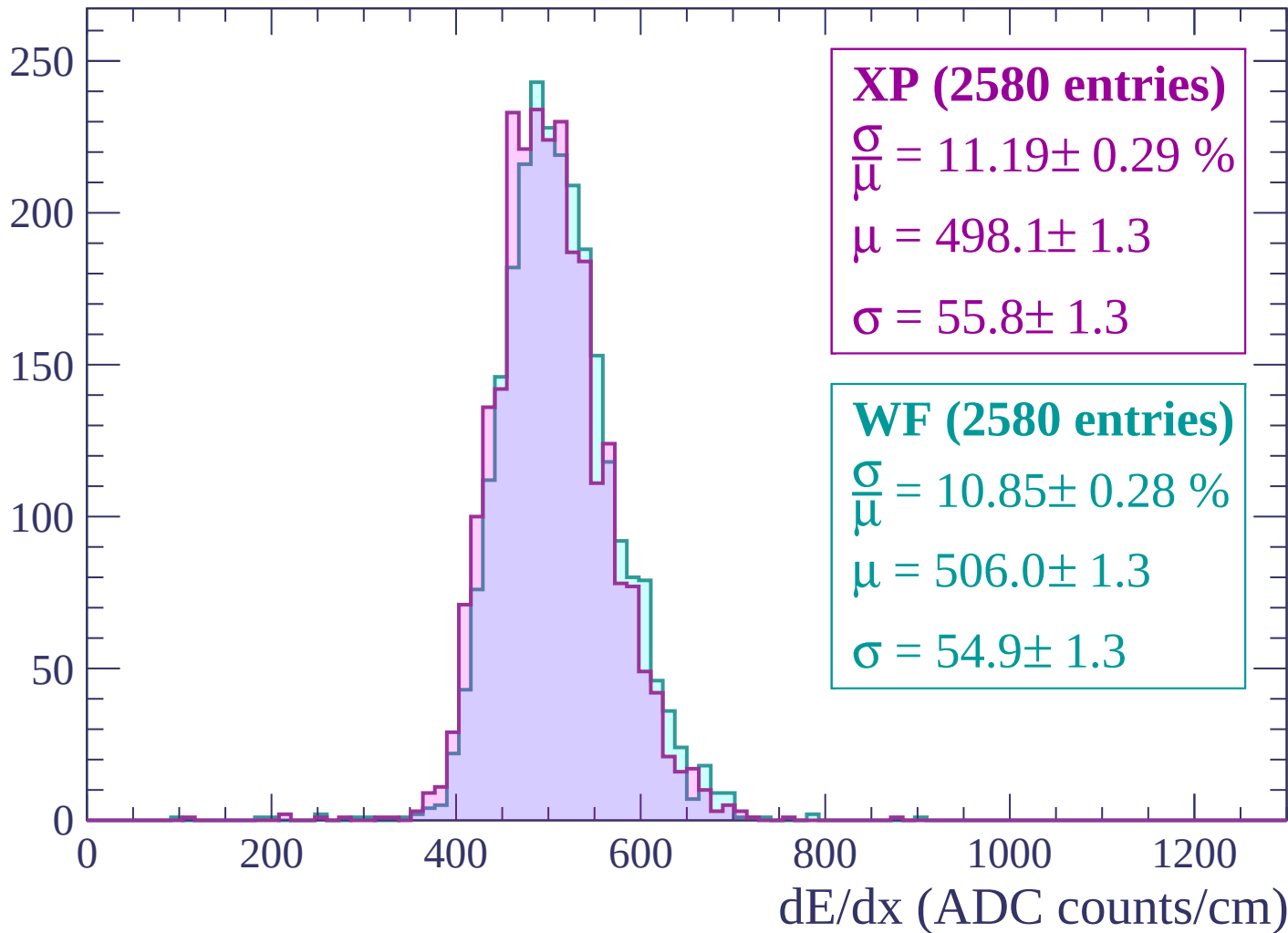
Energy loss | -1860 < p < -1800

Count



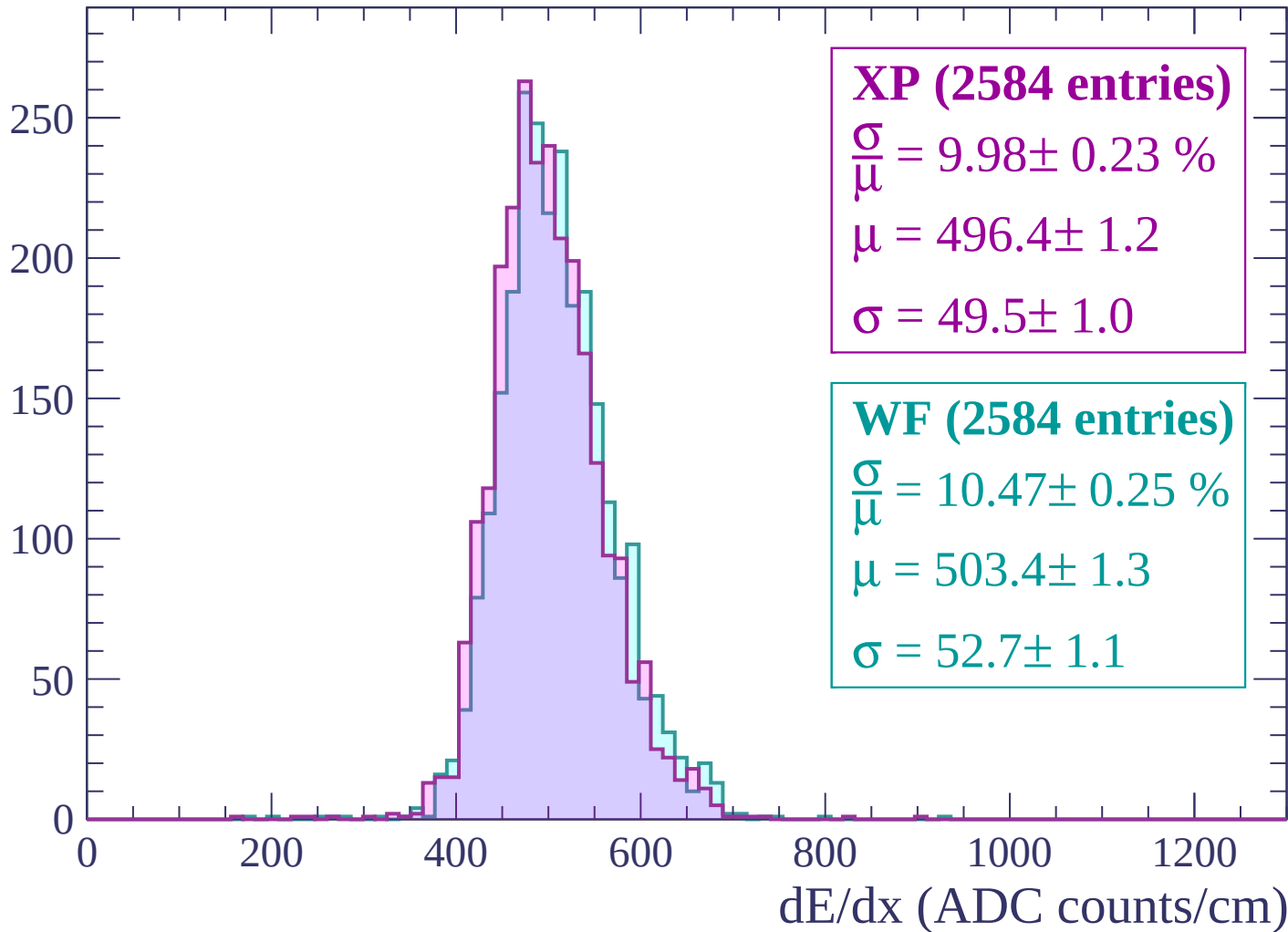
Energy loss | -1800 < p < -1740

Count



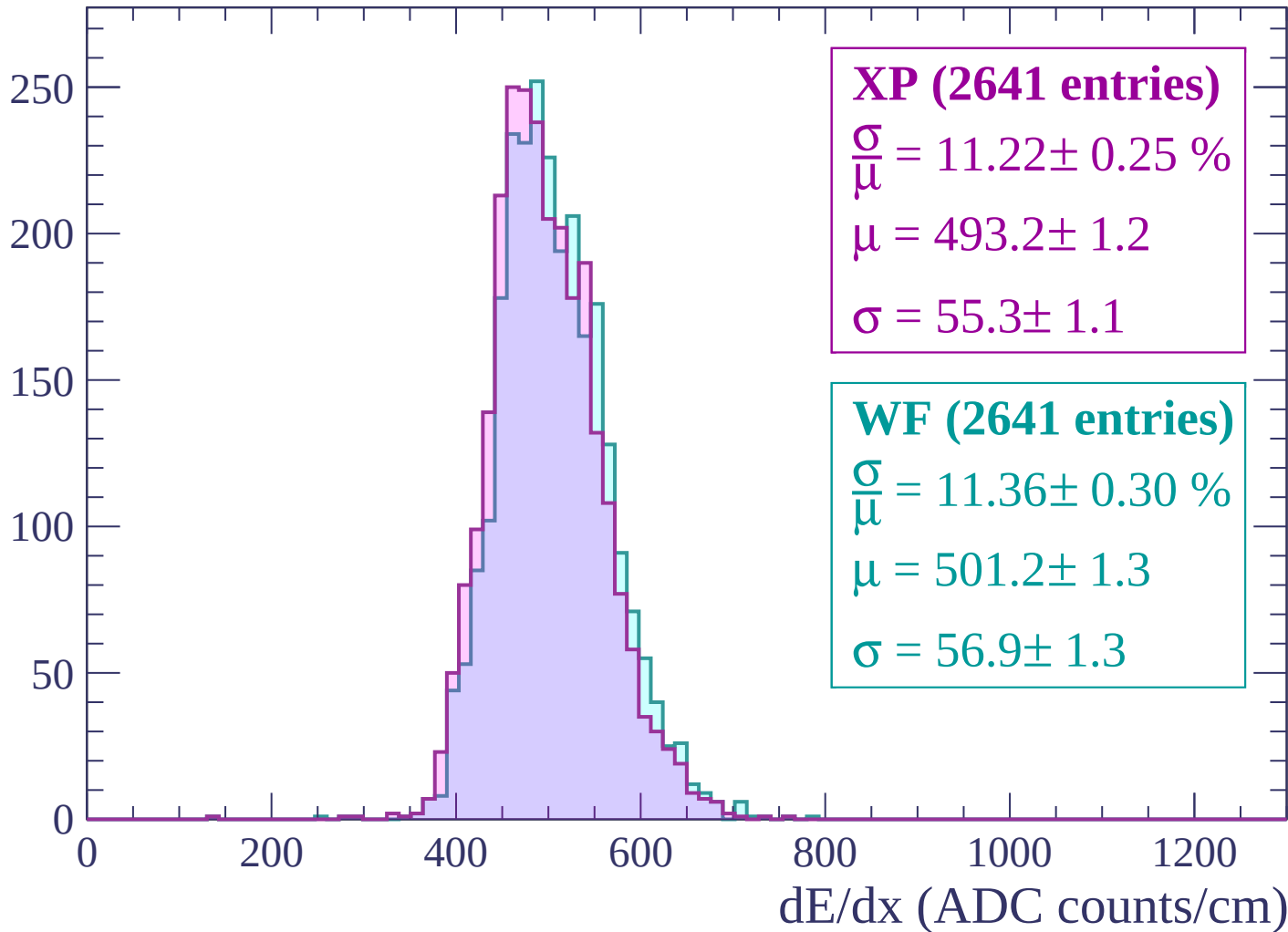
Energy loss | $-1740 < p < -1680$

Count



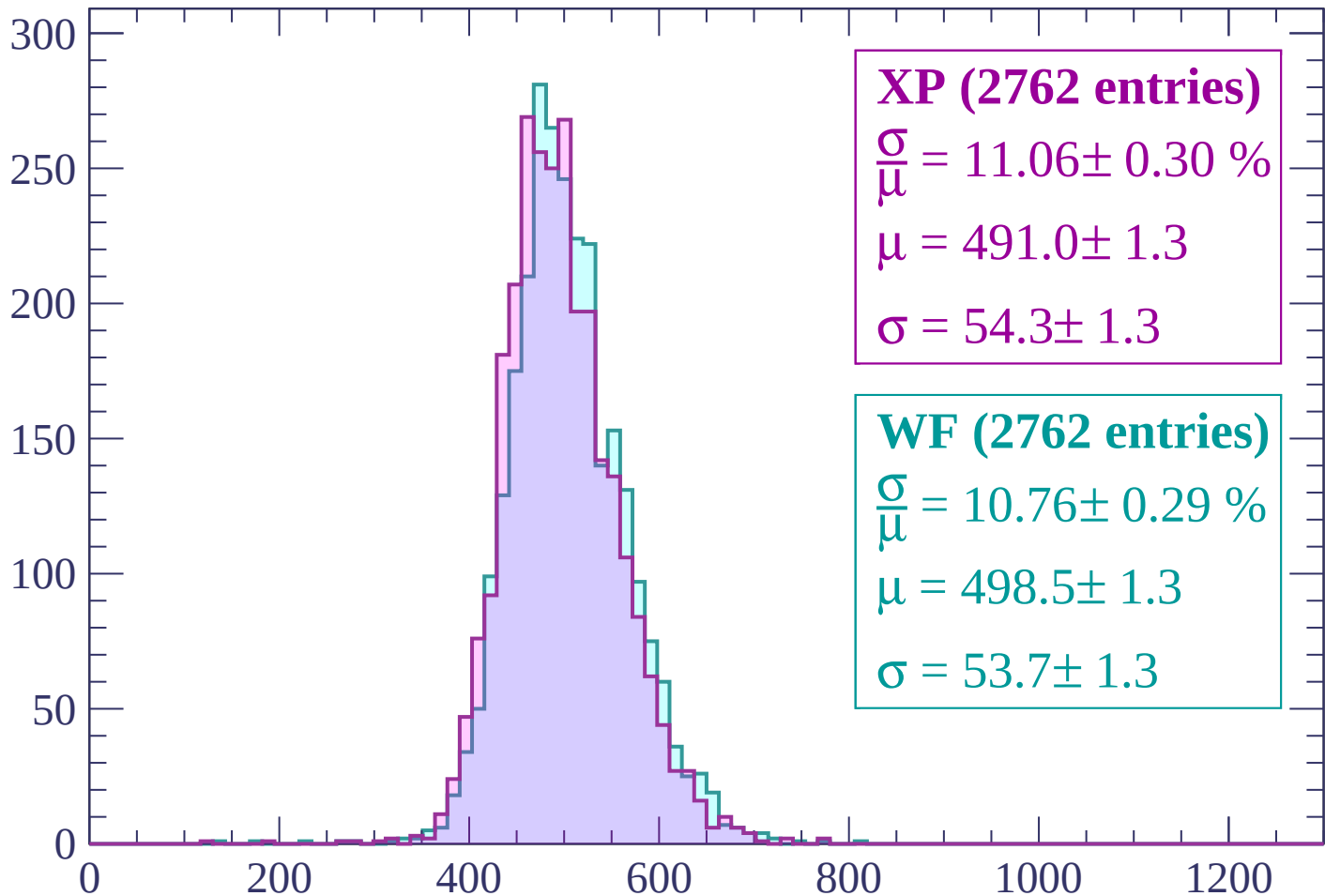
Energy loss | -1680 < p < -1620

Count



Energy loss | -1620 < p < -1560

Count



XP (2762 entries)

$$\frac{\sigma}{\mu} = 11.06 \pm 0.30 \%$$

$$\mu = 491.0 \pm 1.3$$

$$\sigma = 54.3 \pm 1.3$$

WF (2762 entries)

$$\frac{\sigma}{\mu} = 10.76 \pm 0.29 \%$$

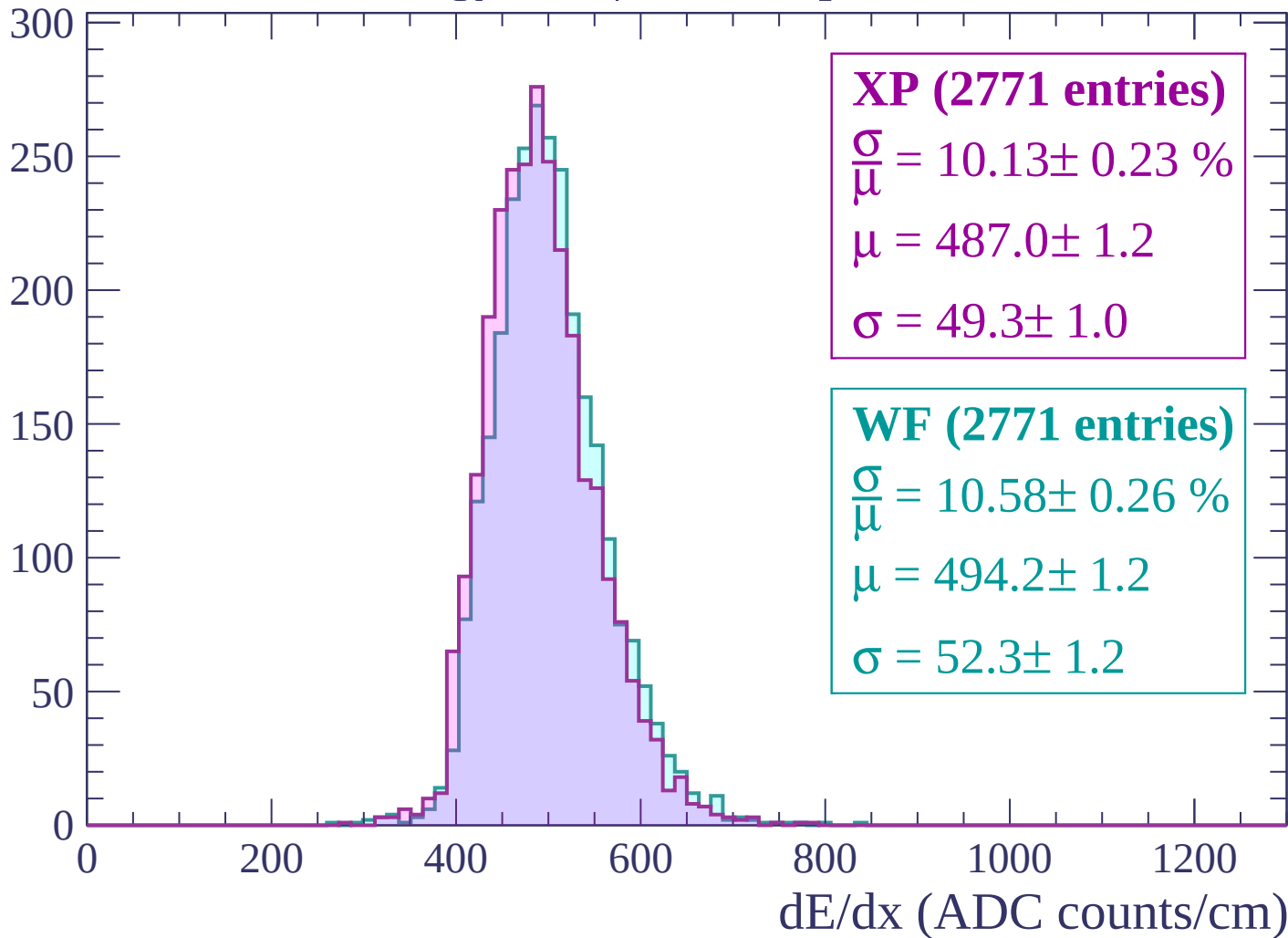
$$\mu = 498.5 \pm 1.3$$

$$\sigma = 53.7 \pm 1.3$$

dE/dx (ADC counts/cm)

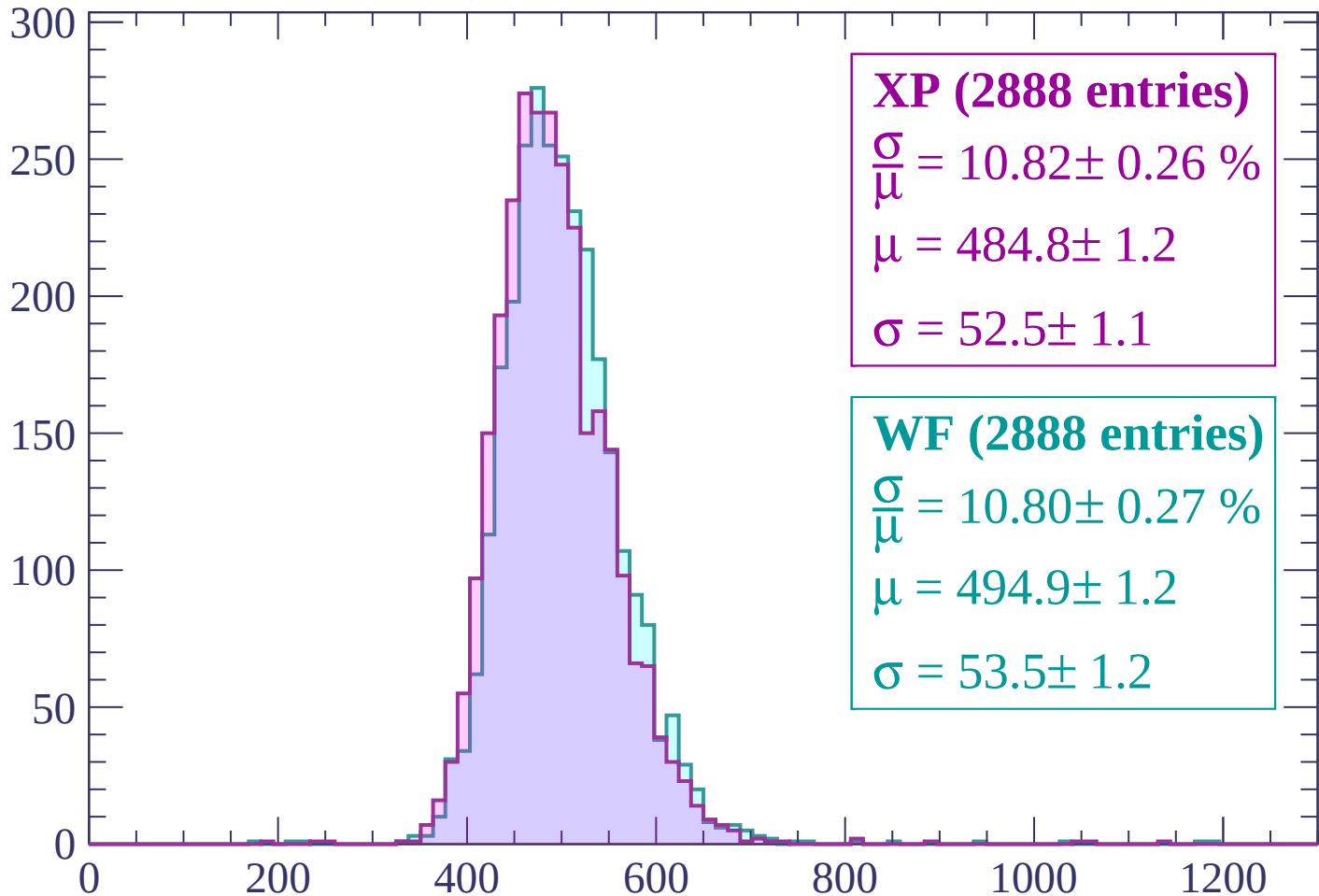
Energy loss | -1560 < p < -1500

Count



Energy loss | -1500 < p < -1440

Count



XP (2888 entries)

$$\frac{\sigma}{\mu} = 10.82 \pm 0.26 \%$$

$$\mu = 484.8 \pm 1.2$$

$$\sigma = 52.5 \pm 1.1$$

WF (2888 entries)

$$\frac{\sigma}{\mu} = 10.80 \pm 0.27 \%$$

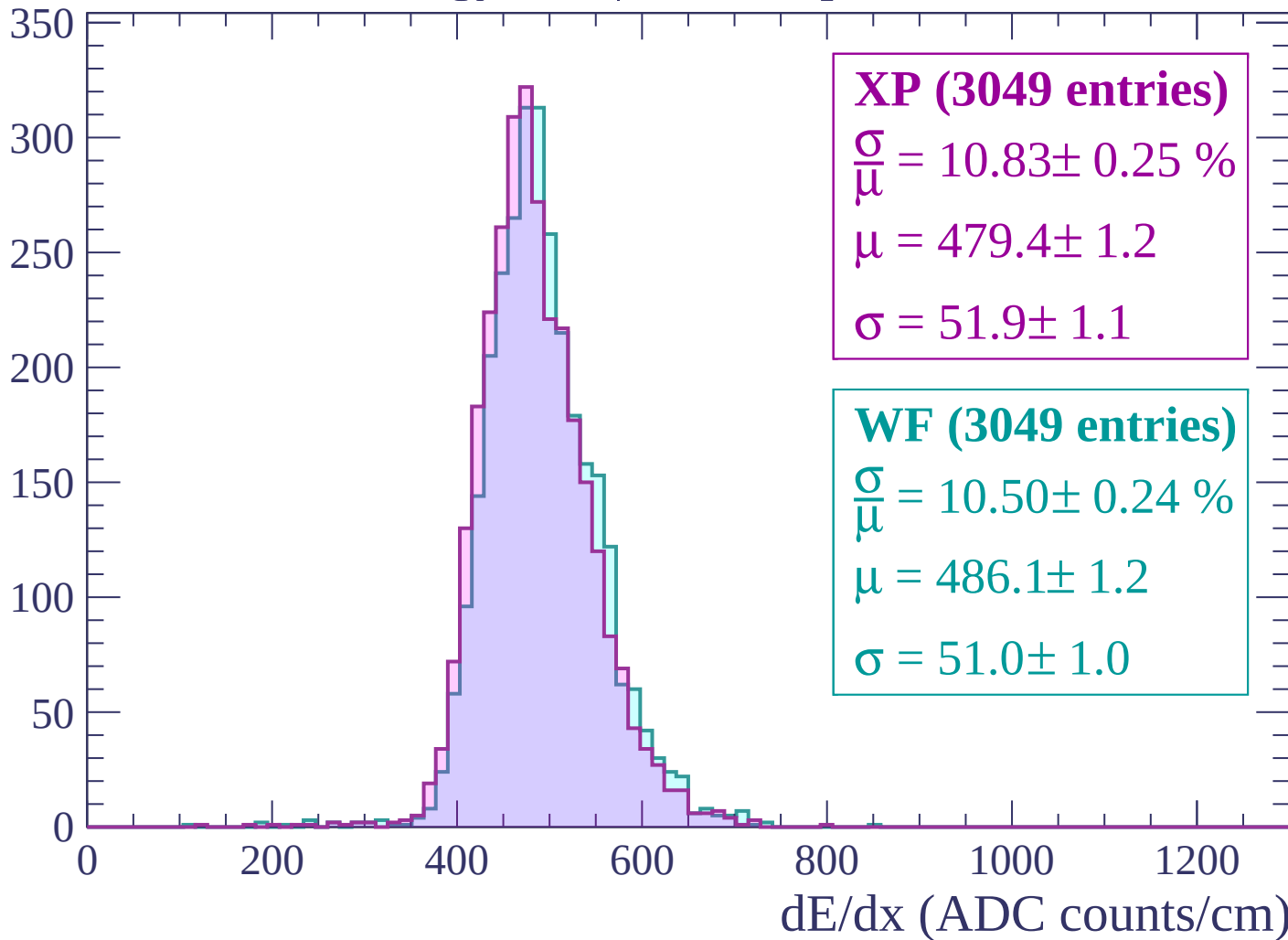
$$\mu = 494.9 \pm 1.2$$

$$\sigma = 53.5 \pm 1.2$$

dE/dx (ADC counts/cm)

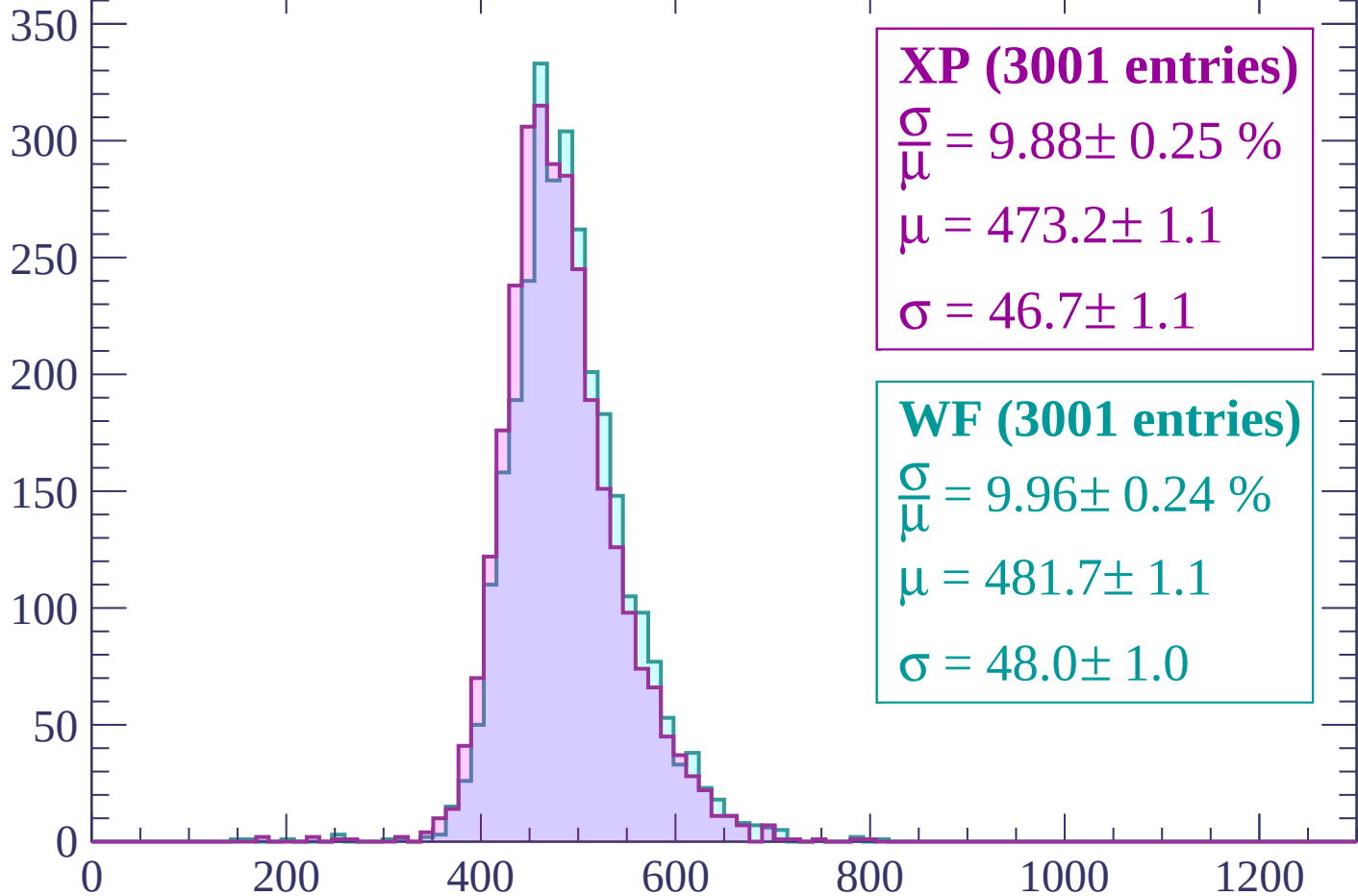
Energy loss | $-1440 < p < -1380$

Count



Energy loss | -1380 < p < -1320

Count



XP (3001 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.25 \%$$

$$\mu = 473.2 \pm 1.1$$

$$\sigma = 46.7 \pm 1.1$$

WF (3001 entries)

$$\frac{\sigma}{\mu} = 9.96 \pm 0.24 \%$$

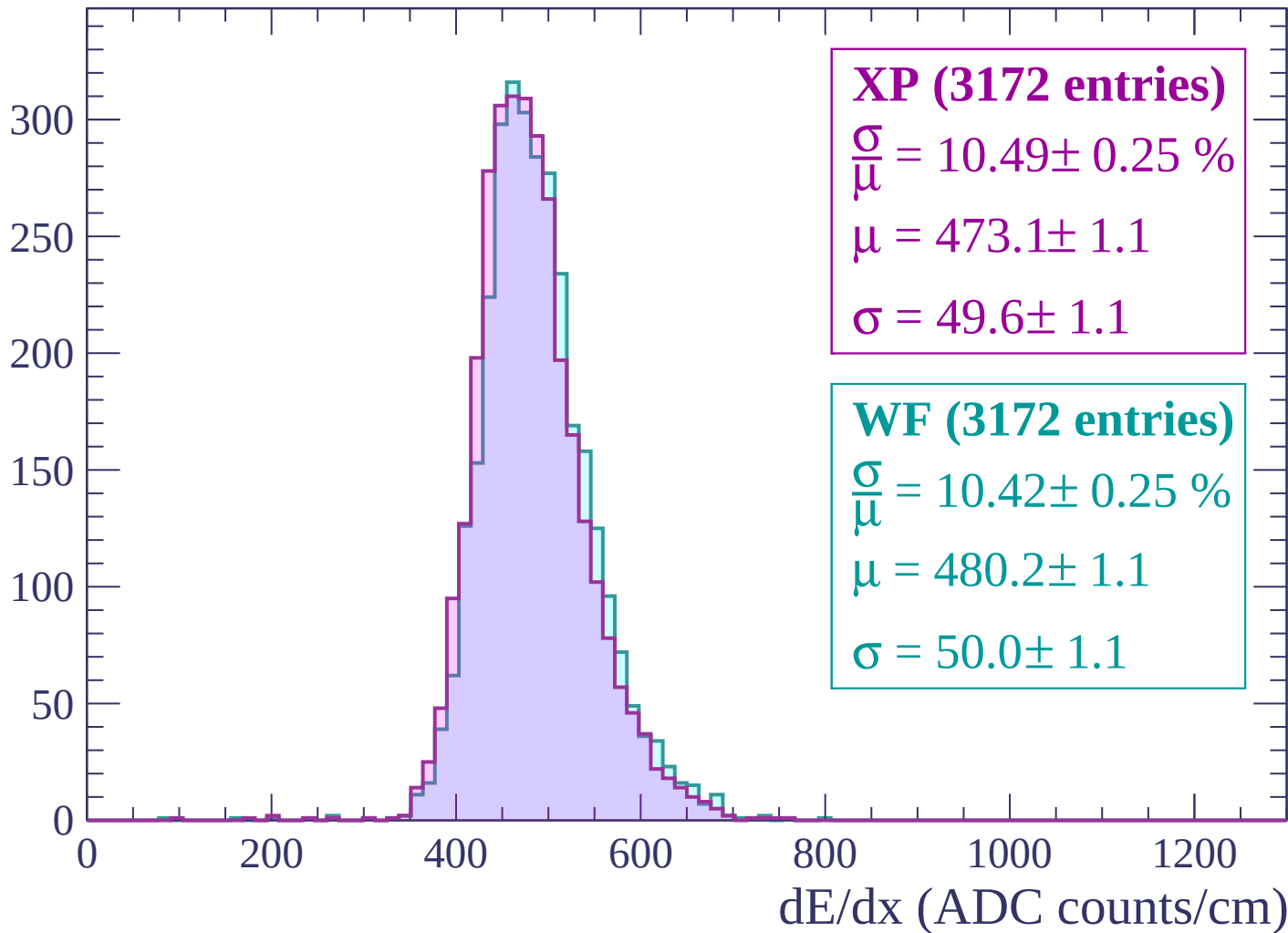
$$\mu = 481.7 \pm 1.1$$

$$\sigma = 48.0 \pm 1.0$$

dE/dx (ADC counts/cm)

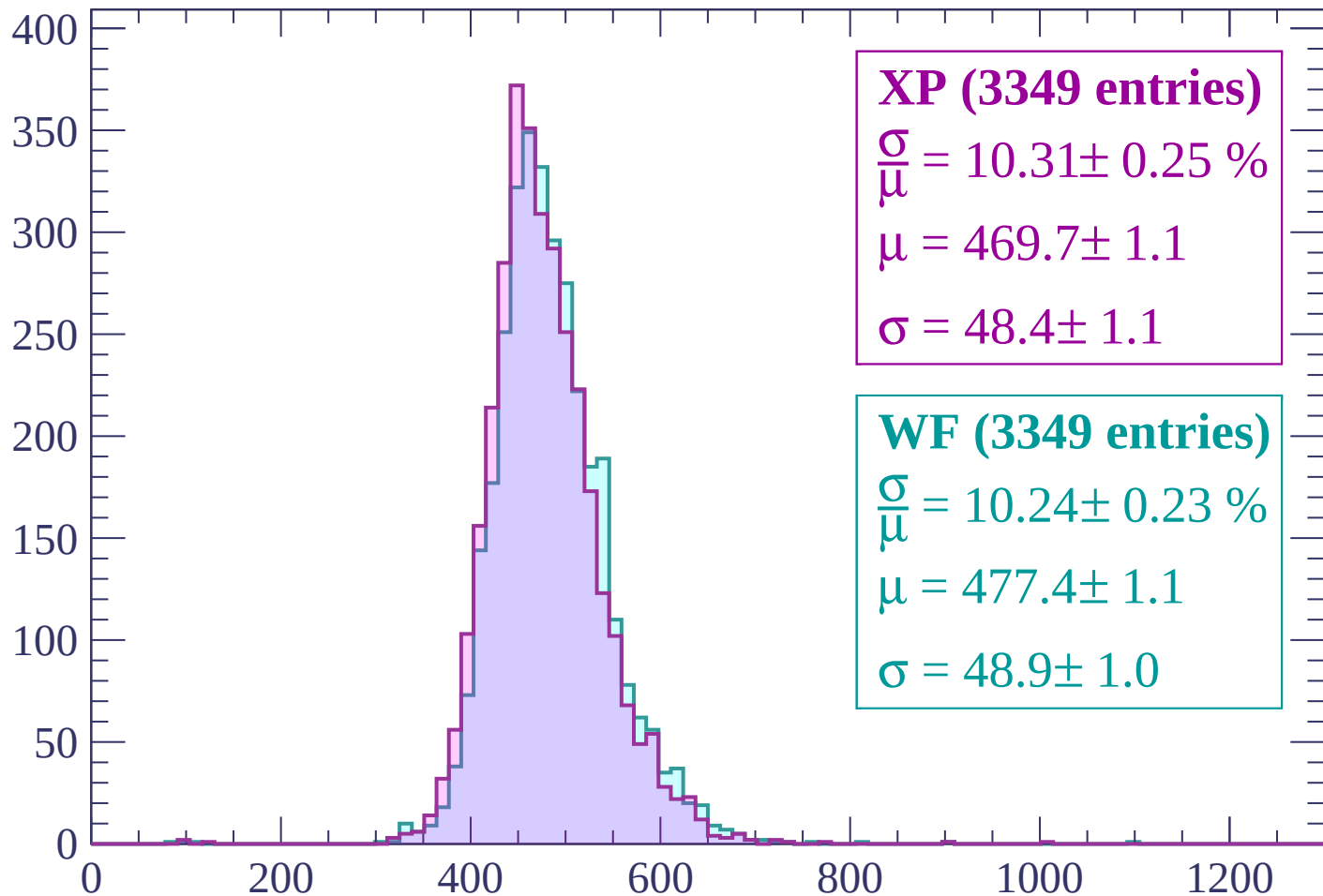
Energy loss | $-1320 < p < -1260$

Count



Energy loss | -1260 < p < -1200

Count



XP (3349 entries)

$\frac{\sigma}{\mu} = 10.31 \pm 0.25 \%$

$\mu = 469.7 \pm 1.1$

$\sigma = 48.4 \pm 1.1$

WF (3349 entries)

$\frac{\sigma}{\mu} = 10.24 \pm 0.23 \%$

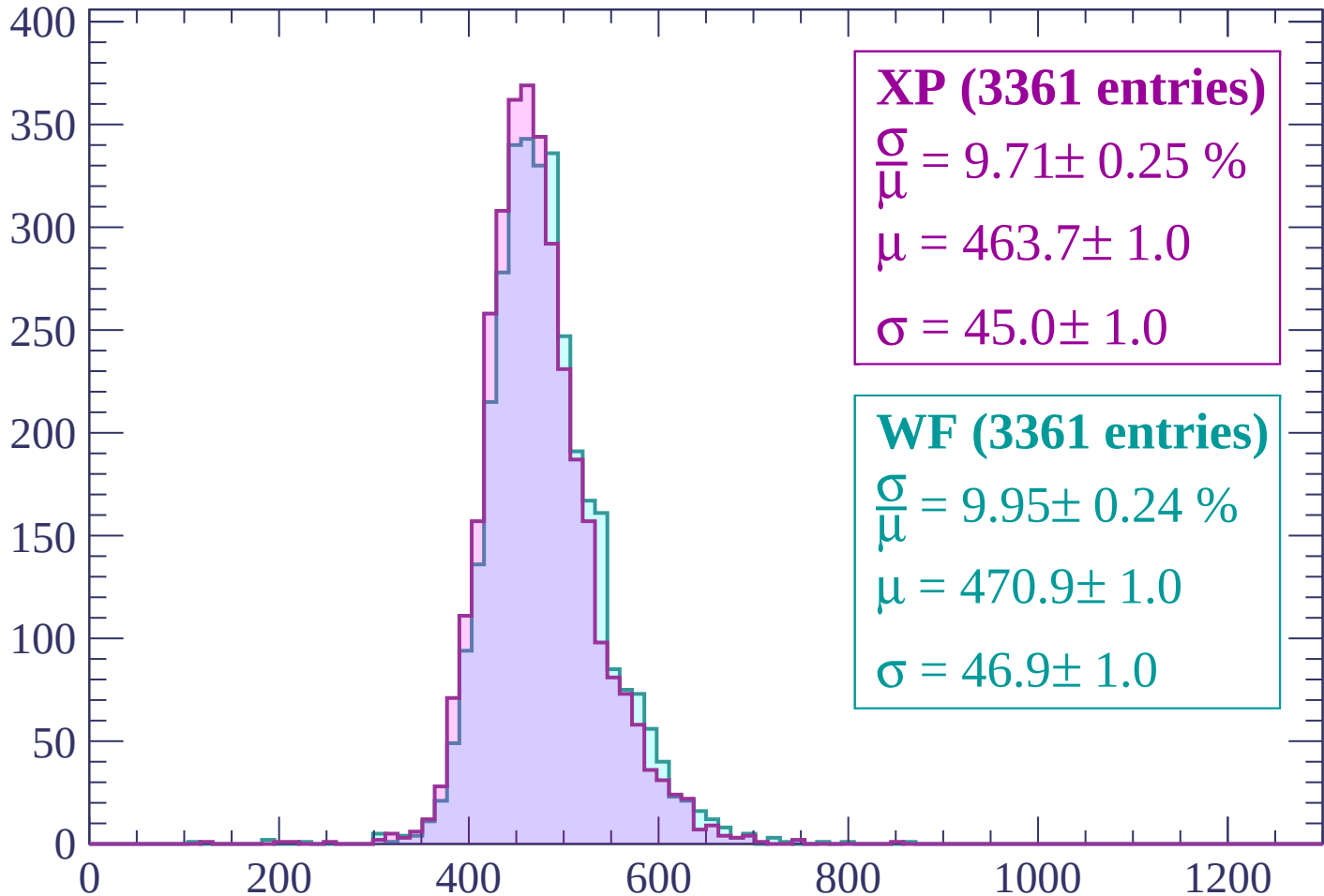
$\mu = 477.4 \pm 1.1$

$\sigma = 48.9 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1140

Count



XP (3361 entries)

$$\frac{\sigma}{\mu} = 9.71 \pm 0.25 \%$$

$$\mu = 463.7 \pm 1.0$$

$$\sigma = 45.0 \pm 1.0$$

WF (3361 entries)

$$\frac{\sigma}{\mu} = 9.95 \pm 0.24 \%$$

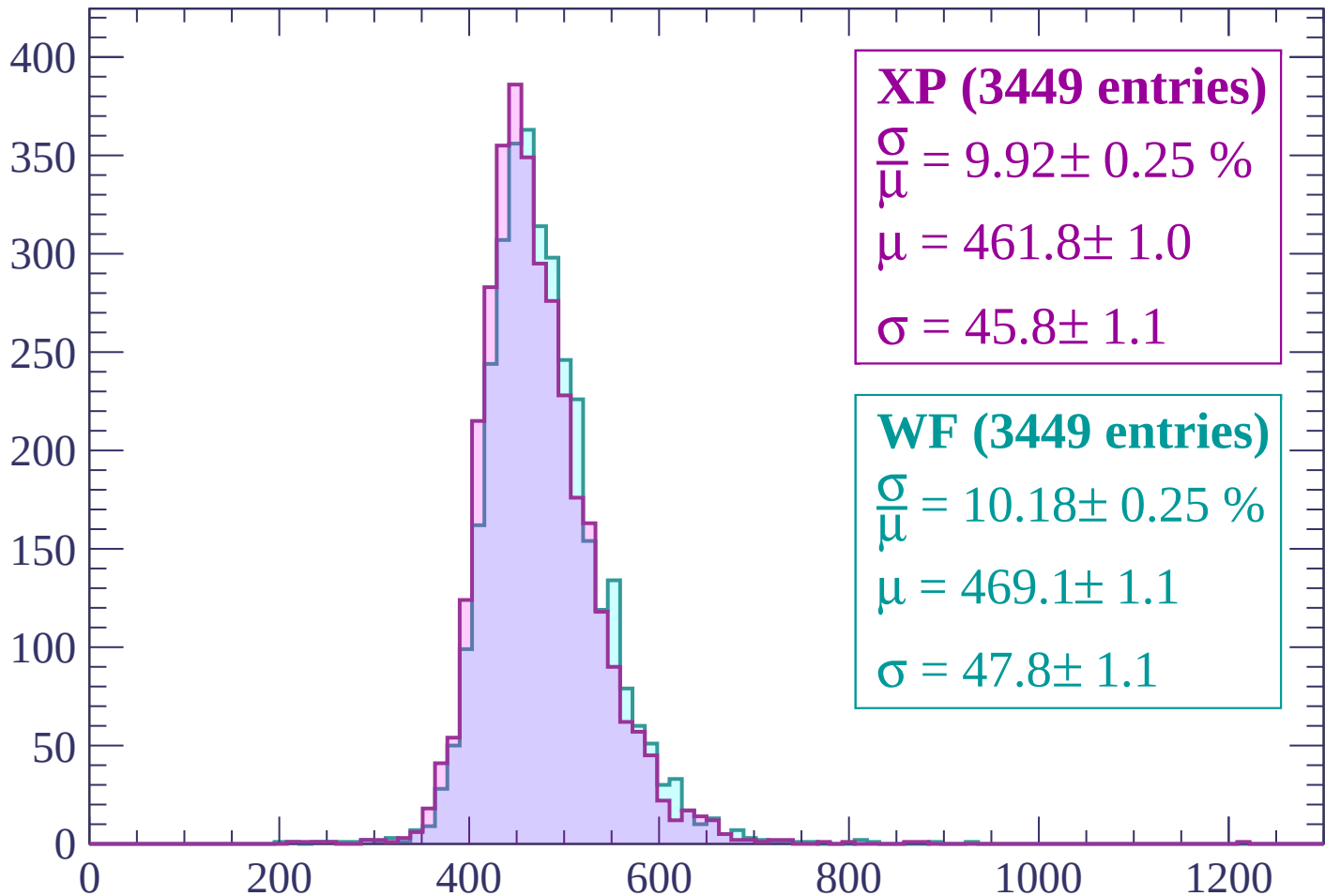
$$\mu = 470.9 \pm 1.0$$

$$\sigma = 46.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1140 < p < -1080$

Count



XP (3449 entries)

$$\frac{\sigma}{\mu} = 9.92 \pm 0.25 \%$$

$$\mu = 461.8 \pm 1.0$$

$$\sigma = 45.8 \pm 1.1$$

WF (3449 entries)

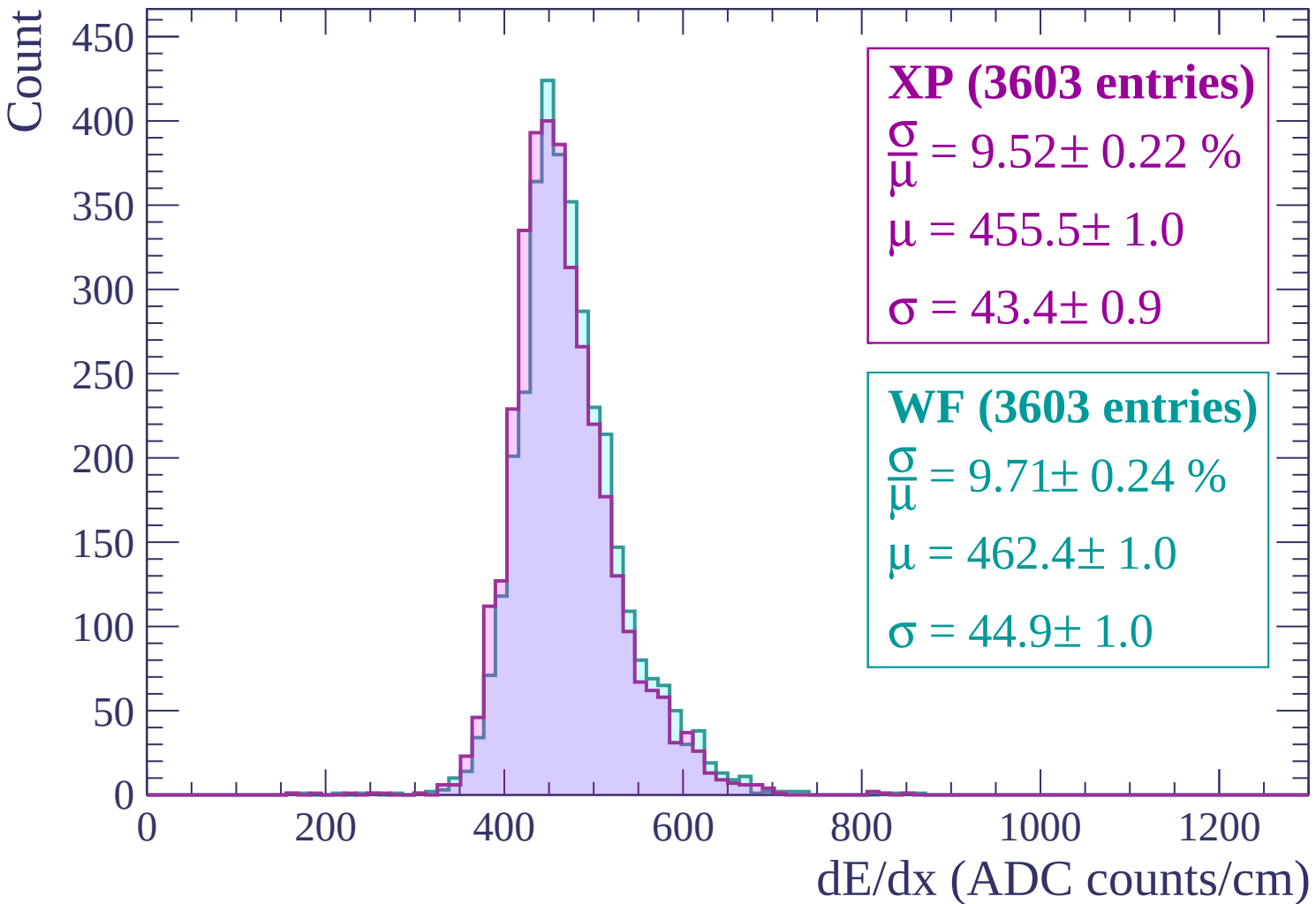
$$\frac{\sigma}{\mu} = 10.18 \pm 0.25 \%$$

$$\mu = 469.1 \pm 1.1$$

$$\sigma = 47.8 \pm 1.1$$

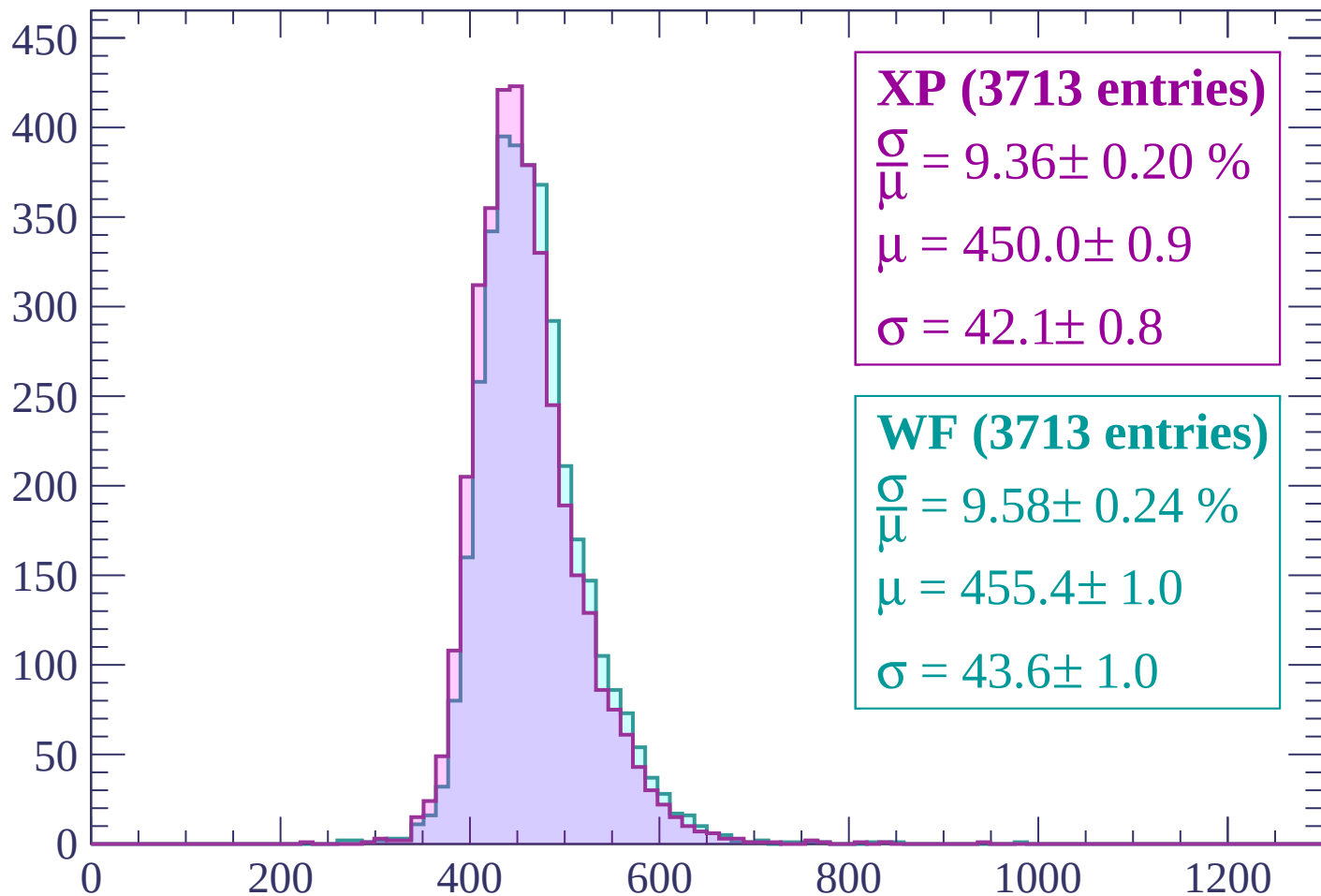
dE/dx (ADC counts/cm)

Energy loss | -1080 < p < -1020



Energy loss | $-1020 < p < -960$

Count



XP (3713 entries)

$$\frac{\sigma}{\mu} = 9.36 \pm 0.20 \%$$

$$\mu = 450.0 \pm 0.9$$

$$\sigma = 42.1 \pm 0.8$$

WF (3713 entries)

$$\frac{\sigma}{\mu} = 9.58 \pm 0.24 \%$$

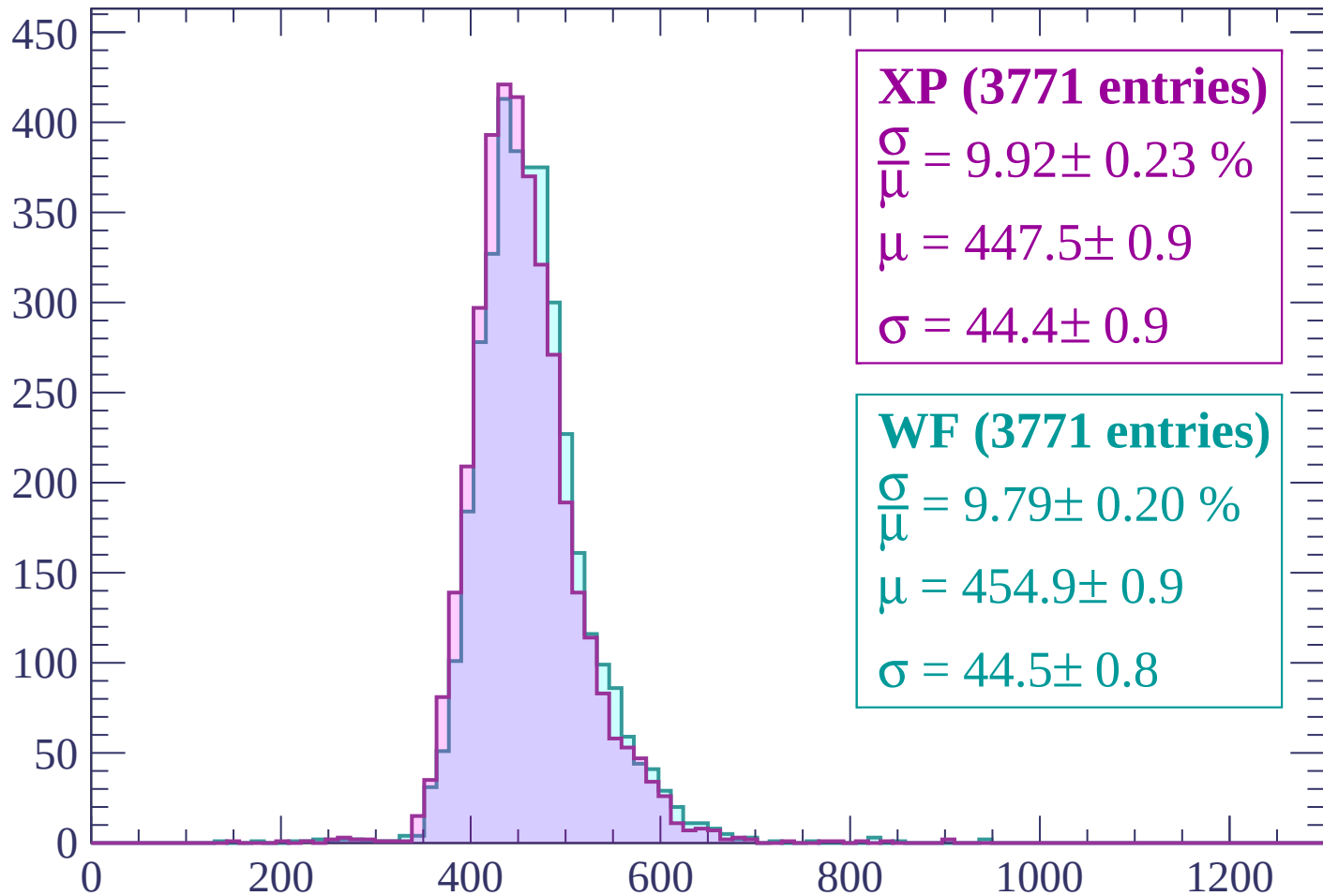
$$\mu = 455.4 \pm 1.0$$

$$\sigma = 43.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (3771 entries)

$$\frac{\sigma}{\mu} = 9.92 \pm 0.23 \%$$

$$\mu = 447.5 \pm 0.9$$

$$\sigma = 44.4 \pm 0.9$$

WF (3771 entries)

$$\frac{\sigma}{\mu} = 9.79 \pm 0.20 \%$$

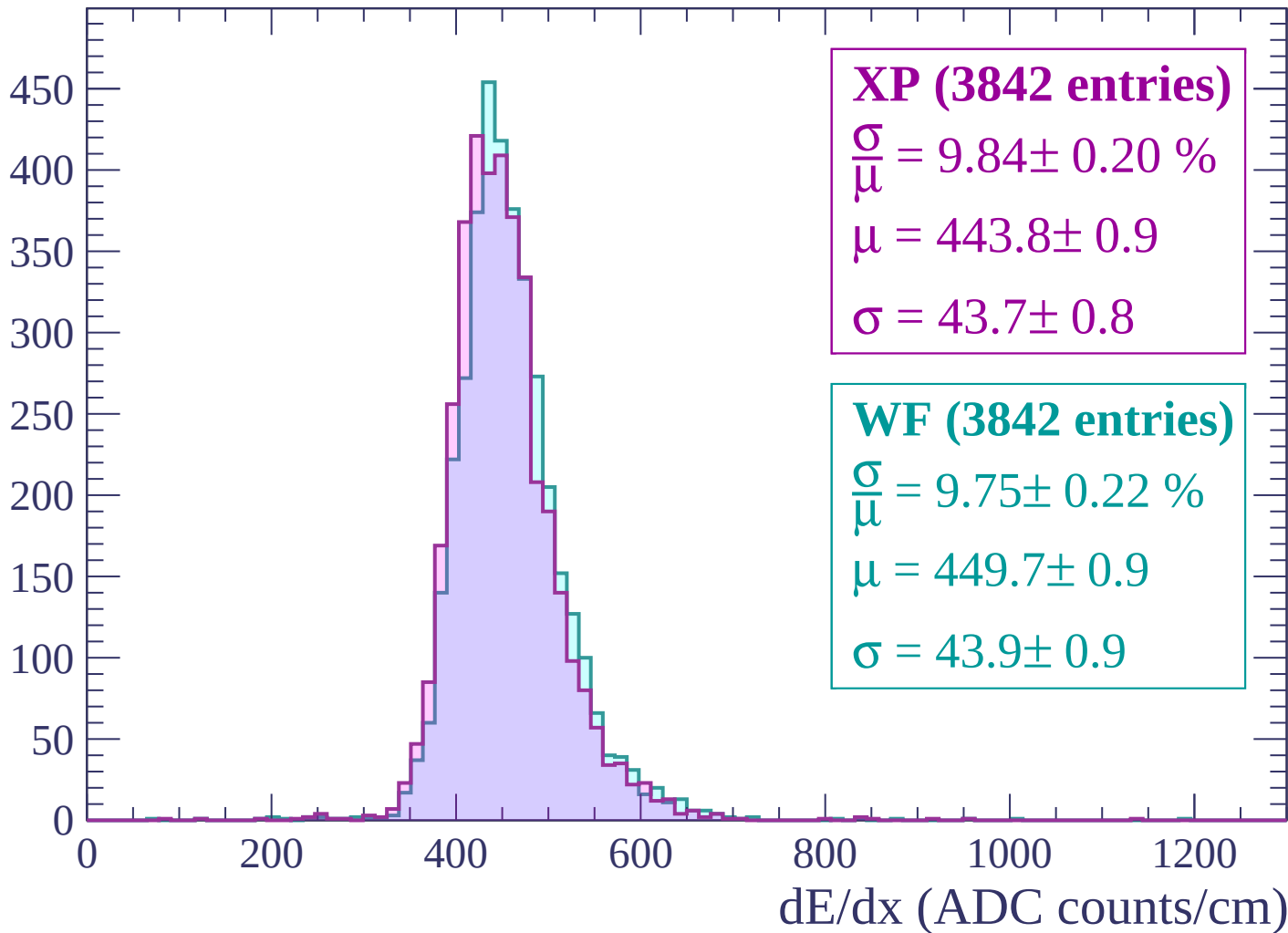
$$\mu = 454.9 \pm 0.9$$

$$\sigma = 44.5 \pm 0.8$$

dE/dx (ADC counts/cm)

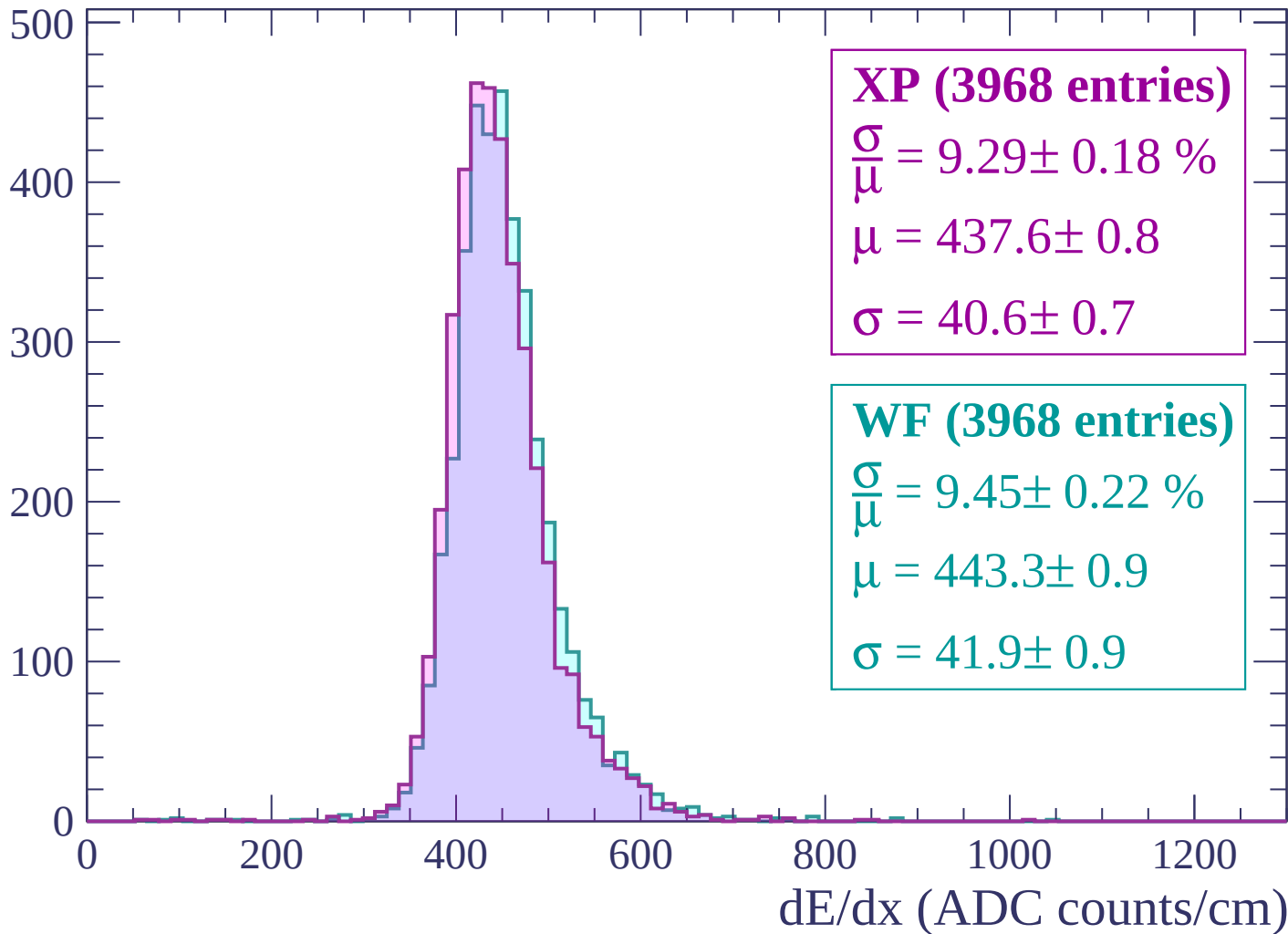
Energy loss | $-900 < p < -840$

Count



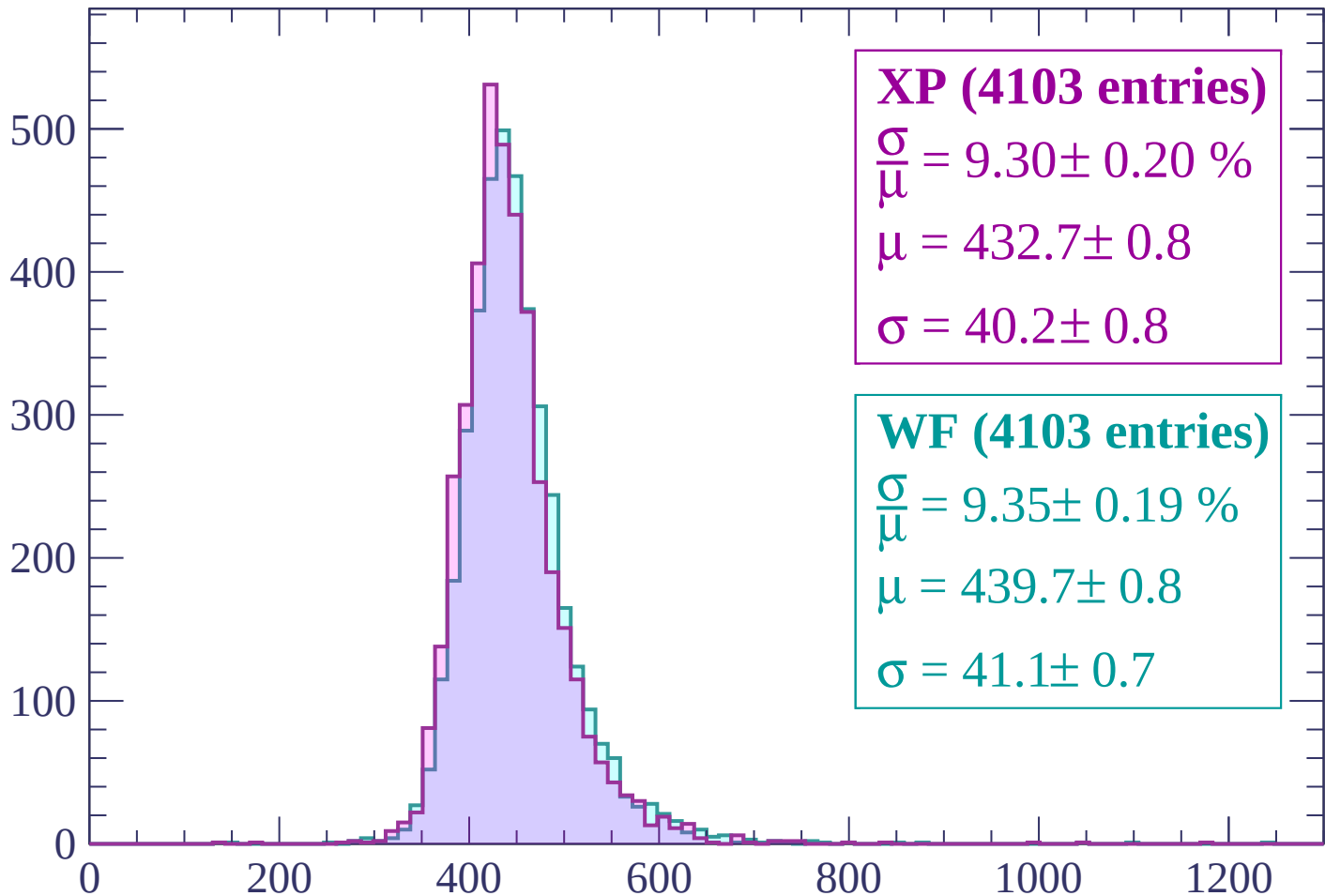
Energy loss | $-840 < p < -780$

Count



Energy loss | $-780 < p < -720$

Count



XP (4103 entries)

$$\frac{\sigma}{\mu} = 9.30 \pm 0.20 \%$$

$$\mu = 432.7 \pm 0.8$$

$$\sigma = 40.2 \pm 0.8$$

WF (4103 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.19 \%$$

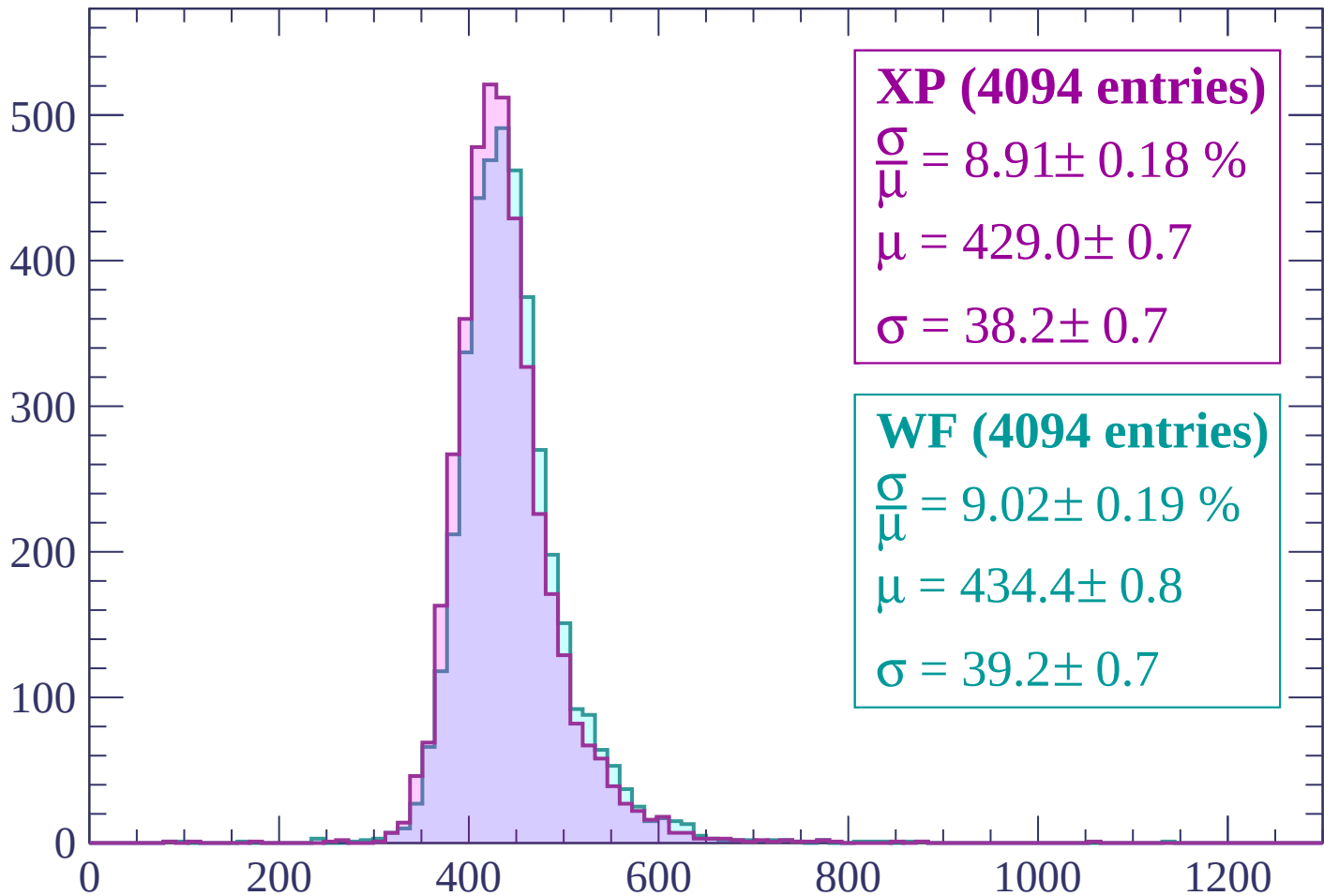
$$\mu = 439.7 \pm 0.8$$

$$\sigma = 41.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (4094 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.18 \%$$

$$\mu = 429.0 \pm 0.7$$

$$\sigma = 38.2 \pm 0.7$$

WF (4094 entries)

$$\frac{\sigma}{\mu} = 9.02 \pm 0.19 \%$$

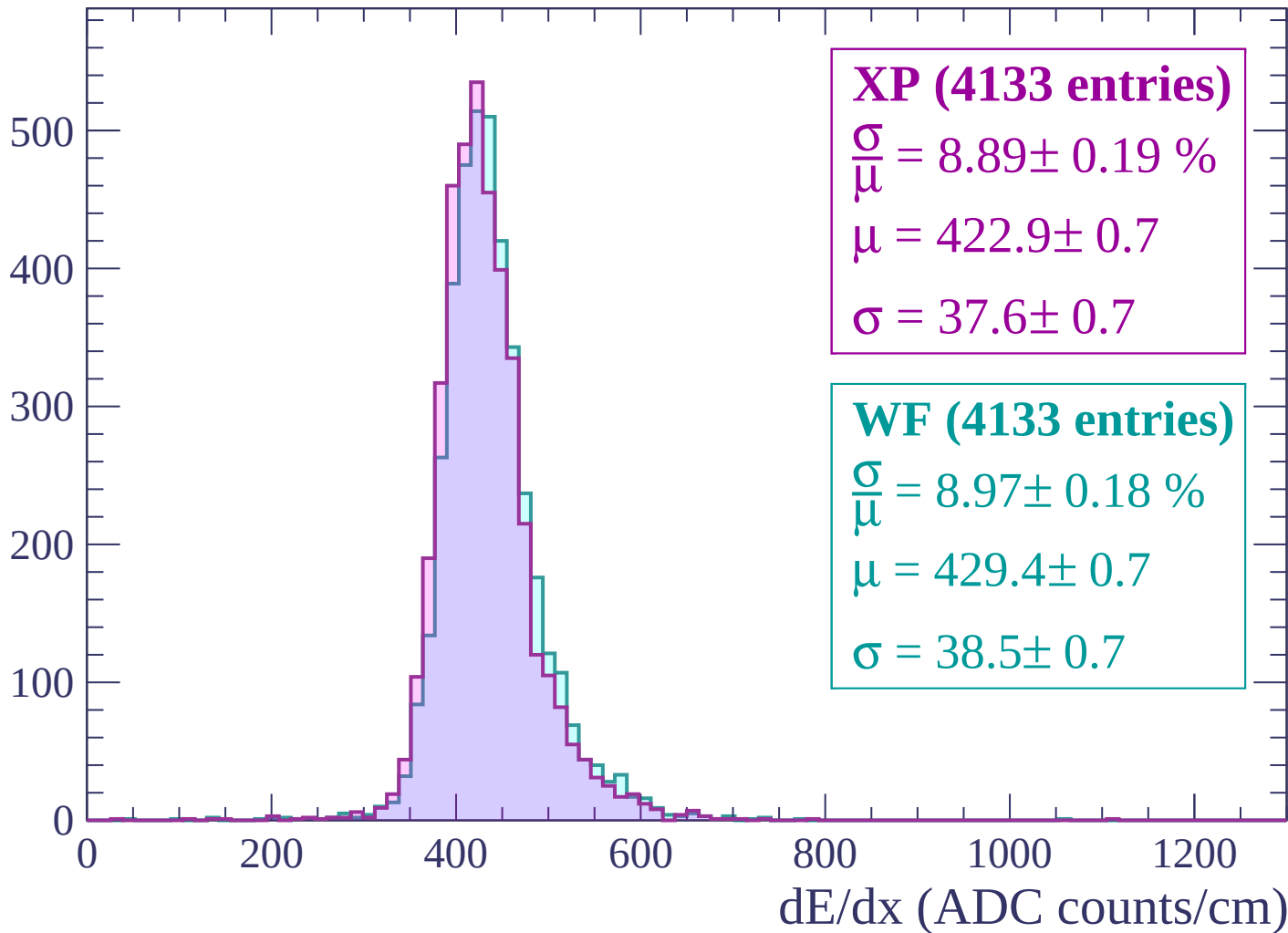
$$\mu = 434.4 \pm 0.8$$

$$\sigma = 39.2 \pm 0.7$$

dE/dx (ADC counts/cm)

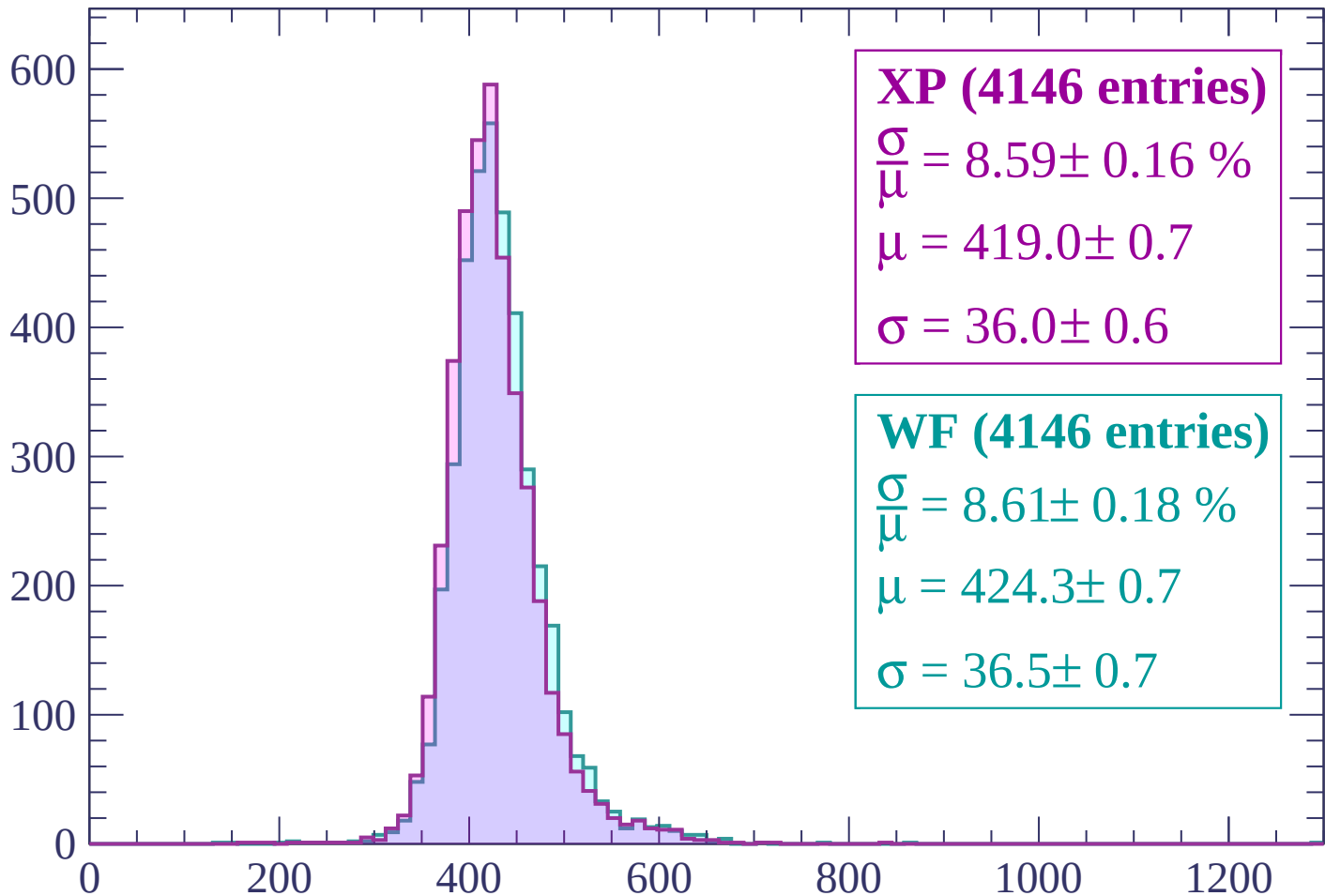
Energy loss | $-660 < p < -600$

Count



Energy loss | $-600 < p < -540$

Count



XP (4146 entries)

$$\frac{\sigma}{\mu} = 8.59 \pm 0.16 \%$$

$$\mu = 419.0 \pm 0.7$$

$$\sigma = 36.0 \pm 0.6$$

WF (4146 entries)

$$\frac{\sigma}{\mu} = 8.61 \pm 0.18 \%$$

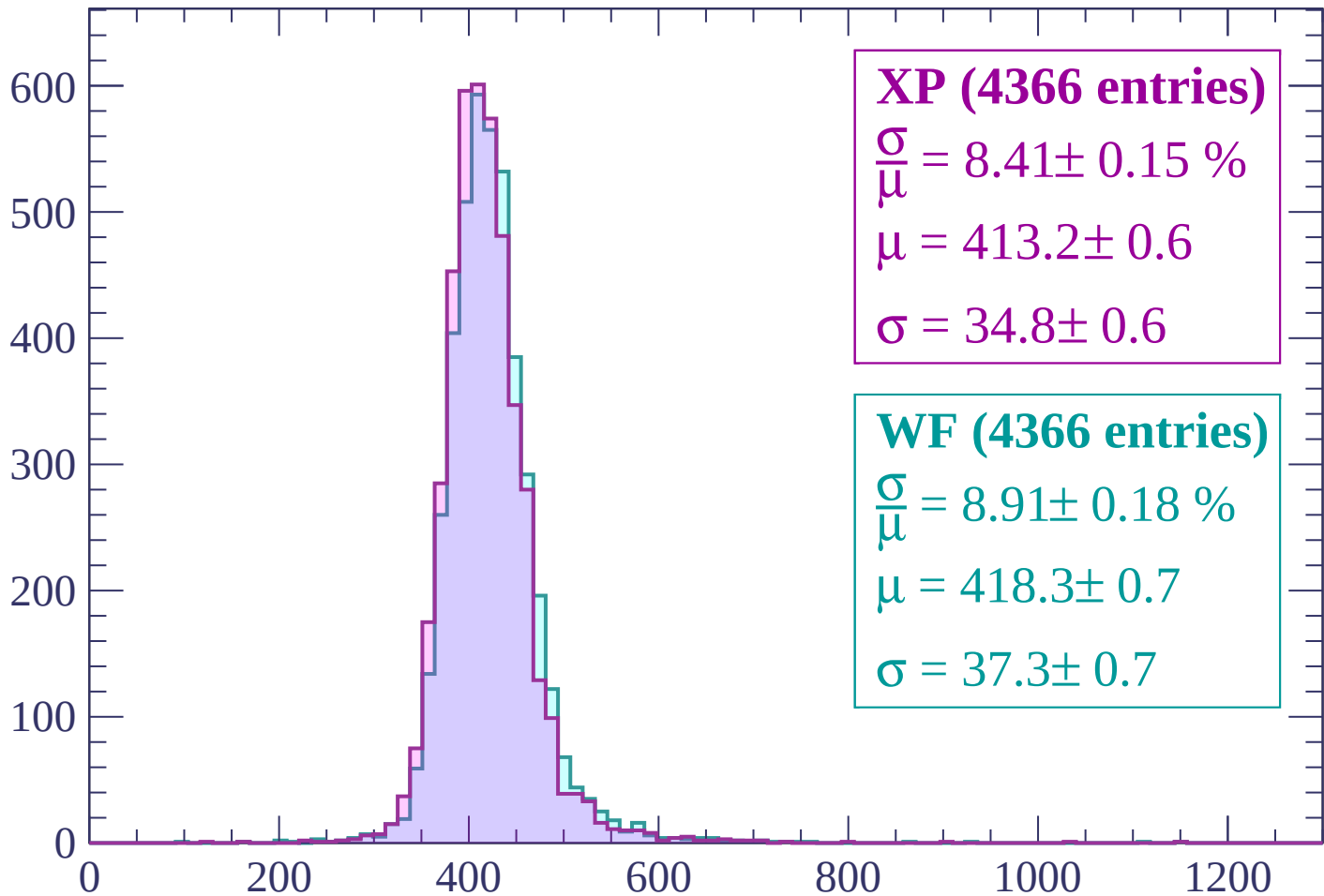
$$\mu = 424.3 \pm 0.7$$

$$\sigma = 36.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP (4366 entries)

$$\frac{\sigma}{\mu} = 8.41 \pm 0.15 \%$$

$$\mu = 413.2 \pm 0.6$$

$$\sigma = 34.8 \pm 0.6$$

WF (4366 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.18 \%$$

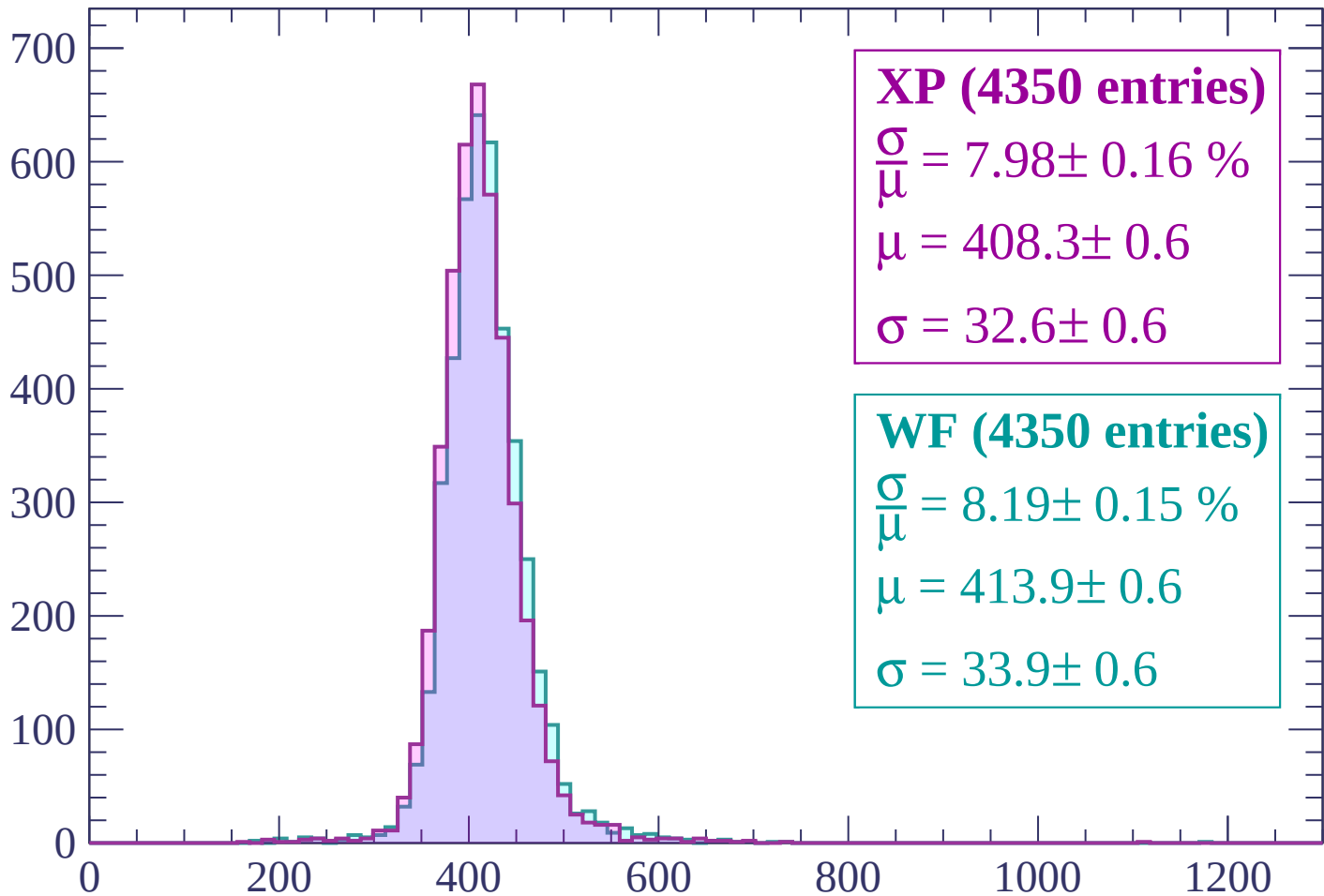
$$\mu = 418.3 \pm 0.7$$

$$\sigma = 37.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



XP (4350 entries)

$$\frac{\sigma}{\mu} = 7.98 \pm 0.16 \%$$

$$\mu = 408.3 \pm 0.6$$

$$\sigma = 32.6 \pm 0.6$$

WF (4350 entries)

$$\frac{\sigma}{\mu} = 8.19 \pm 0.15 \%$$

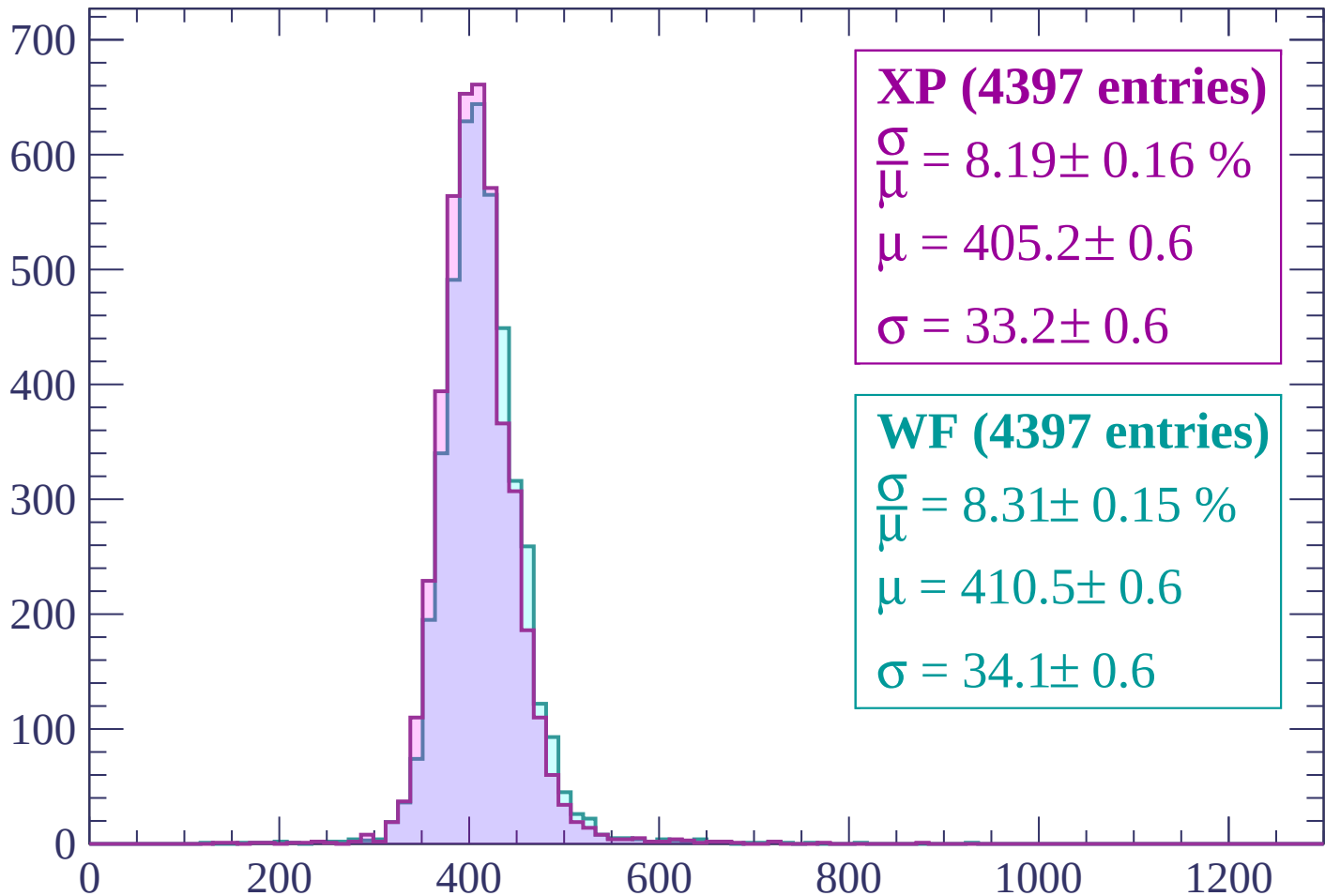
$$\mu = 413.9 \pm 0.6$$

$$\sigma = 33.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count



XP (4397 entries)

$$\frac{\sigma}{\mu} = 8.19 \pm 0.16 \%$$

$$\mu = 405.2 \pm 0.6$$

$$\sigma = 33.2 \pm 0.6$$

WF (4397 entries)

$$\frac{\sigma}{\mu} = 8.31 \pm 0.15 \%$$

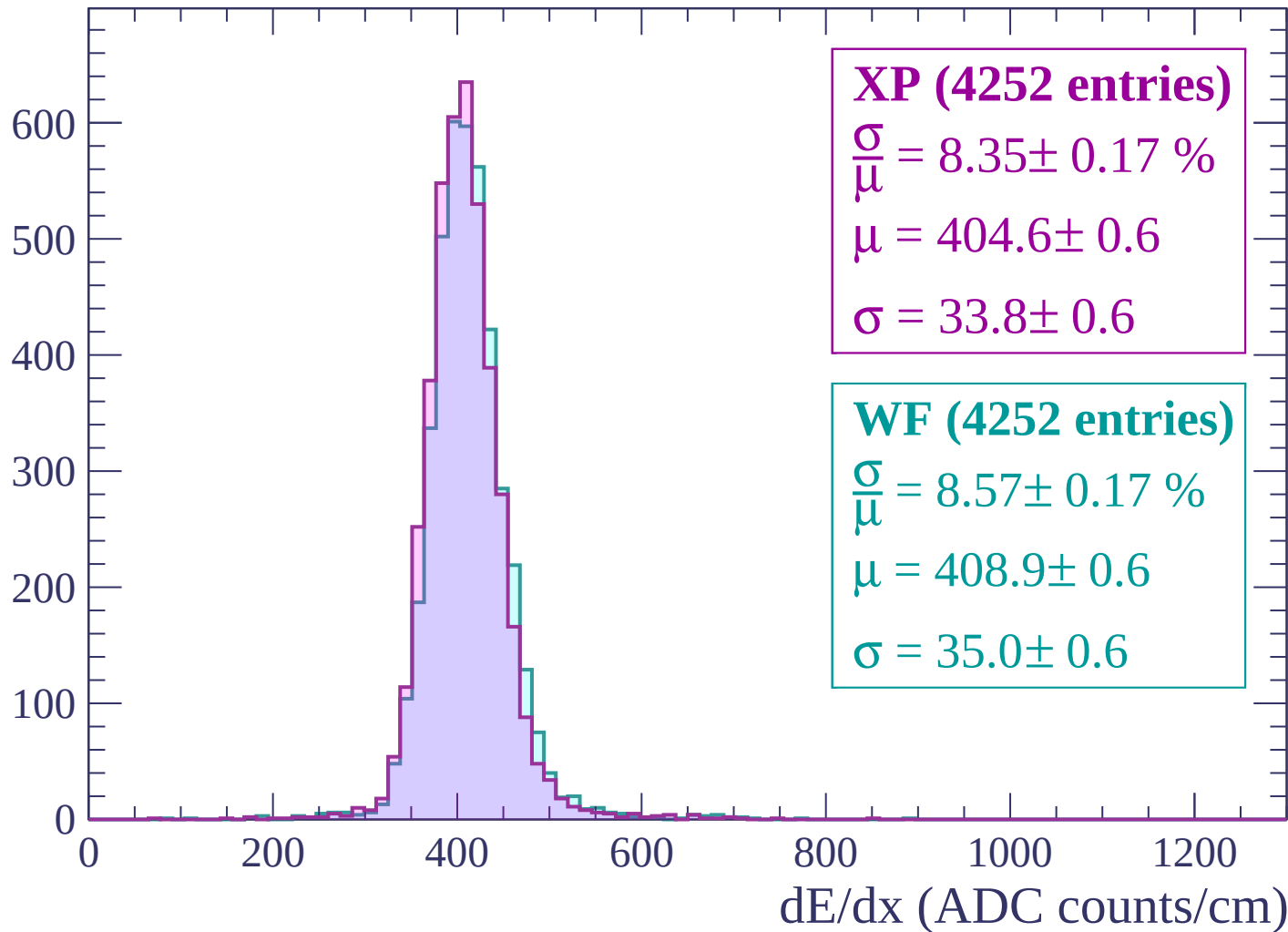
$$\mu = 410.5 \pm 0.6$$

$$\sigma = 34.1 \pm 0.6$$

dE/dx (ADC counts/cm)

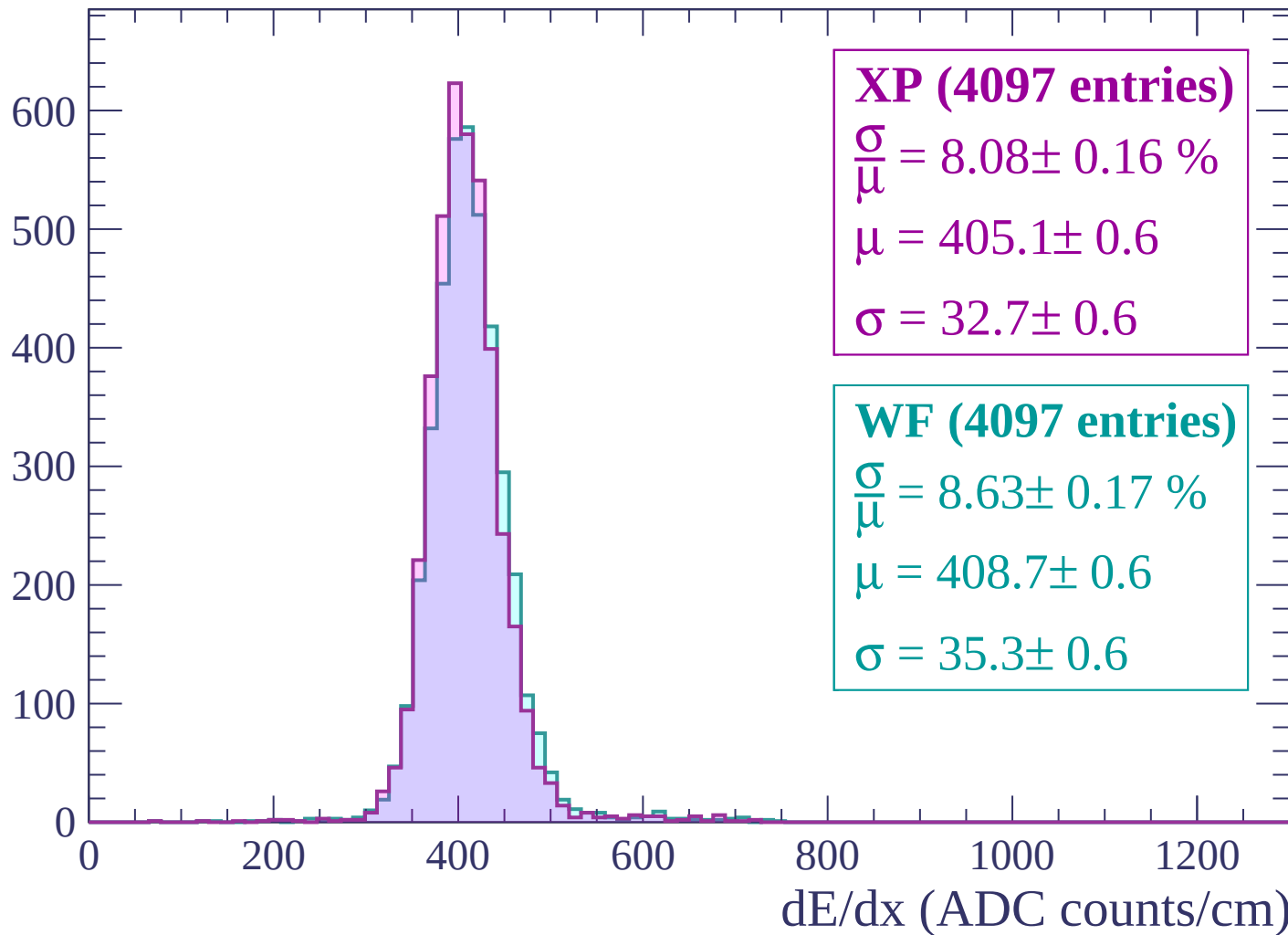
Energy loss | $-360 < p < -300$

Count



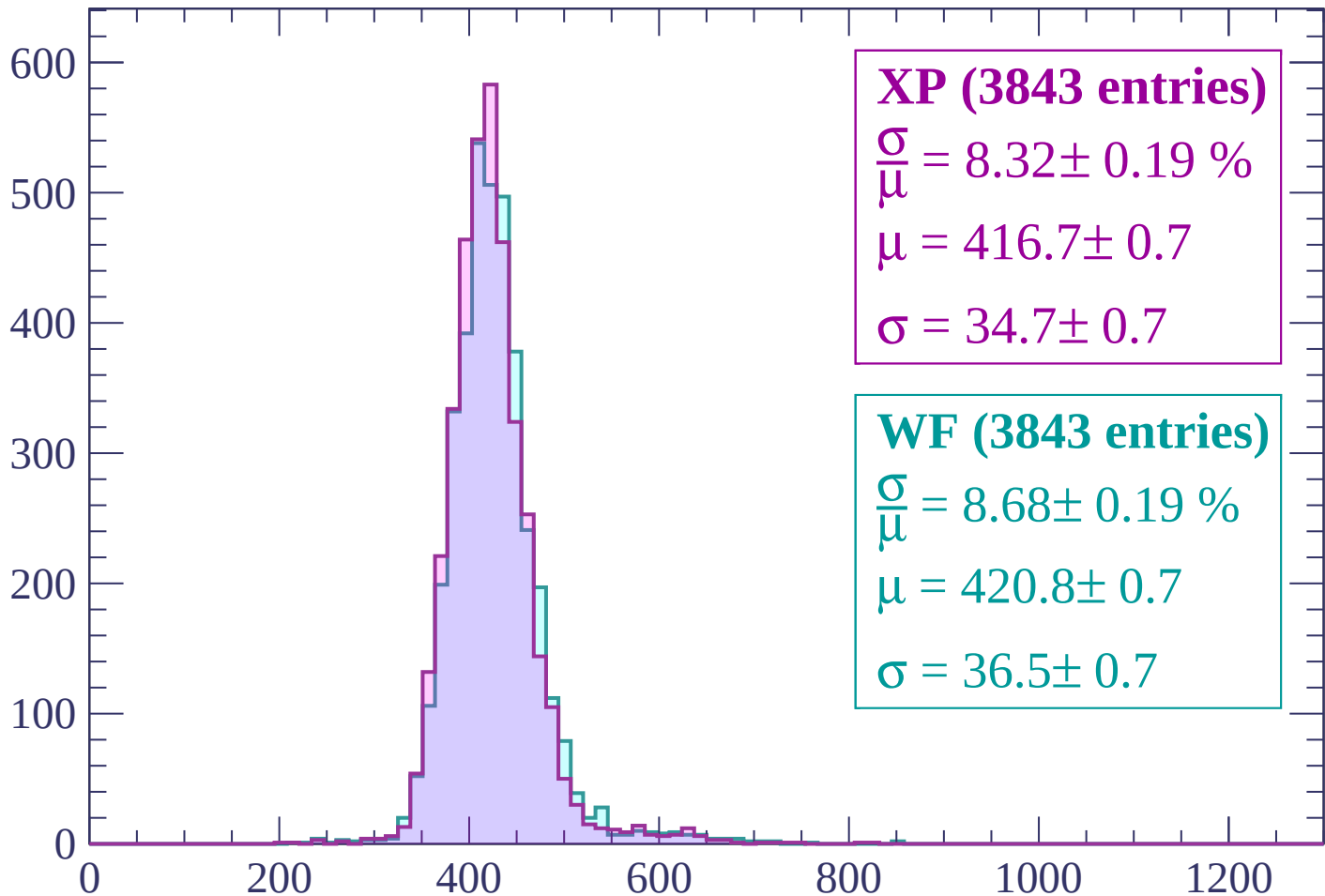
Energy loss | $-300 < p < -240$

Count



Energy loss | $-240 < p < -180$

Count



XP (3843 entries)

$\frac{\sigma}{\mu} = 8.32 \pm 0.19 \%$

$\mu = 416.7 \pm 0.7$

$\sigma = 34.7 \pm 0.7$

WF (3843 entries)

$\frac{\sigma}{\mu} = 8.68 \pm 0.19 \%$

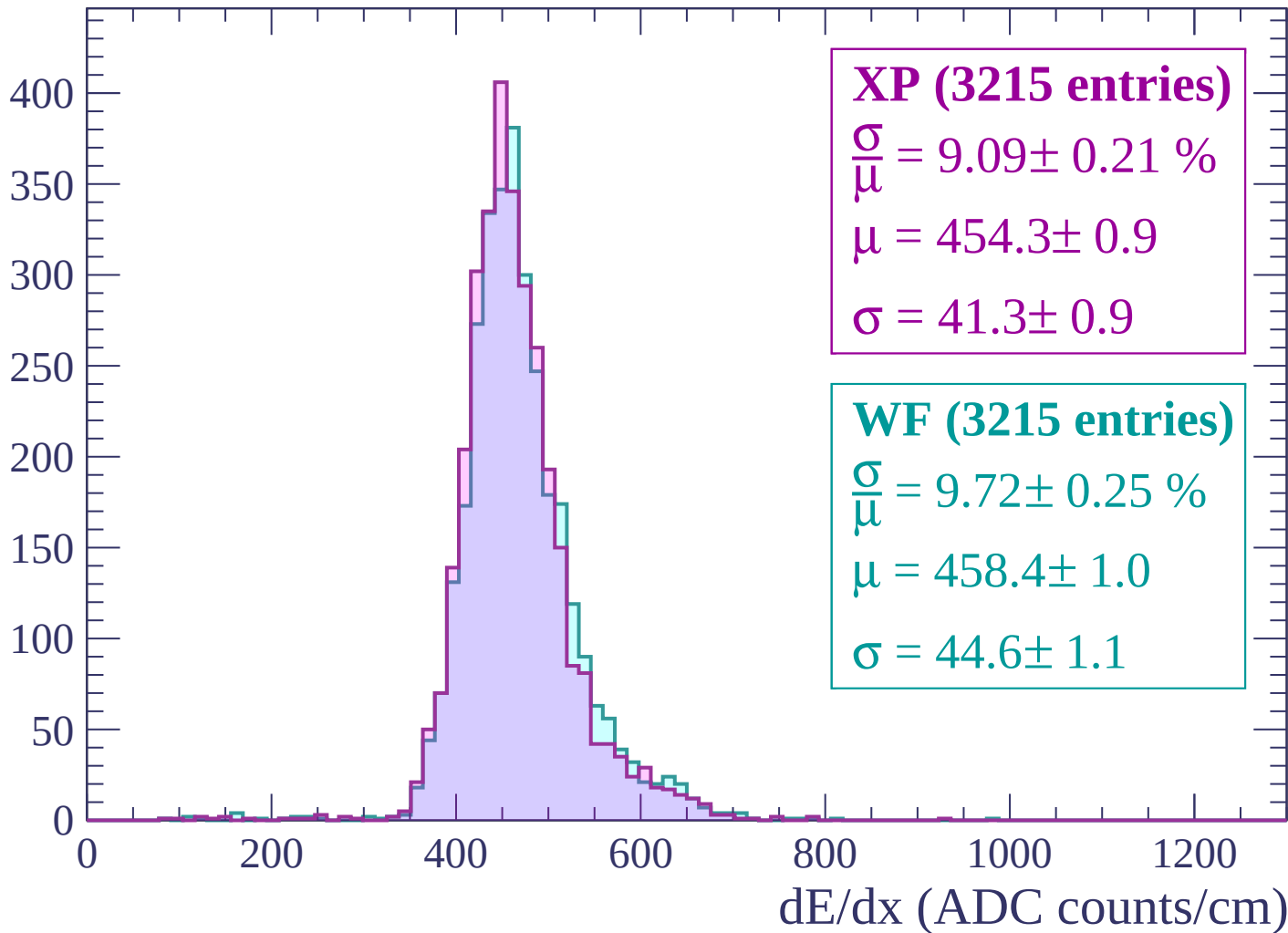
$\mu = 420.8 \pm 0.7$

$\sigma = 36.5 \pm 0.7$

dE/dx (ADC counts/cm)

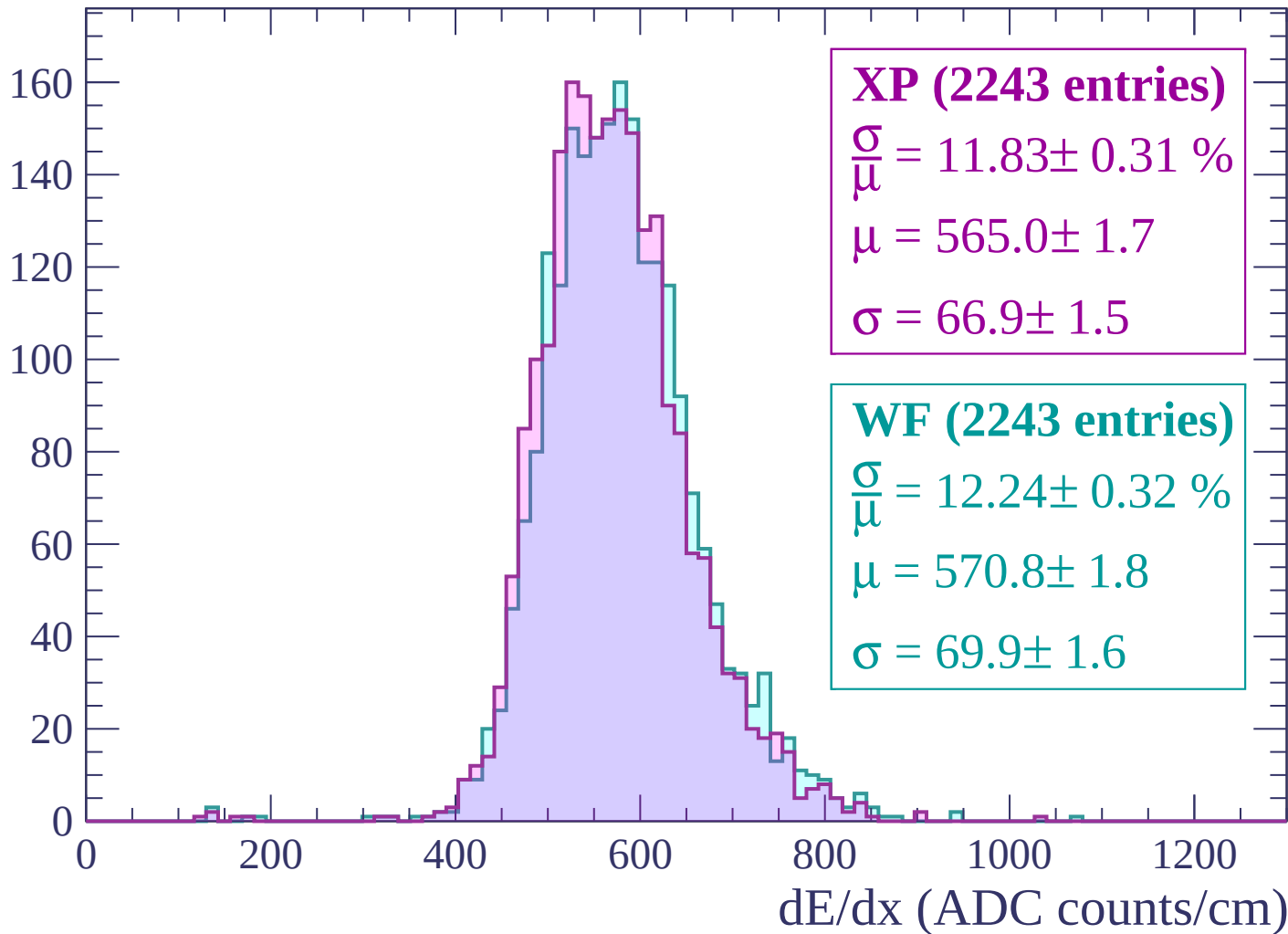
Energy loss | $-180 < p < -120$

Count



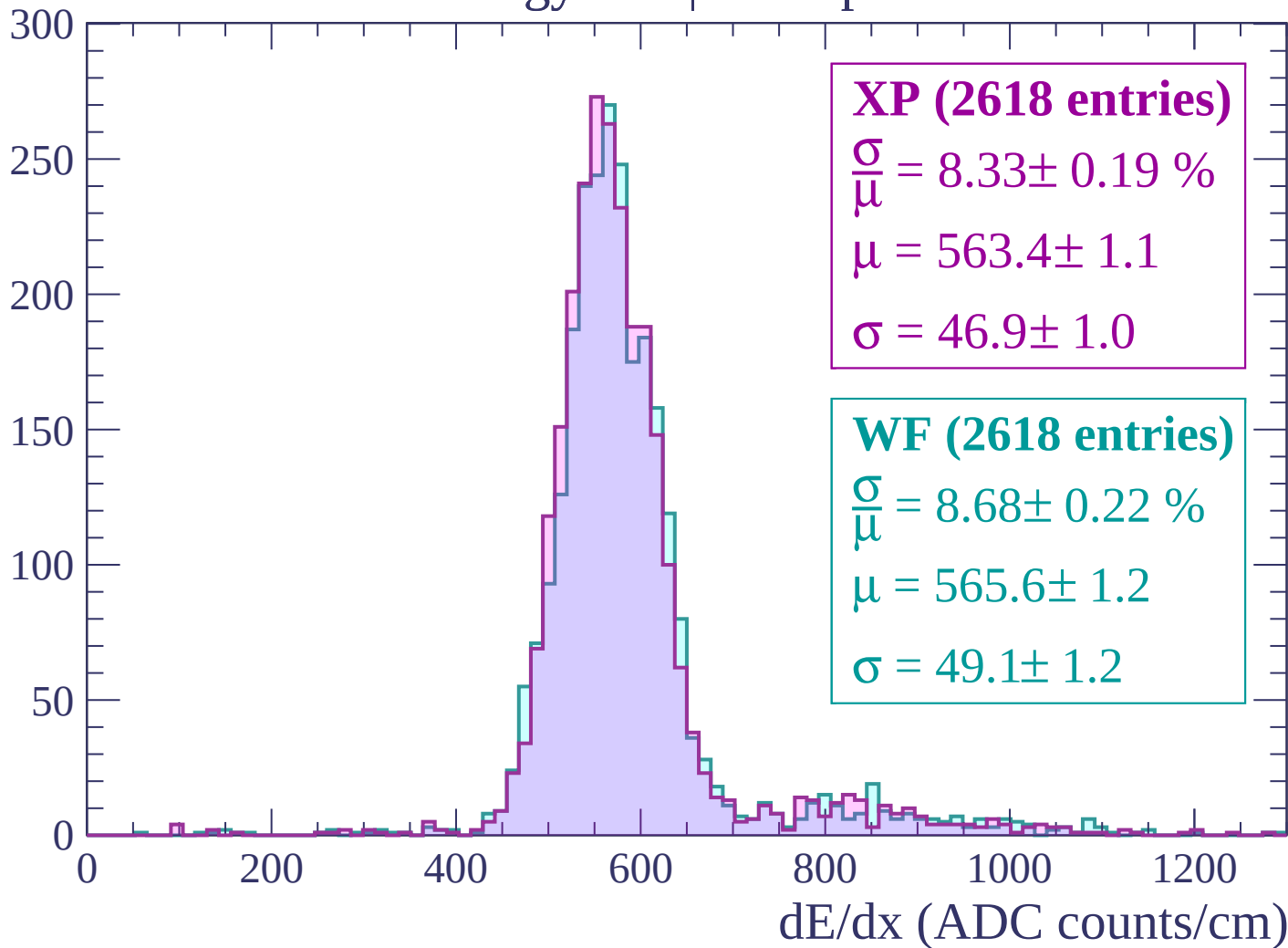
Energy loss | $-120 < p < -60$

Count



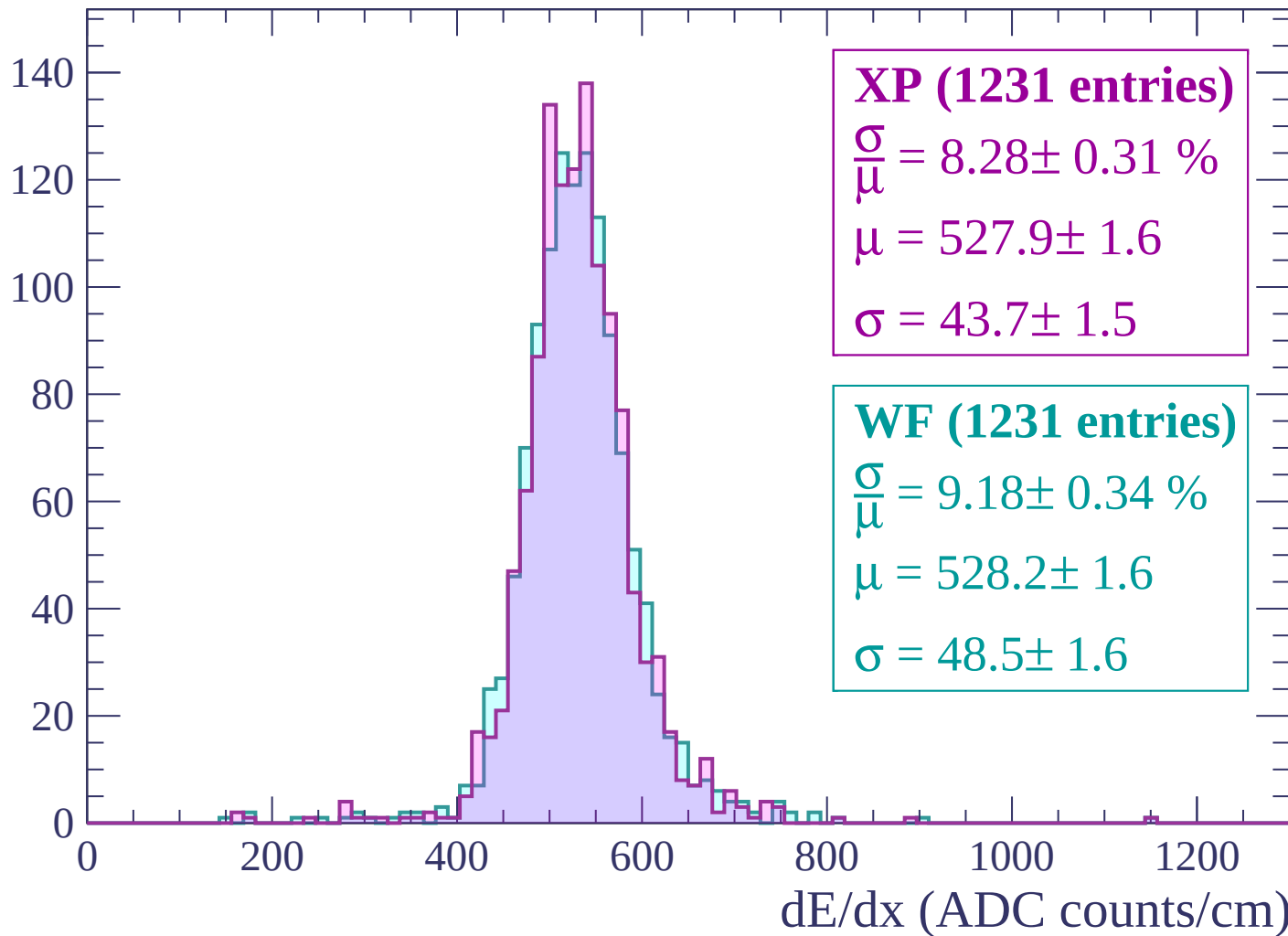
Energy loss | $-60 < p < 0$

Count



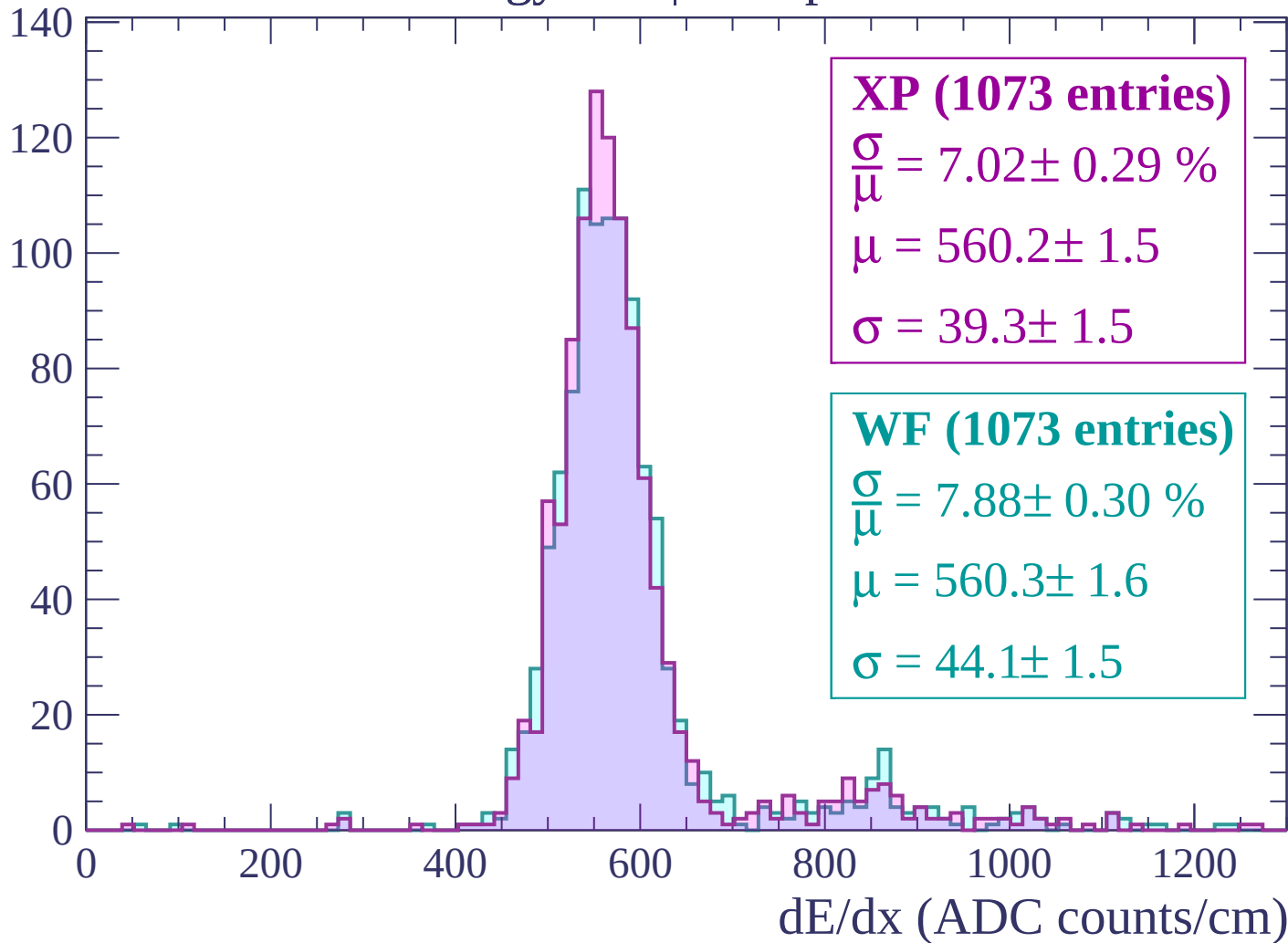
Energy loss | $0 < p < 60$

Count



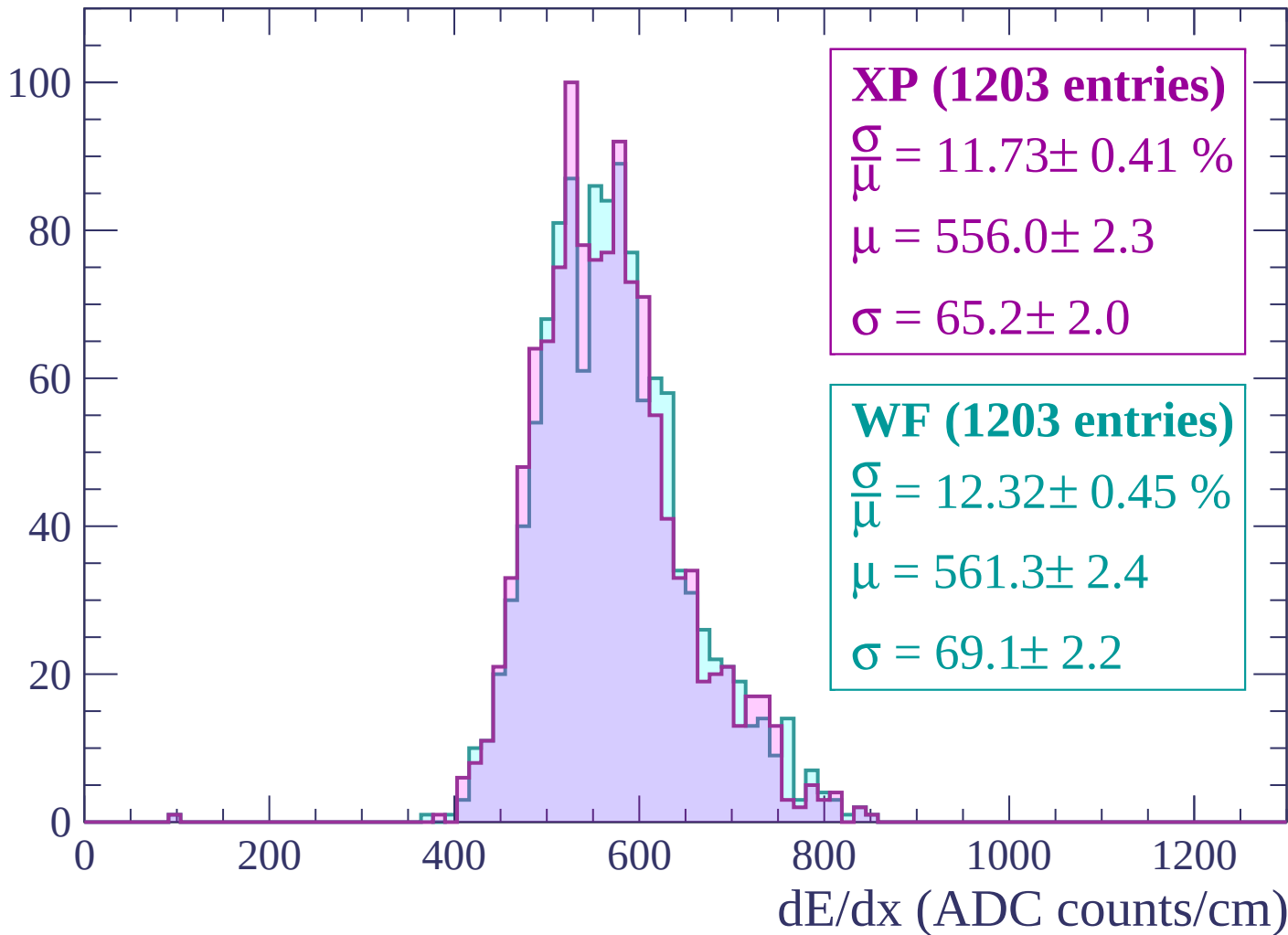
Energy loss | $60 < p < 120$

Count



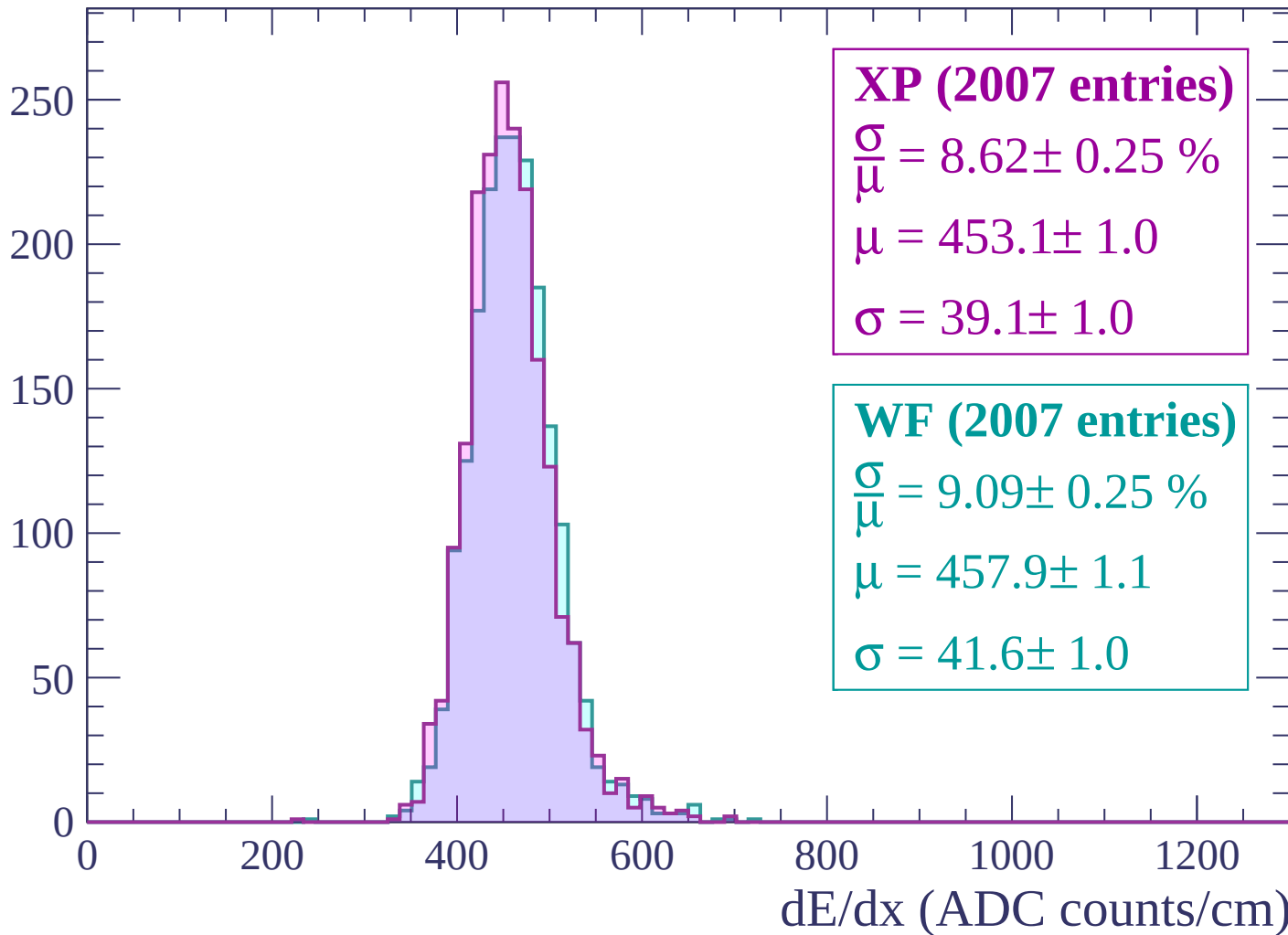
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

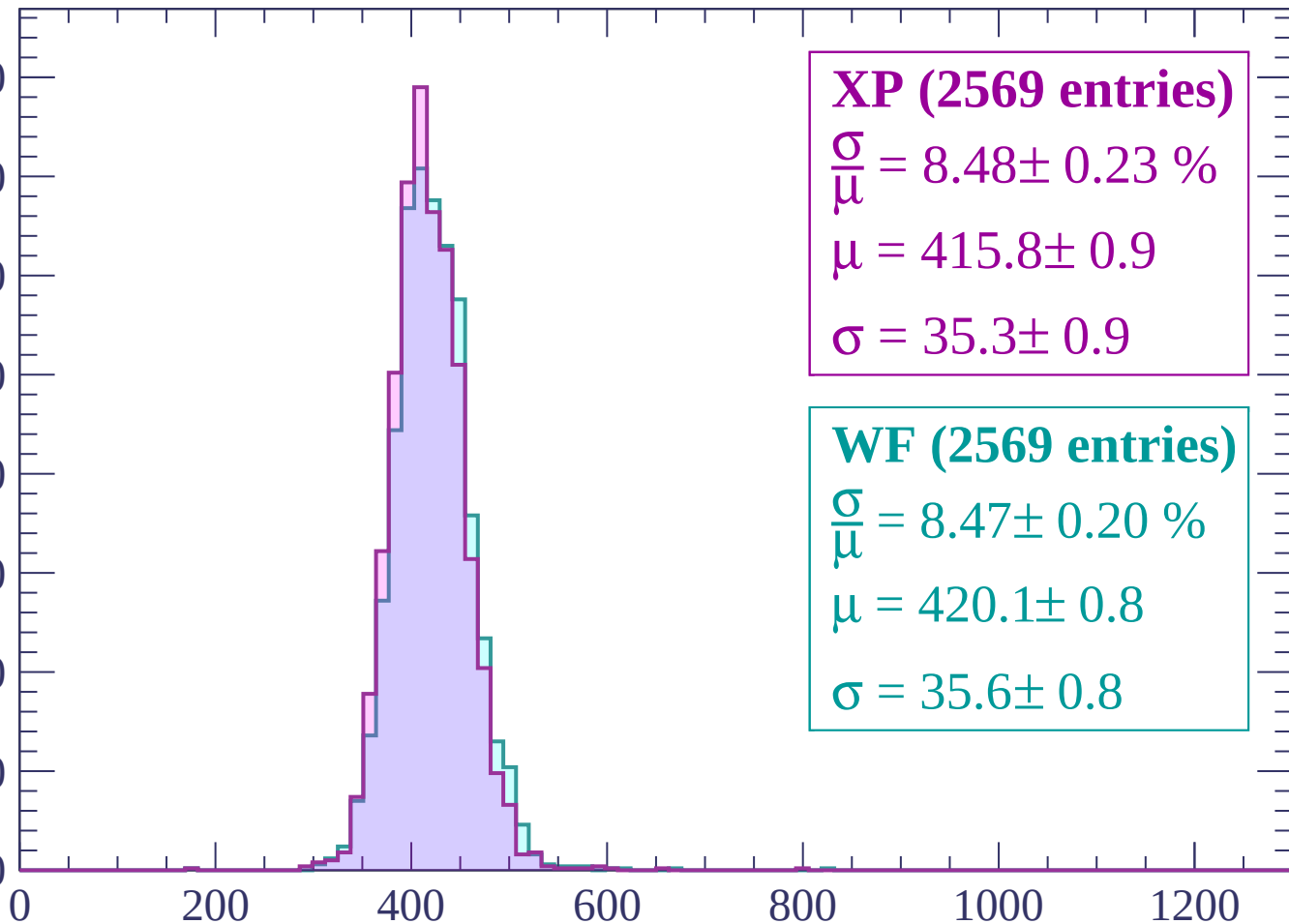
Count



Energy loss | $240 < p < 300$

Count

400
350
300
250
200
150
100
50
0



XP (2569 entries)

$\frac{\sigma}{\mu} = 8.48 \pm 0.23 \%$

$\mu = 415.8 \pm 0.9$

$\sigma = 35.3 \pm 0.9$

WF (2569 entries)

$\frac{\sigma}{\mu} = 8.47 \pm 0.20 \%$

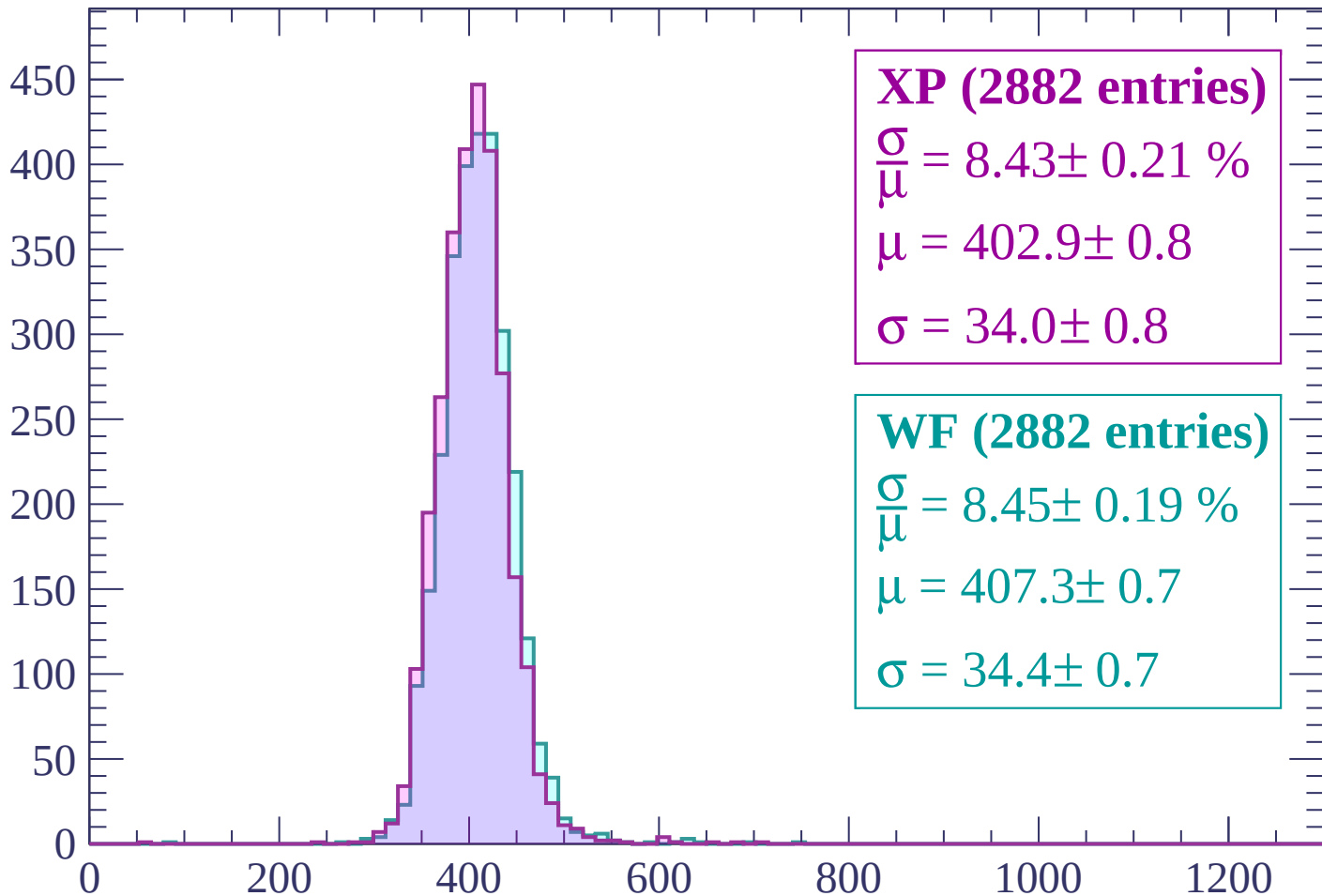
$\mu = 420.1 \pm 0.8$

$\sigma = 35.6 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (2882 entries)

$$\frac{\sigma}{\mu} = 8.43 \pm 0.21 \%$$

$$\mu = 402.9 \pm 0.8$$

$$\sigma = 34.0 \pm 0.8$$

WF (2882 entries)

$$\frac{\sigma}{\mu} = 8.45 \pm 0.19 \%$$

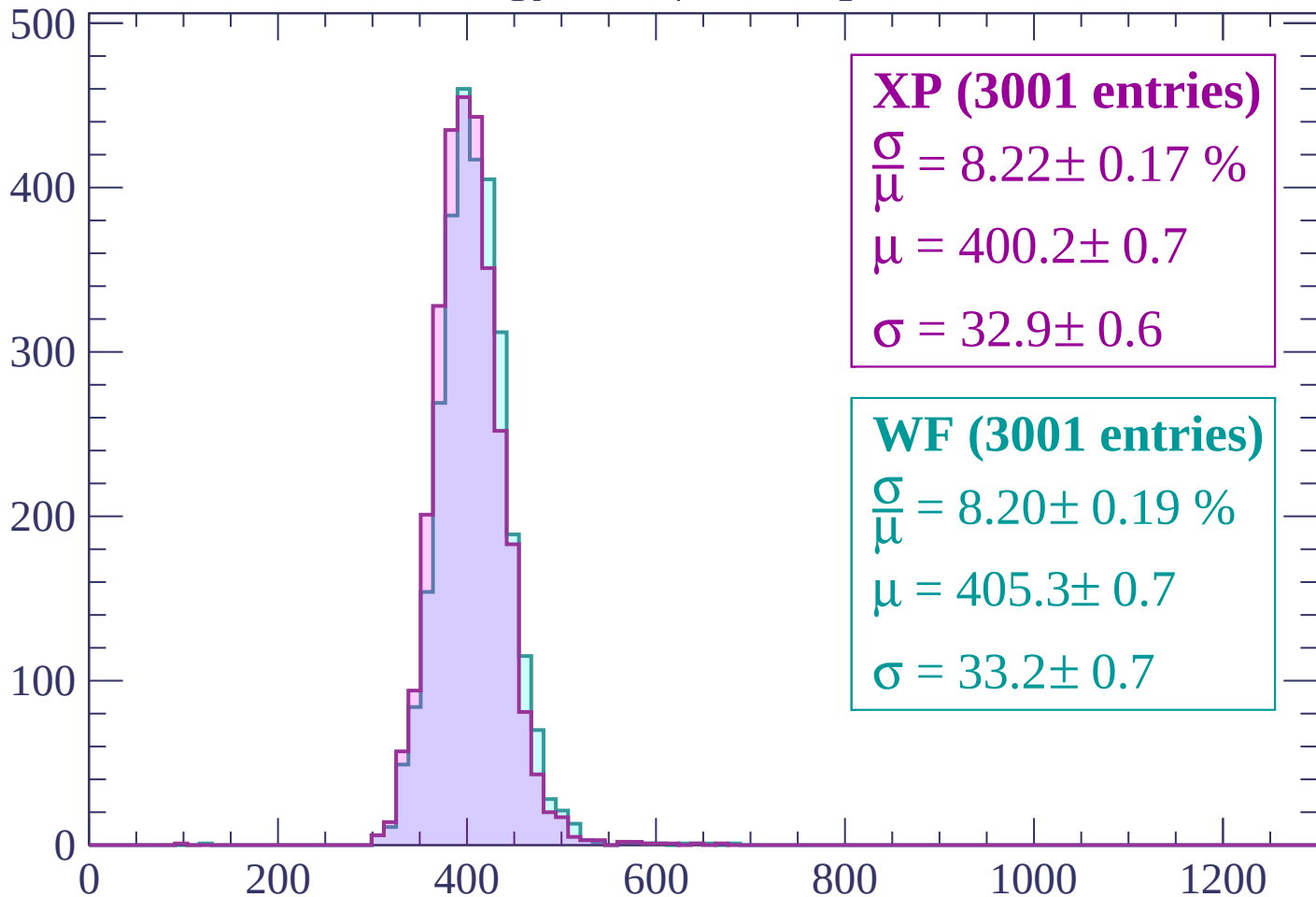
$$\mu = 407.3 \pm 0.7$$

$$\sigma = 34.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (3001 entries)

$$\frac{\sigma}{\mu} = 8.22 \pm 0.17 \%$$

$$\mu = 400.2 \pm 0.7$$

$$\sigma = 32.9 \pm 0.6$$

WF (3001 entries)

$$\frac{\sigma}{\mu} = 8.20 \pm 0.19 \%$$

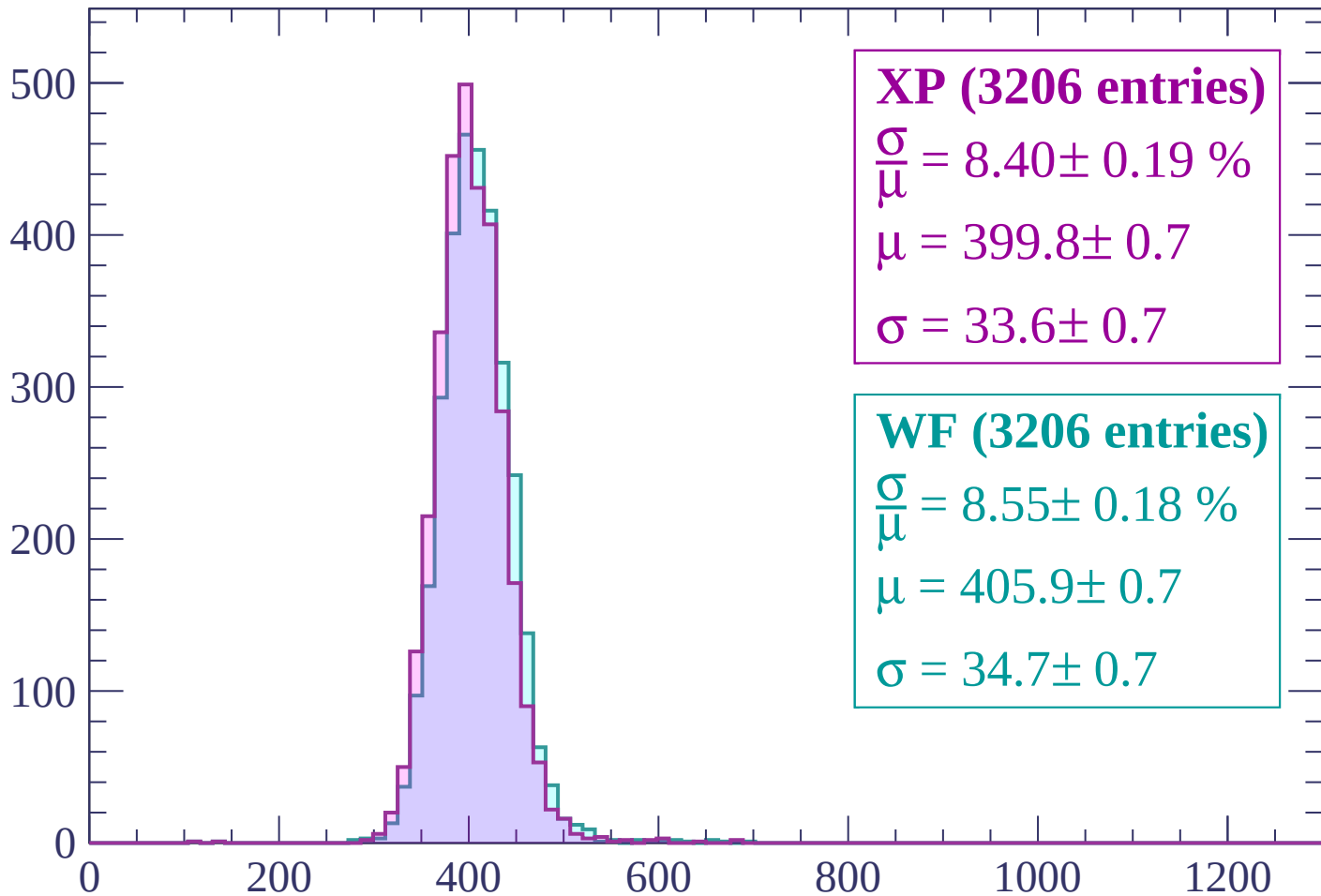
$$\mu = 405.3 \pm 0.7$$

$$\sigma = 33.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (3206 entries)

$$\frac{\sigma}{\mu} = 8.40 \pm 0.19 \%$$

$$\mu = 399.8 \pm 0.7$$

$$\sigma = 33.6 \pm 0.7$$

WF (3206 entries)

$$\frac{\sigma}{\mu} = 8.55 \pm 0.18 \%$$

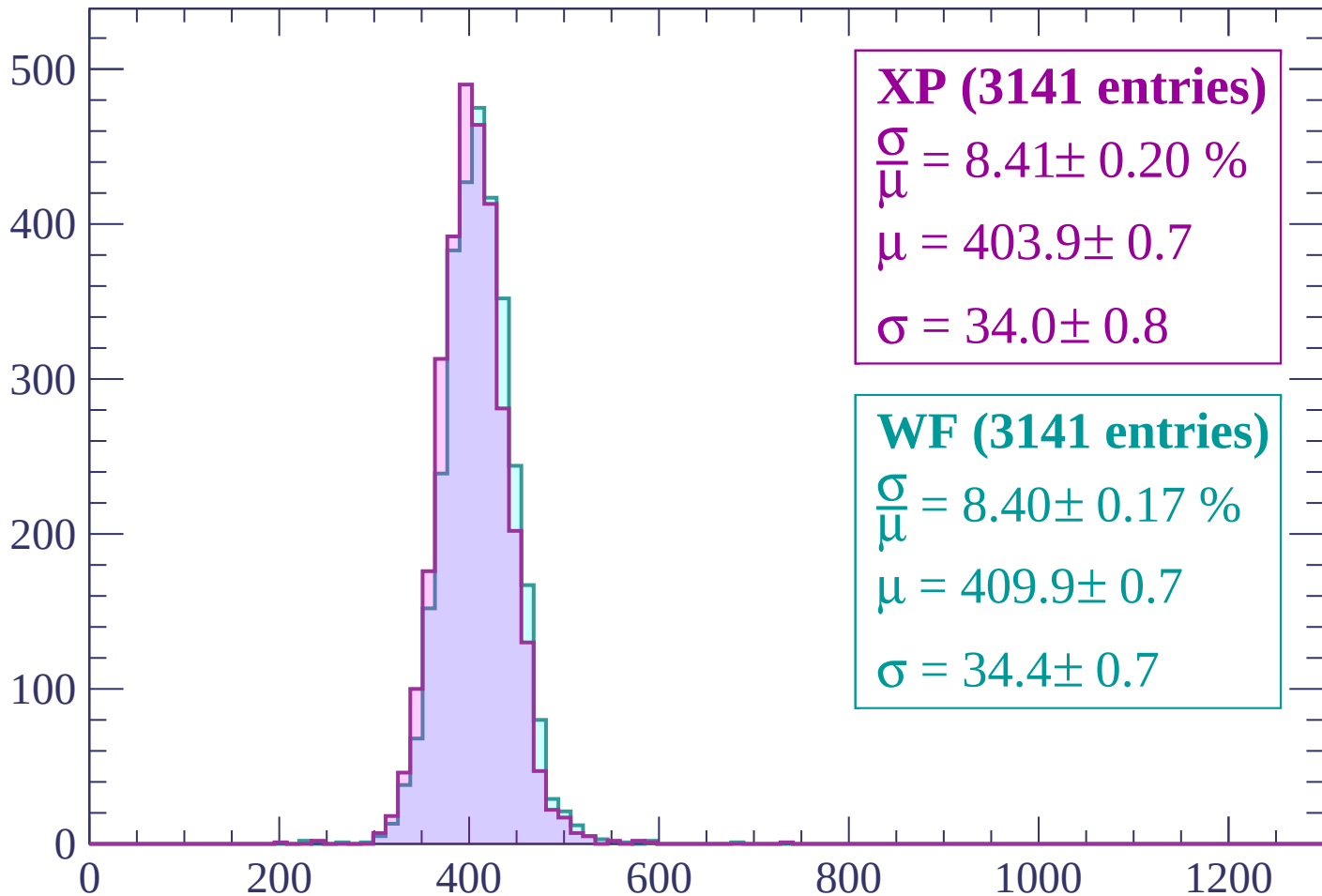
$$\mu = 405.9 \pm 0.7$$

$$\sigma = 34.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (3141 entries)

$$\frac{\sigma}{\mu} = 8.41 \pm 0.20 \%$$

$$\mu = 403.9 \pm 0.7$$

$$\sigma = 34.0 \pm 0.8$$

WF (3141 entries)

$$\frac{\sigma}{\mu} = 8.40 \pm 0.17 \%$$

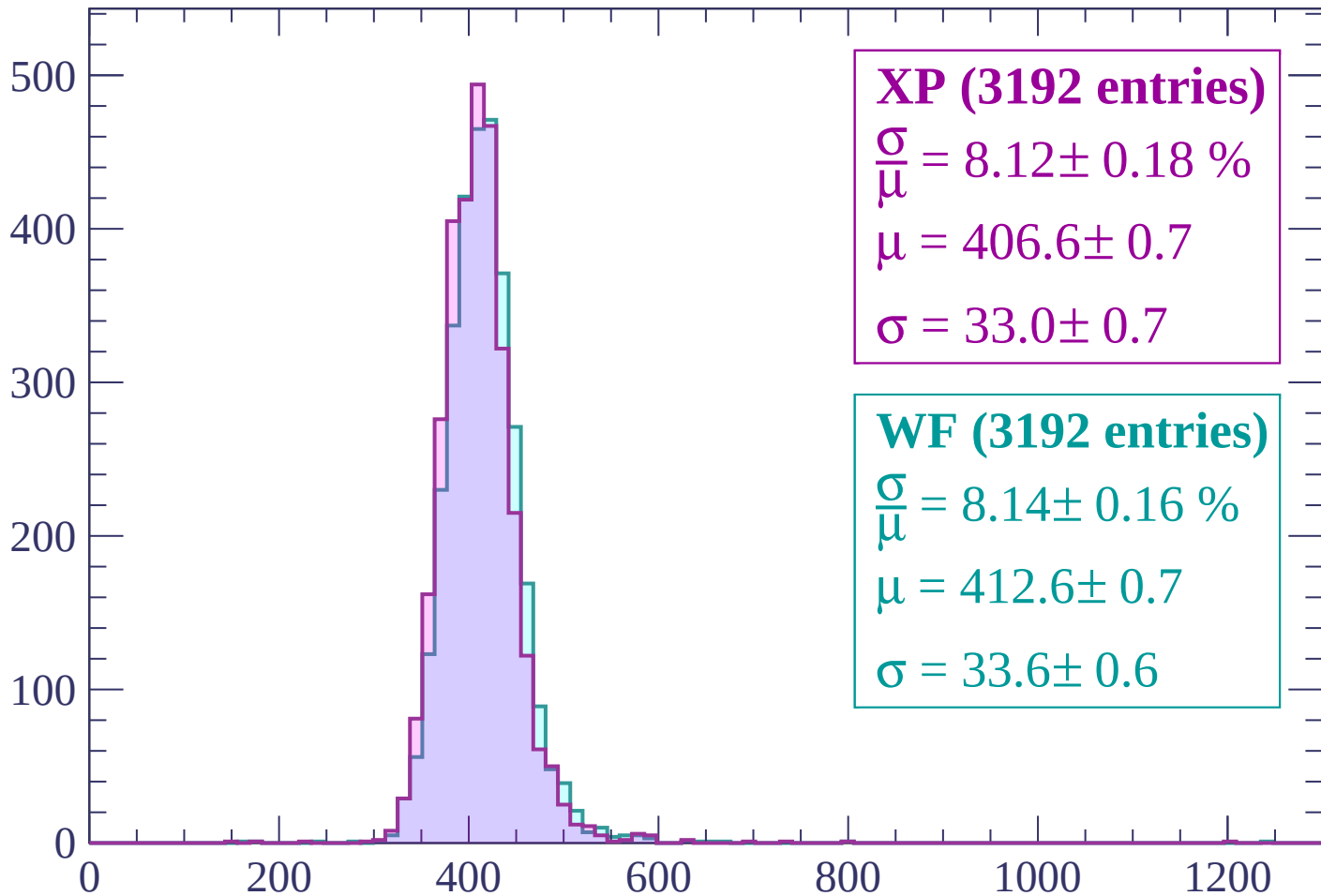
$$\mu = 409.9 \pm 0.7$$

$$\sigma = 34.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (3192 entries)

$$\frac{\sigma}{\mu} = 8.12 \pm 0.18 \%$$

$$\mu = 406.6 \pm 0.7$$

$$\sigma = 33.0 \pm 0.7$$

WF (3192 entries)

$$\frac{\sigma}{\mu} = 8.14 \pm 0.16 \%$$

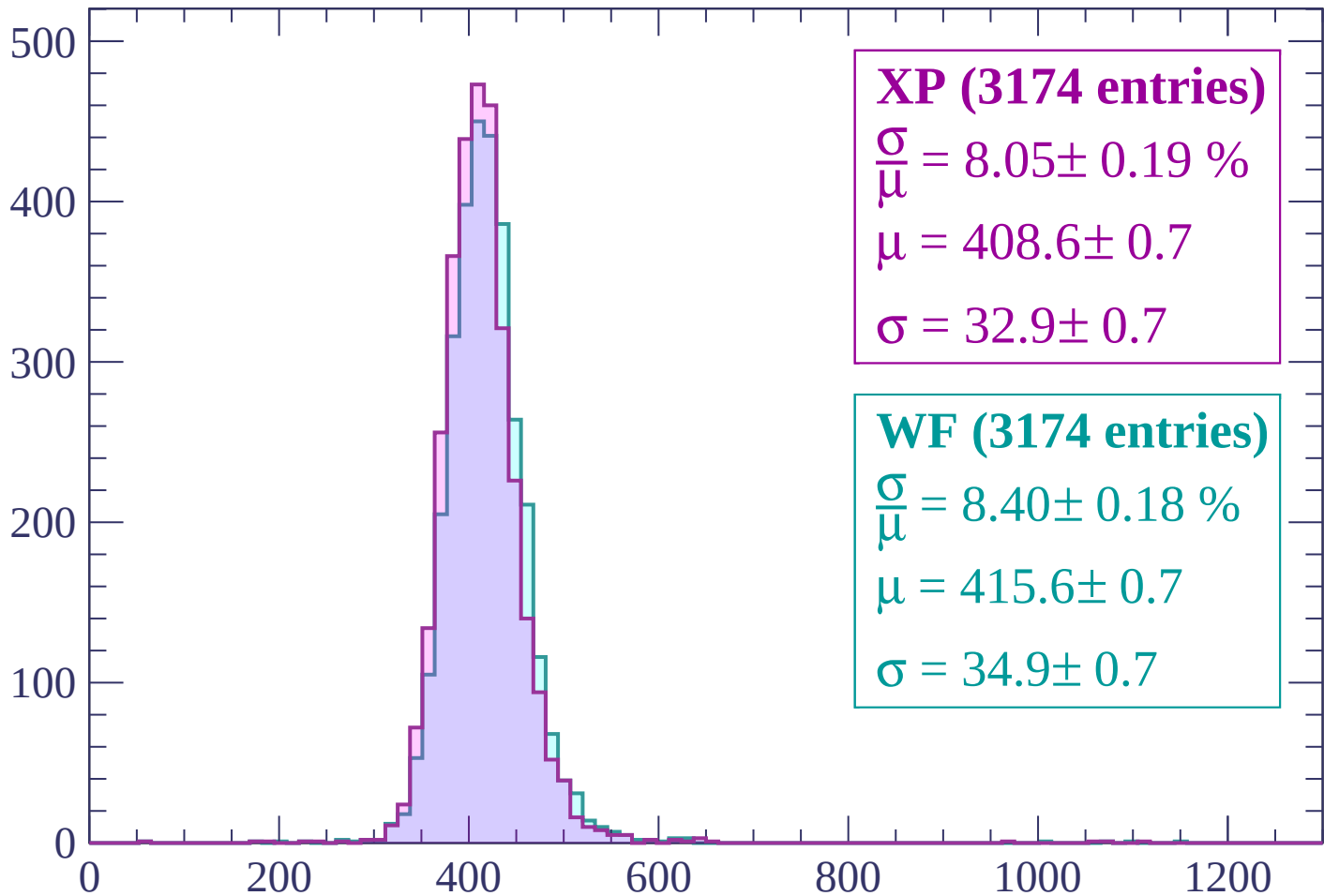
$$\mu = 412.6 \pm 0.7$$

$$\sigma = 33.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (3174 entries)

$\frac{\sigma}{\mu} = 8.05 \pm 0.19 \%$

$\mu = 408.6 \pm 0.7$

$\sigma = 32.9 \pm 0.7$

WF (3174 entries)

$\frac{\sigma}{\mu} = 8.40 \pm 0.18 \%$

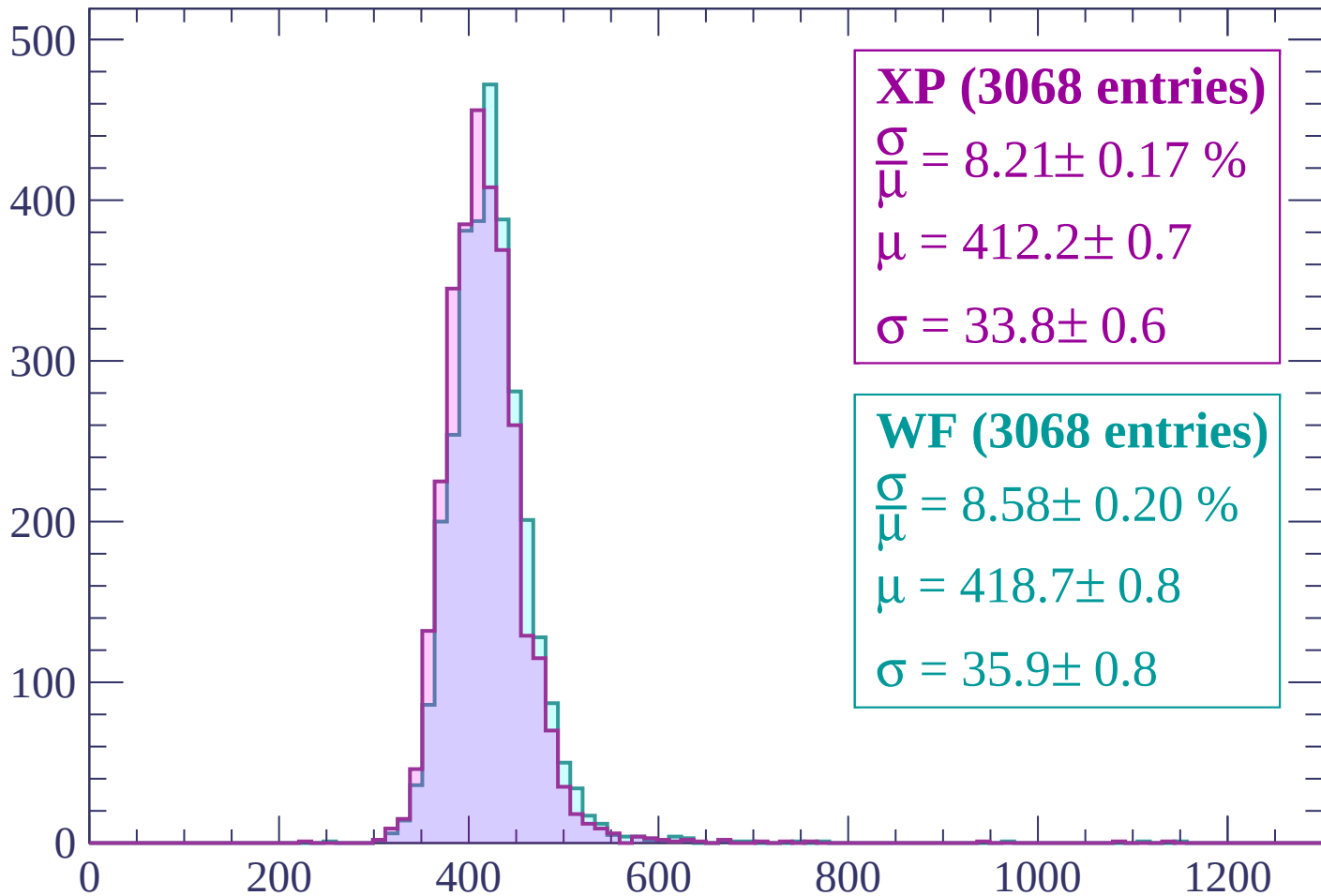
$\mu = 415.6 \pm 0.7$

$\sigma = 34.9 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (3068 entries)

$\frac{\sigma}{\mu} = 8.21 \pm 0.17 \%$

$\mu = 412.2 \pm 0.7$

$\sigma = 33.8 \pm 0.6$

WF (3068 entries)

$\frac{\sigma}{\mu} = 8.58 \pm 0.20 \%$

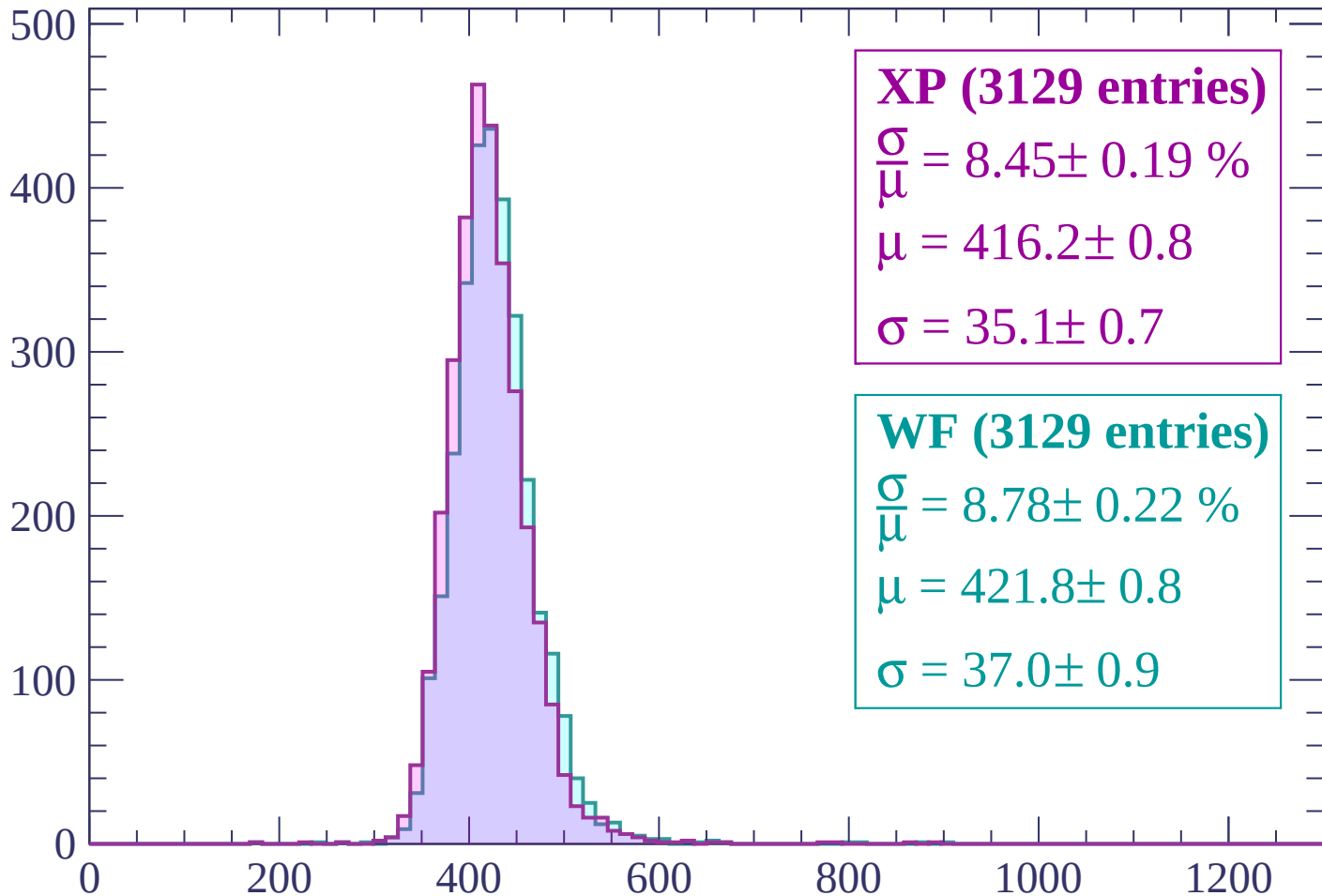
$\mu = 418.7 \pm 0.8$

$\sigma = 35.9 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (3129 entries)

$\frac{\sigma}{\mu} = 8.45 \pm 0.19 \%$

$\mu = 416.2 \pm 0.8$

$\sigma = 35.1 \pm 0.7$

WF (3129 entries)

$\frac{\sigma}{\mu} = 8.78 \pm 0.22 \%$

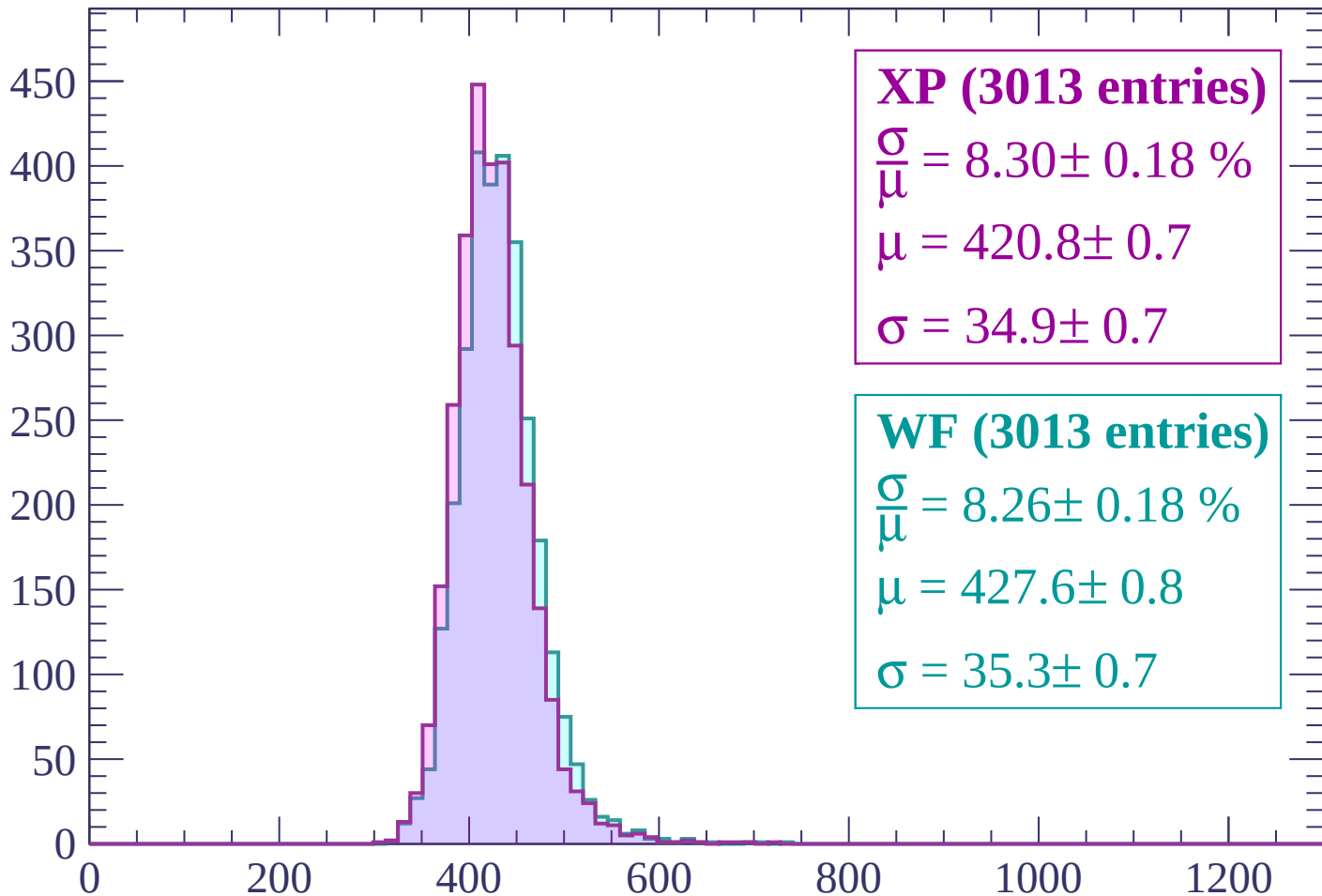
$\mu = 421.8 \pm 0.8$

$\sigma = 37.0 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (3013 entries)

$$\frac{\sigma}{\mu} = 8.30 \pm 0.18 \%$$

$$\mu = 420.8 \pm 0.7$$

$$\sigma = 34.9 \pm 0.7$$

WF (3013 entries)

$$\frac{\sigma}{\mu} = 8.26 \pm 0.18 \%$$

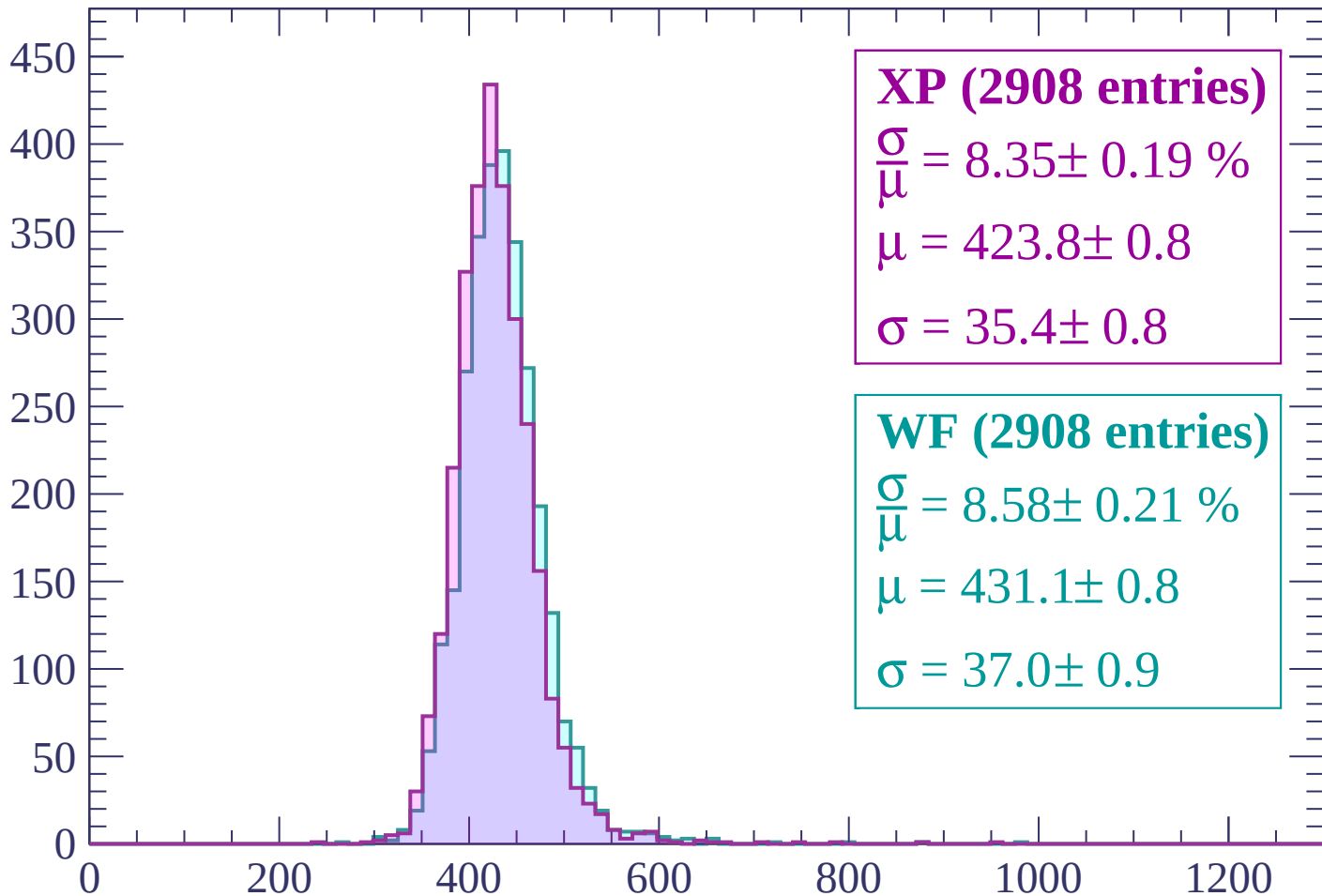
$$\mu = 427.6 \pm 0.8$$

$$\sigma = 35.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (2908 entries)

$$\frac{\sigma}{\mu} = 8.35 \pm 0.19 \%$$

$$\mu = 423.8 \pm 0.8$$

$$\sigma = 35.4 \pm 0.8$$

WF (2908 entries)

$$\frac{\sigma}{\mu} = 8.58 \pm 0.21 \%$$

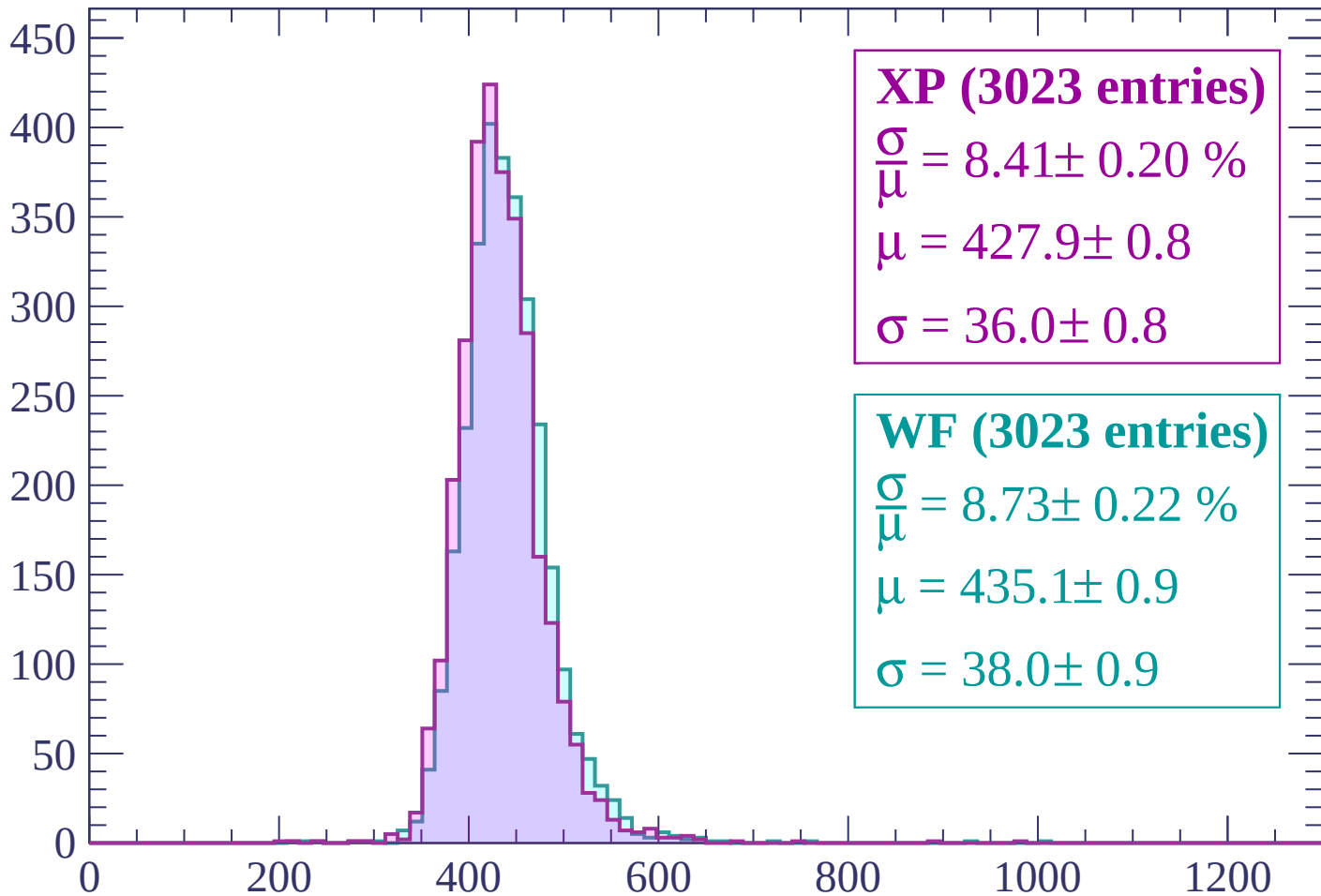
$$\mu = 431.1 \pm 0.8$$

$$\sigma = 37.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (3023 entries)

$$\frac{\sigma}{\mu} = 8.41 \pm 0.20 \%$$

$$\mu = 427.9 \pm 0.8$$

$$\sigma = 36.0 \pm 0.8$$

WF (3023 entries)

$$\frac{\sigma}{\mu} = 8.73 \pm 0.22 \%$$

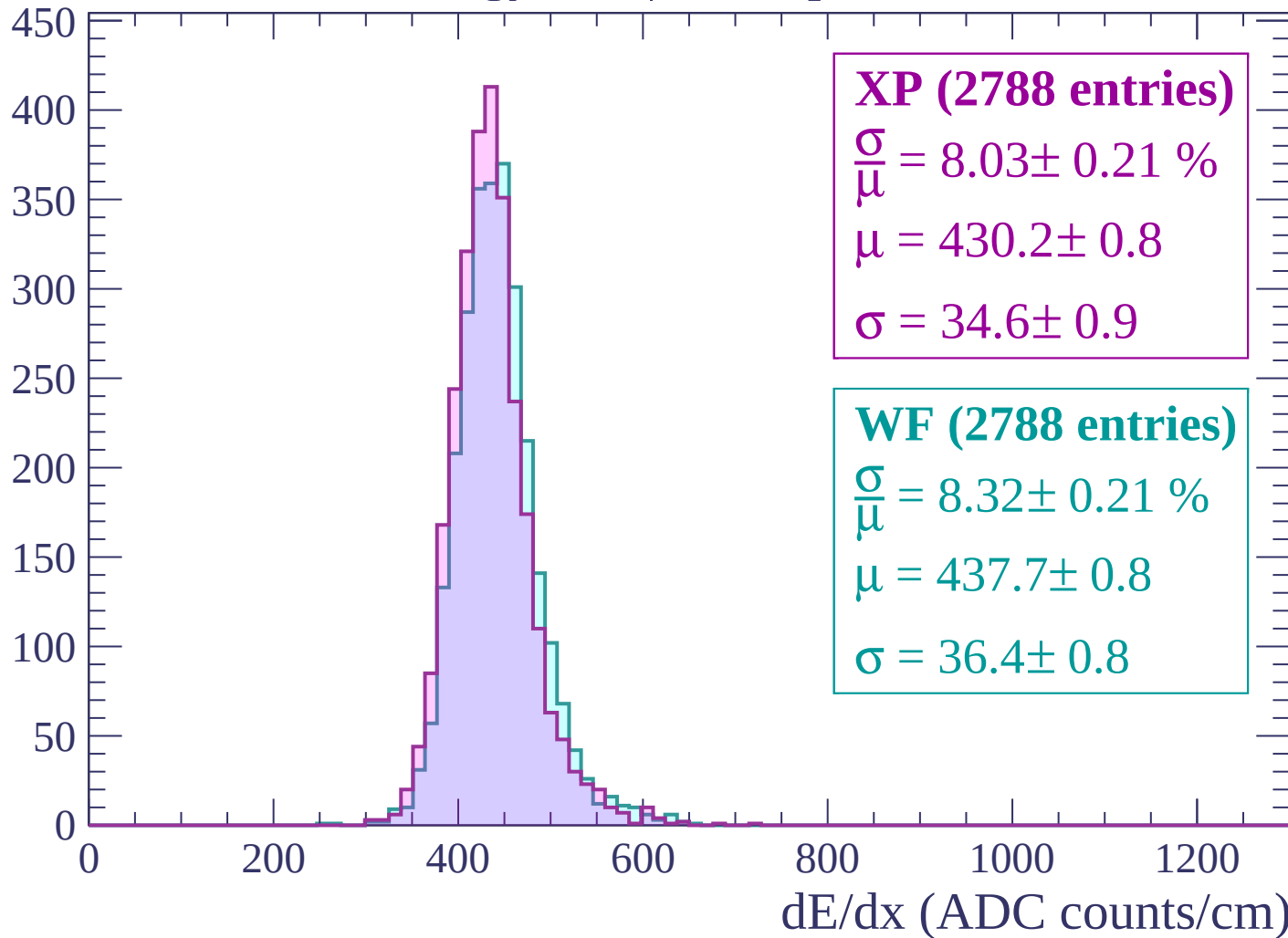
$$\mu = 435.1 \pm 0.9$$

$$\sigma = 38.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

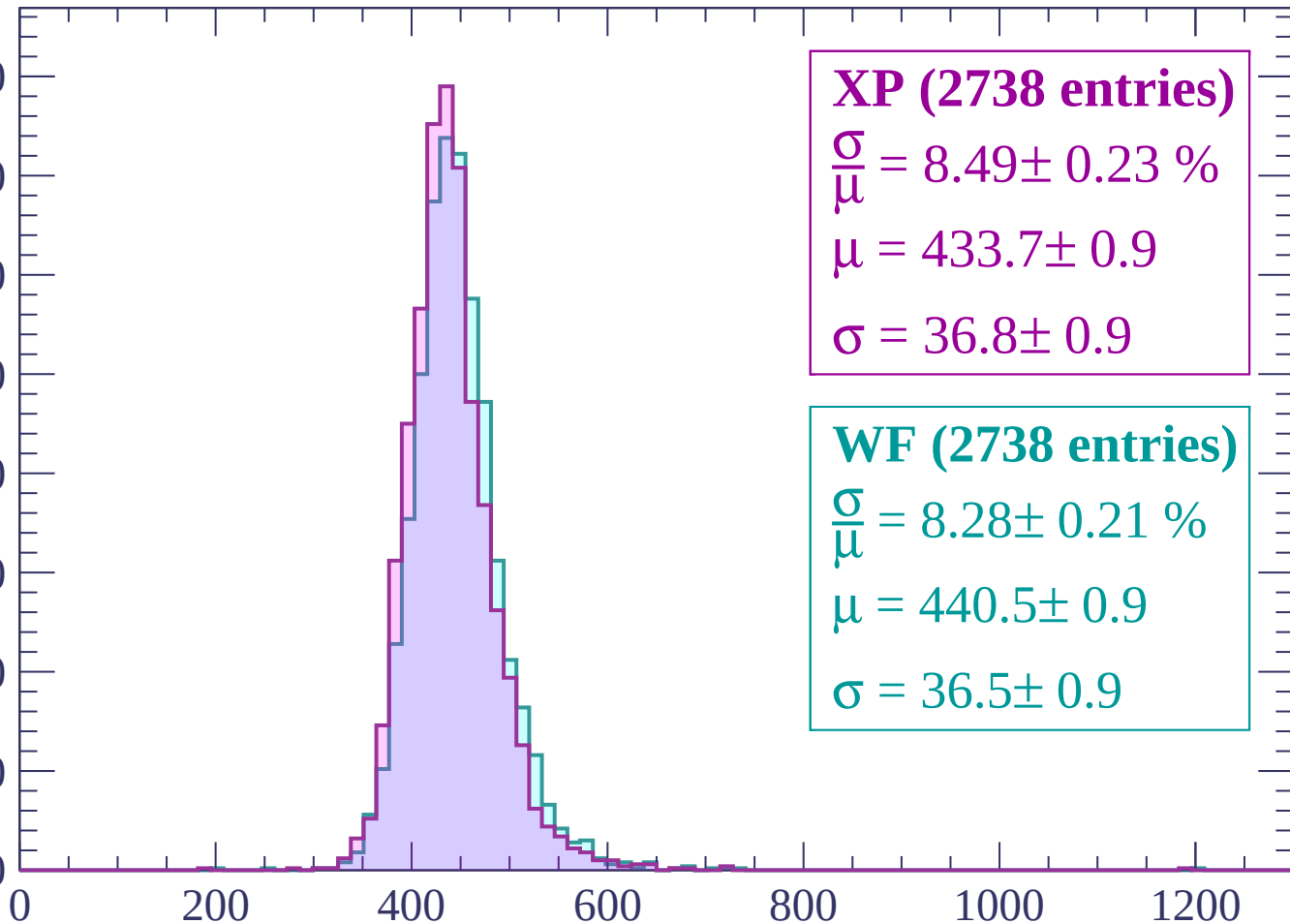
Count



Energy loss | $1020 < p < 1080$

Count

400
350
300
250
200
150
100
50
0



XP (2738 entries)

$\frac{\sigma}{\mu} = 8.49 \pm 0.23 \%$

$\mu = 433.7 \pm 0.9$

$\sigma = 36.8 \pm 0.9$

WF (2738 entries)

$\frac{\sigma}{\mu} = 8.28 \pm 0.21 \%$

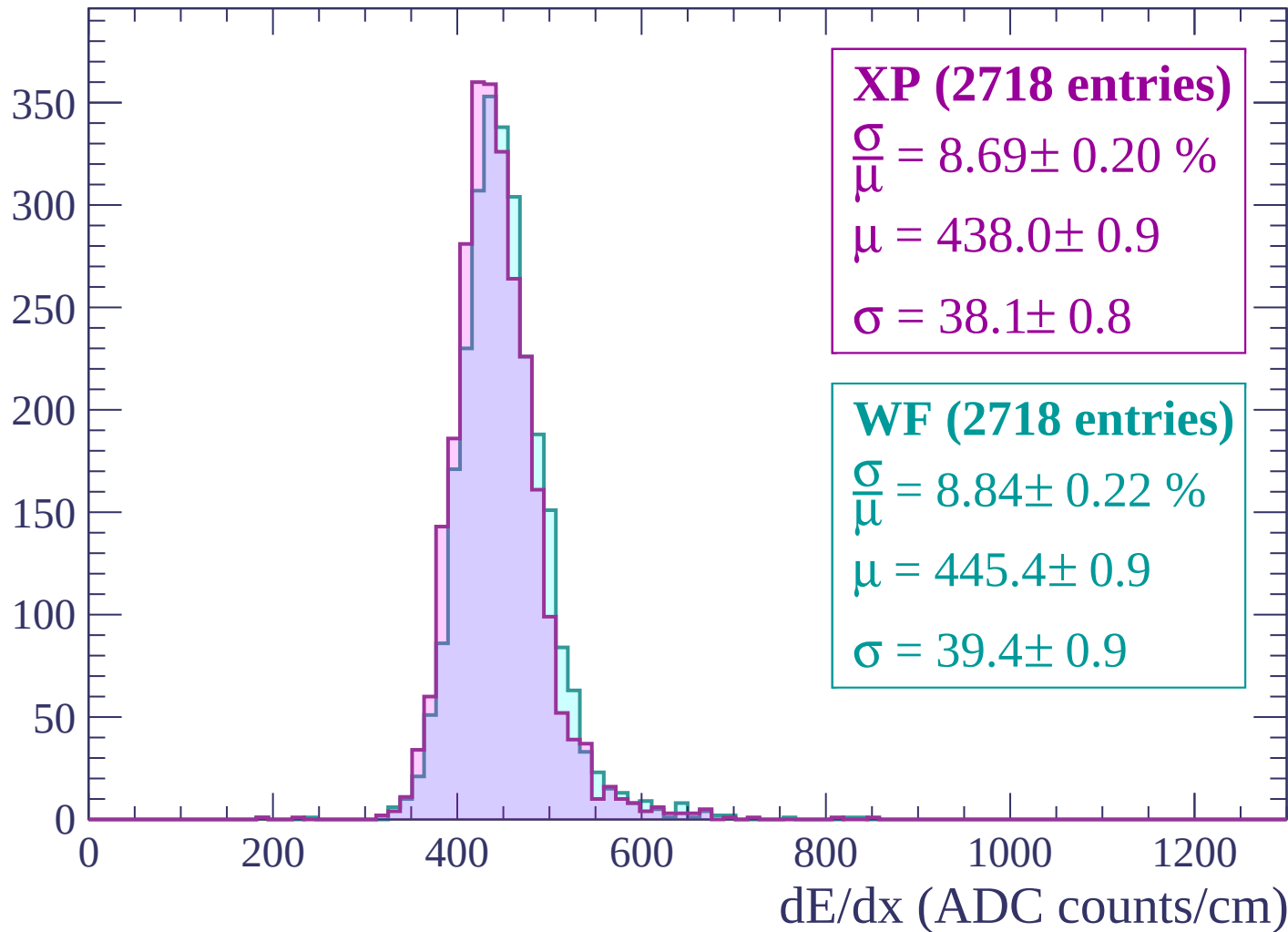
$\mu = 440.5 \pm 0.9$

$\sigma = 36.5 \pm 0.9$

dE/dx (ADC counts/cm)

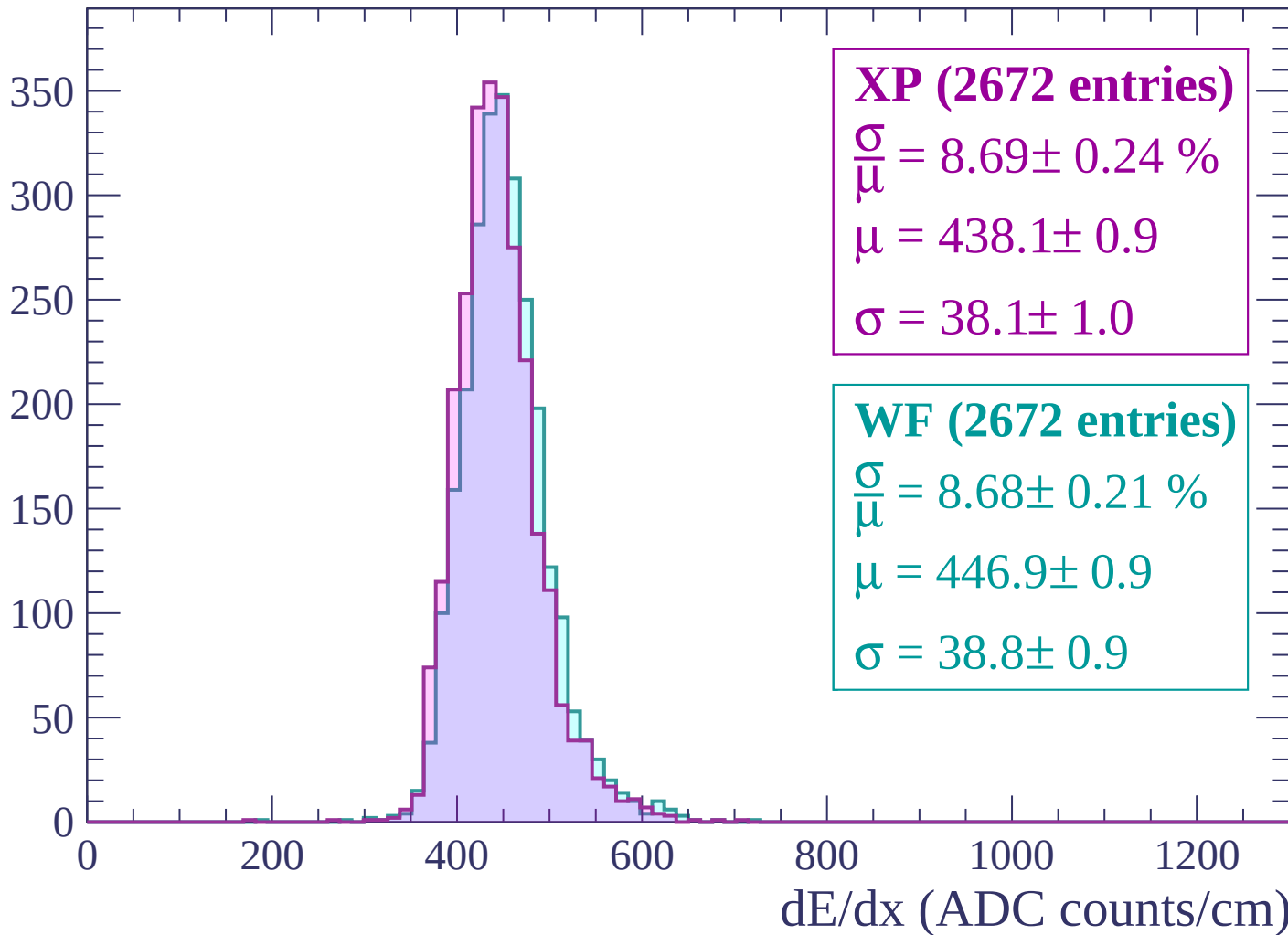
Energy loss | $1080 < p < 1140$

Count



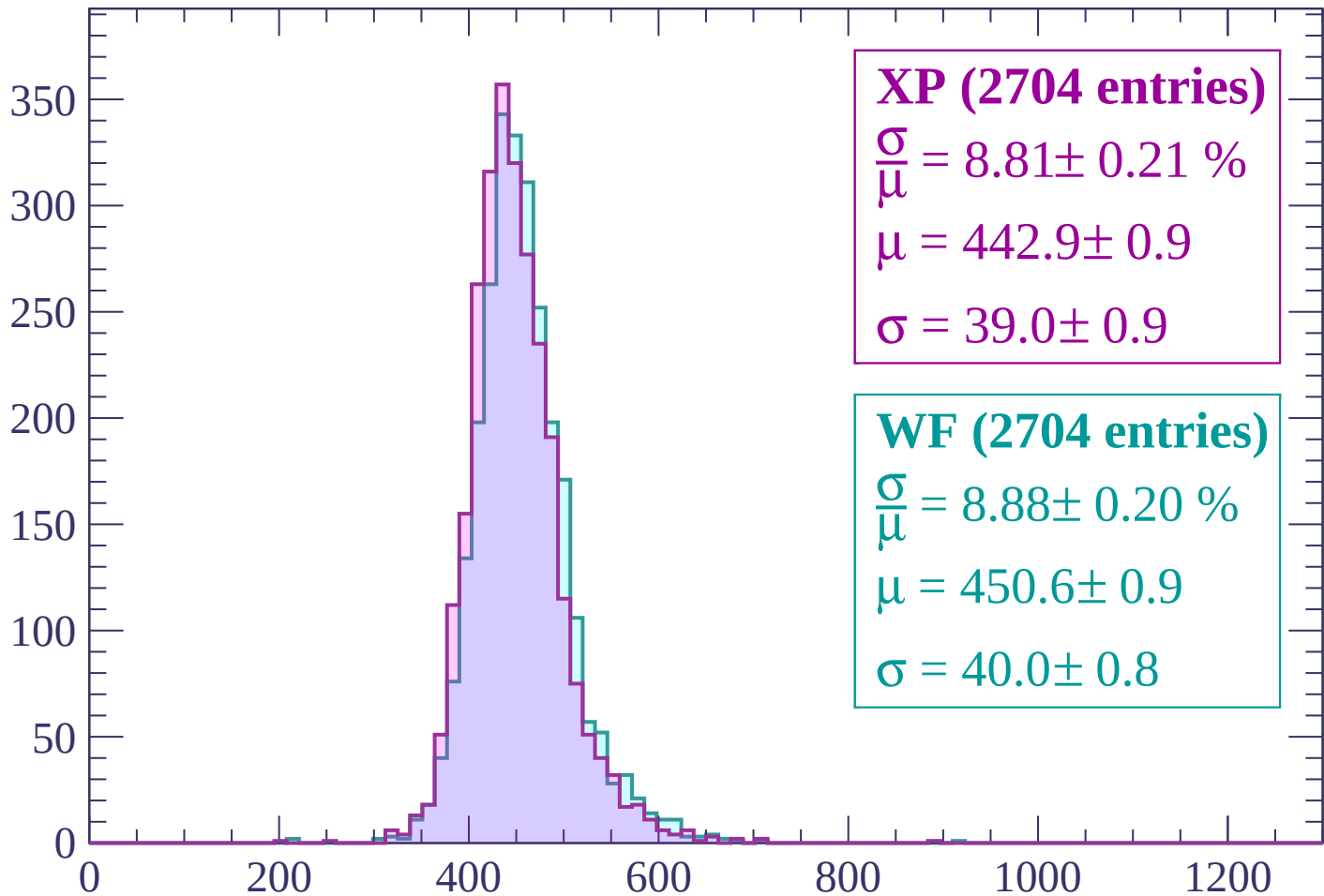
Energy loss | $1140 < p < 1200$

Count



Energy loss | $1200 < p < 1260$

Count



XP (2704 entries)

$$\frac{\sigma}{\mu} = 8.81 \pm 0.21 \%$$

$$\mu = 442.9 \pm 0.9$$

$$\sigma = 39.0 \pm 0.9$$

WF (2704 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.20 \%$$

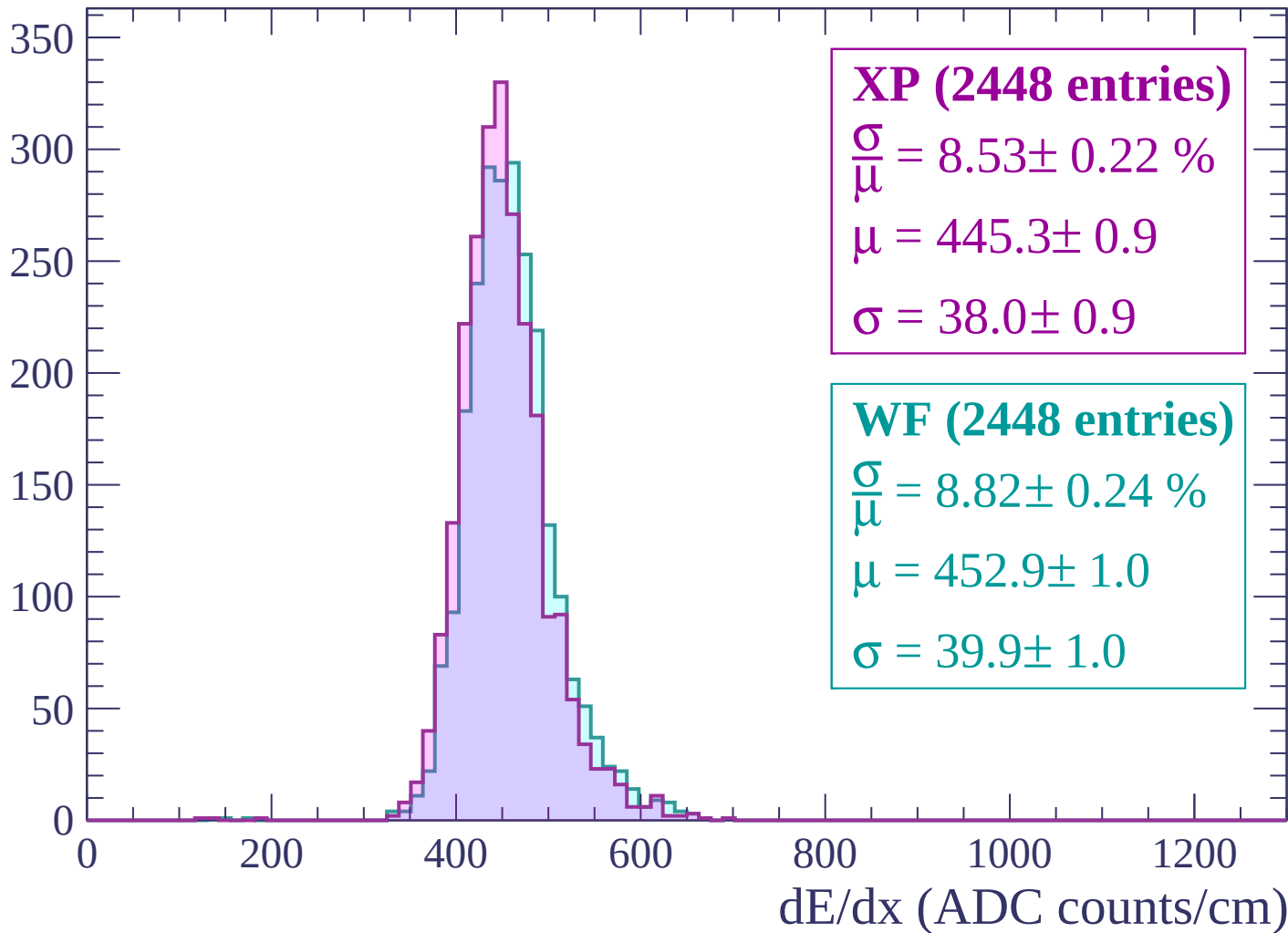
$$\mu = 450.6 \pm 0.9$$

$$\sigma = 40.0 \pm 0.8$$

dE/dx (ADC counts/cm)

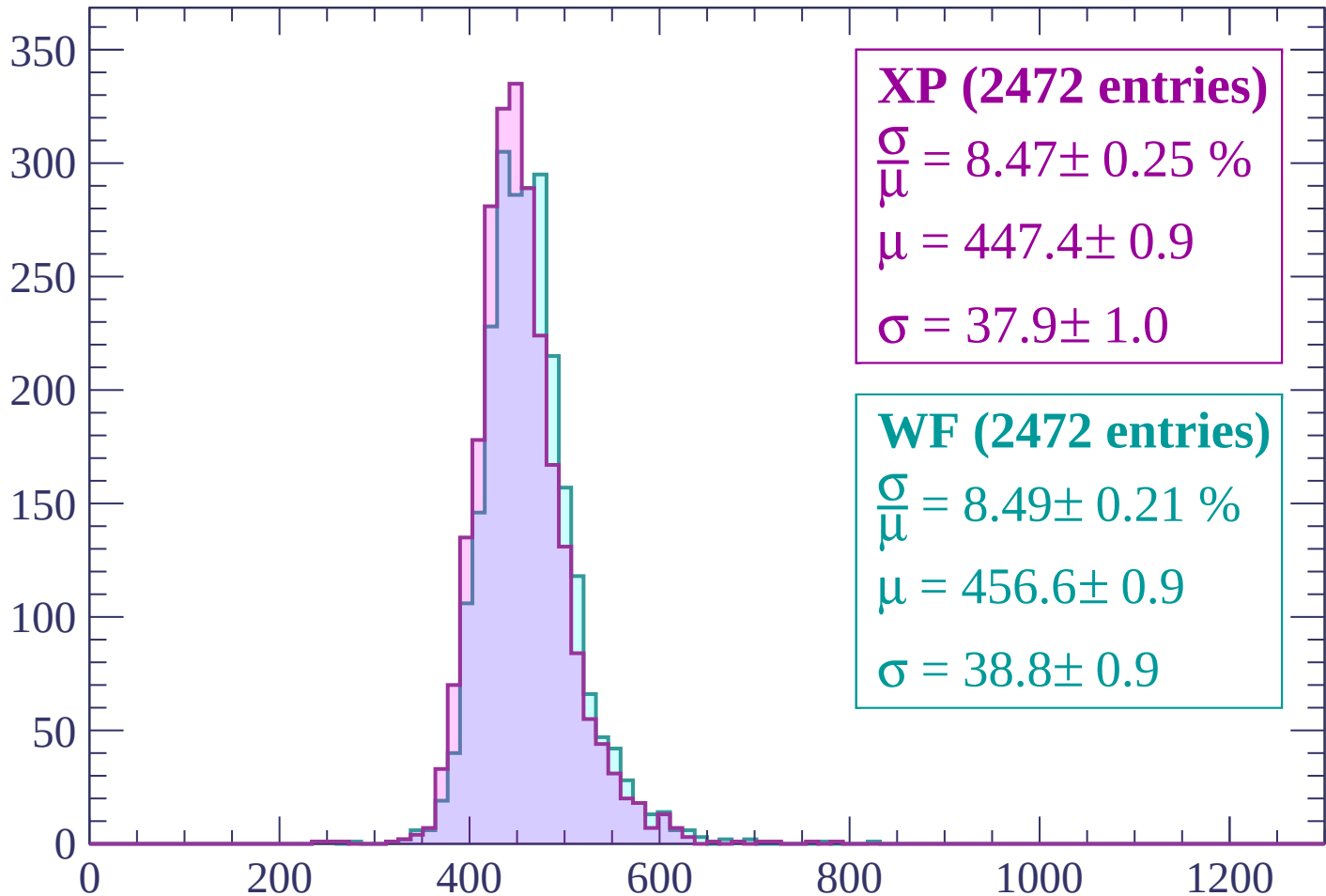
Energy loss | $1260 < p < 1320$

Count



Energy loss | $1320 < p < 1380$

Count



XP (2472 entries)

$$\frac{\sigma}{\mu} = 8.47 \pm 0.25 \%$$

$$\mu = 447.4 \pm 0.9$$

$$\sigma = 37.9 \pm 1.0$$

WF (2472 entries)

$$\frac{\sigma}{\mu} = 8.49 \pm 0.21 \%$$

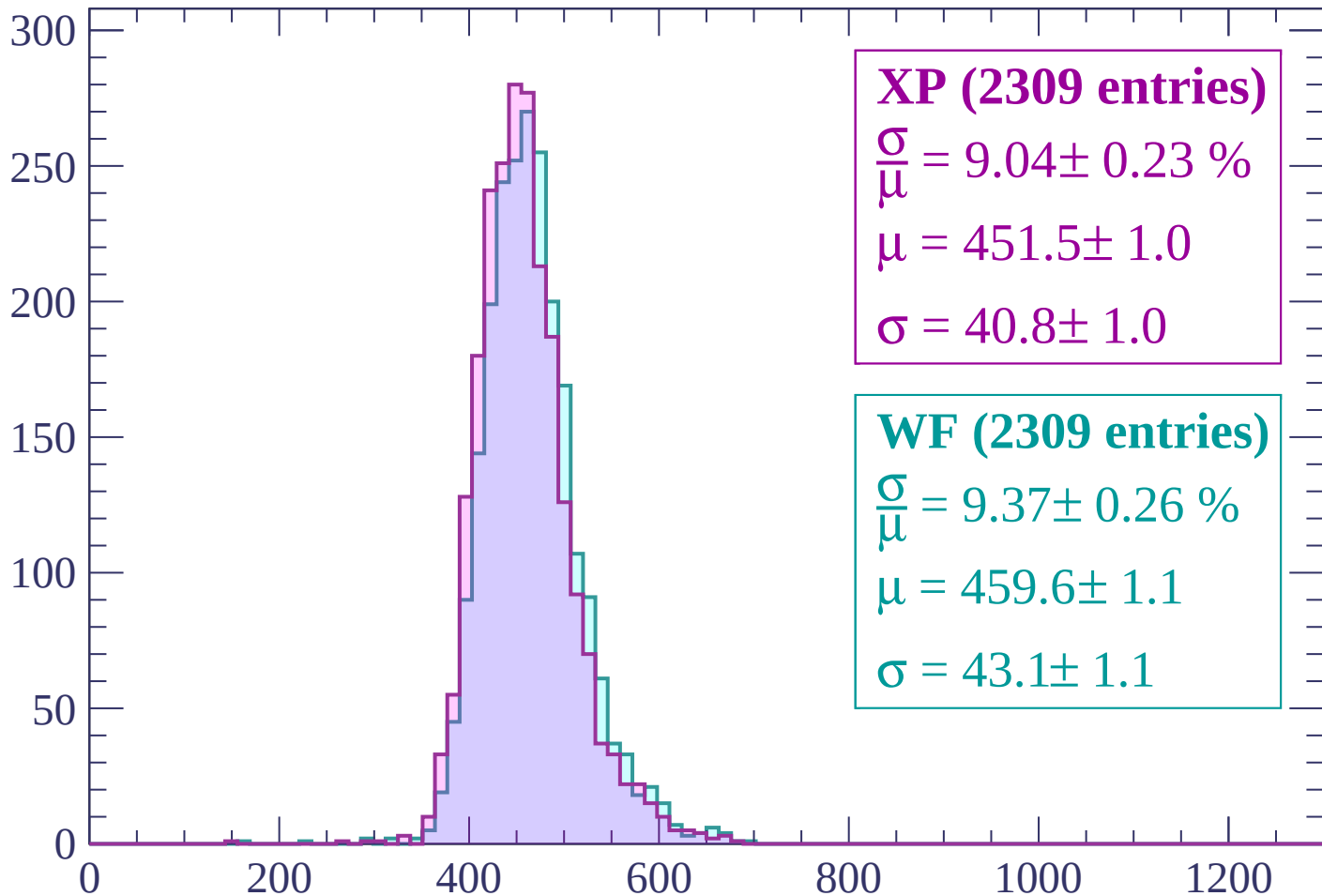
$$\mu = 456.6 \pm 0.9$$

$$\sigma = 38.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (2309 entries)

$\frac{\sigma}{\mu} = 9.04 \pm 0.23 \%$

$\mu = 451.5 \pm 1.0$

$\sigma = 40.8 \pm 1.0$

WF (2309 entries)

$\frac{\sigma}{\mu} = 9.37 \pm 0.26 \%$

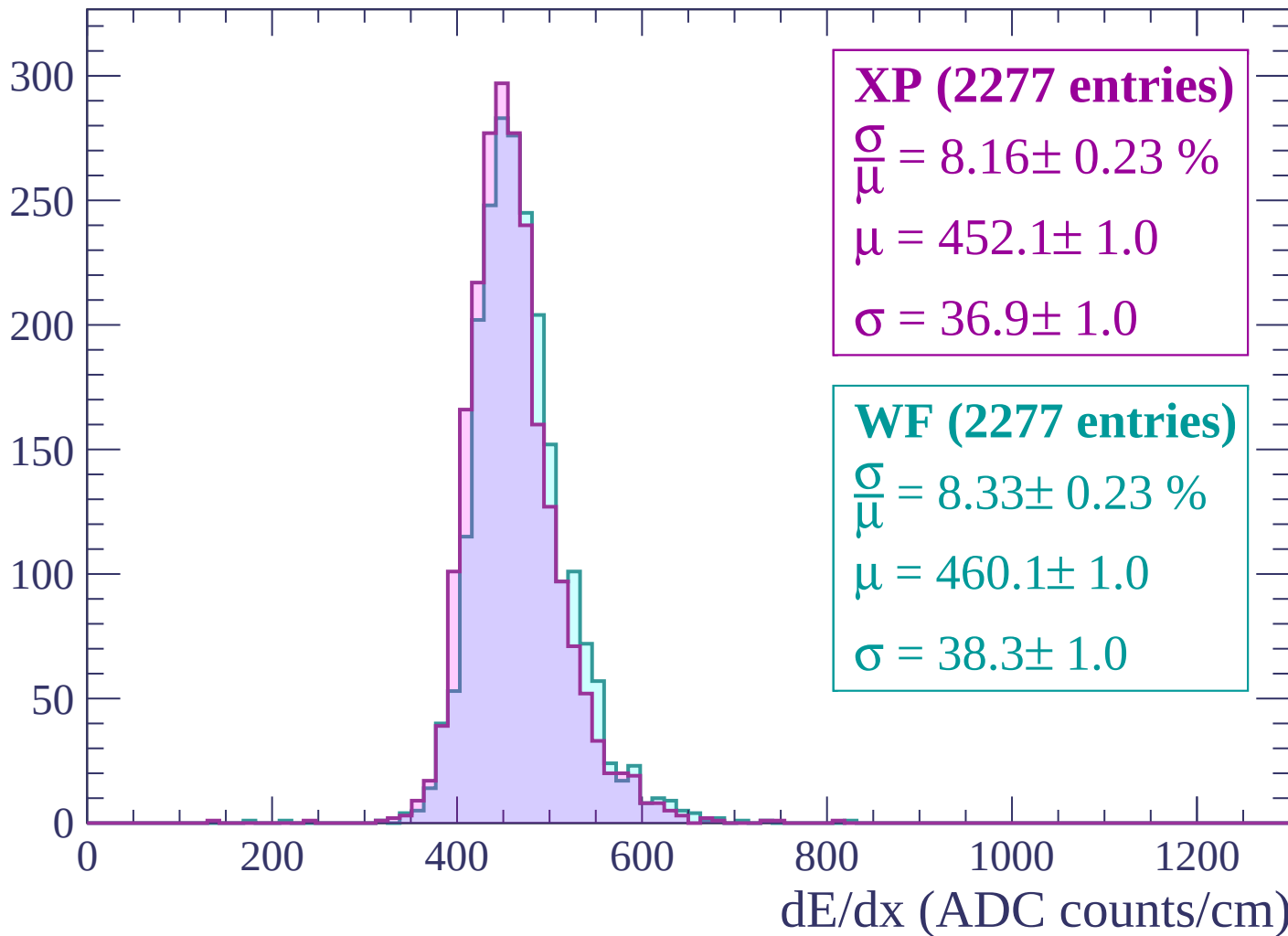
$\mu = 459.6 \pm 1.1$

$\sigma = 43.1 \pm 1.1$

dE/dx (ADC counts/cm)

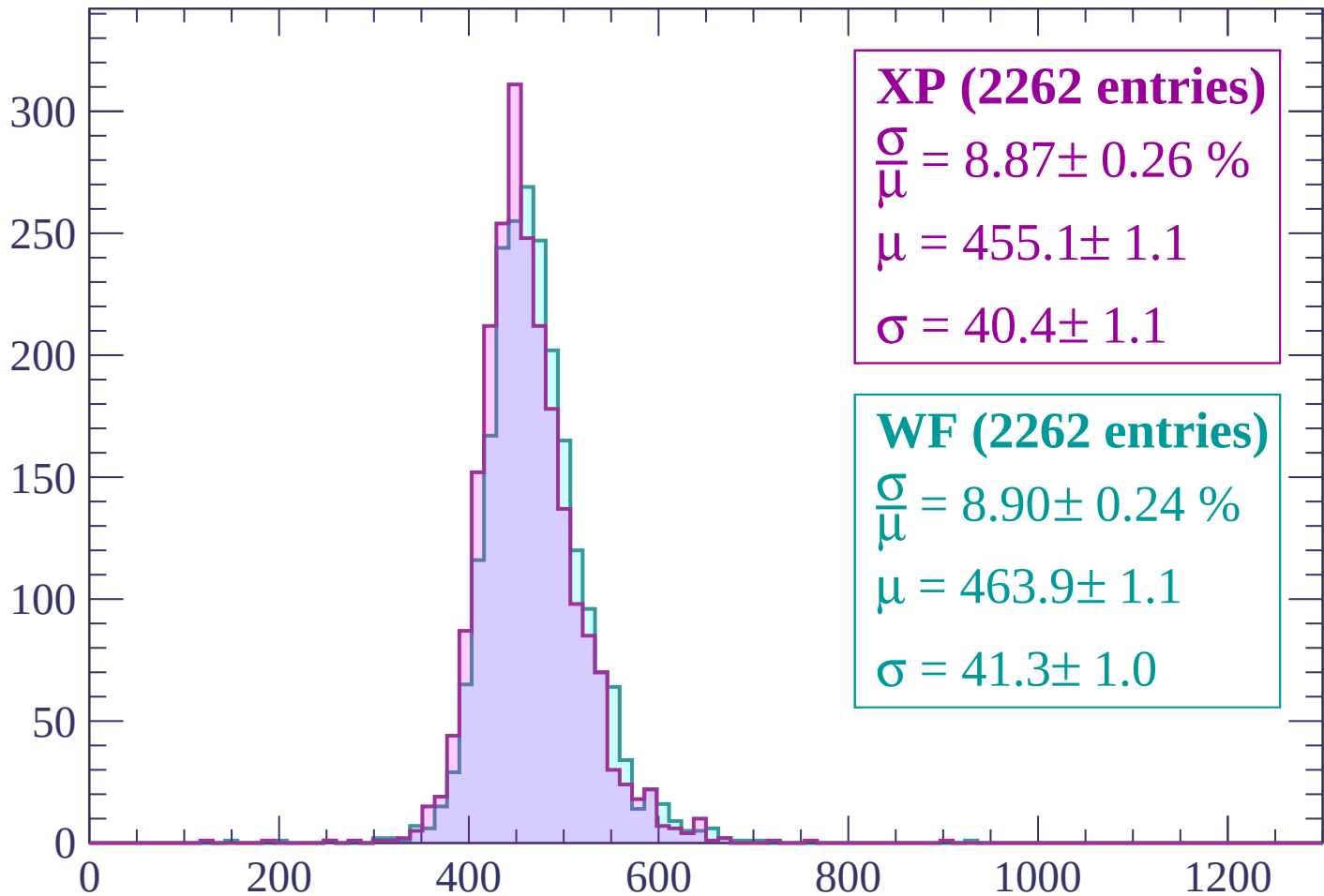
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



XP (2262 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.26 \%$$

$$\mu = 455.1 \pm 1.1$$

$$\sigma = 40.4 \pm 1.1$$

WF (2262 entries)

$$\frac{\sigma}{\mu} = 8.90 \pm 0.24 \%$$

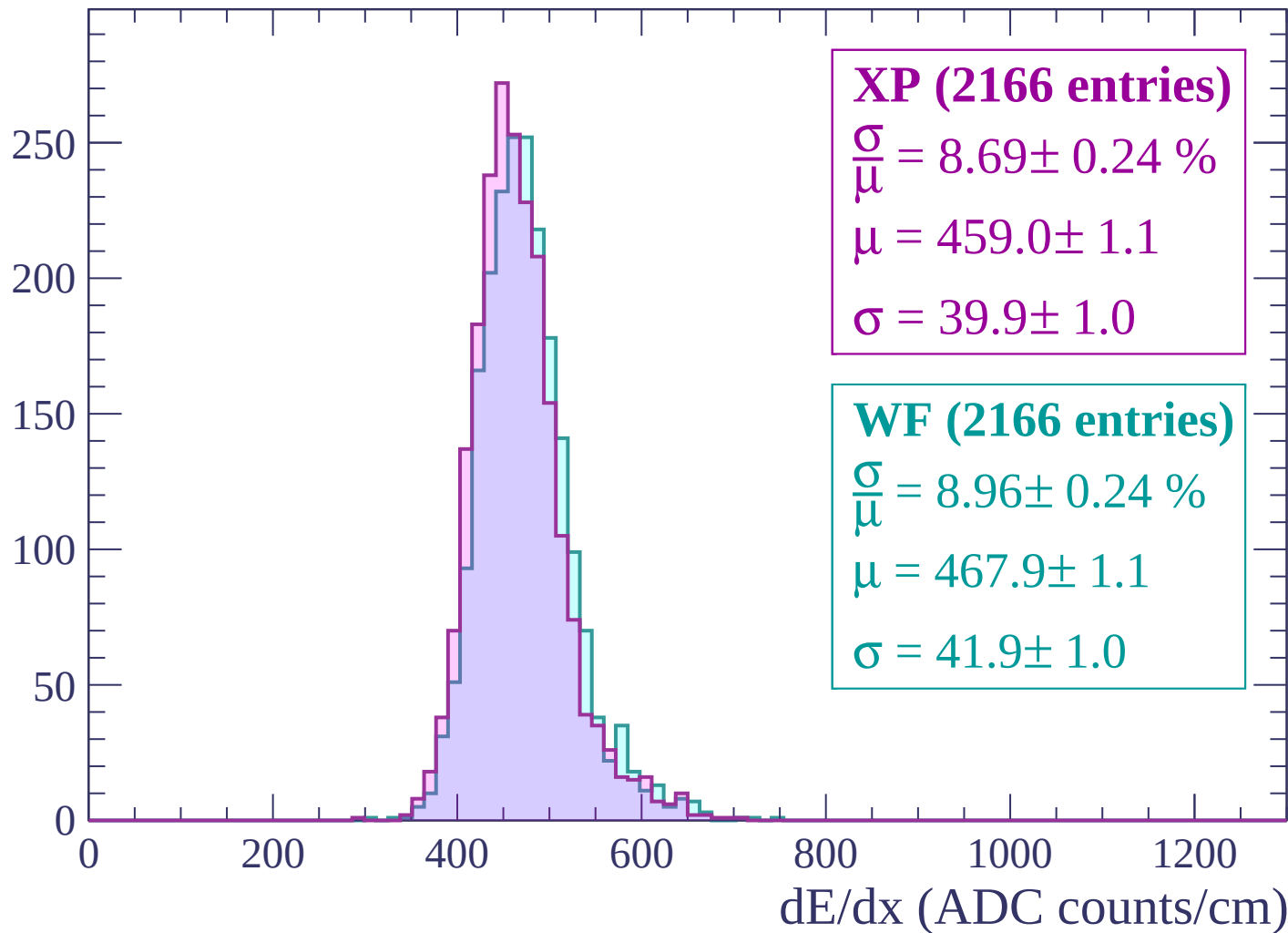
$$\mu = 463.9 \pm 1.1$$

$$\sigma = 41.3 \pm 1.0$$

dE/dx (ADC counts/cm)

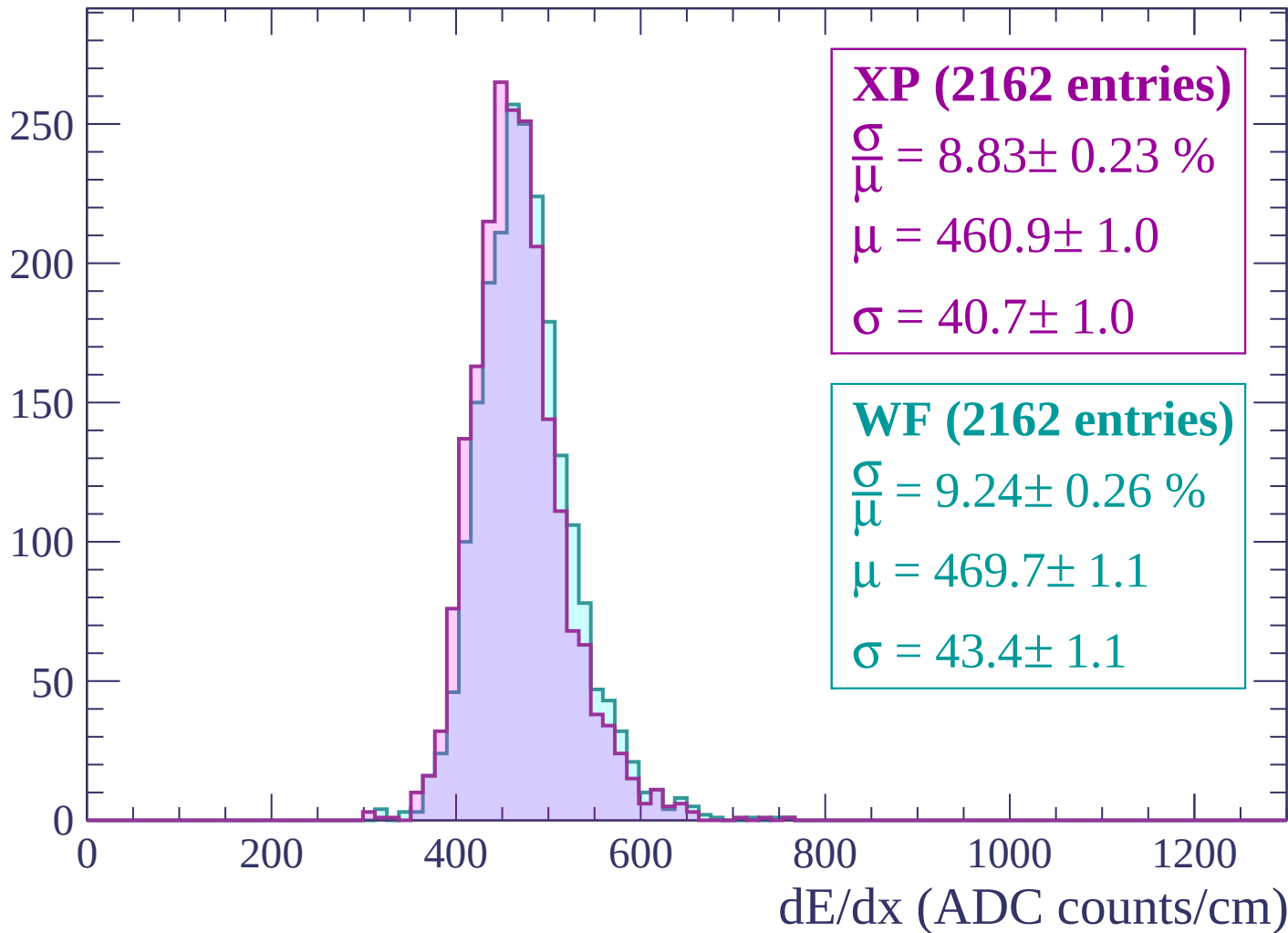
Energy loss | $1560 < p < 1620$

Count



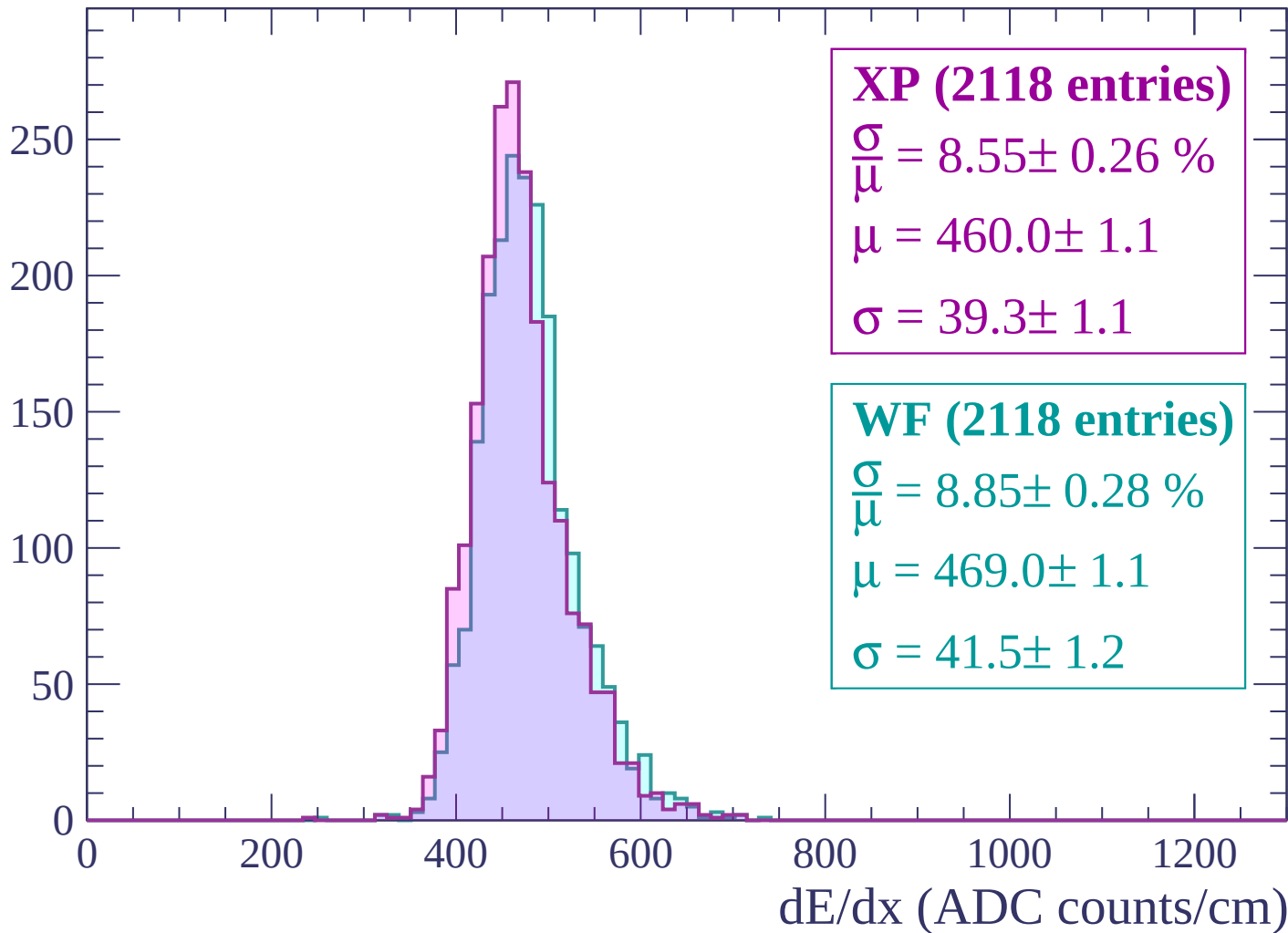
Energy loss | $1620 < p < 1680$

Count



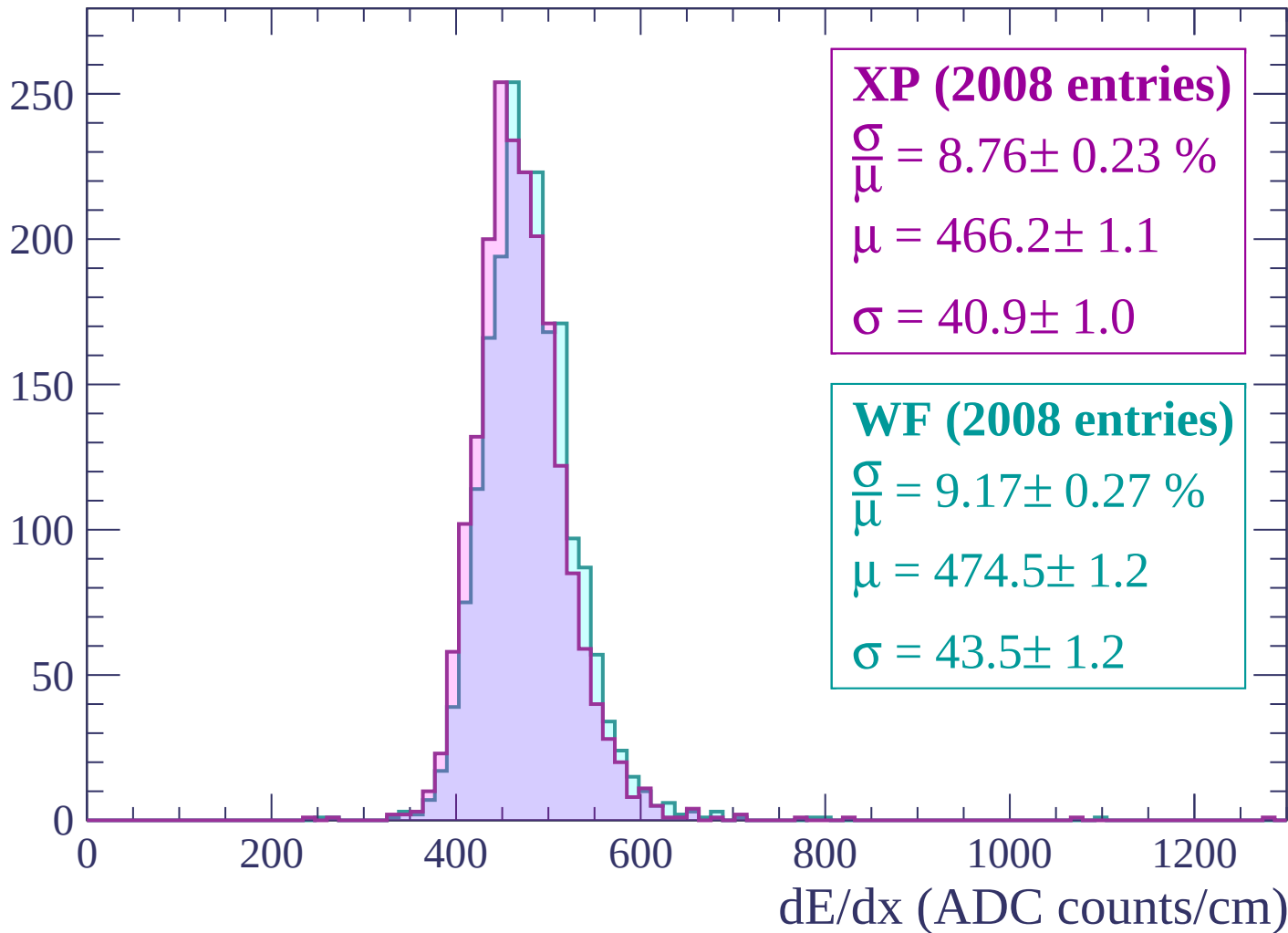
Energy loss | $1680 < p < 1740$

Count



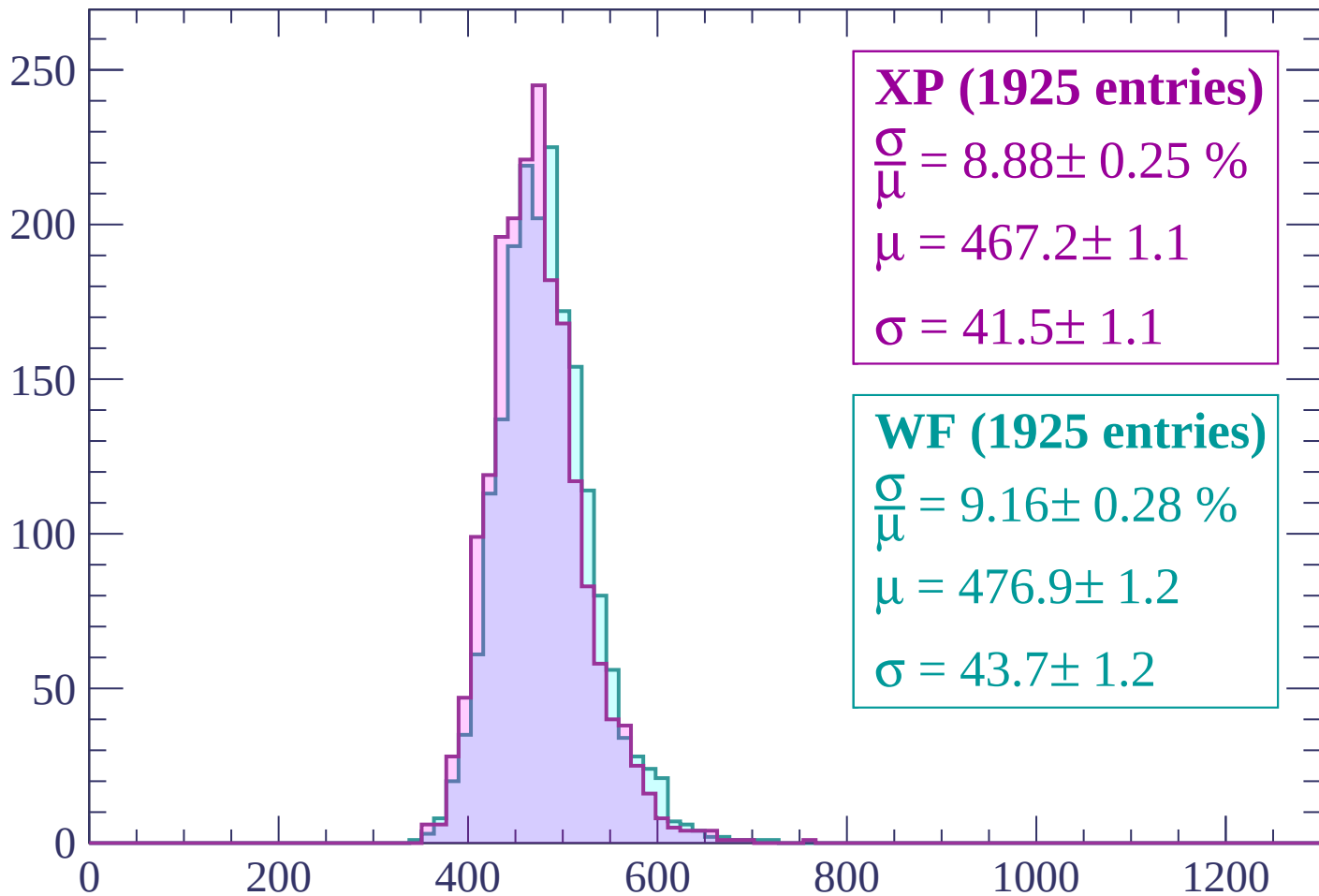
Energy loss | $1740 < p < 1800$

Count



Energy loss | $1800 < p < 1860$

Count



XP (1925 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.25 \%$$

$$\mu = 467.2 \pm 1.1$$

$$\sigma = 41.5 \pm 1.1$$

WF (1925 entries)

$$\frac{\sigma}{\mu} = 9.16 \pm 0.28 \%$$

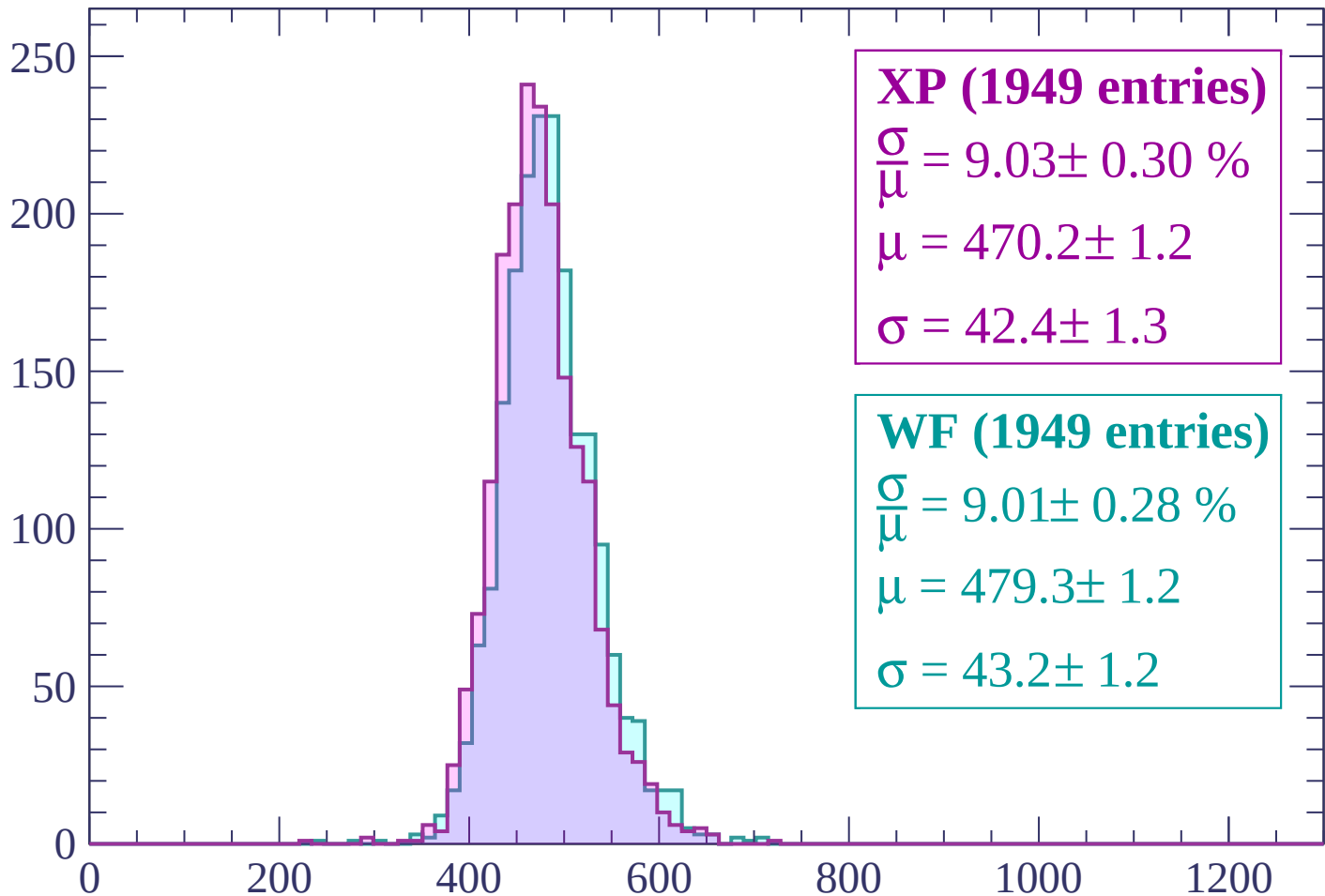
$$\mu = 476.9 \pm 1.2$$

$$\sigma = 43.7 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1860 < p < 1920$

Count



XP (1949 entries)

$$\frac{\sigma}{\mu} = 9.03 \pm 0.30 \%$$

$$\mu = 470.2 \pm 1.2$$

$$\sigma = 42.4 \pm 1.3$$

WF (1949 entries)

$$\frac{\sigma}{\mu} = 9.01 \pm 0.28 \%$$

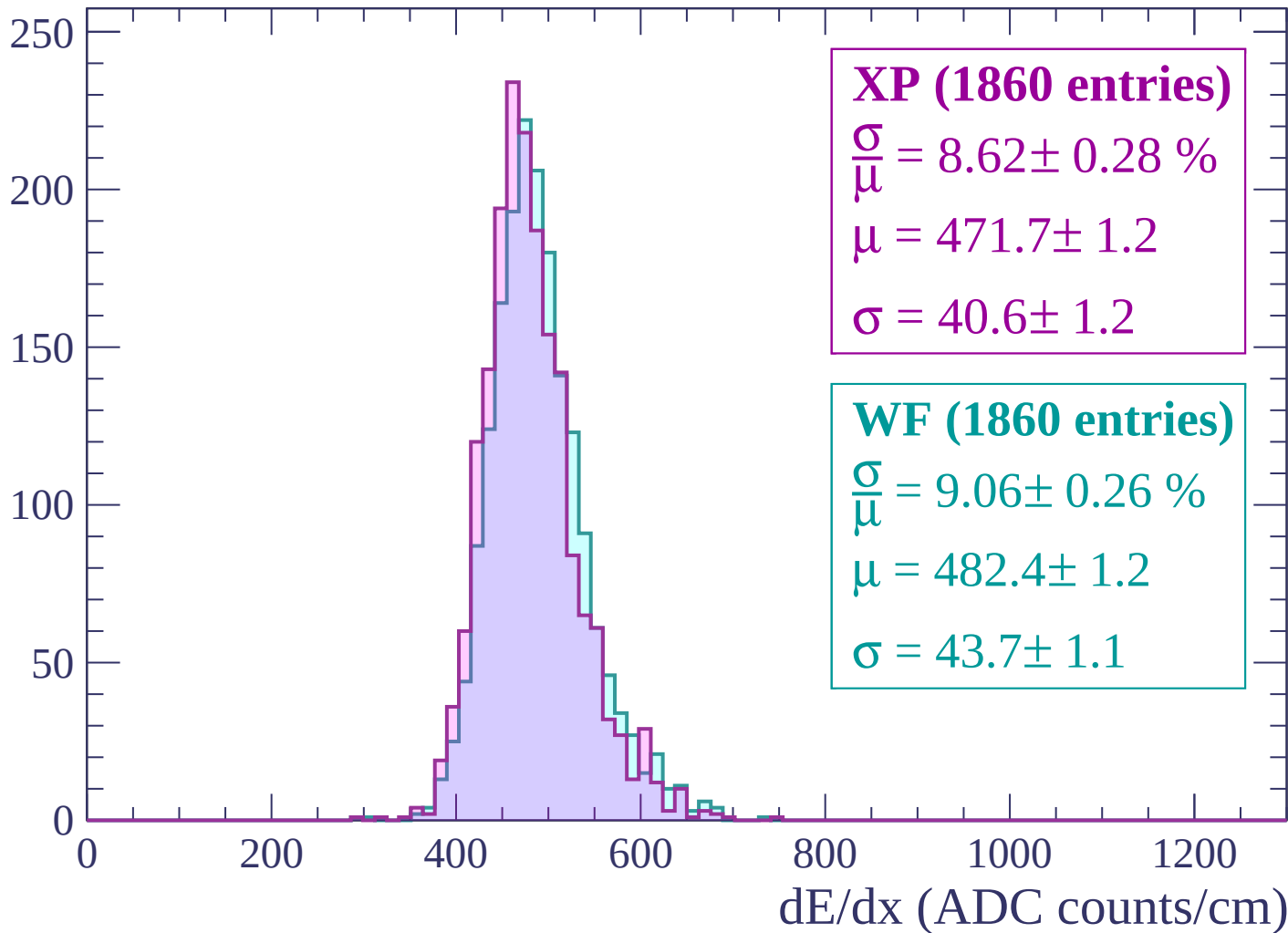
$$\mu = 479.3 \pm 1.2$$

$$\sigma = 43.2 \pm 1.2$$

dE/dx (ADC counts/cm)

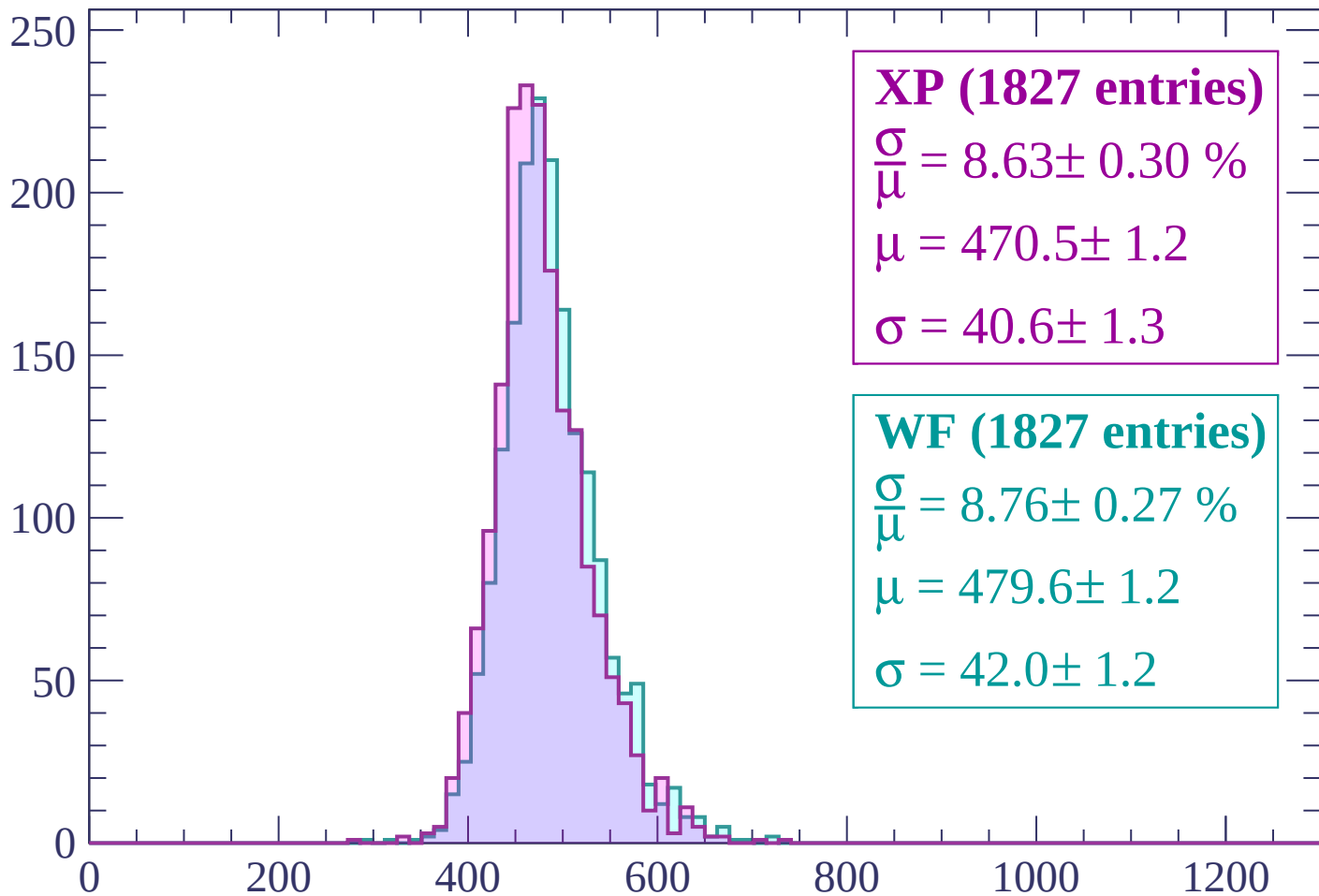
Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

Count



XP (1827 entries)

$$\frac{\sigma}{\mu} = 8.63 \pm 0.30 \%$$

$$\mu = 470.5 \pm 1.2$$

$$\sigma = 40.6 \pm 1.3$$

WF (1827 entries)

$$\frac{\sigma}{\mu} = 8.76 \pm 0.27 \%$$

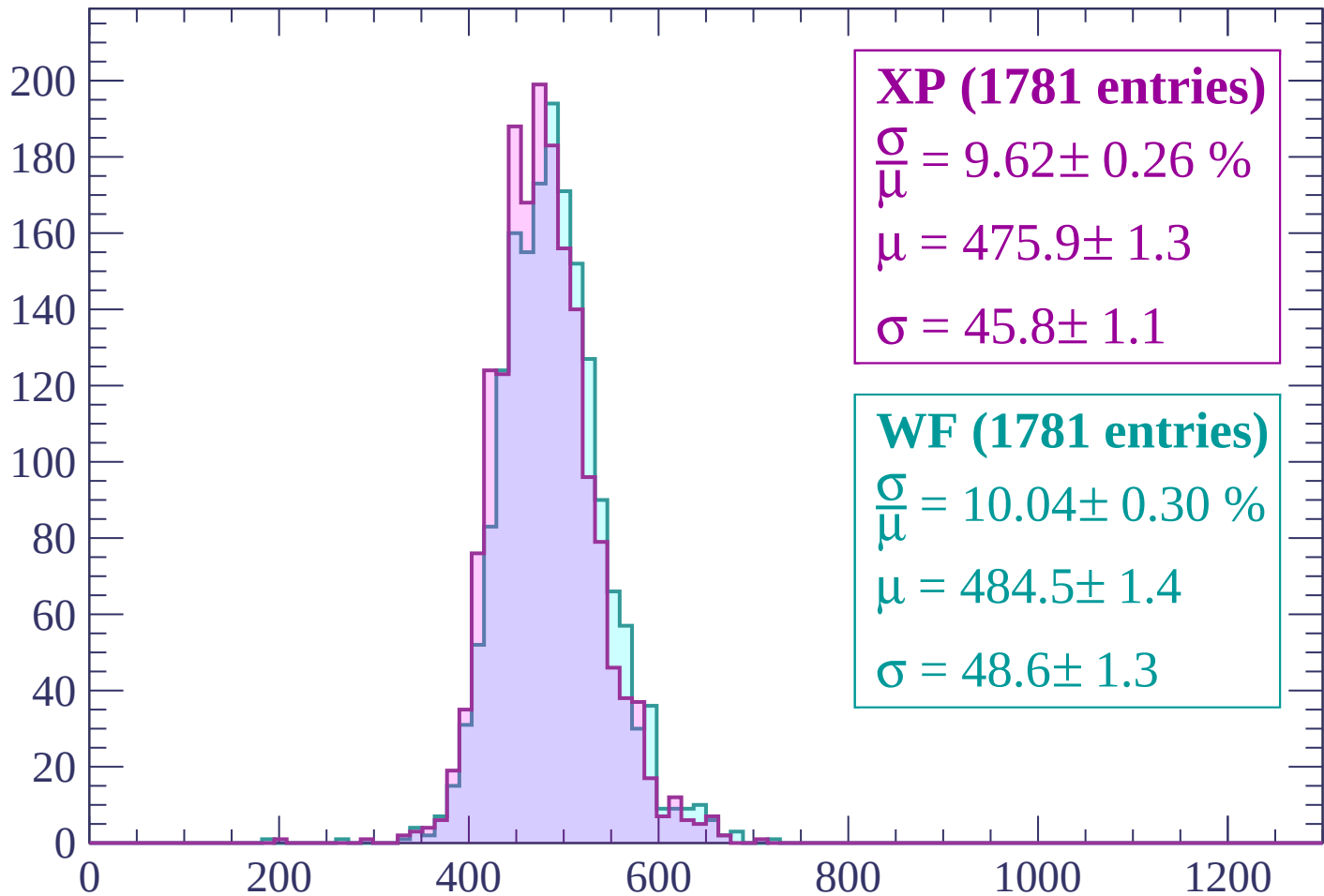
$$\mu = 479.6 \pm 1.2$$

$$\sigma = 42.0 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP (1781 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.26 \%$$

$$\mu = 475.9 \pm 1.3$$

$$\sigma = 45.8 \pm 1.1$$

WF (1781 entries)

$$\frac{\sigma}{\mu} = 10.04 \pm 0.30 \%$$

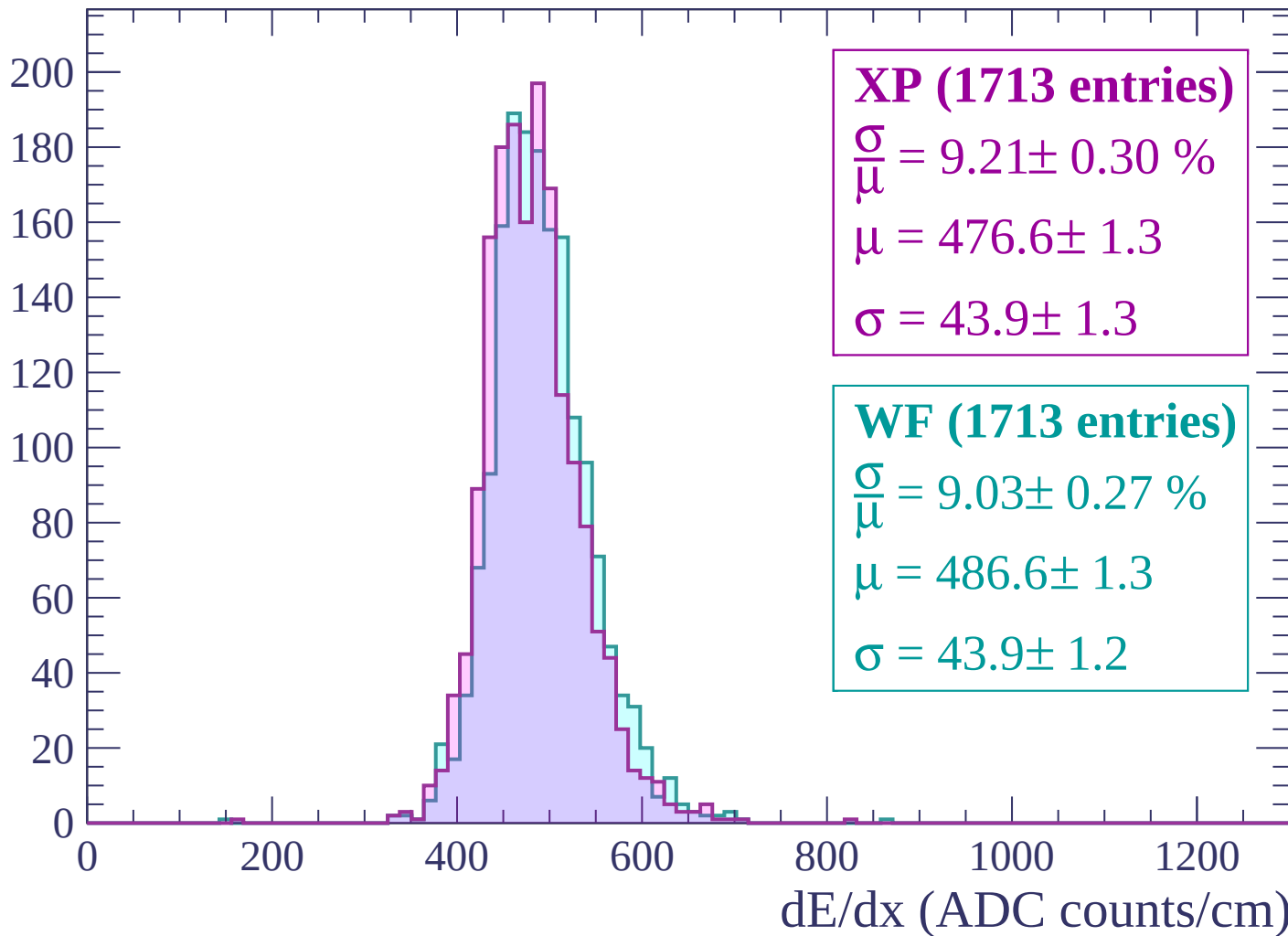
$$\mu = 484.5 \pm 1.4$$

$$\sigma = 48.6 \pm 1.3$$

dE/dx (ADC counts/cm)

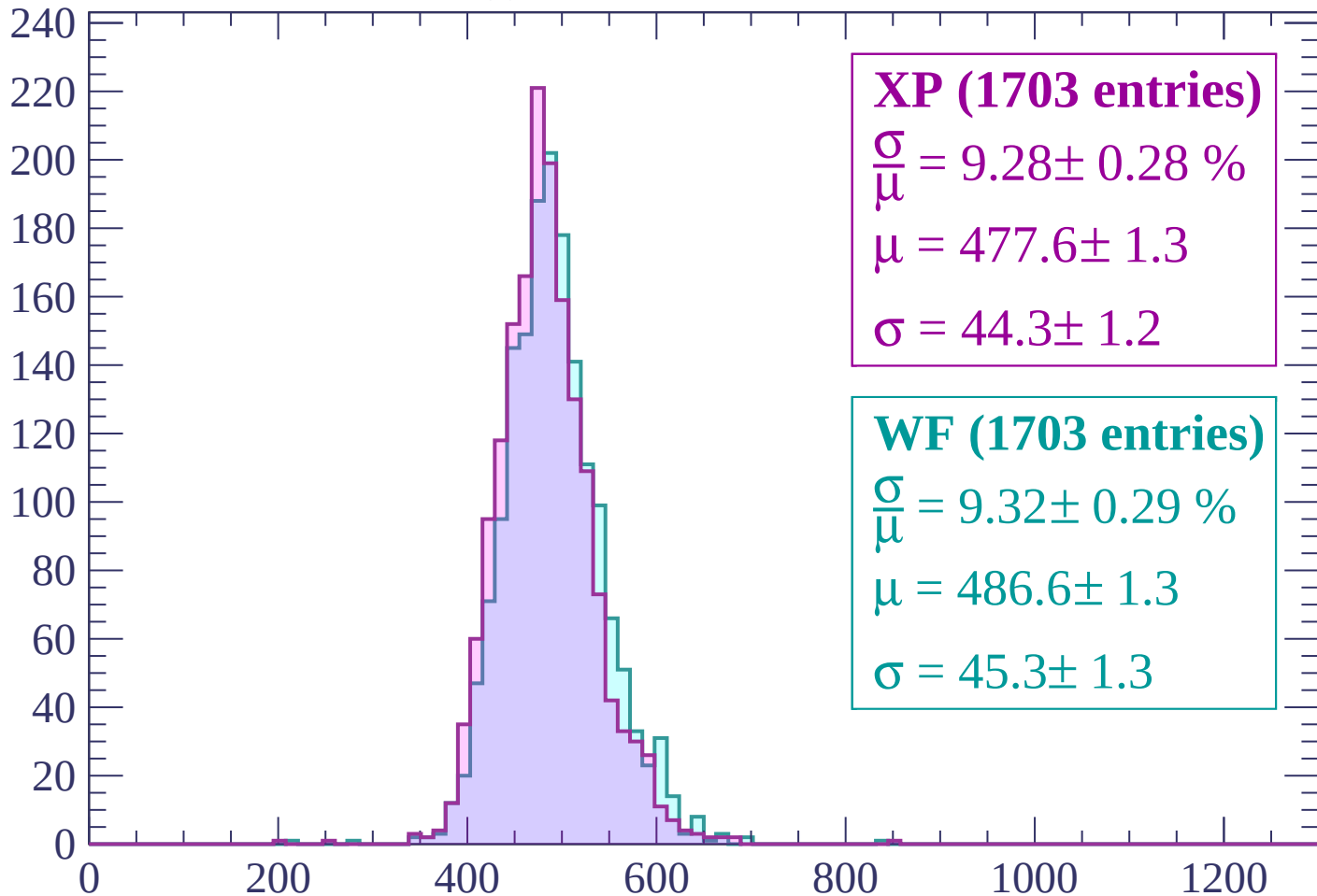
Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count



XP (1703 entries)

$\frac{\sigma}{\mu} = 9.28 \pm 0.28 \%$

$\mu = 477.6 \pm 1.3$

$\sigma = 44.3 \pm 1.2$

WF (1703 entries)

$\frac{\sigma}{\mu} = 9.32 \pm 0.29 \%$

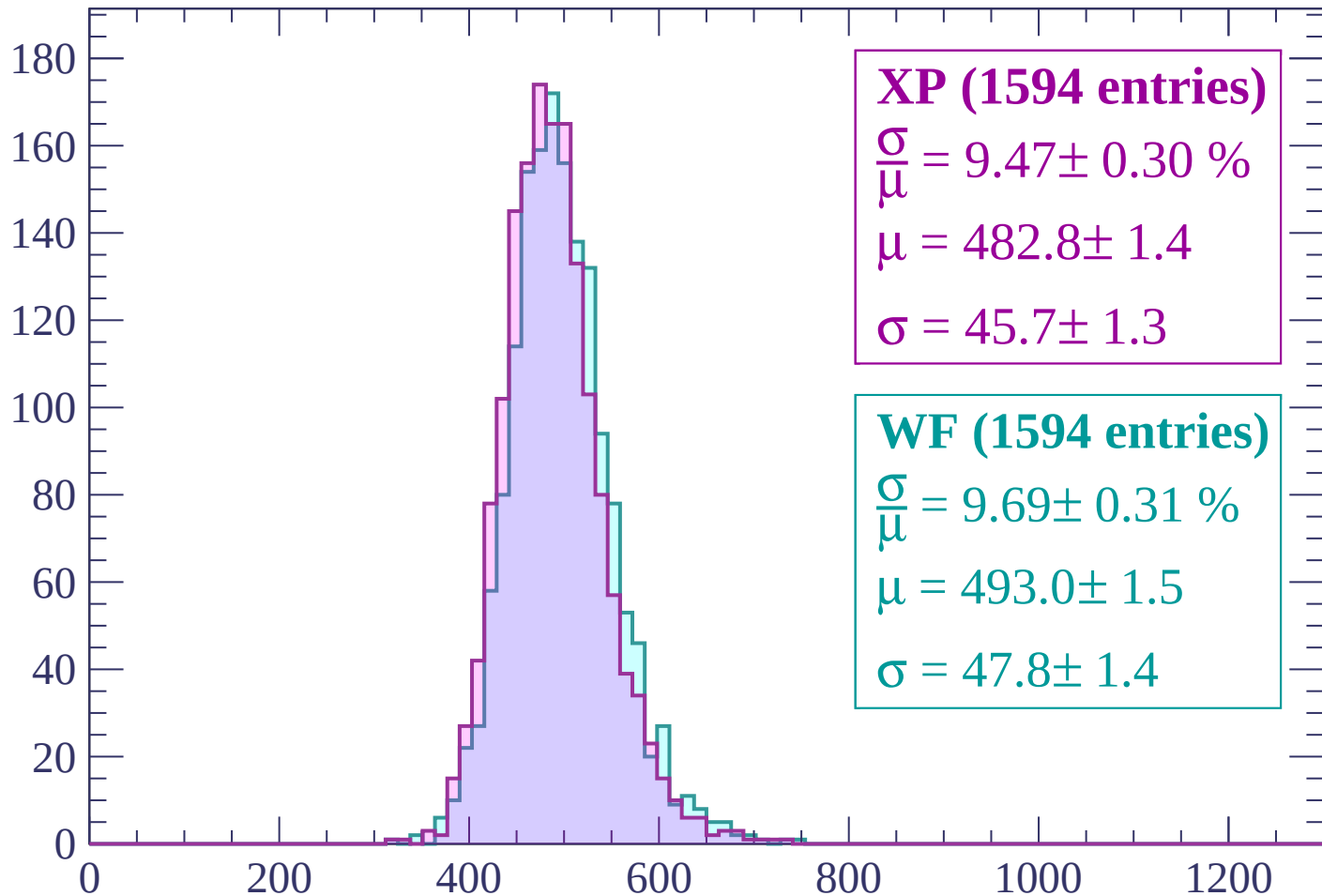
$\mu = 486.6 \pm 1.3$

$\sigma = 45.3 \pm 1.3$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (1594 entries)

$$\frac{\sigma}{\mu} = 9.47 \pm 0.30 \%$$

$$\mu = 482.8 \pm 1.4$$

$$\sigma = 45.7 \pm 1.3$$

WF (1594 entries)

$$\frac{\sigma}{\mu} = 9.69 \pm 0.31 \%$$

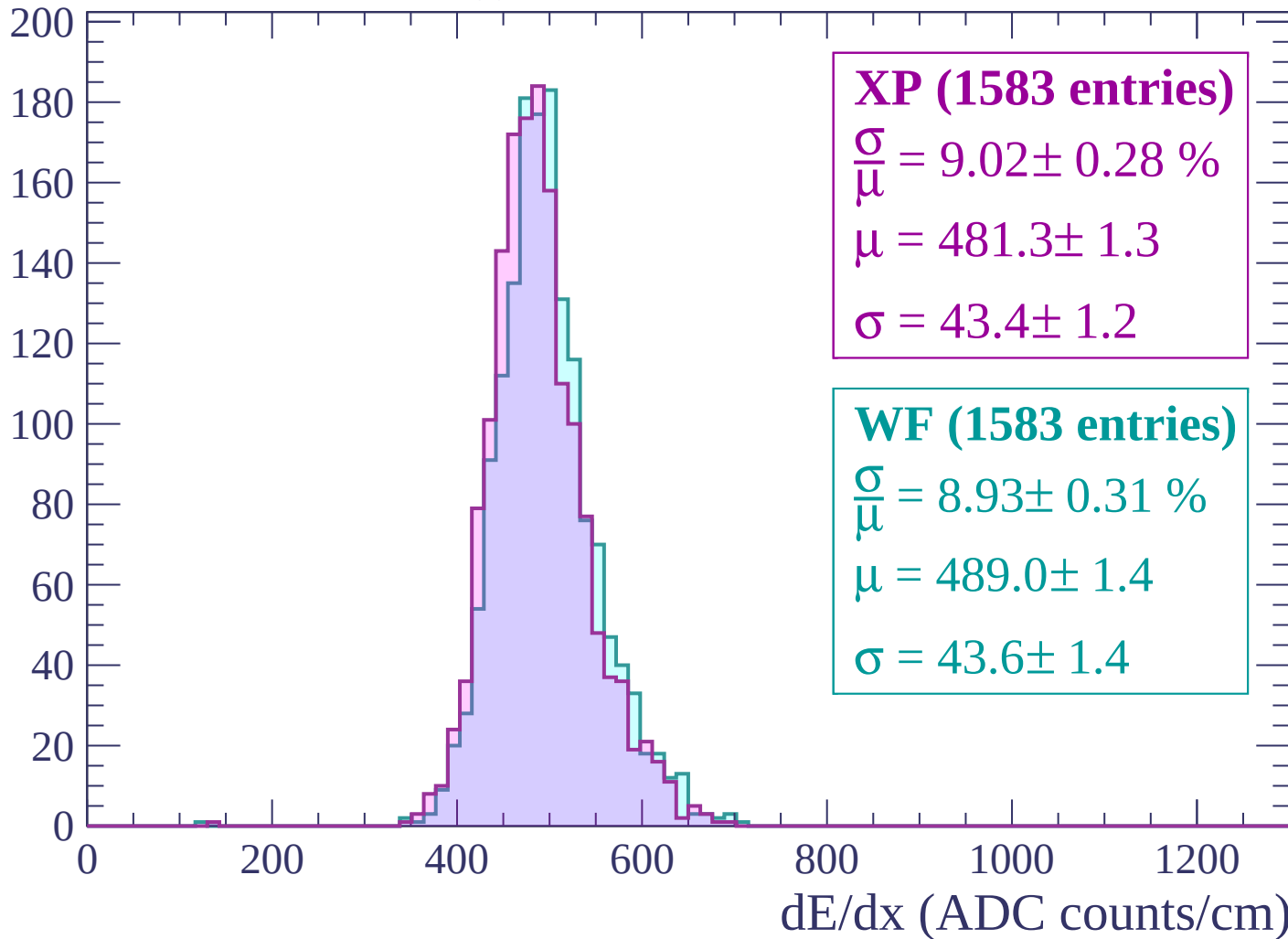
$$\mu = 493.0 \pm 1.5$$

$$\sigma = 47.8 \pm 1.4$$

dE/dx (ADC counts/cm)

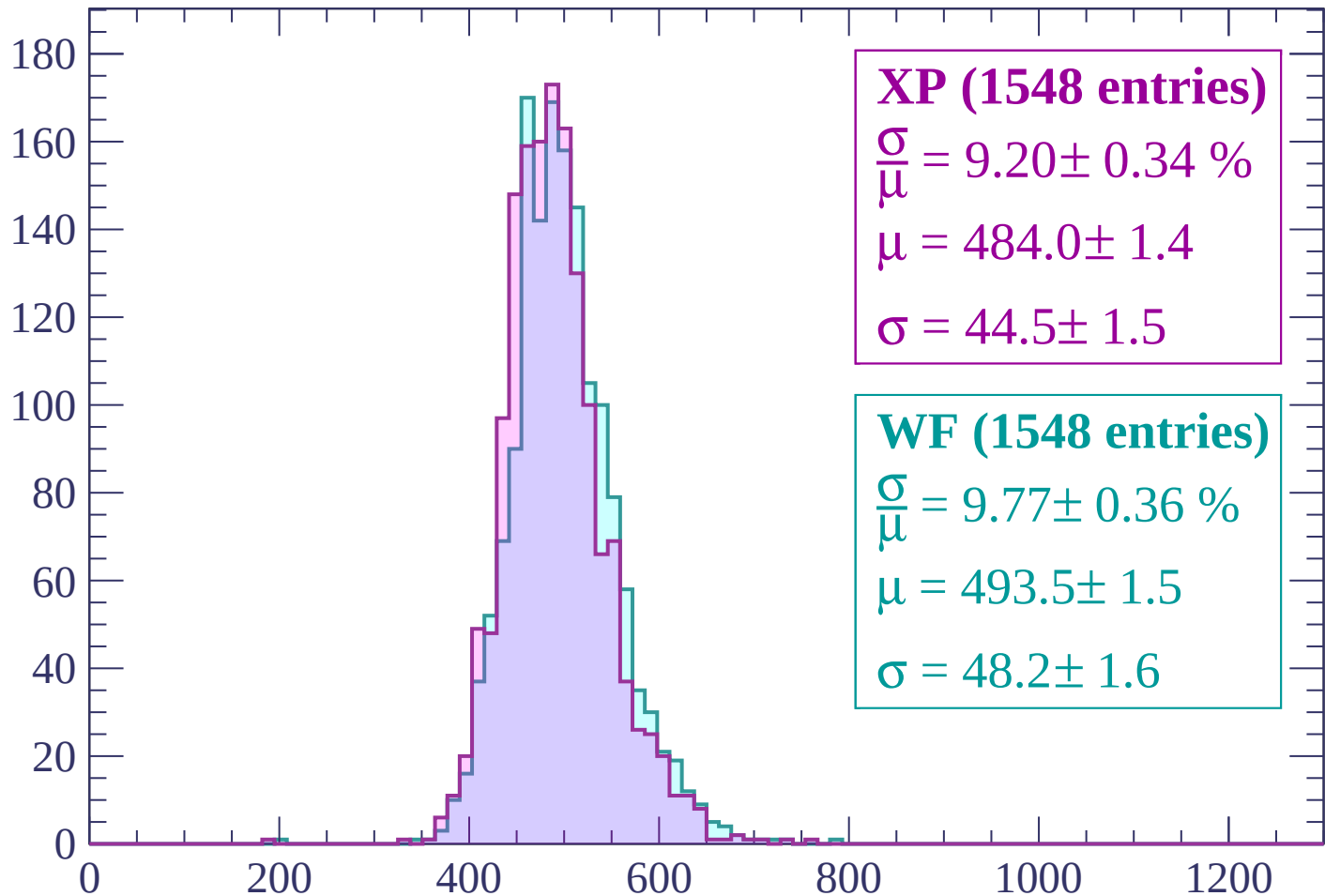
Energy loss | $2280 < p < 2340$

Count



Energy loss | $2340 < p < 2400$

Count



XP (1548 entries)

$$\frac{\sigma}{\mu} = 9.20 \pm 0.34 \%$$

$$\mu = 484.0 \pm 1.4$$

$$\sigma = 44.5 \pm 1.5$$

WF (1548 entries)

$$\frac{\sigma}{\mu} = 9.77 \pm 0.36 \%$$

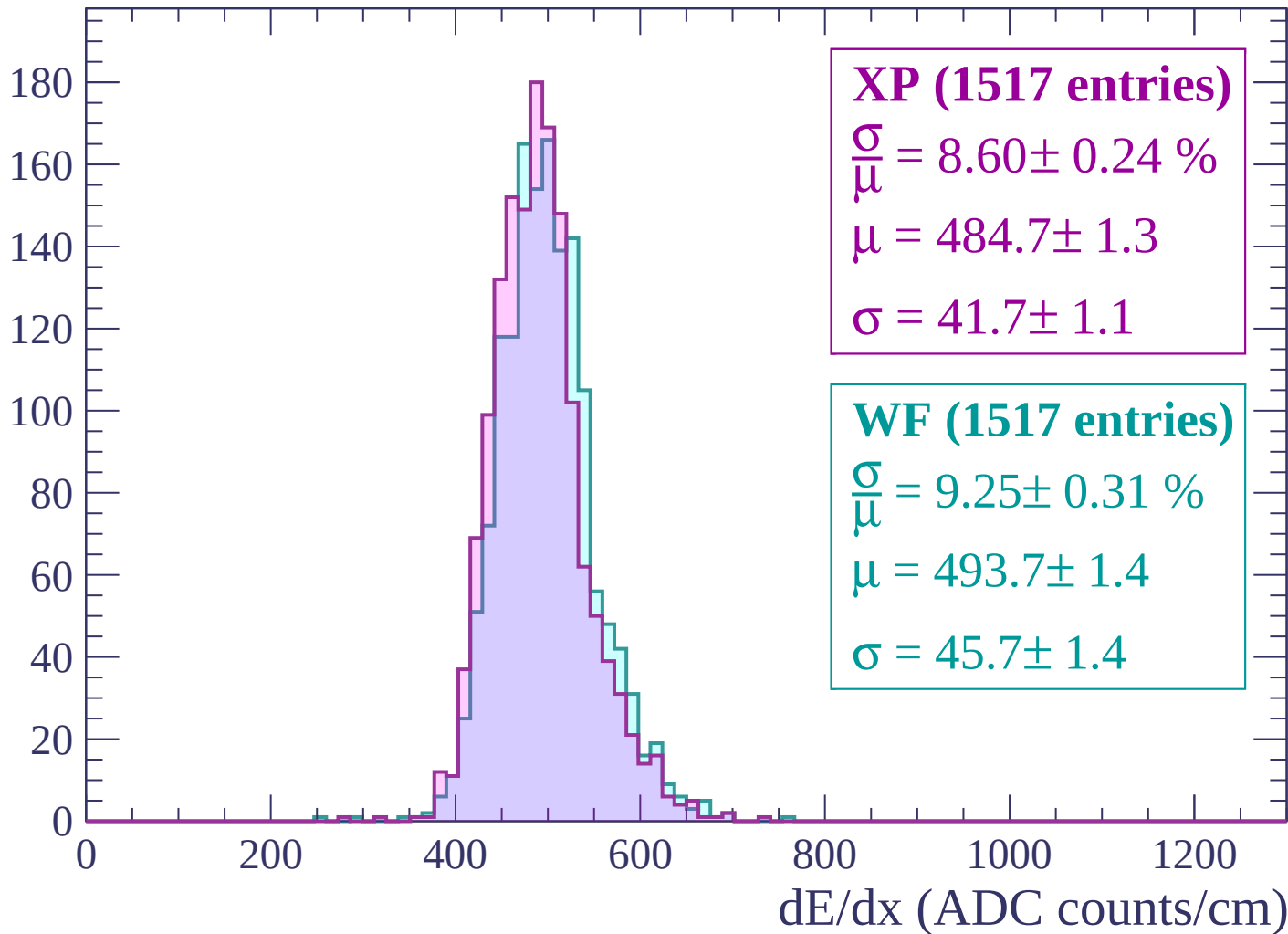
$$\mu = 493.5 \pm 1.5$$

$$\sigma = 48.2 \pm 1.6$$

dE/dx (ADC counts/cm)

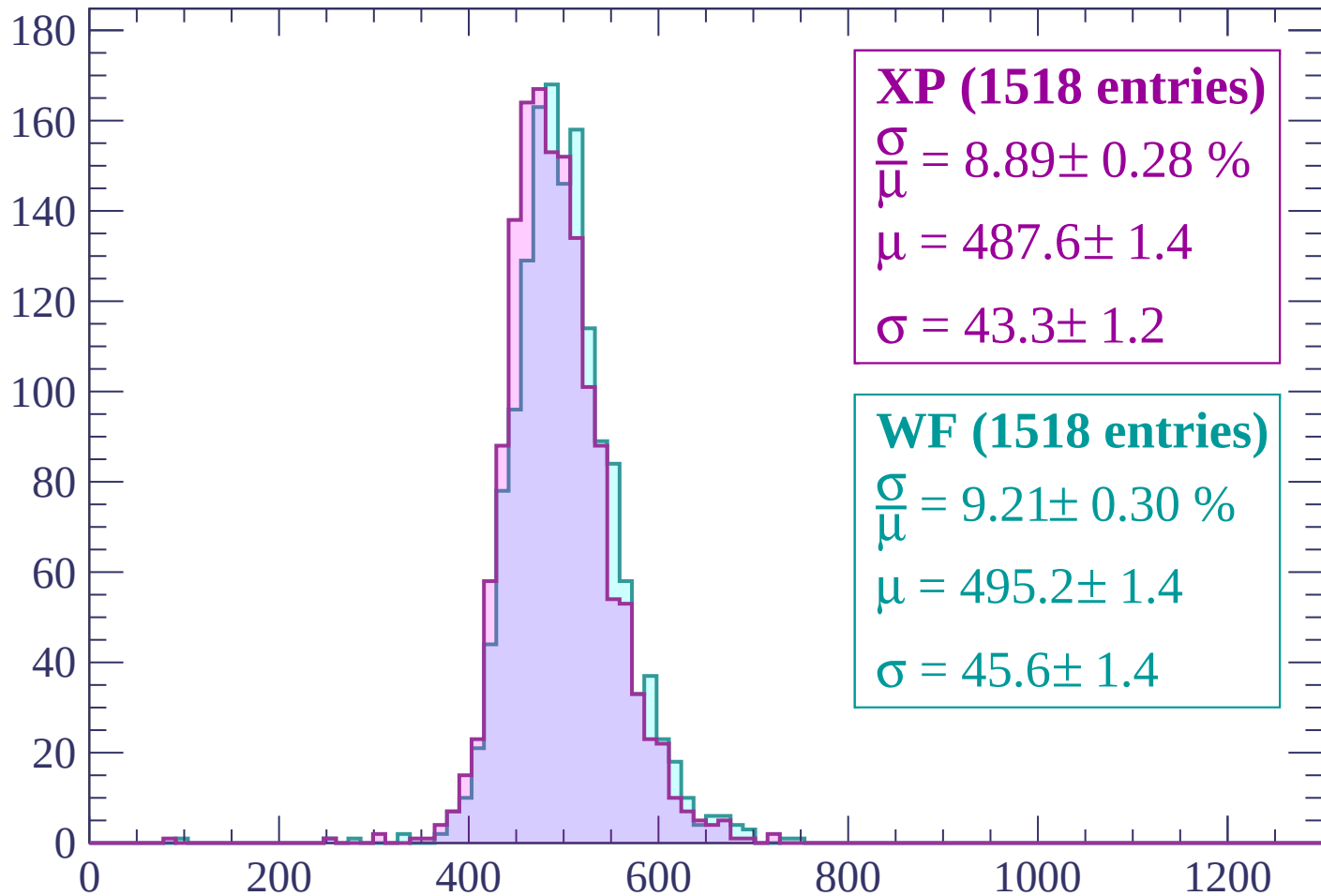
Energy loss | $2400 < p < 2460$

Count



Energy loss | $2460 < p < 2520$

Count



XP (1518 entries)

$$\frac{\sigma}{\mu} = 8.89 \pm 0.28 \%$$

$$\mu = 487.6 \pm 1.4$$

$$\sigma = 43.3 \pm 1.2$$

WF (1518 entries)

$$\frac{\sigma}{\mu} = 9.21 \pm 0.30 \%$$

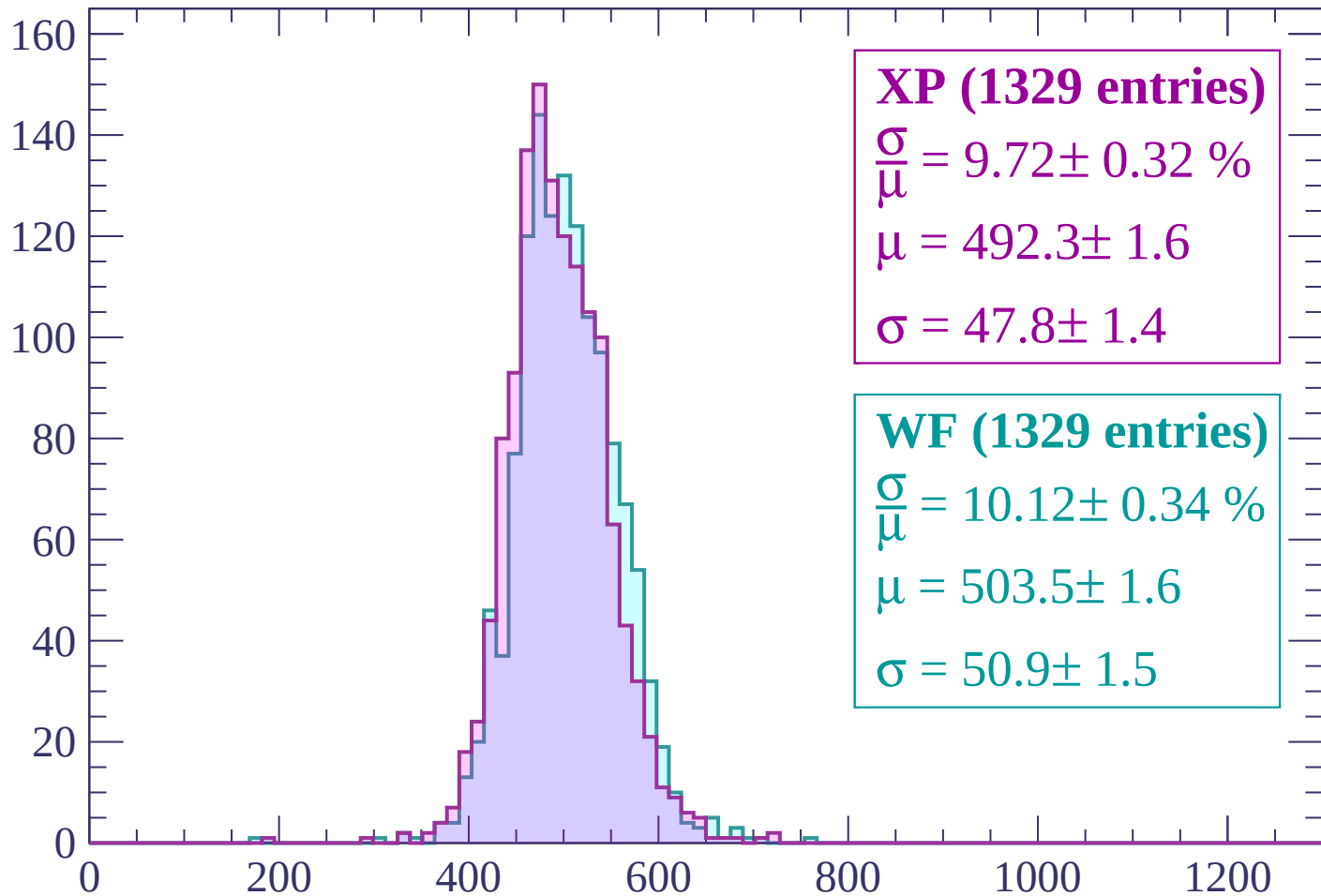
$$\mu = 495.2 \pm 1.4$$

$$\sigma = 45.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2520 < p < 2580$

Count



XP (1329 entries)

$\frac{\sigma}{\mu} = 9.72 \pm 0.32 \%$

$\mu = 492.3 \pm 1.6$

$\sigma = 47.8 \pm 1.4$

WF (1329 entries)

$\frac{\sigma}{\mu} = 10.12 \pm 0.34 \%$

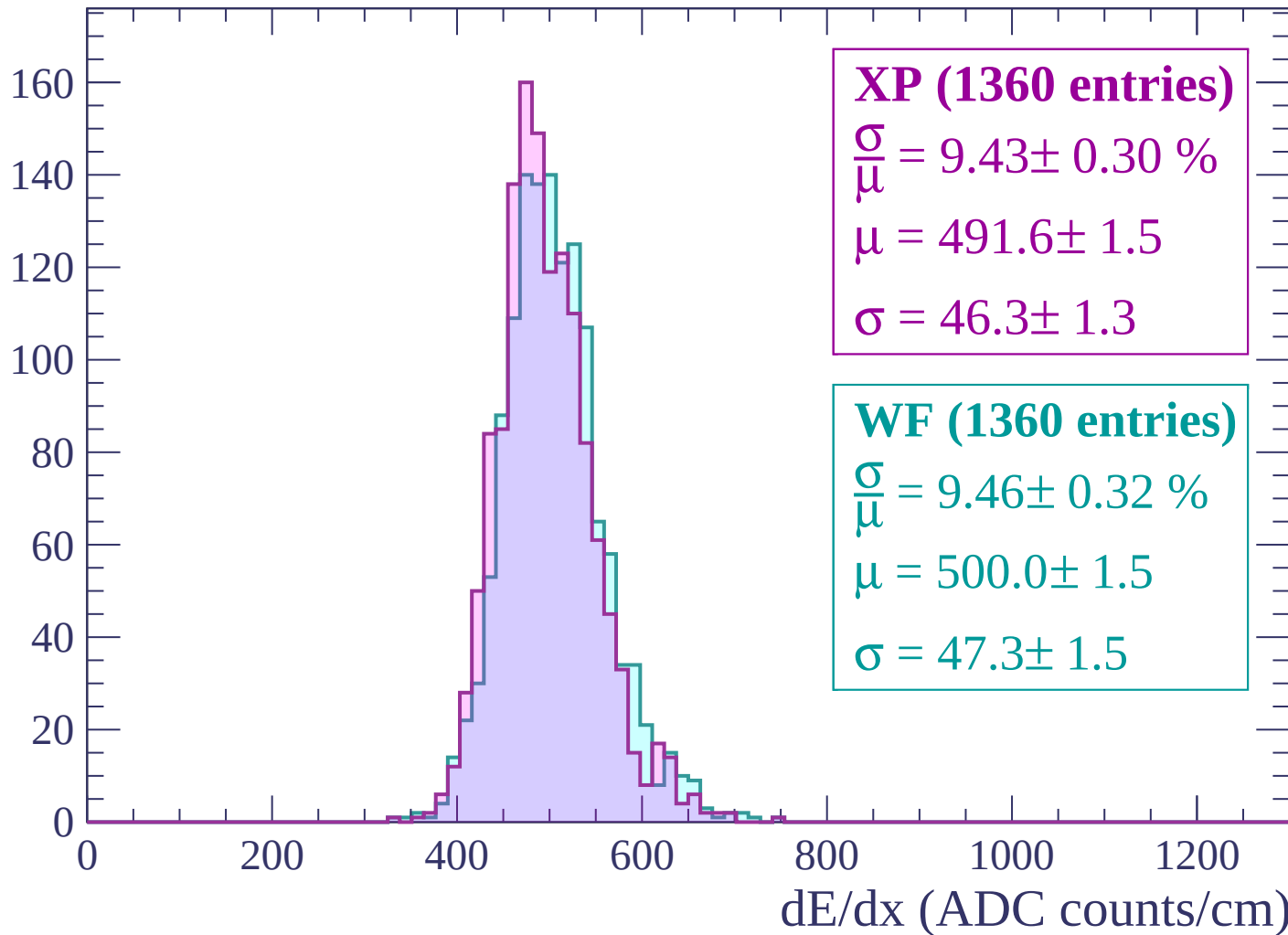
$\mu = 503.5 \pm 1.6$

$\sigma = 50.9 \pm 1.5$

dE/dx (ADC counts/cm)

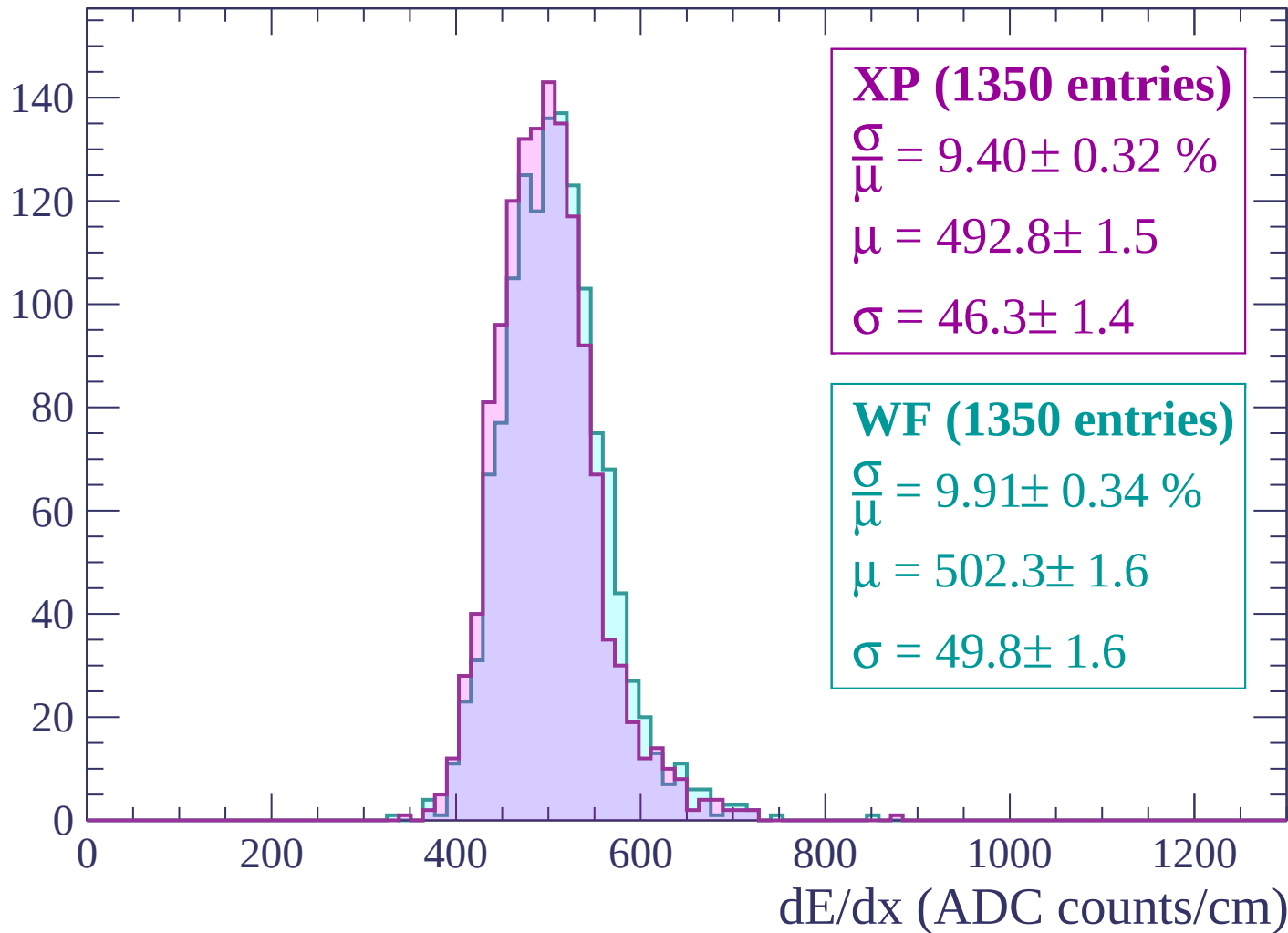
Energy loss | $2580 < p < 2640$

Count



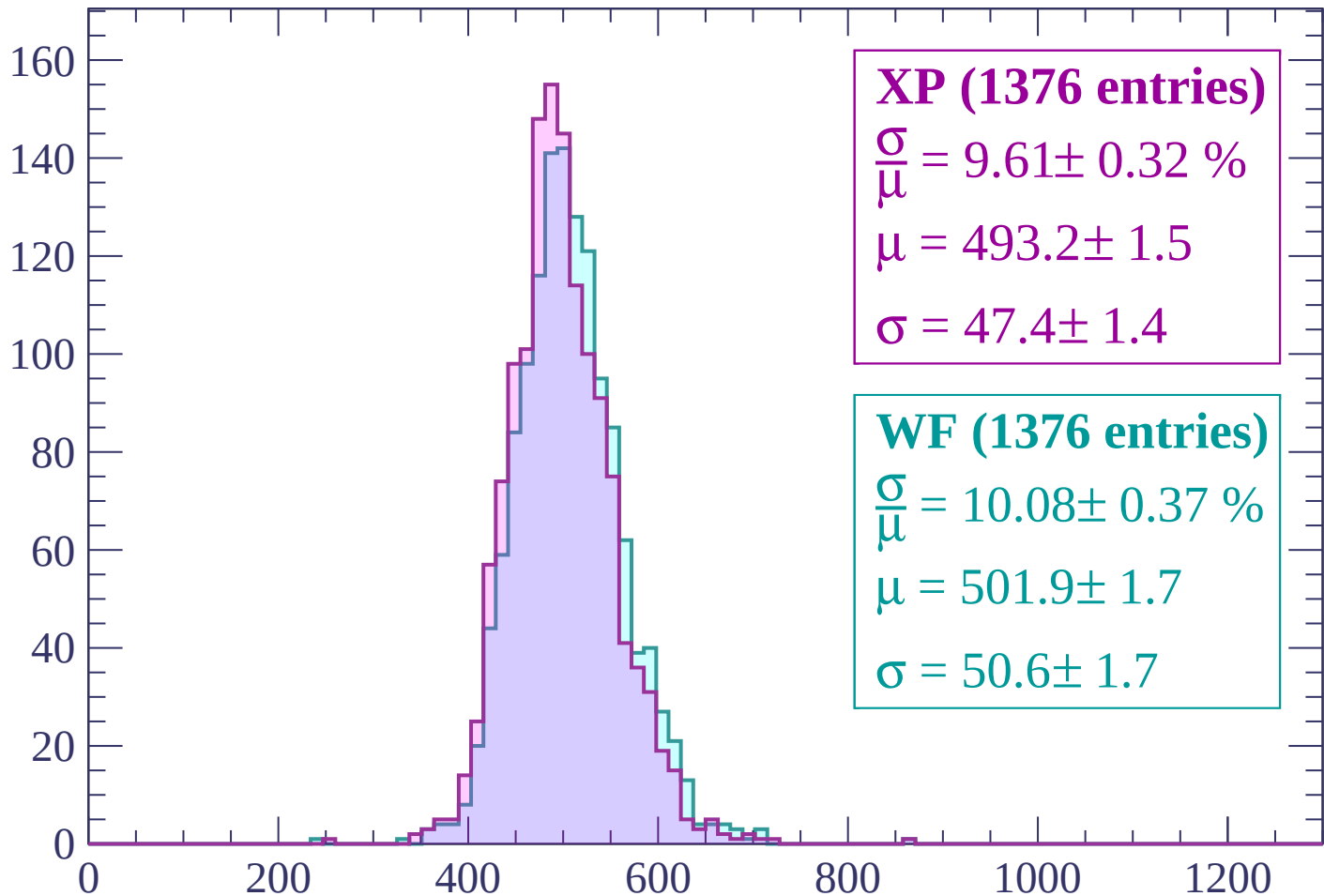
Energy loss | $2640 < p < 2700$

Count



Energy loss | $2700 < p < 2760$

Count



XP (1376 entries)

$$\frac{\sigma}{\mu} = 9.61 \pm 0.32 \%$$

$$\mu = 493.2 \pm 1.5$$

$$\sigma = 47.4 \pm 1.4$$

WF (1376 entries)

$$\frac{\sigma}{\mu} = 10.08 \pm 0.37 \%$$

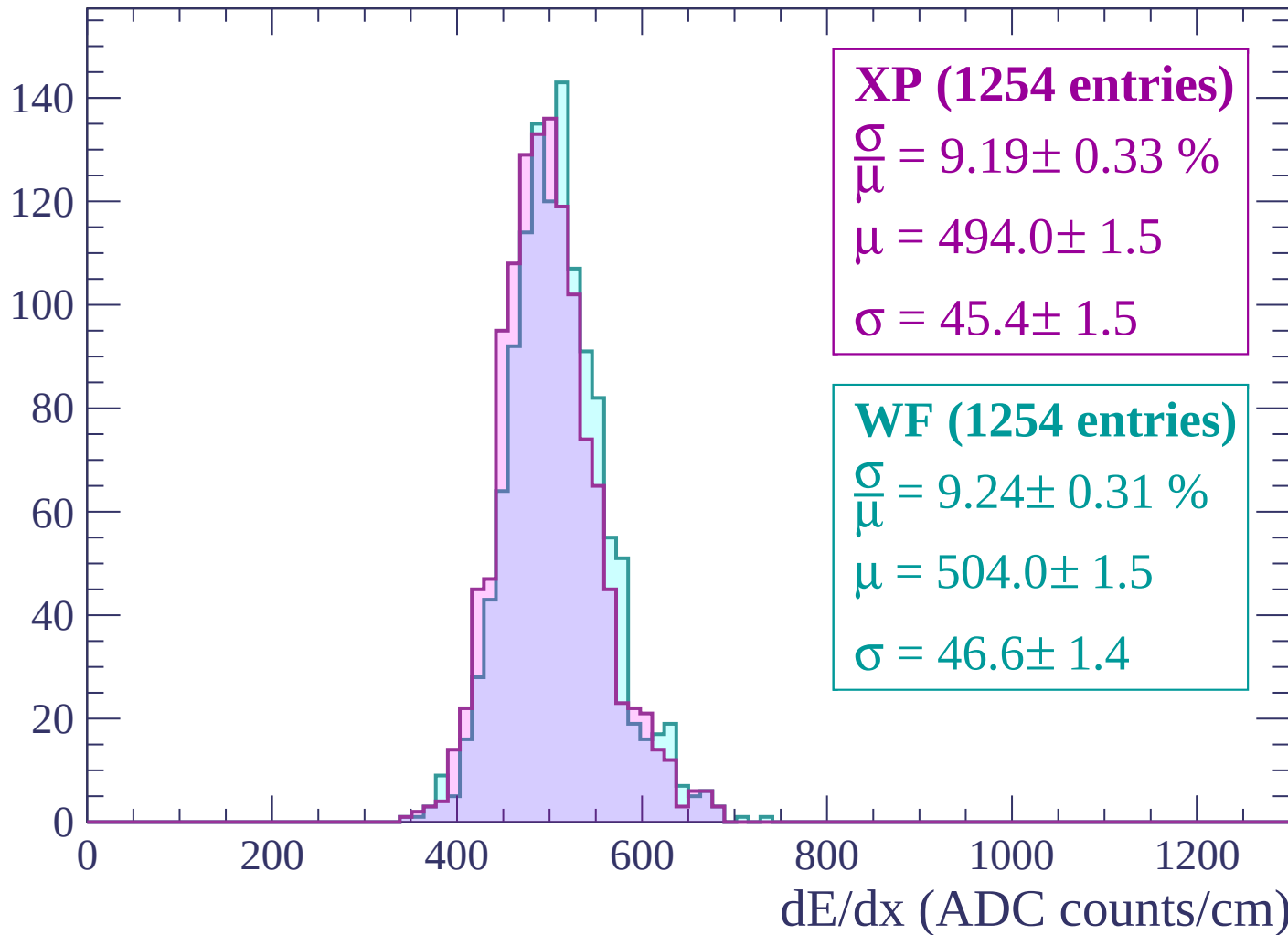
$$\mu = 501.9 \pm 1.7$$

$$\sigma = 50.6 \pm 1.7$$

dE/dx (ADC counts/cm)

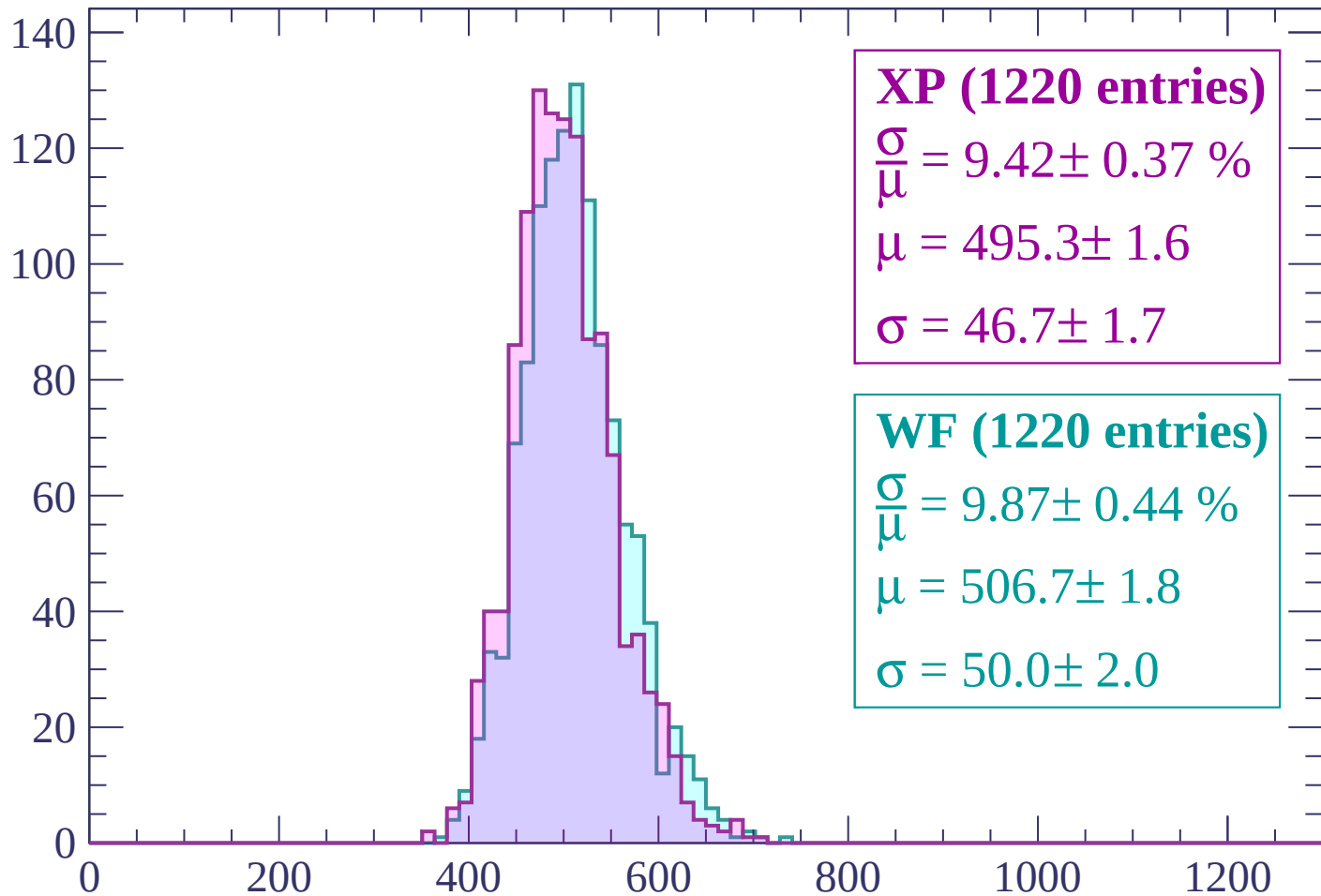
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (1220 entries)

$$\frac{\sigma}{\mu} = 9.42 \pm 0.37 \%$$

$$\mu = 495.3 \pm 1.6$$

$$\sigma = 46.7 \pm 1.7$$

WF (1220 entries)

$$\frac{\sigma}{\mu} = 9.87 \pm 0.44 \%$$

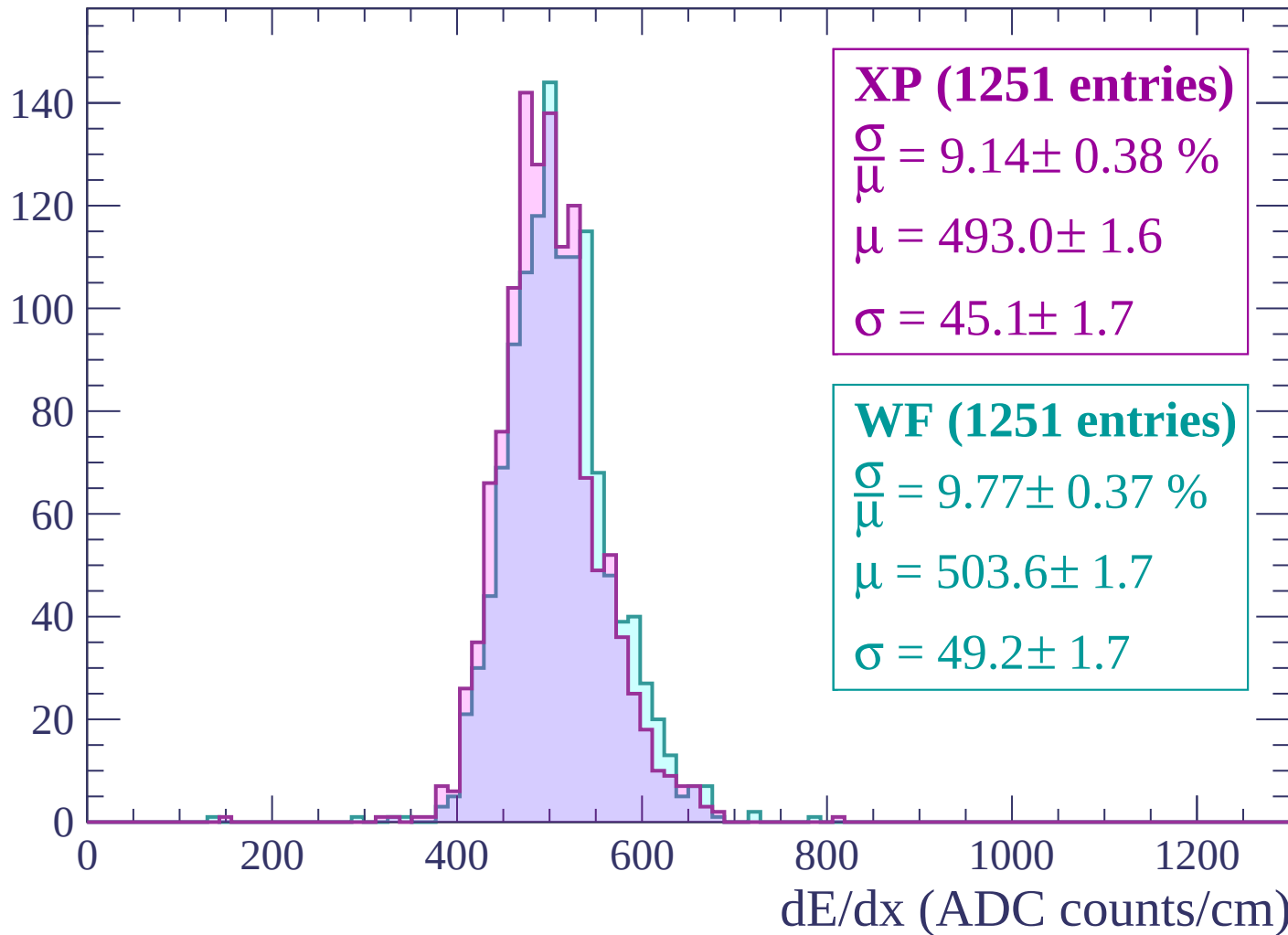
$$\mu = 506.7 \pm 1.8$$

$$\sigma = 50.0 \pm 2.0$$

dE/dx (ADC counts/cm)

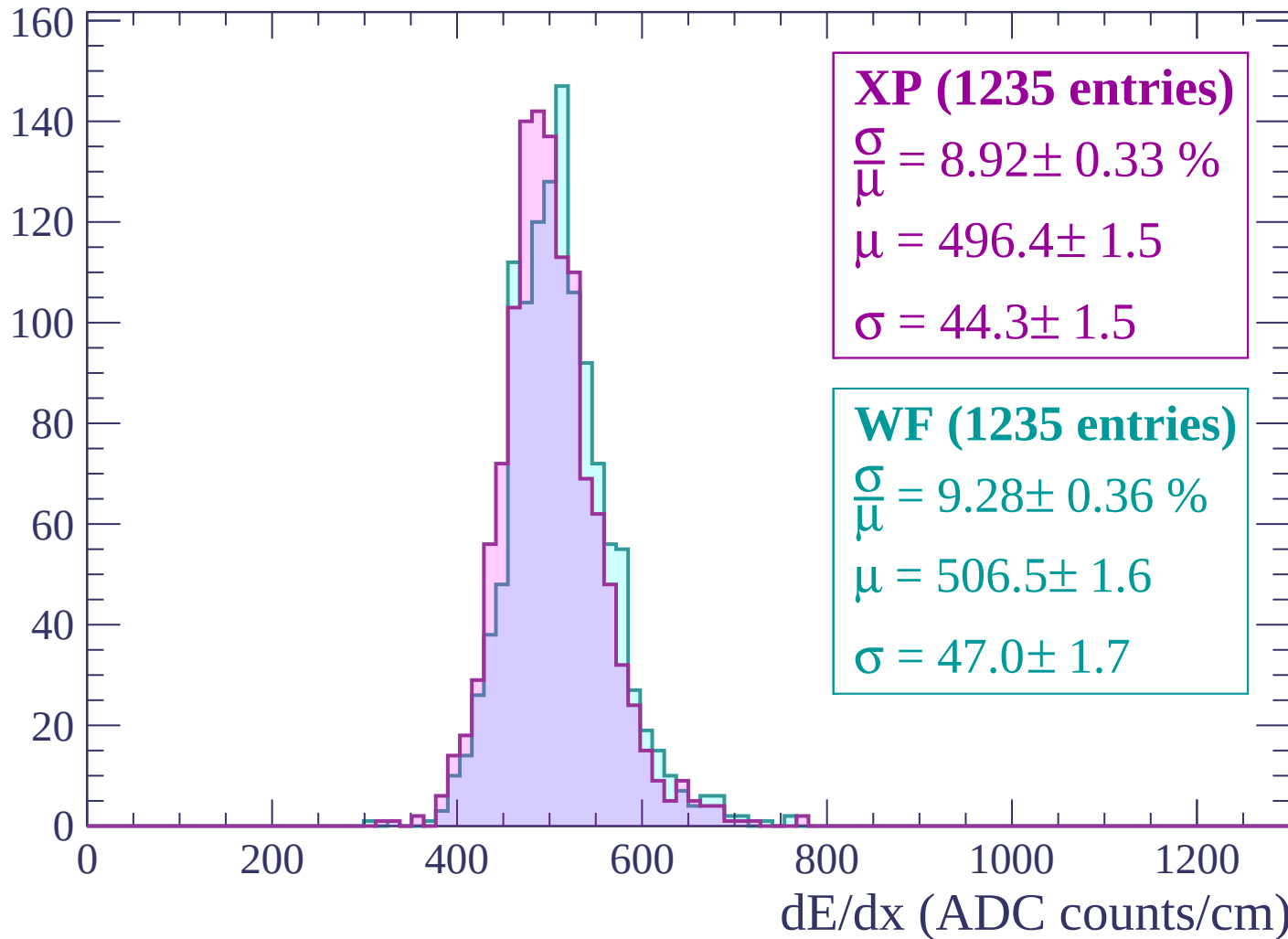
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count

